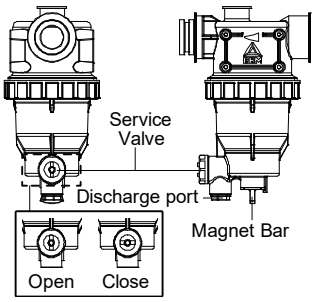
12.2.9 Maintenance

- In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

12.2.9.1 Maintenance for Magnetic Water Filter Set ⑨

1 Turn OFF power supply.
2 Place a container below Magnetic Water Filter Set ⑨.
3 Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑨.
4 By using Allen key (8mm), remove the Cap of Discharge Port.
5 By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
6 Reinstall the Cap of Discharge Port and Magnet Bar.
7 Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 12.2.5 for details.)
8 Turn ON power supply.



12.2.9.2 Maintenance for Safety Relief Valve ㉕

- It is strongly recommended to operate the valve by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

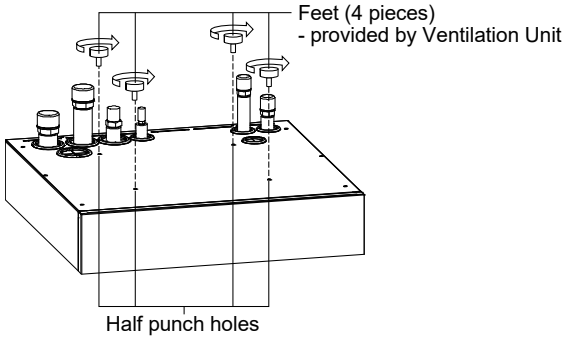
12.2.9.3 Proper Pump Down Procedure

⚠ WARNING
Strictly follow the steps below for proper pump down procedure. Explosion may occur if the steps are not followed as per sequence. 1. When the Tank Unit is not in operation (standby), enter the Service setup menu in the Remote Controller and select Pump down operation to turn it ON. (See APPENDIX for detail) 2. After 10~15 minutes, (after 1 or 2 minutes in case very low ambient temperatures (< 10°C)), fully close 2 way valve on Outdoor Unit. 3. After 3 minutes, fully close 3 way valve on Outdoor Unit. 4. Press the “OFF/ON” switch on the Remote Controller ① to stop pump down operation. 5. Remove the refrigerant piping.

12.2.9.4 Installation of Ventilation Unit on Top of Tank Unit (Optional)

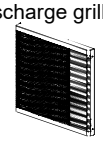
- For installation works of Ventilation Unit on top of Tank Unit, refer to the Ventilation Unit Installation Manual.

⚠ CAUTION
Before install Ventilation Unit, fix the Feet that provided by Ventilation Unit to the half punch holes on Top Panel of Tank Unit, Otherwise, heavy Ventilation Unit may fall and cause injury.



12.3 Outdoor Unit

Attached accessories

No.	Accessories part	Qty.
1	Drain elbow 	1
2	Rubber cap 	7 (For WH-UDZ03KE5*) 3 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
3	Discharge grille 	1 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
4	Metal plate 	2 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
5	Screw 	8 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)

Optional accessories

No.	Accessories part	Qty.
6	Base Pan Heater CZ-NE2P	1

- It is strongly recommended to install a Base Pan Heater (optional) if the outdoor unit is install in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.
- Applicable Piping Kit (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*) CZ-52F5,7,10BP
- Applicable Piping Kit (For WH-UDZ03KE5*) CZ-4F5,7,10BP

12.3.1 Select the Best Location

- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- For WH-UDZ03KE5*, avoid installation in areas where the ambient temperature may drop below -20°C.
- For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*, avoid installations in areas where the ambient temperature may drop below -25°C.
- Keep the spaces indicated by arrows from wall, ceiling, fence or other obstacles.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- If outdoor unit installed near sea, region with high content of sulphur or oily location (e.g. machinary oil, etc), it lifespan maybe shorten.
- If piping length is over 10 m, additional refrigerant should be added as shown in the table.

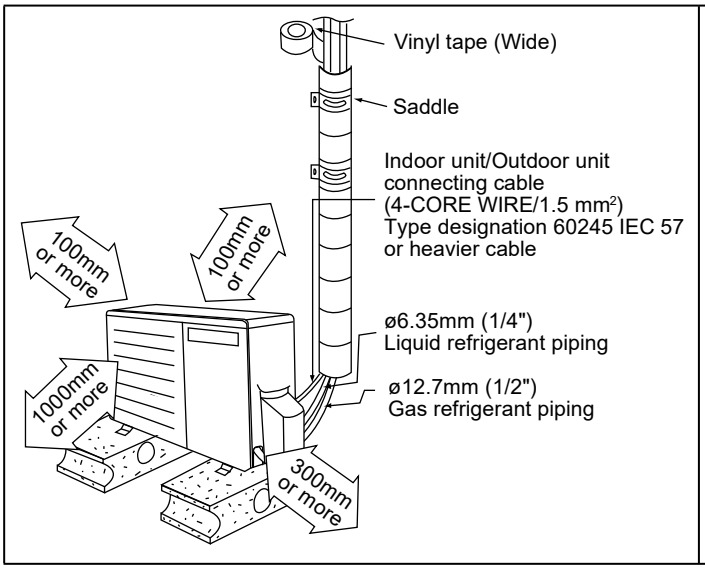
Model	Piping size		Pre-charged Refrigerant (kg)	Rated Length (m)		Max. Elevation (m)	Min. Piping Length (m)	Max. Piping Length (m)	Additional Refrigerant (g/m)
	Gas	Liquid		For Heat Pump Indoor Unit	For Hydromodule + Tank				
WH-UDZ03KE5*	ø12.7mm (1/2")	ø6.35mm (1/4")	0.90	7	7	20	3	25	20
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8")	ø6.35mm (1/4")	1.30	7	7	30	3	50	25

Example: WH-UDZ03KE5*
If piping length is 15m, the quantity of additional refrigerant should be 100g. [(15-10)m x 20 g/m = 100g]

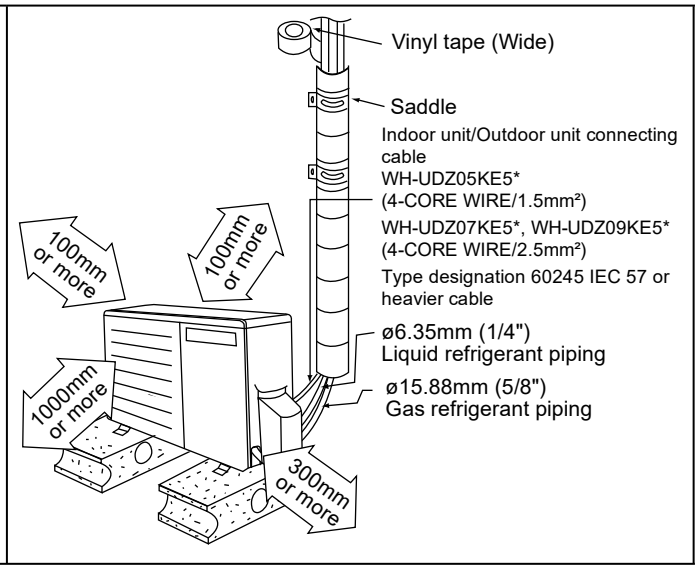
12.3.2 Install the Outdoor Unit

12.3.2.1 Installation Diagram

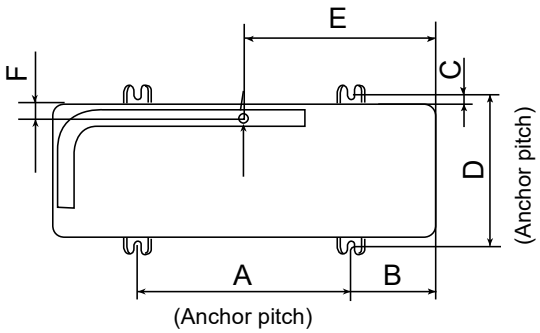
- It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.
- This illustration is for explanation purposes only.



For WH-UDZ03KE5*



For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*



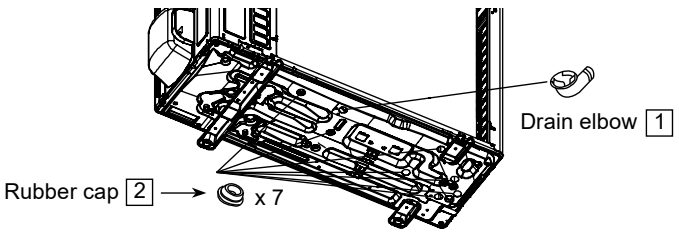
Model	A	B	C	D	E	F
WH-UDZ03KE5*	540	160	20	330	430	46
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	613	130	24	360.5	543	32

(Unit : mm)

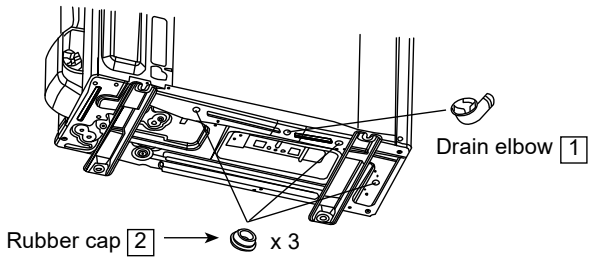
- After selecting the best location, start installation according to Installation Diagram.
  1 Fix the unit on concrete or rigid frame firmly and horizontally by bolt nut (ø10 mm).
  2 When installing at roof, please consider strong wind and earthquake. Please fasten the installation stand firmly with bolt or nails.

12.3.2.2 Disposal of Outdoor Unit Drain Water

- When a Drain elbow [1] is used, please ensure to follow below:
  - the unit should be placed on a stand which is taller than 50 mm.
  - cover the ø20mm holes with Rubber cap [2] (refer to illustration below).
  - use a tray (field supply) when necessary to dispose the outdoor unit drain water.
- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow [1] and Rubber cap [2], for the drain water freezes and the fan will not rotate.



WH-UDZ03KE5*



WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*

12.3.3 Install the Discharge Grille

- This section is for WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5* only.
- Make sure to install the outdoor unit on concrete or rigid frame by bold nut before install the discharge grille 3.
- Hook the metal plate 4 on the cabinet front plate at 2 locations and tighten with screw 5. (Figure a)
- Hook the discharge grille 3 to the cabinet front plate at 4 locations. (Figure b)
- Tighten the discharge grille to the cabinet front plate at 6 locations with screw 5. (Figure c)

⚠ CAUTION
Make sure to install the discharge grille to the outdoor unit before powering ON to protect againts a rotating fan

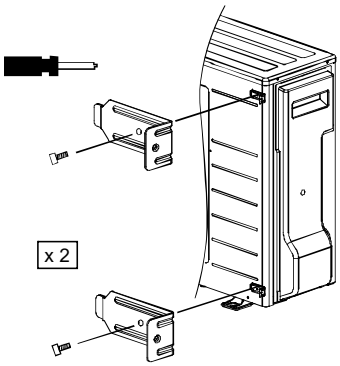


Figure a

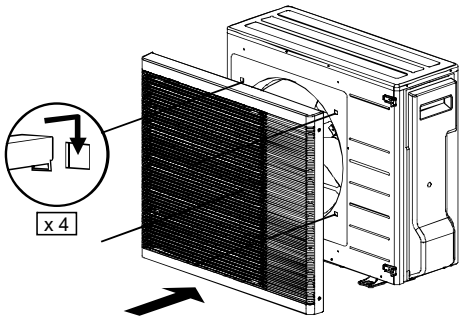


Figure b

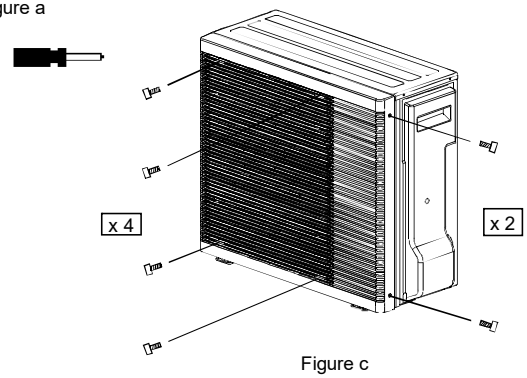
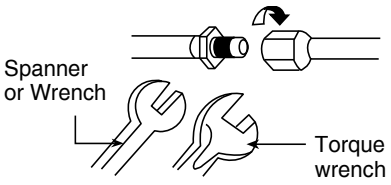


Figure c

12.3.4 Connecting the Piping

12.3.4.1 Connecting the Piping to Outdoor Unit

Decide piping length and then cut by using pipe cutter. Remove burrs from cut edge. Make flare after inserting the flare nut (locate at valve) onto the copper pipe. Align center of piping to valves and then tighten with torque wrench to the specified torque as stated in the table.

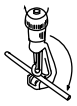


Be sure to use two spanners to tighten.
(If the nuts are overtightened, it may cause the flares to break or leak.)

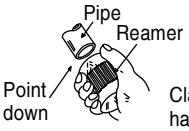
Model	Piping size (Torque)	
	Gas	Liquid
WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]

12.3.4.1.1 Cutting and Flaring the Piping

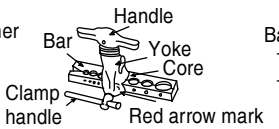
1 Please cut using pipe cutter and then remove the burrs.
2 Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3 Please make flare after inserting the flare nut onto the copper pipes.



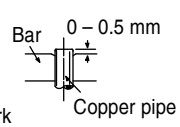
1. To cut

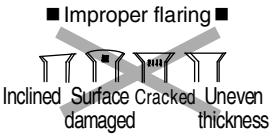


2. To remove burrs



3. To flare



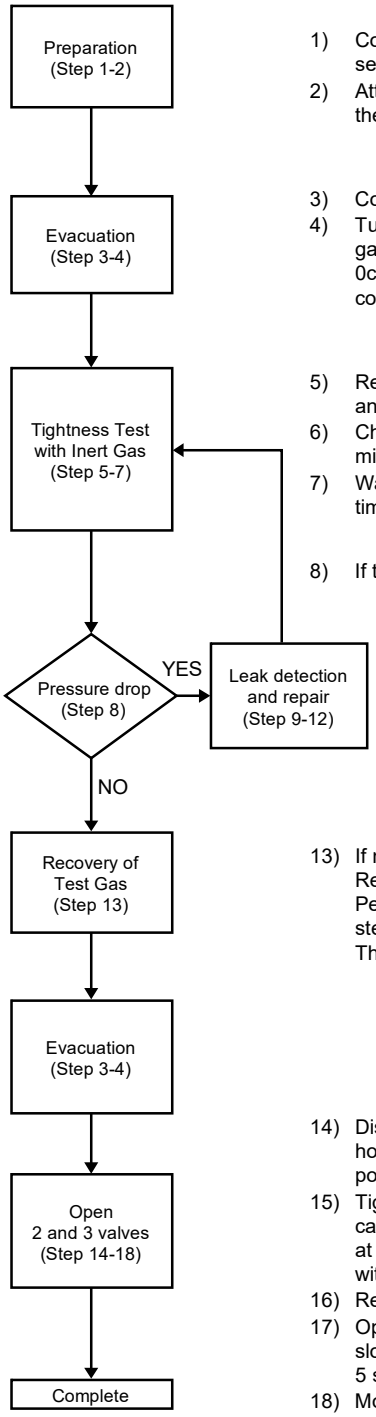


When properly flared, the internal surface of the flare will evenly shine and be of even thickness. Since the flare part comes into contact with the connections, carefully check the flare finish.

12.3.5 Air Tightness Test on the Refrigerating System

⊘	Do not purge the air with refrigerants but use a vacuum pump to vacuum the installation.
❗	There is no extra refrigerant in the outdoor unit for air purging.

- Before system is charged with refrigerant and before the refrigerating system is put into operation, below site test procedure and acceptance criteria shall be vertified by the certified technicians, and/or the installer.
- Be sure to check whole system for gas leakage.



1) Connect a charging hose with a push pin to the Low side of a charging set and the service port of the 3-way valve.
2) Attach the gauge manifold set correctly and tightly. Make sure that both valves of the manifold gauge (low pressure and high pressure) is in close position.

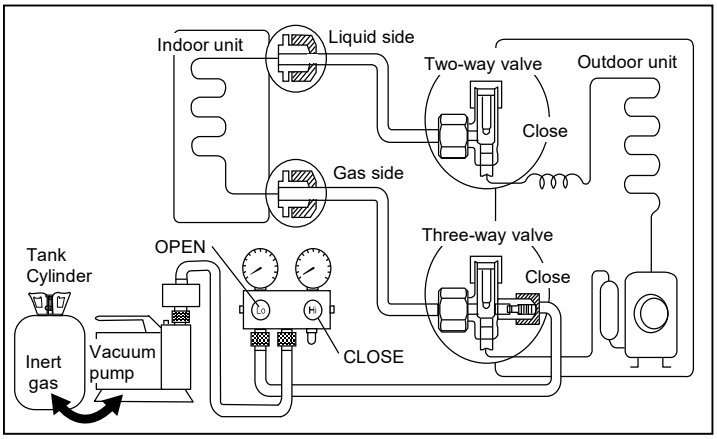
3) Connect the center hose of the manifold gauge to a vacuum pump.
4) Turn on the power switch of the vacuum pump, then turn open the low side manifold gauge valve and make sure that the needle in the gauge moves from 0cmHg (0 MPa) to –76 cmHg (–0.1 MPa) or vacuum until 500 microns is achieved. This process continues for approximately ten minutes. Then close the low side manifold gauge valve.

5) Remove the vacuum pump from the centre hose and connect the center hose to cylinder of any applicable inert gas as test gas.
6) Charge test gas into the system and wait until the pressure within the system to reach min. 1.04MPa (10.4barg).
7) Wait and monitor the pressure reading on the gauges. Check if there is any pressure drop. Waiting time depends on the size of the system.

8) If there is any pressure drop, perform step 9-12. If there is no pressure drop, perform step 13.

9) Use Gas Leak Detector to check for leaks. Must use the detection equipment with a sensitivity of 5 grams per year of test gas or better.
10) Move the probe along the Air-to-Water Heatpump system to check for leaks, and mark for repair.
11) Any leak detected and marked shall be repaired.
12) After repair, repeat evacuation steps 3-4 and tightness test steps 5-7. Check the pressure drop as in step 8.

13) If no leak,
Recover the test gas.
Perform evacuation of steps 3-4.
Then proceed to step 14.



14) Disconnect the charging hose from the service port of the 3-way valve.
15) Tighten the service port caps of the 3-way valve at a torque of 18 N•m with a torque wrench.
16) Remove the valve caps of both of the 2-way valve and 3-way valve.
17) Open both of the valves, using a hexagonal wrench (4mm). It is recommended to allow refrigerant slowly flow into the refrigerant system to prevent refrigerant freezing. Slightly open 2-way valve for 5 seconds then close the valve. Repeat this action for 3 cycles then fully open the valve.
18) Mount back the valve caps onto the 2-way valve and the 3-way valve to complete this process.

Notes:

Recommended use of any of the following leak detector,
I) Universal Sniffer leak detector
II) Electronic halogen leak detector
III) Ultrasonic Leak Detector

12.3.6 Connect the Cable to the Outdoor Unit

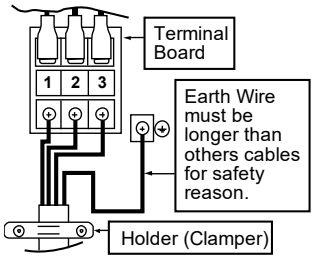
(FOR DETAIL REFER TO WIRING DIAGRAM AT UNIT)

1 Remove the control board cover from the unit by loosening the screw.

2 Connecting cable between indoor unit and outdoor unit shall be approved polychloroprene sheathed flexible cable (see below table), type designation 60245 IEC 57 or heavier cable.

Model	Flexible cable specification
WH-UDZ03KE5*, WH-UDZ05KE5*	4 x (1.5 mm^2)
WH-UDZ07KE5*, WH-UDZ09KE5*	4 x (2.5 mm^2)

Terminals on the indoor unit	1	2	3	⏚
Colour of wires				
Terminals on the outdoor unit	1	2	3	⏚



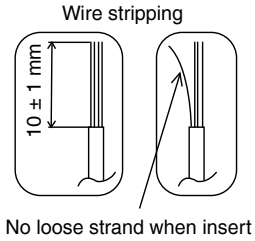
⚠ WARNING

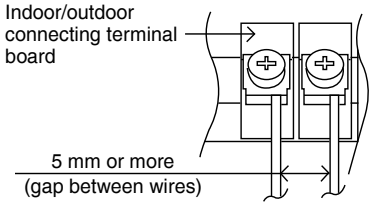
⏚ This equipment must be properly earthed.

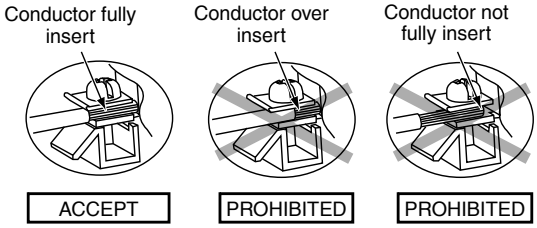
3 Secure the cable onto the control board with the holder (clamper).

4 Attach the control board cover back to the original position with screw.

12.3.6.1 Wire Stripping and Connecting Requirement







12.3.7 Pipe Insulation

1 Please carry out insulation at pipe connection portion as mentioned in Indoor/Outdoor Unit Installation Diagram. Please wrap the insulated piping end to prevent water from going inside the piping.

2 If drain hose or connecting piping is in the room (where dew may form), please increase the insulation by using POLY-E FOAM with thickness 6 mm or above.

⚠ CAUTION

If cleaning of the outdoor unit is necessary during installation or servicing, do not clean the outdoor unit with any hydrocarbon based solvent.

12.4 Appendix

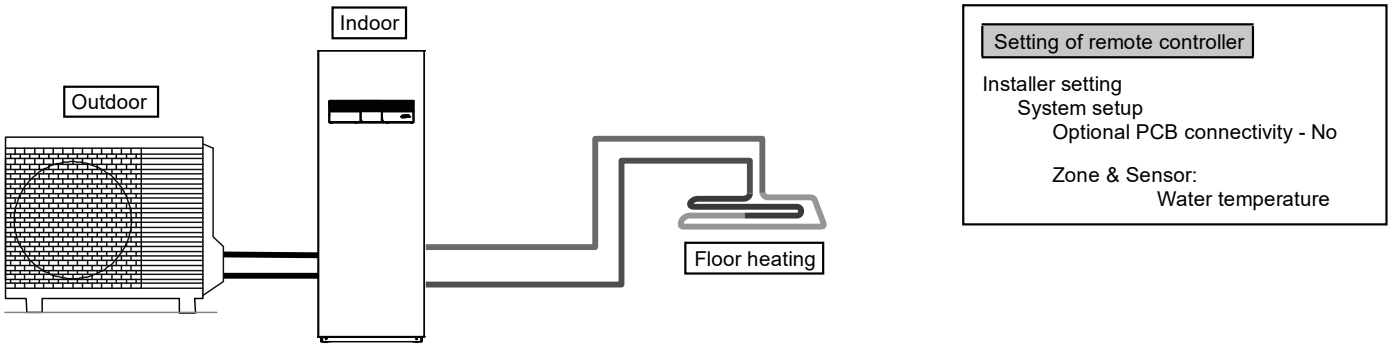
12.4.1 Variation of System

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.
(NOTE) : For this model, both external room thermistor of Zone 1 and external room thermostat of Zone 1 must always be connected to main indoor PCB only regardless of Optional PCB (CZ-NS5P) connection.

12.4.1.1 Introduce Application Related to Temperature Setting

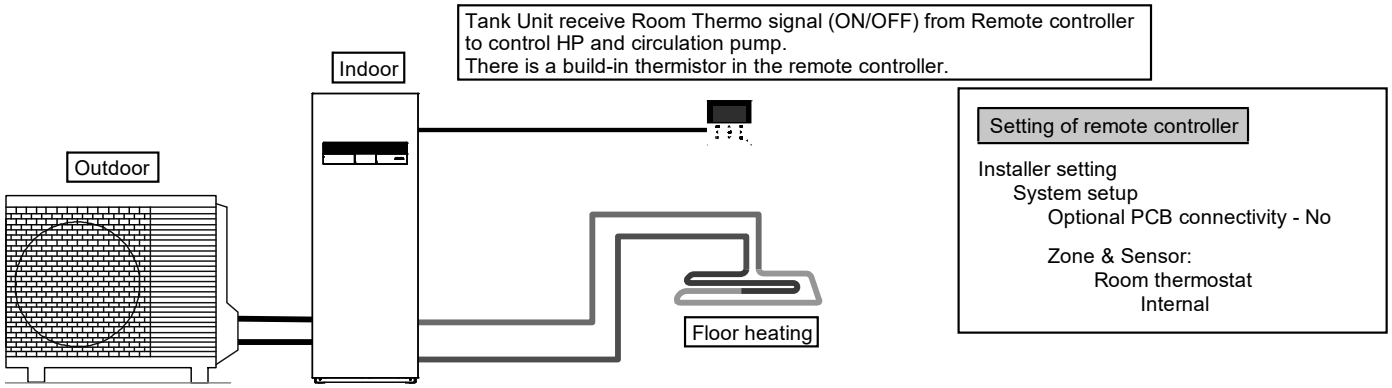
12.4.1.1.1 Temperature Setting Variation for Heating

1. Remote Controller



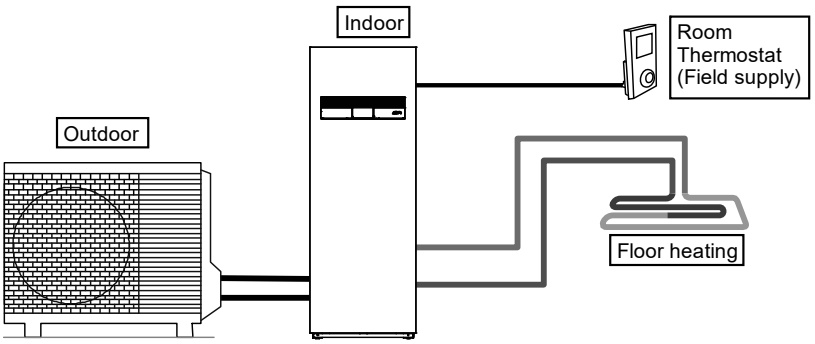
Connect floor heating or radiator directly to the Tank Unit.
Remote controller is installed on Tank Unit.
This is the basic form of the most simple system.

2. Room Thermostat



Connect floor heating or radiator directly to the Tank Unit.
Remove remote controller from Tank Unit and install it in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat



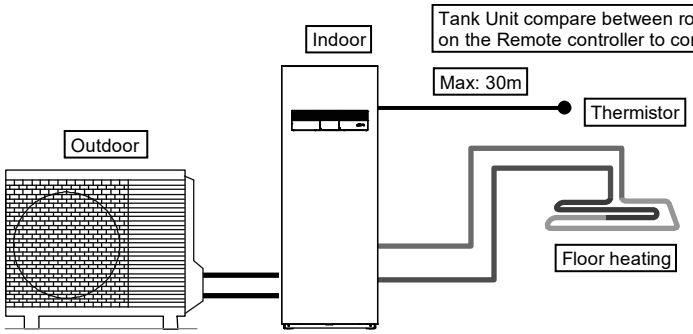
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
(External)

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermistor

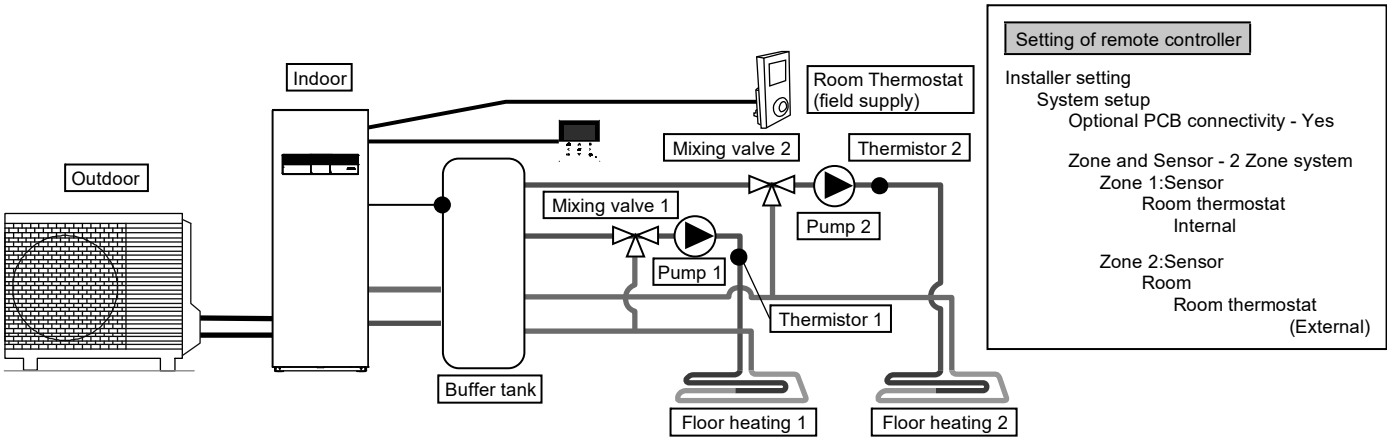
Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, shift compensation curve according to the thermo ON/OFF situation.

- (Example) If room temperature increasing speed is;
  very slow → shift up the compensation curve
  very fast → shift down the compensation curve

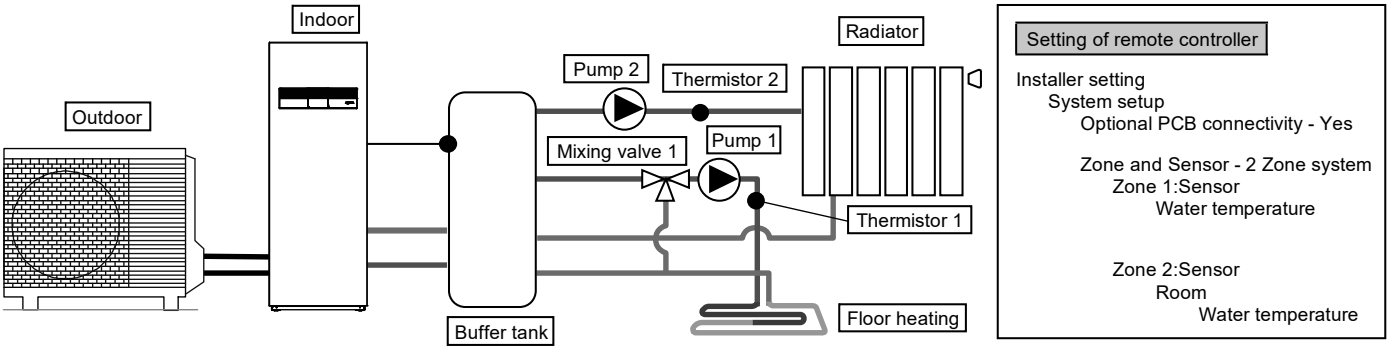
12.4.1.1.2 Examples of Installations

Floor heating 1 + Floor heating 2



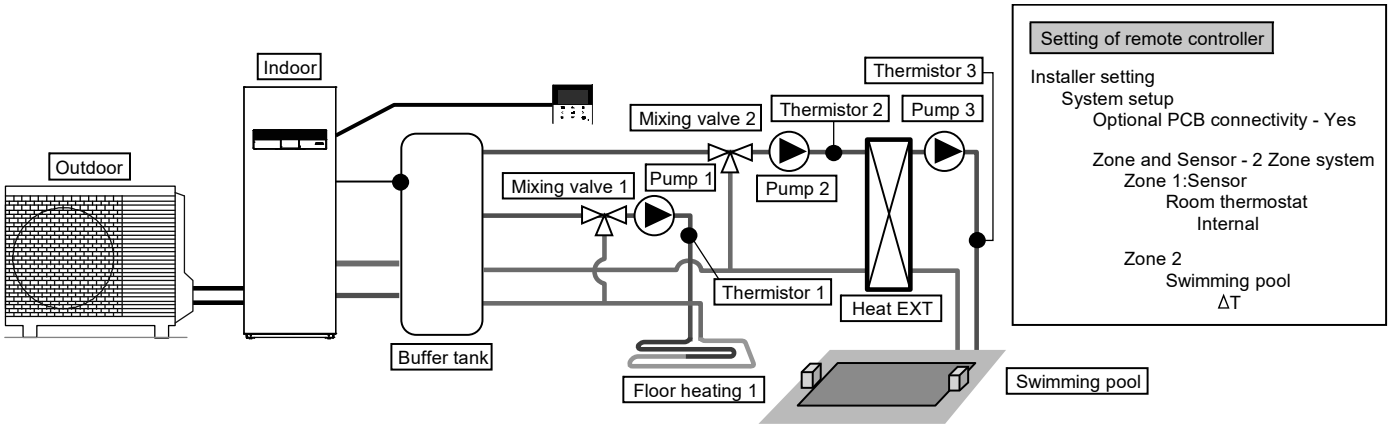
Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Remove remote controller from Tank Unit, install it in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Floor heating + Radiator



Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
Remote controller is installed on Tank Unit.
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the Optional PCB (CZ-NS5P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

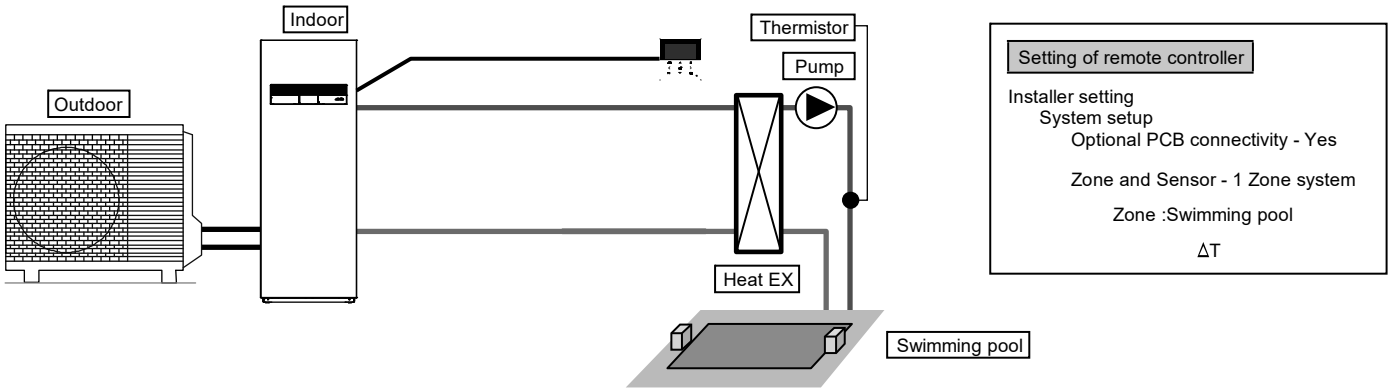
Floor heating + Swimming pool



Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Remove remote controller from Tank Unit and install in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the Optional PCB (CZ-NS5P).

❇ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

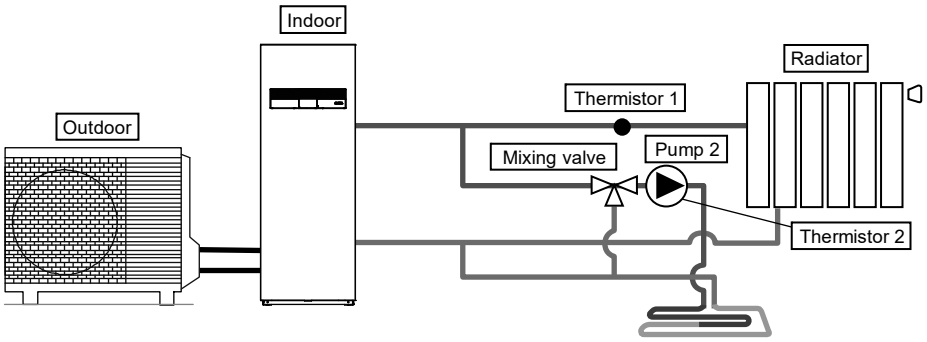
Swimming pool only



This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Tank Unit without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Remove remote controller from Tank Unit and install in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the Optional PCB (CZ-NS5P).

In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor - 2 Zone system
Zone 1:Sensor
Water temperature

Zone 2:Sensor
Room
Water temperature

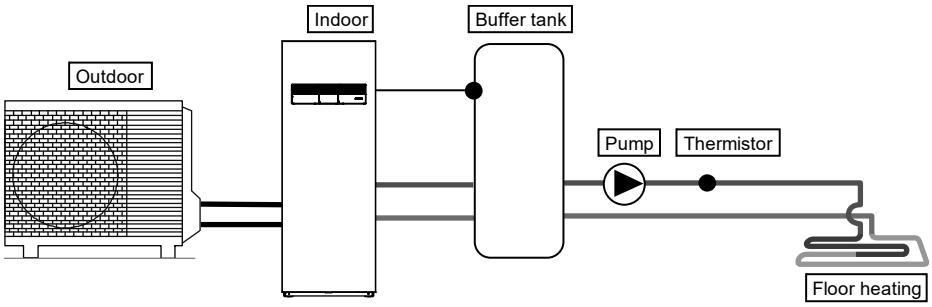
Operation setup
Heat
ΔT for heating ON – 1°C

Cool
ΔT for cooling ON – 1°C

This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Tank Unit served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the Optional PCB (CZ-NS5P).

(NOTE)
- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
- Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

Buffer tank connection



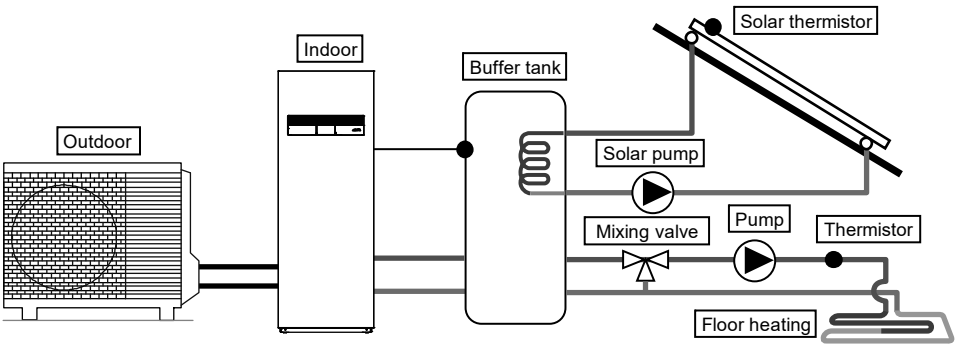
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Buffer Tank connection - Yes
ΔT for buffer tank

This is an application that connects the buffer tank to the Tank Unit.
Buffer tank’s temperature is detected by buffer tank thermistor (specified by Panasonic).
Without connection of Optional PCB, external pump can be used for circulation in the floor heating circuit.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Buffer tank + Solar

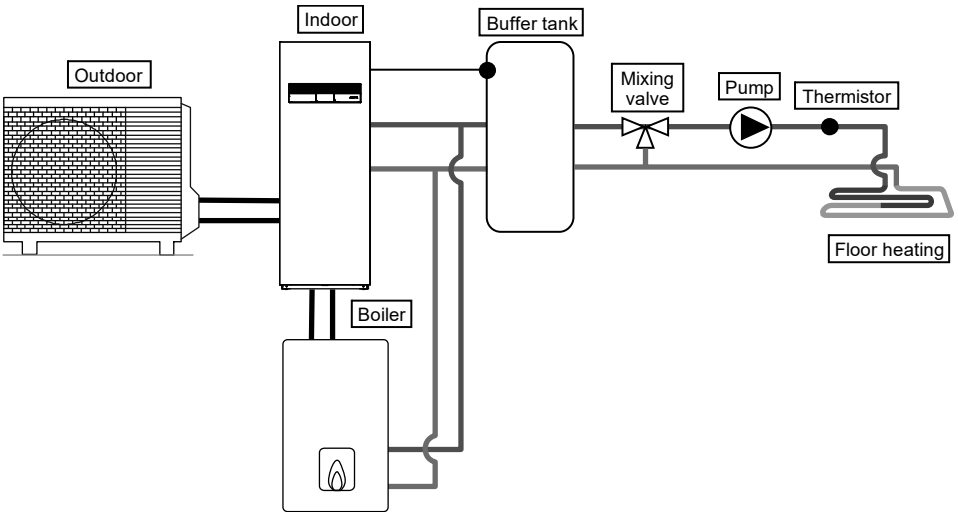


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank
Solar connection - Yes
Buffer tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the buffer tank to the Tank Unit before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Boiler connection



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Bivalent - Yes
Turn ON: outdoor temp.
Control pattern

This is an application that connects the boiler to the Tank Unit, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
Besides that, an application that connects to the DHW tank's circuit to heat up tank 's hot water is also possible.
Boiler output can be control by either SG ready input from optional PCB or Auto control by 3 modes selection pattern. (Operation setting of boiler shall be responsible by installer.)
This system requires Optional PCB (CZ-NS5P) for SG ready input control.
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when select Advanced Parallel setting.)
Note : Buffer tank thermistor must be connected to main indoor PCB only.

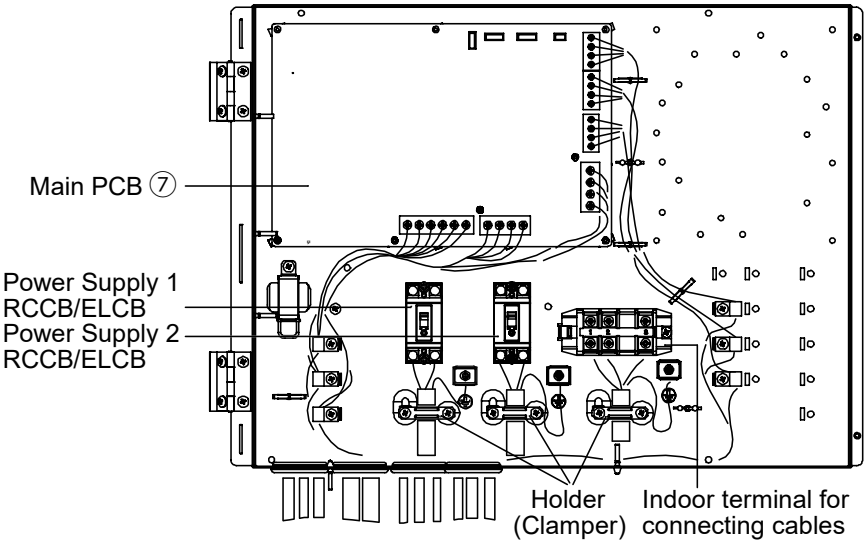
⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the Tank Unit does NOT exceed 55°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

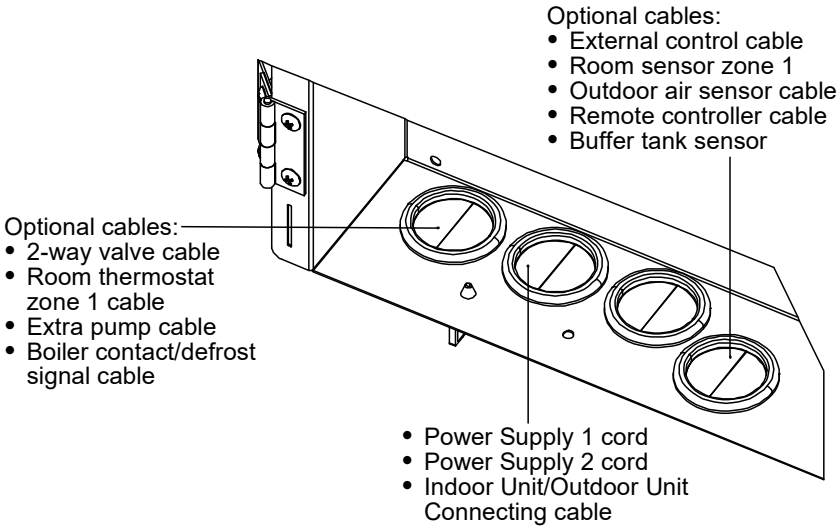
12.4.2 How to Fix Cable

12.4.2.1 Connecting with External Device (Optional)

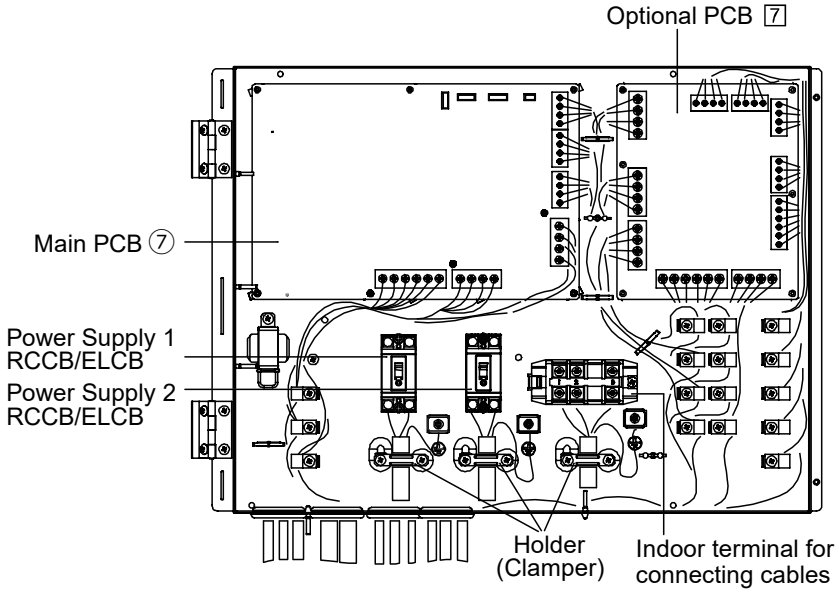
- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB ⑤
  1 Two-way valve shall be spring and electronic type, refer to “Field Supply Accessories” table for details. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
  *note: - Two-way Valve shall be CE marking compliance component.
  - Maximum load for the valve is 9.8VA.
  2 Room thermostat zone 1 cable must be (4 or 3 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
  3 Extra pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  4 Boiler contact/defrost signal cable shall be (2 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
  5 External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  *note: - Switch used shall be CE compliance component.
  - Maximum operating current shall be less than $3A_{rms}$.
  6 Room sensor zone 1, outdoor air sensor and buffer tank sensor cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.



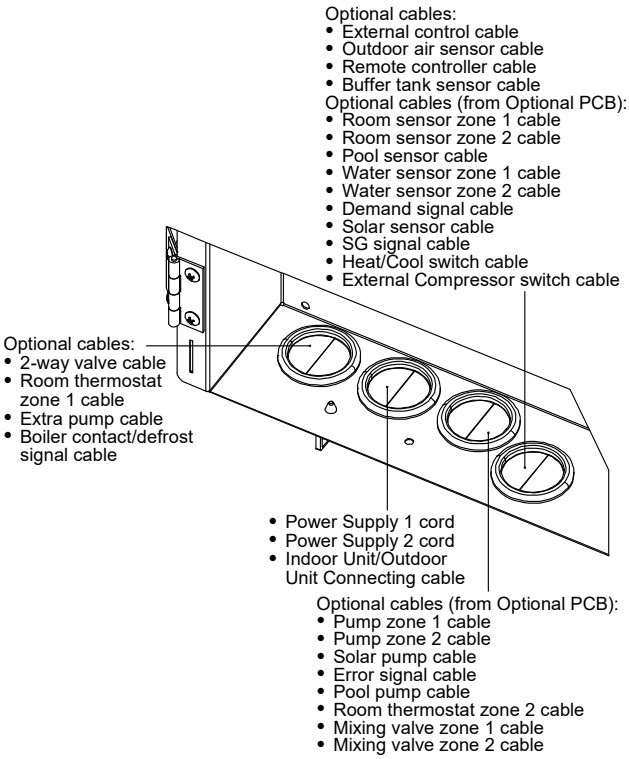
How to guide the optional cables and power supply cord
(view without internal wiring)



- For connection to Optional PCB [7]
  1 By connecting Optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in Optional PCB. Temperature of each zone can be controlled independently by remote controller.
  2 Pump zone 1 and zone 2 cable shall be (2 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  3 Solar pump cable shall be (2 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  4 Pool pump cable shall be (2 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  5 Room thermostat zone 2 cable shall be (4 x min 0.5 mm²), of type designation 60245 IEC 57 or heavier.
  6 Mixing valve zone 1 and zone 2 cable shall be (3 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  7 Room sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm²), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  8 Pool water sensor and solar sensor cable shall be (2 x min 0.3 mm²), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  9 Water sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  10 Demand signal cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  11 SG signal cable shall be (3 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  12 Heat/Cool switch cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  13 External compressor switch cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide the optional cables and power supply cord
(view without internal wiring)

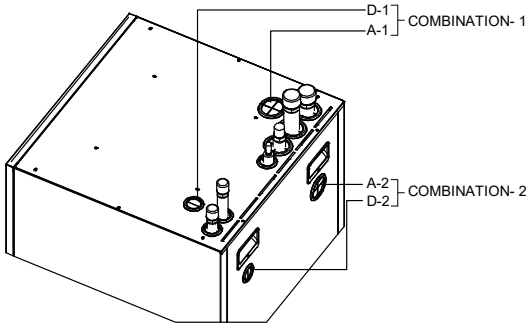


Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

12.4.2.2 Guide Optional Cables and Power Supply Cords to Bushings

⚠ CAUTION
Wire guiding shall be free from hot surfaces. Else, cable insulator damage and electrical shock may happen.
Wire ways shall be smooth and free from sharp edges. Else, cable insulator damage and electrical shock may happen.

■ Use either “COMBINATION-1” or “COMBINATION-2” for guiding Optional Cables and Power Supply Cords to Bushings.



■ A-1 and A-2 Bushings are for:

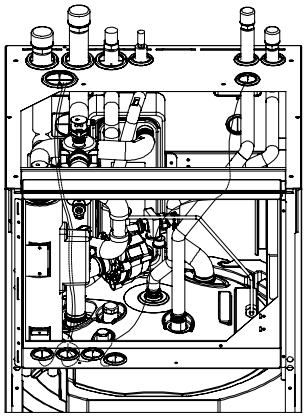
- Power Supply 1 cord
- Power Supply 2 cord
- Indoor Unit/Outdoor Unit Connecting cable
- Pump zone 1 cable
- Pump zone 2 cable
- Solar pump cable
- Room thermostat zone 1 cable
- Room thermostat zone 2 cable
- Mixing valve zone 1 cable
- Mixing valve zone 2 cable
- 2-way valve cable
- Extra pump cable
- Boiler contact cable

■ D-1 and D-2 Bushings are for:

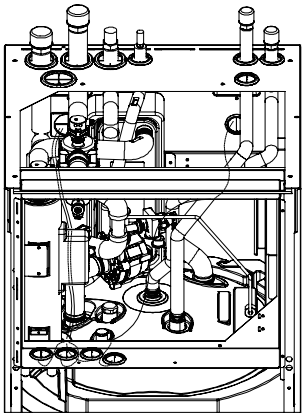
- External control cable
- Outdoor air sensor cable
- Remote controller cable
- Room sensor zone 1 cable
- Room sensor zone 2 cable
- Buffer tank sensor cable
- Pool sensor cable
- Water sensor zone 1 cable
- Water sensor zone 2 cable
- Demand signal cable
- Solar sensor cable
- SG signal cable
- Heat/Cool switch cable
- External Compressor switch cable

■ Ensure all sensor cables are not touching with Front Panel ⑯

■ Guide the wiring inside the unit like below figure.
Once all wiring work done, tie the cable / cord with the banding strap (field supply), to prevent them touching with hot surfaces such as Heater Assembly, bare copper pipes and etc.



Wiring for “COMBINATION-1”



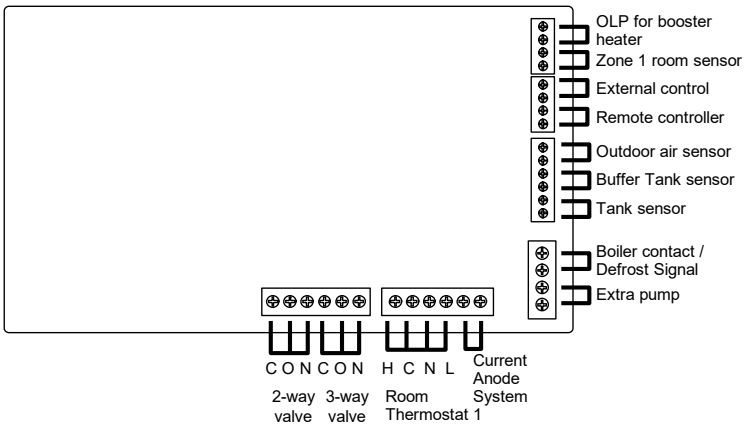
Wiring for “COMBINATION-2”

12.4.2.3 Connecting Cables Length

When connecting cables between Tank Unit and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Mixing valve	50
Room thermostat	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Room sensor	30
Outdoor air sensor	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

12.4.2.4 Connection of the Main PCB



- Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

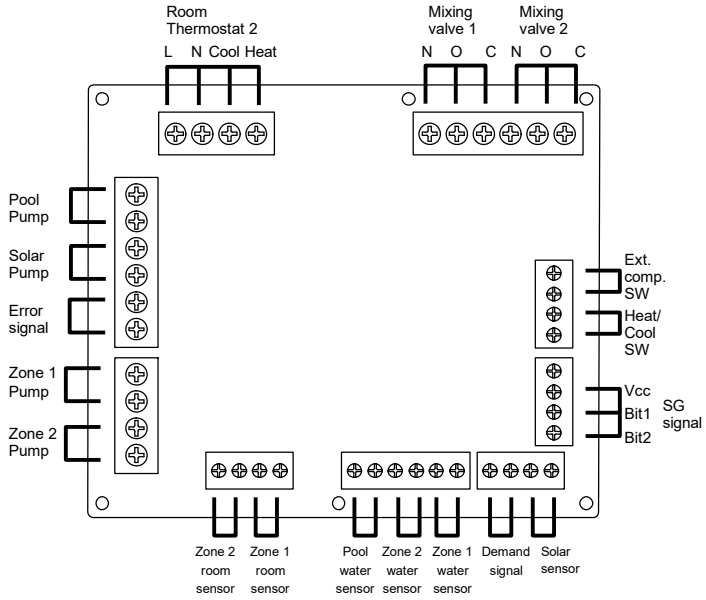
- Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)	AC230V, 12 VA
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)	AC230V, 12 VA
Extra pump	AC230V (Used when Tank Unit pump capacity is insufficient)	AC 230V, 0.6 A max
Boiler contact / Defrost signal	Dry contact (System setup necessary)	

- Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT ❊It does not work when using the Optional PCB
Outdoor air sensor	AW-A2W-TSOD (Total cable length shall be 30m or less)
Buffer tank sensor	PAW-A2W-TSBU

12.4.2.5 Connection of Optional PCB (CZ-NS5P)



- Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.OFF, Short=Comp.ON (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

- Outputs

Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s	AC230V, 6VA
Pool pump	AC230V	AC 230V, 0.6 A max
Solar pump	AC230V	AC 230V, 0.6 A max
Zone pump	AC230V	AC 230V, 0.6 A max

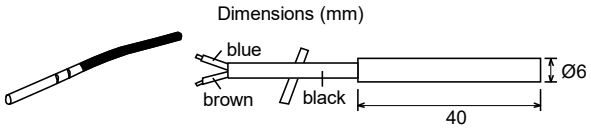
- Thermistor inputs

Zone room sensor	PAW-A2W-TSRT
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

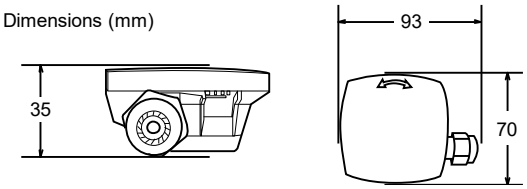
12.4.2.6 Recommended External Device Specification

- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.

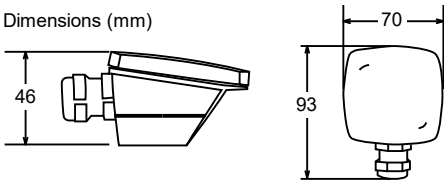
1 Buffer tank sensor: PAW-A2W-TSBU
Use for measurement of the buffer tank temperature.
Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



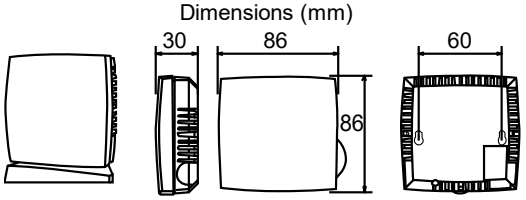
2 Zone water sensor: PAW-A2W-TSHC
Use to detect the water temperature of the control zone.
Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



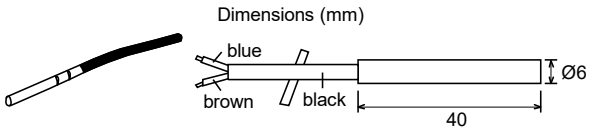
3 Outdoor sensor: PAW-A2W-TSOD
If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



4 Room sensor: PAW-A2W-TSRT
Install the room temperature sensor to the room which requires room temperature control.



5 Solar sensor: PAW-A2W-TSSO
Use for measurement of the solar panel temperature.
Insert the sensor into the sensor pocket and paste it on the solar panel surface.

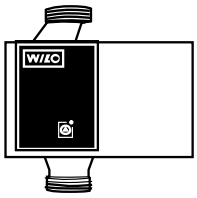


6 Please refer to the table below for sensor characteristic of the sensors mentioned above.

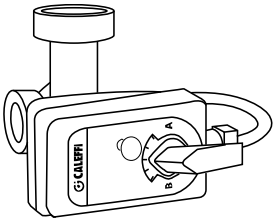
Temperature (°C)	Resistance (kΩ)
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo



- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

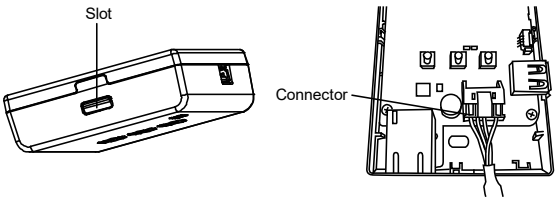


⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

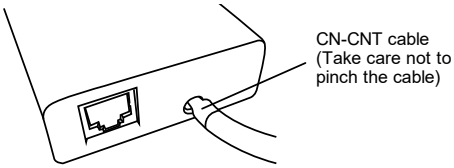
12.4.2.7 Network Adaptor 6 Installation (Optional)

1 **Remove the Control Board Cover ⑤, then connect the cable included with this adaptor to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Tank Unit so that there is no pinching.
   - If an Optional PCB has been installed in the Tank Unit, connect to the CN-CNT connector of the Optional PCB.

2 **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**

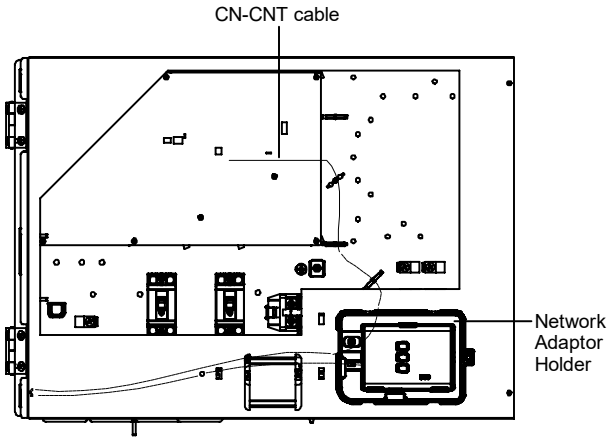


3 **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**

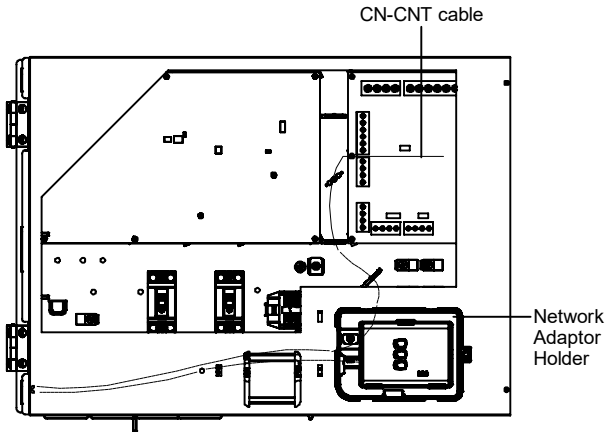


4 **Fix the Network Adaptor 6 to Network Adaptor Holder.**
Guide the cable as shown in the diagram so that external forces cannot act on the connector in the adaptor.

Connection examples:



Without Optional PCB

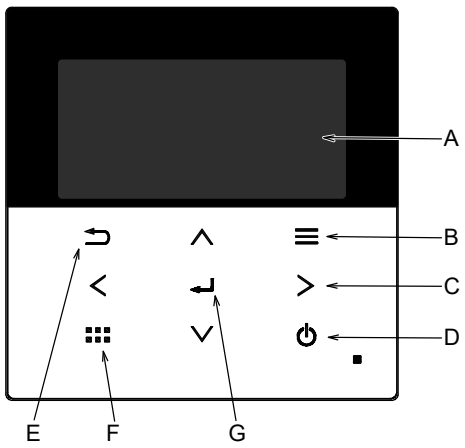


With Optional PCB

12.4.3 System Installation

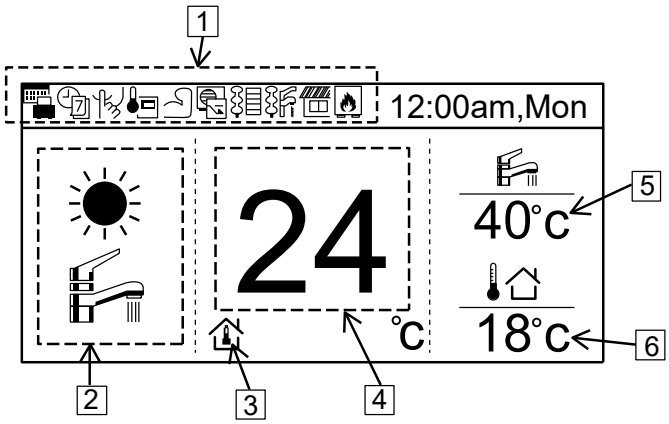
12.4.3.1 Remote Controller Outline

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.



LCD Display
(Actual - Dark background with white icons)

	Name	Function
A:	Main screen	Display information
B:	Menu	Open/Close main menu
C:	Triangle (Move)	Select or change item
D:	Operate	Start/Stop operation
E:	Back	Back to previous item
F:	Quick Menu	Open/Close Quick menu
G:	OK	Confirm



	Name	Function
1:	Function icon	Display set function/status

	Holiday mode		Demand control
	Weekly timer		Room heater
	Quiet mode		Tank heater
	Remote controller room thermostat		Solar
	Powerful mode		Boiler

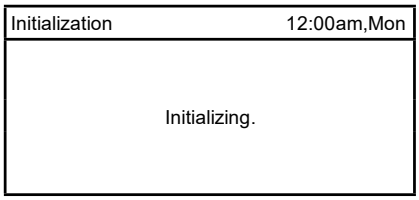
2:	Mode	Display set mode/current status of mode

	Heating		Cooling					
	Auto		Hot water supply		Auto heating		Auto cooling	
	Heat pump operating							

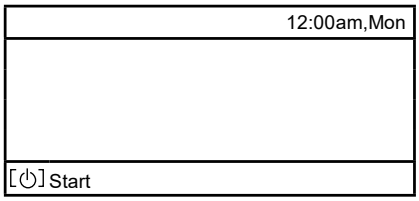
3:	Temp setting	Set room temp		Compensation curve		Set direct water temp		Set pool temp

4:	Display Heat temp	Display current heating temperature (it is set temperature when enclosed by line)
5:	Display tank temp	Display current tank temperature (it is set temperature when enclosed by line)
6:	Outdoor temp	Display outdoor temp

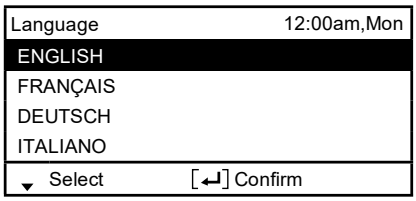
First time of power ON (Start of installation)



When power is ON, firstly initialization screen appears (10 sec)



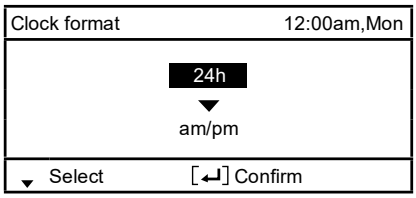
When initialization screen ends, it turns to normal screen.



When any button is pressed, language setting screen appears.
(NOTE) If initial setting is not performed, it does not go into menu.

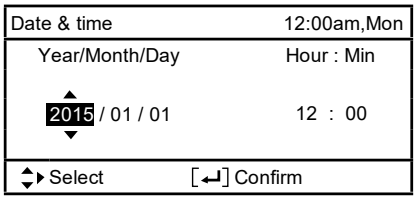
When there are two remote controllers installed from the beginning, the first remote controller to set and confirm language will be recognised as main remote controller.

Set language & confirm



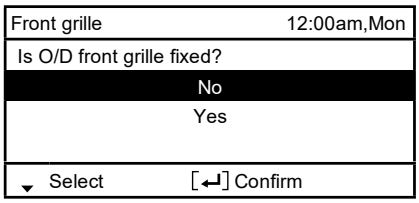
When language is set, setting screen of time display appears (24h/am/pm)

Set time display & confirm

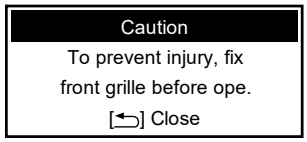


YY/MM/DD/Time setup screen appears

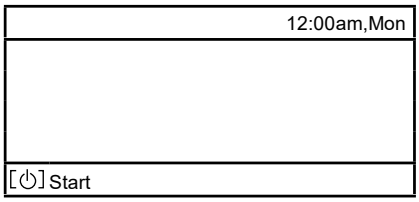
Set YY/MM/DD/Time & confirm



If set No & confirm, a caution message will be displayed to ensure outdoor front grille is installed before proceed to operate the unit.

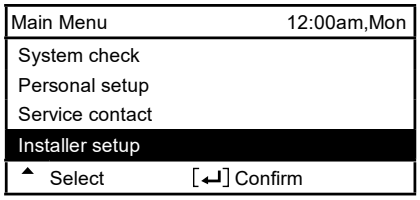


Set Yes & confirm if outdoor front grille has been installed



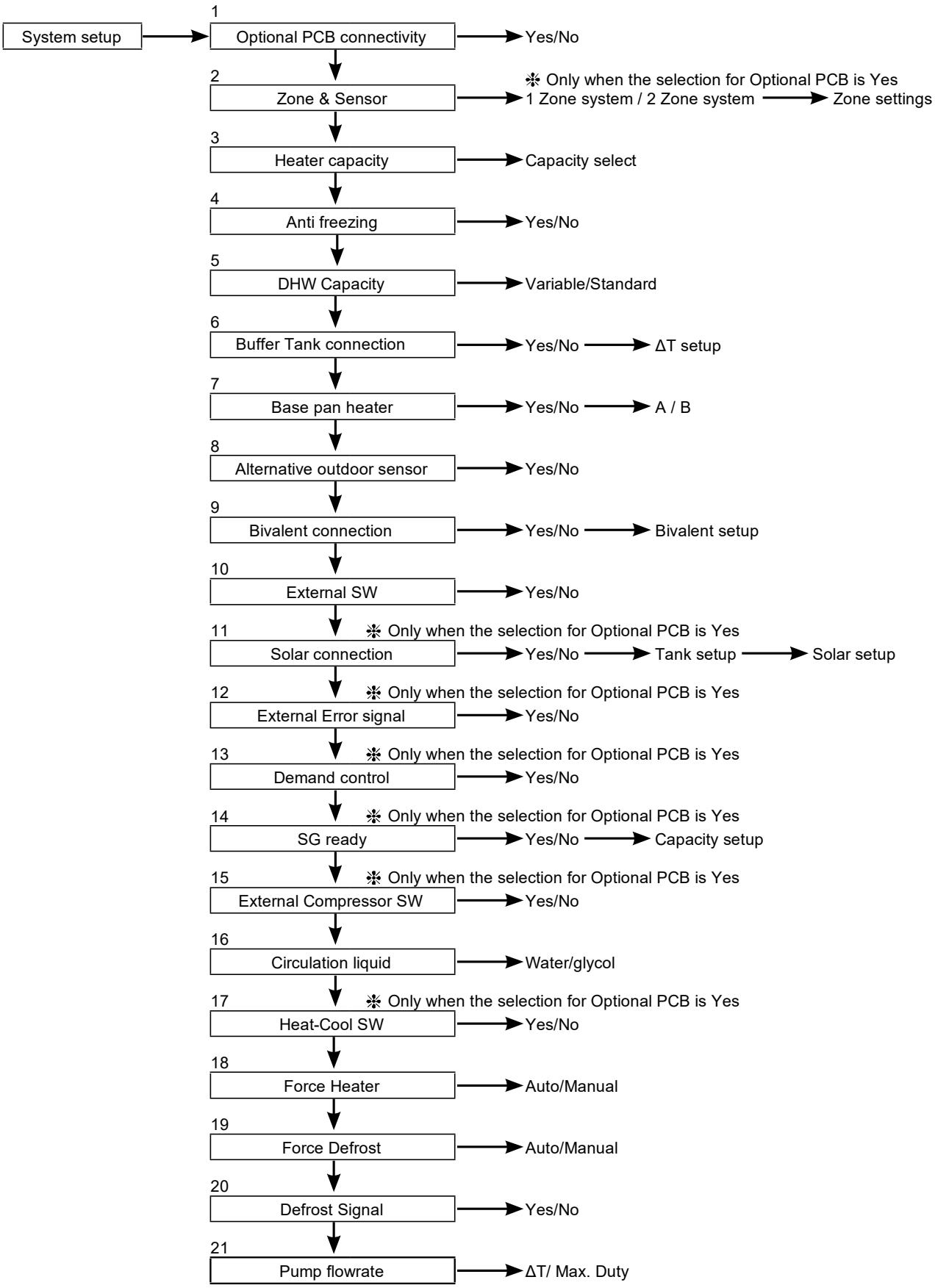
Back to initial screen

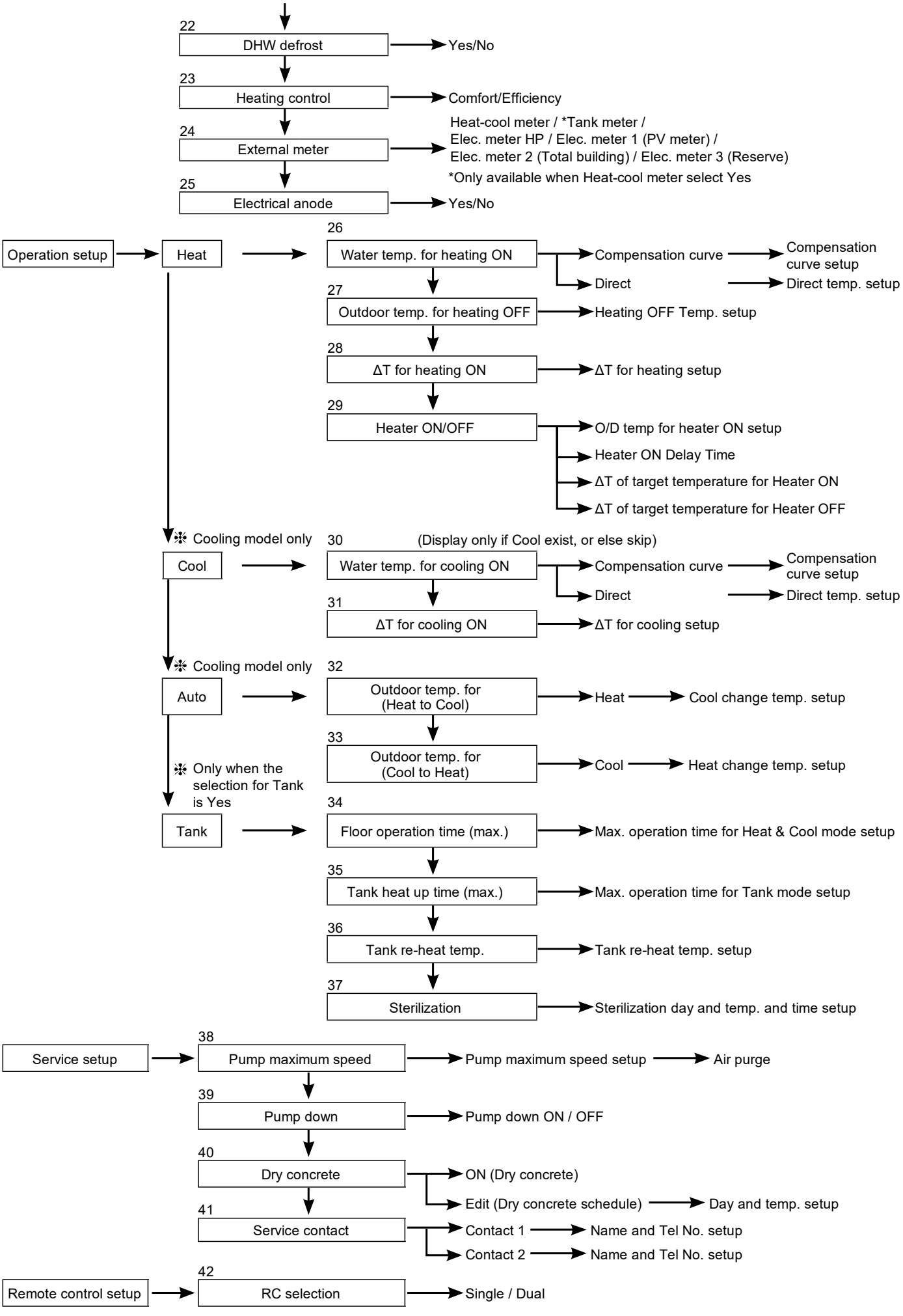
Press menu, select Installer setup



Confirm to go into Installer setup

12.4.3.2 Installer Setup



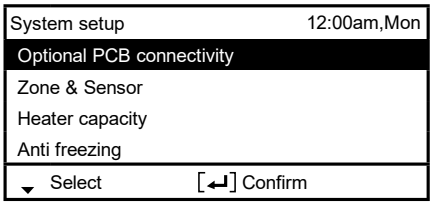

22
DHW defrost
Yes/No
23
Heating control
Comfort/Efficiency
24
External meter
Heat-cool meter / *Tank meter /
Elec. meter HP / Elec. meter 1 (PV meter) /
Elec. meter 2 (Total building) / Elec. meter 3 (Reserve)
*Only available when Heat-cool meter select Yes
25
Electrical anode
Yes/No
Operation setup
Heat
26
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
27
Outdoor temp. for heating OFF
Heating OFF Temp. setup
28
ΔT for heating ON
ΔT for heating setup
29
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
Cooling model only
Cool
30
(Display only if Cool exist, or else skip)
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
31
ΔT for cooling ON
ΔT for cooling setup
Cooling model only
Auto
32
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
33
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
Only when the selection for Tank is Yes
Tank
34
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
35
Tank heat up time (max.)
Max. operation time for Tank mode setup
36
Tank re-heat temp.
Tank re-heat temp. setup
37
Sterilization
Sterilization day and temp. and time setup
Service setup
38
Pump maximum speed
Pump maximum speed setup
Air purge
39
Pump down
Pump down ON / OFF
40
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
41
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup
Remote control setup
42
RC selection
Single / Dual

12.4.3.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install Optional PCB.
Please select Yes after installing Optional PCB.

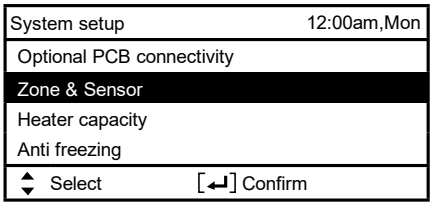
- 2-zone control
- Pool
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

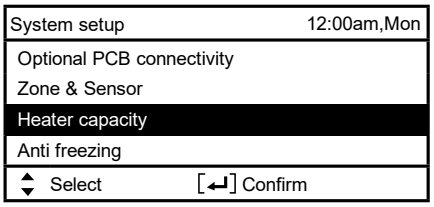
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

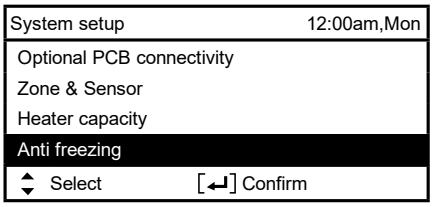
(NOTE) There are models which cannot select Heater capacity.



4. Anti freezing | Initial setting: Yes

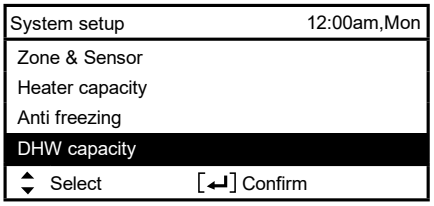
Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



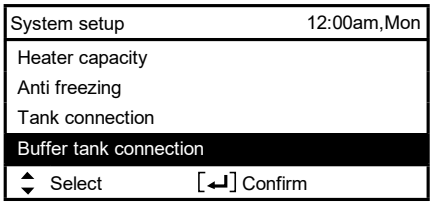
5. DHW Capacity | Initial setting: Variable

Variable DHW capacity setting normally run with efficient boiling which is energy saving heating. But while hot water usage high and tank water temperature low, variable DHW mode will run with fast heat up which heat up the tank with high heating capacity.
If standard DHW capacity setting is selected, heat pump run with heating rated capacity at tank heat up operation.



6. Buffer Tank connection | Initial setting: No

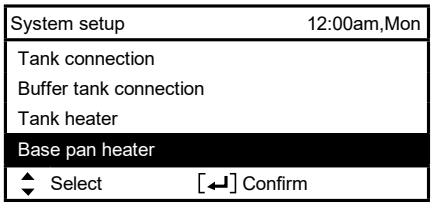
Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
If the buffer tank capacity is not so large, please set larger value for ΔT.



7. Base pan heater | Initial setting: No

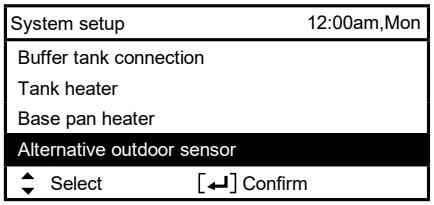
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



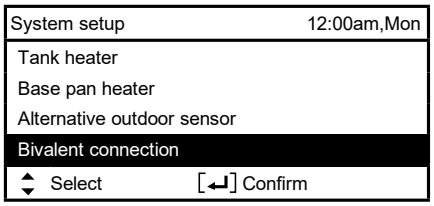
8. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



9. Bivalent connection | Initial setting: No

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.



After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)
1) SG ready (Only available to set when optional PCB set to YES)
- SG Ready input from optional PCB terminal control ON/OFF of boiler and heat pump as below condition

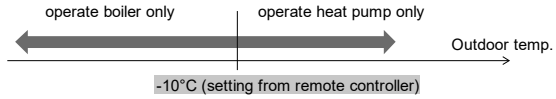
SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

* This bivalent SG ready input is sharing same terminal as [14. SG ready] connection. Only one of these two setting can be set at the same time.
When one is set, another setting will reset to not set.

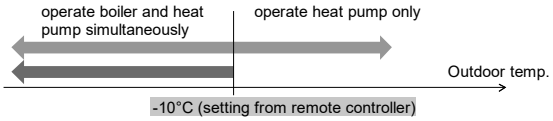
2) Auto
There are 3 different modes in the boiler auto pattern operation. Movement of each modes are shown below.
② Alternative (switch to boiler operation when drops below setting temperature)
③ Parallel (allow boiler operation when drops below setting temperature)
④ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is “ON”, “boiler contact” is “ON”, “_”(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

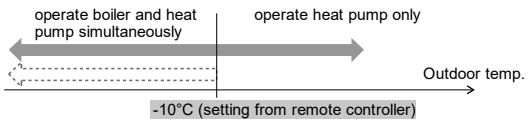


Parallel mode

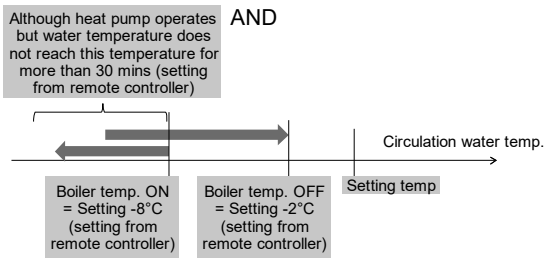


Advanced Parallel mode

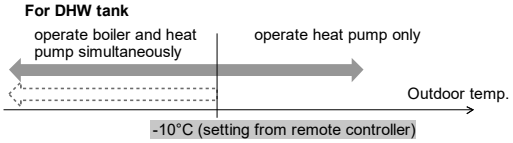
For heating



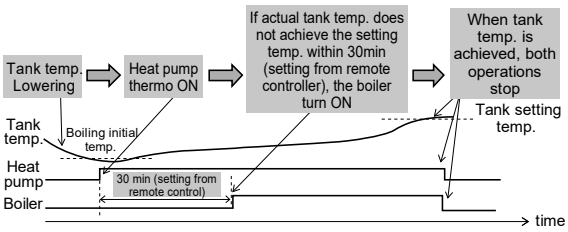
AND



For DHW tank



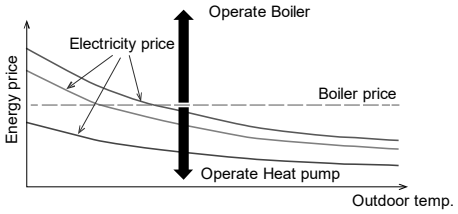
AND



In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of “Heating/Tank” mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

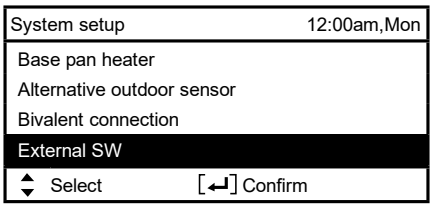
3) Smart

There are Energy price (both electricity and boiler) and Schedule to be set on remote controller.
Operation setting of Energy price and Schedule shall be responsible by installer.
Based on these settings, system will calculate the final price for both electricity and boiler.
When final price of Electricity is lower than Boiler's, heat pump will operate.
When final price of Electricity is higher than Boiler's, boiler will operate.



10. External SW | Initial setting: No

Able to turn ON/OFF the operation by external switch.

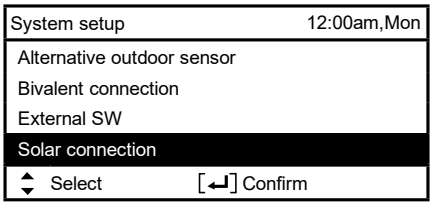


11. Solar connection | Initial setting: No

Set when solar water heater is installed.

Setting include items below.

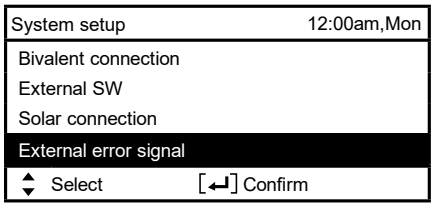
① Set either buffer tank or DHW tank for connection with solar water heater.
② Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to operate the solar pump.
③ Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to stop the solar pump.
④ Anti-freezing operation start temperature (please change setting based on usage of glycol.)
⑤ Solar pump stop operation when it exceeds high limit temperature (when tank temperature exceed designated temperature (70~90°C))



12. External Error Signal | Initial setting: No

Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

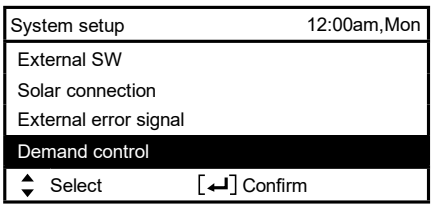
(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off “close” from the display, error signal will still remain ON.



13. Demand control | Initial setting: No

Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.



Analog input [v]	Rate [%]
0.0	not activate
0.1 ~ 0.6	not activate
0.7	not activate
0.8	not activate
0.9 ~ 1.1	10
1.2	10
1.3	10
1.4 ~ 1.6	15
1.7	15
1.8	15
1.9 ~ 2.1	20
2.2	20
2.3	20
2.4 ~ 2.6	25
2.7	25
2.8	25
2.9 ~ 3.1	30
3.2	30
3.3	30
3.4 ~ 3.6	35
3.7	35
3.8	35

Analog input [v]	Rate [%]
3.9 ~ 4.1	40
4.2	40
4.3	40
4.4 ~ 4.6	45
4.7	45
4.8	45
4.9 ~ 5.1	50
5.2	50
5.3	50
5.4 ~ 5.6	55
5.7	55
5.8	55
5.9 ~ 6.1	60
6.2	60
6.3	60
6.4 ~ 6.6	65
6.7	65
6.8	65
6.9 ~ 7.1	70
7.2	70
7.3	70

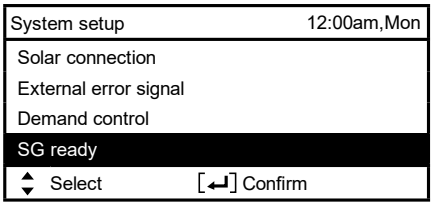
Analog input [v]	Rate [%]
7.4 ~ 7.6	75
7.7	75
7.8	75
7.9 ~ 8.1	80
8.2	80
8.3	80
8.4 ~ 8.6	85
8.7	85
8.8	85
8.9 ~ 9.1	90
9.2	90
9.3	90
9.4 ~ 9.6	95
9.7	95
9.8	95
9.9 ~	100

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

14. SG ready | Initial setting: No

Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2



Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Set by SG ready setting of remote controller

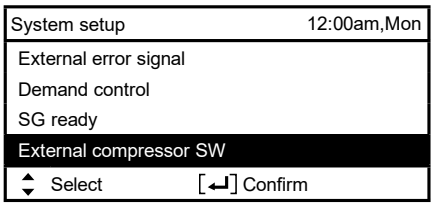
(When SG ready set to YES, Bivalent control pattern will set to Auto.)
(NOTE) Does not display if there is no Optional PCB.

15. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

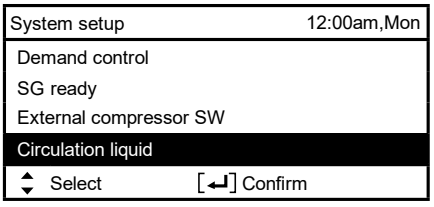


16. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

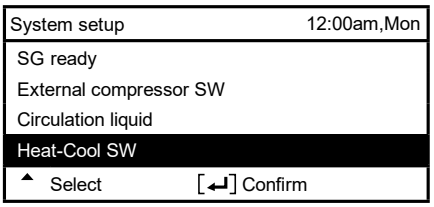


17. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

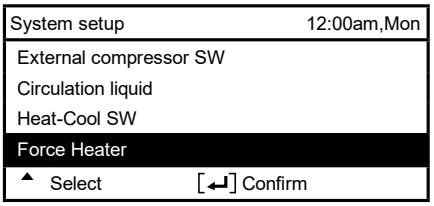


18. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

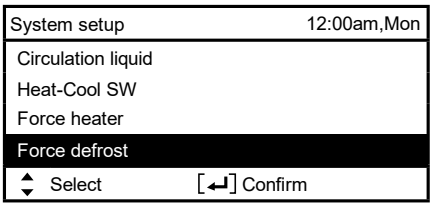
Heater source will ON during force heater mode.



19. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

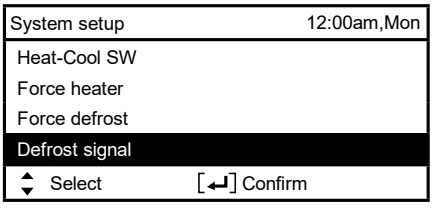
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition. (Even auto is selected, user still can turn on force defrost through quick menu)



20. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).

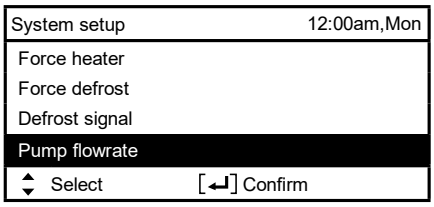


21. Pump flowrate

Initial setting: ΔT

If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

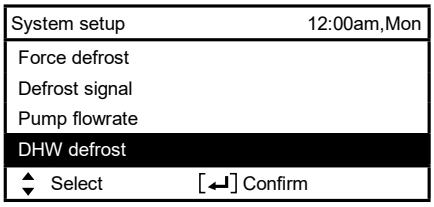
If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



22. DHW defrost

Initial setting: Yes

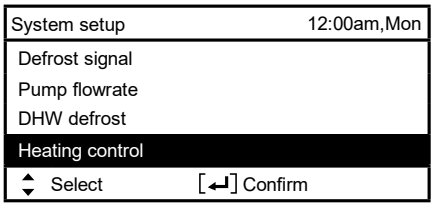
When DHW defrost set to YES, hot water of domestic hot water tank will be used during defrost cycle.
When DHW defrost set to NO, hot water of floor heating circuit will be used during defrost cycle.



23. Heating control

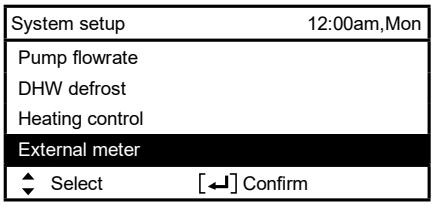
Initial setting : Comfort

There are two modes to select for compressor frequency control : Comfort or Efficiency.
When set to Comfort mode, compressor will run at zone limit maximum frequency to reach set temperature faster.
When set to Efficiency mode, compressor will run at part load frequency at initial stage for energy saving.



24. External meter

Initial setting : [Heat-cool meter : No]
[Tank meter : No] *only available when Heat-cool meter select Yes
[Elec. meter HP : No]
[Elec. meter 1 (PV meter) : No]
[Elec. meter 2 (Total building) : No]
[Elec. meter 3 (Reserve) : No]



There are two systems for generation meter connection : one generation meter system (Heat-cool meter) or two generation meter system (Heat-cool meter and Tank meter)
Both systems can provide all generation data of heating, cooling and DHW directly from external meter.

If Heat-cool meter is set to Yes, it will read from external meter for heat pump's energy generation data during heating, cooling and DHW operation *1.
If Heat-cool meter is set to No, it will base on unit's calculation for heat pump's energy generation data during heating, cooling and DHW operation.
If Tank meter is set to Yes, it will read from external meter for heat pump's energy generation data during DHW operation *1.

If Elec. meter HP is set to Yes, it will read from external meter for heat pump's energy consumption data.
If Elec. meter HP is set to No, it will base on unit's calculation for heat pump's energy consumption data.

If Elec. meter 1 (PV meter) is set to Yes, it will read from external meter for energy generation data of solar system and display it on Cloud system.
If Elec. meter 2 (Building) is set to Yes, it will read from external meter for energy consumption data of the building and display it on Cloud system.
If Elec. meter 3 (Reserve) is set to Yes, it will read from external meter for energy consumption data obtained from reserved electricity meter and display it on Cloud system.

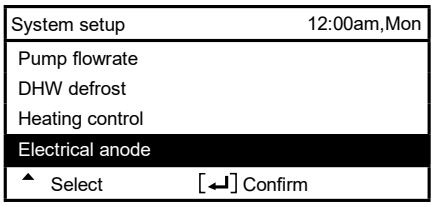
*1 Set Heat-cool meter to Yes and set Tank meter to No when 1 generation meter system is installed.
Set Heat-cool meter to Yes and set Tank meter to Yes when 2 generation meter system is installed.

Remark : Elec. meter HP refers to Electricity meter that measures Heat Pump unit's consumption.
Elec. meter 1 / 2 / 3 refers to Electricity meter no. 1 / no. 2 / no. 3

25. Electrical anode

For WH-ADC0309K3E5AN, WH-ADC0309K6E5AN model, initial setting : Yes
For other models, initial setting : No

When Electrical anode set to YES, anode will be turned on.
When Electrical anode set to NO, anode will not be turned on.



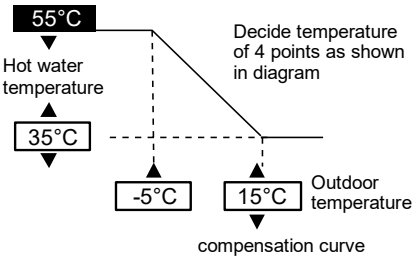
12.4.3.4 Operation Setup

Heat

26. Water temp. for heating ON | Initial setting: compensation curve

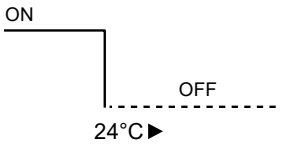
Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



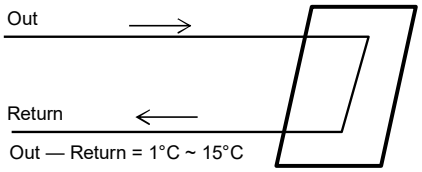
27. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C



28. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

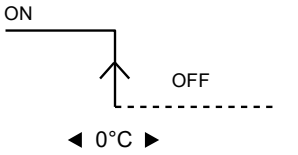


29. Heater ON/OFF

a. Outdoor temp. for heater ON | Initial setting: 0°C

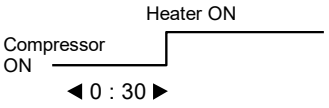
Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.



b. Heater ON delay time | Initial setting: 30 minutes

Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

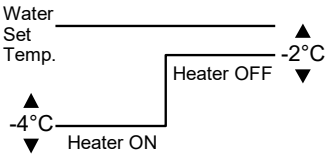


c. Heater ON: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp | Initial setting: -2°C

Set water temperature for heater to turn off at heat mode.
Setting range is -8°C ~ 0°C

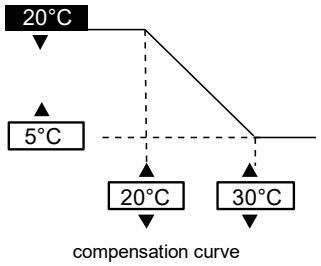


Cool

30. Water temp. for cooling ON | Initial setting: compensation curve

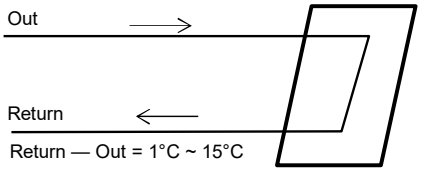
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



31. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

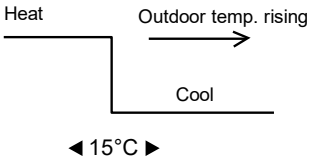


Auto

32. Outdoor temp. for (Heat to Cool) | Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 5°C ~ 25°C

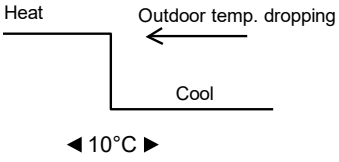
Timing of judgement is every 1 hour



33. Outdoor temp. for (Cool to Heat) | Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 25°C

Timing of judgement is every 1 hour

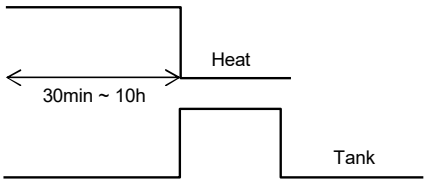


Tank

34. Floor operation time (max.) Initial setting: 8h

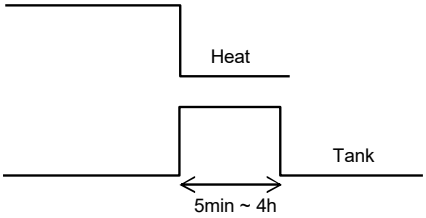
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



35. Tank heat up time (max.) Initial setting: 60min

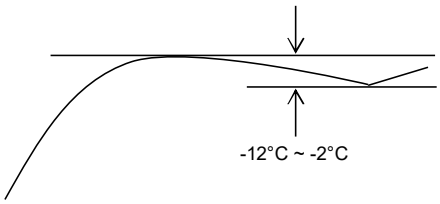
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



36. Tank re-heat temp. Initial setting: -8°C

Set temp to perform reboil of tank water.
(When boiled by heat pump only, (51°C – Tank re-heat temp) shall become max temp.)

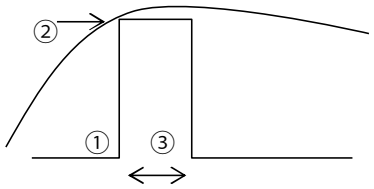
Setting range is -12°C ~ -2°C



37. Sterilization Initial setting: 65°C 10min

Set timer to perform sterilization.
① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~75°C ❋ If use back-up heater, it is 65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.

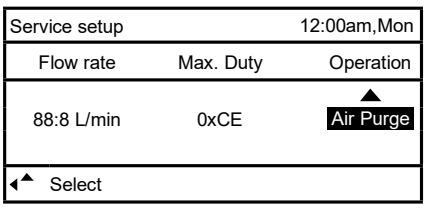


12.4.3.5 Service Setup

38. Pump maximum speed | Initial setting: Depend on model

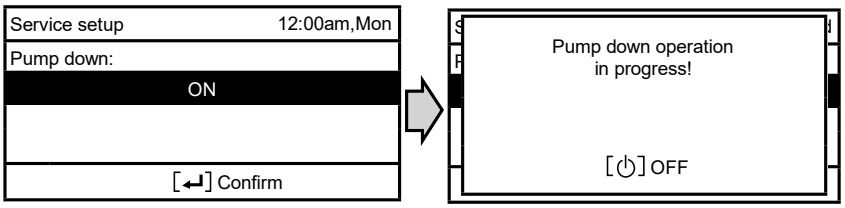
Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.



39. Pump down

Operate pump down operation

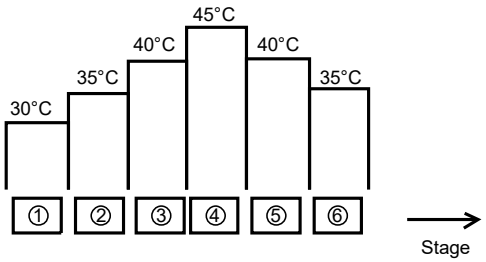


40. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

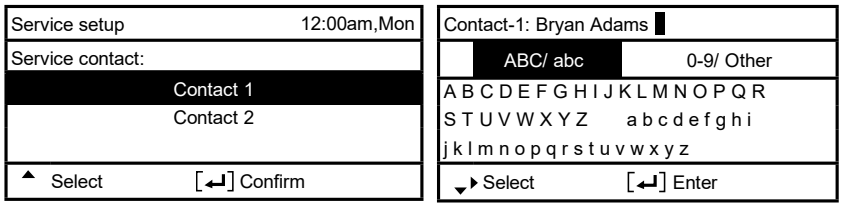
When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



41. Service contact

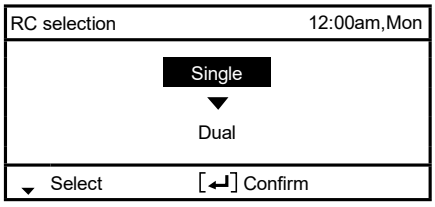
Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)



12.4.3.6 Remote Control Setup

42. RC selection | Initial setting : Single

Set to "Single" when only one remote controller is installed.
Set to "Dual" when two remote controllers are installed.



12.5 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

12.5.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

12.5.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it. In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer. When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

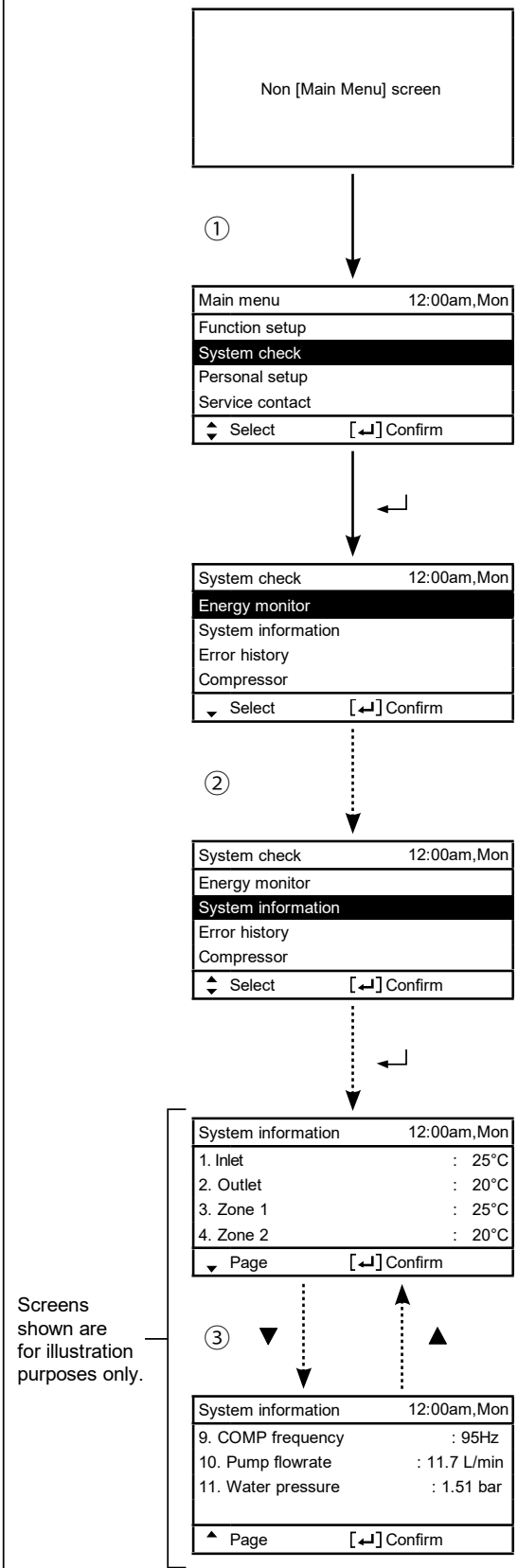
③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

12.5.3 Check Water Pressure from Remote Controller

1. Press ☰ SW and scroll to "System check".
2. Press ↵ and scroll to "System information".
3. Press ↵ and search for "Water pressure".



13. Installation Instruction

(For WH-ADC0309K3E5B only)

13.1 Indoor Floor Area Requirement

- **If the total refrigerant charge in the system is <1.84 kg**, no additional minimum floor area is required.
- **If the total refrigerant charge in the system is ≥1.84 kg**, additional minimum floor area requirements is complied as described below:

Symbol	Description	Unit
m_c	Total refrigerant charge in system	kg
m_{max}	Maximum refrigerant charge allowed	kg
m_{excess}	m_c - m_{max}	kg
H	Installation height	m
VA_{min}	Minimum ventilation opening area	cm^2

Total refrigerant charge in system, m_c (kg)
= Pre-charged refrigerant amount in unit (kg)
+ Additional refrigerant amount after installation (kg)

A) Determine **Maximum refrigerant charge allowed, m_{max}**

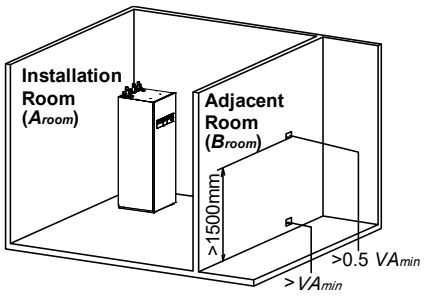
1. Calculate Installation Room Area, A_{room}.
2. Based on Table I, select m_{max} which corresponds to the calculated A_{room} value.
3. If $m_{max} \geq m_c$, the unit can be installed in the installation room with the specified installation height (***H*=600mm**) in Table I and without additional room area or any additional ventilation.
4. Else, proceed to B) and C).

B) Determine **Total Floor Area of A_{room} and B_{room} compliance to $A_{min\ total}$**

1. Calculate the B_{room} area adjacent to the A_{room}.
2. Determine the $A_{min\ total}$ based on the Total Refrigerant Charge, m_c from Table II.
3. The total floor area of both A_{room} and B_{room} must exceed $A_{min\ total}$.

C) Determine **Minimum Venting Opening Area**, **VA_{min}** for natural ventilation

1. From Table III, calculate m_{excess}.
2. Then determine VA_{min} corresponding to the calculated m_{excess} for natural ventilation between A_{room} and B_{room}.
3. The unit can be installed at specific room only when the following conditions are fulfilled:
   - Two permanent openings (cannot be closed), one at bottom, another at top, for ventilation purposes are made between A_{room} and B_{room}.
   - **Bottom opening:**
     - Must comply to the minimum area requirement of VA_{min}.
     - Opening must be located ≤300mm from the floor.
     - At least 50% of required opening area must be ≤200mm from the floor.
     - The bottom of the opening shall not be higher than the point of release when the unit is installed and must be situated ≤100mm above the floor.
     - Must be as close as possible to the floor and lower than H.
   - **Top opening:**
     - The total size of the Top opening must be more than 50% of VA_{min}.
     - Opening must be located ≥1500mm above the floor.



   - The height of the openings must more than 20mm.
   - A direct ventilation opening to outside is **NOT** encouraged for ventilation opening (the user can block the opening when it is cold).
   - The value of H is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.

Table I - Maximum refrigerant charge allowed in a room

A_{room} (m²)	Maximum refrigerant charge in a room (m_{max}) (kg)
	***H*=0.6m**
1	0.138
2	0.276
3	0.414
4	0.553
5	0.691
6	0.829
7	0.907
8	0.970
9	1.028
10	1.084
11	1.137
12	1.187
13	1.236
14	1.283
15	1.328
16	1.371
17	1.413
18	1.454
19	1.494
20	1.533
21	1.571
22	1.608
23	1.644
24	1.679
25	1.714
26	1.748
27	1.781
28	1.814
29	1.846
30	1.877
31	1.909
32	1.939
33	1.969
34	1.999
35	2.028
36	2.057
37	2.085
38	2.113
39	2.141
40	2.168
41	2.195
42	2.221
43	2.248
44	2.274
45	2.299

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate A_{room} values, the value that corresponds to the lower A_{room} value from the table is considered.
  Example:
  For A_{room} = 10.5 m², the value that corresponds to "A_{room} = 10 m²" is considered.

Table II - Minimum floor area

m_c (kg)	Minimum floor area ($A_{min\ total}$ (m²))
	***H*=0.6m**
1.84	28.81
1.86	29.44
1.88	30.08
1.90	30.72
1.92	31.37
1.94	32.03
1.96	32.70
1.98	33.37
2.00	34.04
2.02	34.73
2.04	35.42
2.06	36.12
2.08	36.82
2.10	37.53
2.12	38.25
2.14	38.98
2.16	39.71
2.18	40.45
2.20	41.19
2.22	41.94
2.24	42.70
2.26	43.47
2.28	44.24
2.30	45.02

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate m_c values, the value that corresponds to the higher m_c value from the table is considered.
  Example:
  If m_c = 1.85 kg, the value that corresponds to "m_c = 1.86 kg" is considered.
- Systems with total refrigerant charge lower than 1.84 kg are not subjected to any room area requirements.
- Charges above 2.30 kg are not allowed in the unit.

Table III - Minimum venting opening area for natural ventilation

m_c (kg)	m_{max} (kg)	m_{excess} (kg) = m_c - m_{max}	Minimum venting opening area (VA_{min}) (cm²)
			***H*=0.6m**
2.3	0.1	2.20	890
2.3	0.3	2.00	809
2.3	0.5	1.80	728
2.3	0.7	1.60	647
2.3	0.9	1.40	583
2.3	1.1	1.20	552
2.3	1.3	1.00	500
2.3	1.5	0.80	430
2.3	1.7	0.60	343
2.3	1.9	0.40	242
2.3	2.1	0.20	127
2.3	2.3	0.00	0

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate m_{excess} values, the value that corresponds to the higher m_{excess} value from the table is considered.
  Example:
  m_{excess} = 1.45 kg, the value that corresponds to "m_{excess} = 1.6 kg" is considered.

Attached accessories

No.	Accessory part	Qty.	No.	Accessory part	Qty.
1	Adjustable Feet 	4	3	Drain Elbow 	1
2	Reducing Adapter 	1	4	Packing 	1

Optional Accessories

No.	Accessories part	Qty.
5	Remote Controller Case	1
6	Network Adaptor (CZ-TAW1B) and Extension Cable (CZ-TAW1-CBL)	1

Field Supply Accessories (Optional)

No.	Part		Model	Specifications	Maker
i	2-way valve kit *Cooling model	Electromotoric Actuator	SFA21/18	AC230V, 12 VA	Siemens
		2-port Valve	VXI46/25		Siemens
ii	Room thermostat	Wired	PAW-A2W-RTWIRED	AC230V	-
		Wireless	PAW-A2W-RTWIRELESS		
iii	Pump	-	Yonos 25/6	AC 230V, 0.6 A max	Wilo
iv	Outdoor sensor	-	PAW-A2W-TSOD	-	-
v	Zone water sensor	-	PAW-A2W-TSHC	-	-
vi	Zone room sensor	-	PAW-A2W-TSRT	-	-

- It is recommended to purchase the field supply accessories listed in above table.

13.2 Indoor Unit

13.2.1 Select the Best Location

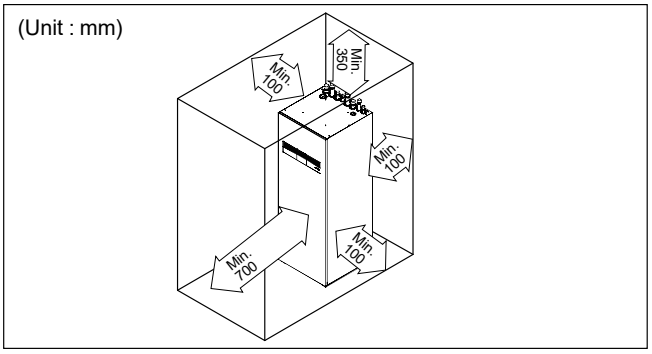
Before choosing the installation site, obtain user approval.

- Install the Tank Unit in indoors with frost free weather proof location only.
- Must install on a flat horizontal and solid hard surface.
- There should not be any heat source or steam near the Tank Unit.
- A place where air circulation in the room is good.
- A place where drainage can be easily done (e.g. Utility room).
- A place where Tank Unit's operation noise will not cause discomfort to the user.
- A place where Tank Unit is far from door way.
- A place where accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated below from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- Secure the Tank Unit to prevent it being knocked over accidentally or during earthquakes.

Please avoid installations which expose the Tank Unit to any of the following conditions:

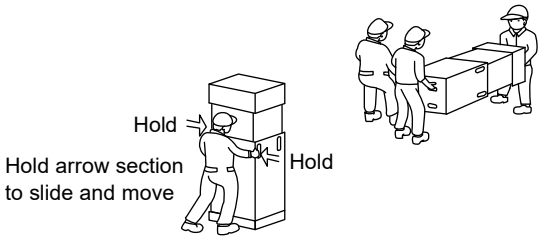
- Extraordinary environment conditions; installation in frost or exposure to unfavorable weather conditions.
- Voltage input exceeding the specified voltage.

13.2.1.1 Required Space for Installation



13.2.1.2 Transport and Handling

- Be careful during transportating the unit so that it is not damaged by impact.
- Only remove the packaging material once it has reached it is desired installation location.
- It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.
- The Tank Unit can be transported either in vertical or horizontal.
  - If it transported in horizontal, make sure Front of packaging material (printed with "FRONT") must facing upwards.
  - If it transported in vertical, use the hand holes on sides, slide and move to the desired location.
- Fix the Adjustable Feet [1], if the Tank unit installed on a uneven surface.

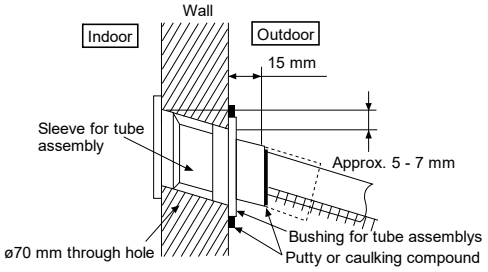


13.2.2 To Drill a Hole in the Wall and Install a Sleeve of Piping

1 Make a Ø70 mm through hole.
2 Insert the piping sleeve to the hole.
3 Fix the bushing to the sleeve.
4 Cut the sleeve until it extrudes about 15 mm from the wall.

⚠ CAUTION
When the wall is hollow, please be sure to use the sleeve for tube assembly to prevent dangers caused by mice biting the connection cable.

5 Finish by sealing the sleeve with putty or caulking compound at the final stage.

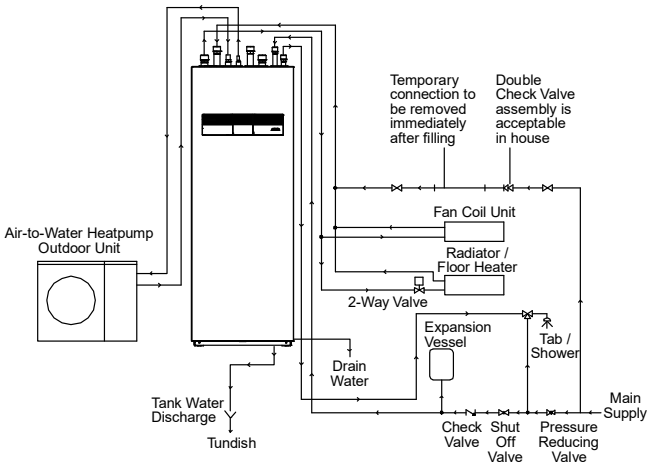


13.2.3 Piping Installation

13.2.3.1 Water Quality Requirement

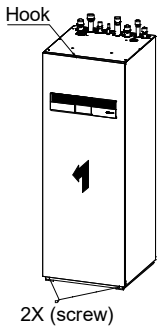
Must use water that complies with European water quality standard 98/83 EC. The lifespan of the Tank Unit will be shorter if groundwater (include spring water and well water) is used.
The Tank Unit shall not be used with the tap water containing contaminants such as salt, acid, and other impurities which may corrode the tank and its component.

13.2.3.2 Typical Piping Installation



13.2.3.3 Access to Internal Components

⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

Open and Close Front Plate ⑱

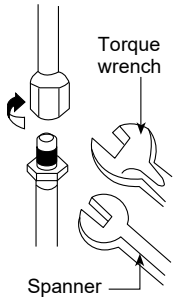
1 Remove the 2 mounting screws of Front Plate ⑱.
2 Slide it upwards to unhook the Front Plate ⑱ hook.
3 Reverse above steps 1~2 for close it.

13.2.3.4 Refrigerant Piping Installation

This Tank Unit is designed for combination with Panasonic Air-to-Water Heat Pump Outdoor Unit. If Outdoor Unit from other manufacturer are being used in combination with Panasonic Tank Unit, optimum operation and reliability of the system is not guaranteed. Thus warranty cannot be given in such case.

1 Connect Tank Unit to Air-to-Water Heatpump Outdoor Unit with correct piping size. Use Reducing Adapter [2] for Outdoor Unit WH-UDZ03KE5* Refrigerant Gas ⓔ piping connection.

Model		Piping size (Torque)		Use Reducing Adapter [2]
Tank Unit	Outdoor Unit	Gas	Liquid	
WH-ADC0309K3E5B	WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]	Yes
	WH-UDZ05KE5*, WH-UDZ07KE5*, WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]	No



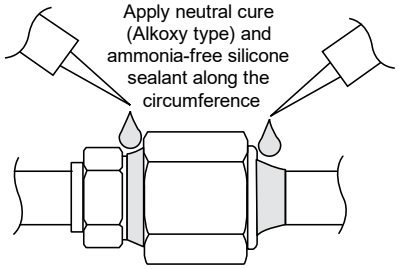
⚠ CAUTION
Do not overtighten, overtightening may cause gas leakage.
Do not pull and push refrigerant piping excessively, deformed pipe may cause refrigerant leak.

2 Please make flare after inserting flare nut (located at joint portion of tube assembly) onto the copper pipe. (In case of using long piping)
3 Do not use pipe wrench to open refrigerant piping. Flare nut may be broken and cause leakage. Use proper spanner or ring wrench.
4 Connect the piping:
   - Align the centre of piping and sufficiently tighten the flare nut with fingers.
   - Further tighten the flare nut with torque wrench in specified torque as stated in the table.

<u>Additional Precautions For R32 Models when connecting by flaring at indoor side</u>

❗	Ensure to do the re-flaring of pipes before connecting to units to avoid leaking.
❗	Connections made between components of refrigerant system shall be accessible for ease of maintenance.

Seal sufficiently the flare nut (both gas and liquid sides) with neutral cure (Alkoxy type) & ammonia-free silicone sealant and insulation material to avoid the gas leak caused by freezing.



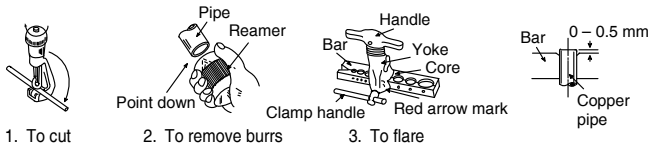
Neutral cure (Alkoxy type) & ammonia-free silicone sealant is only to be applied after pressure testing and cleaning up by following instructions of sealant, only to the outside of the connection. The aim is to prevent moisture from entering the connection joint and possible occurrence of freezing. Curing sealant will take some time. Make sure sealant will not peel off when wrapping the insulation.

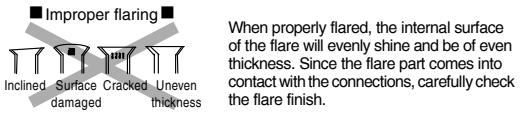
13.2.3.4.1 Checking for Gas Leakage

- Check for leakage of gas after air purging.
- See the in the installation manual for the outdoor.

13.2.3.5 Cutting and Flaring the Piping

1 Please cut using pipe cutter and then remove the burrs.
2 Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3 Please make flare after inserting the flare nut onto the copper pipes.



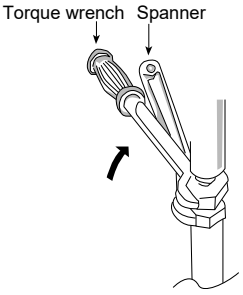


When properly flared, the internal surface of the flare will evenly shine and be of even thickness. Since the flare part comes into contact with the connections, carefully check the flare finish.

13.2.3.6 Water Piping Installation

- Please engage a licensed water circuit installer to install this water circuit.
- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not use worn out tube.
- Do not apply excessive force to pipes that may damage the pipes.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Further tighten the nuts with torque wrench in specified torque as stated in the table.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall.
- If non-brass metallic piping is used for installation, make sure to insulate the pipes to prevent galvanic corrosion.
- Do not connect galvanised pipes, this will cause galvanic corrosion.
- Use correct nut for all Tank Unit tube connections and clean all tubes with tap water before installation. See Tube Position Diagram for detail.

Tube Connector	Nut Size	Torque
ⓐ, ⓑ, ⓙ & ⓚ	RP 1¼"	117.6 N•m
ⓒ & ⓓ	RP ¾"	58.8 N•m



⚠ CAUTION
Do not overtighten, overtightening may cause gas leakage.

- Make sure to insulate the water circuit pipes to prevent reduction of heating capacity.
- After installation, check the water leakage condition in connection area during test run.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Protection From Frost:
  If the Tank Unit is being exposed to frost while power supply failure or pump operating failure, drain the system. When water is idle inside the system, freezing up is very likely to happen which could damage the system. Make sure the power supply is turned off before draining. Heater Assembly ⑩ may be damaged under dry heating.
- Corrosion Resistance:
  Duplex stainless steel is naturally corrosion resistant to mains water supply. No specific maintenance is required to maintain this resistance. However, please note that Tank Unit is not guaranteed for use with a private water supply.
- It is recommended to use a tray (field supply) to collect water from the Tank Unit if water leakage occur.

Recommended piping installation sequence:
(a) → (j) → (k) → (c) → (e) → (f) → (b) → (d)

(A) Space Heating/Cooling Pipework

- Connect Tank Unit Tube Connector ⓐ to outlet connector of Zone 1 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓑ to inlet connector of Zone 1 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓙ to outlet connector of Zone 2 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓚ to inlet connector of Zone 2 Panel/Floor heater.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Refer below table for the rated flow rate of each particular Outdoor Unit.

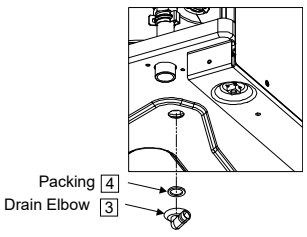
Model		Rated Flow Rate (l/min)	
Tank Unit	Outdoor Unit	Cool	Heat
WH-ADC0309K3E5B	WH-UDZ03KE5*	9.2	9.2
	WH-UDZ05KE5*	14.3	14.3
	WH-UDZ07KE5*	19.2	20.1
	WH-UDZ09KE5*	23.5	25.8

(B) Domestic Hot Water Tank Pipework

- It's strongly recommended to install an expansion vessel (field supply) in the Domestic Hot Water Tank circuit. Refer Typical Piping Installation section to locate the expansion vessel.
  - Recommended pre-charge pressure of the expansion vessel (field supply) = 0.35MPa (3.5 bars)
- In high water pressure or water supply is above 500kPa, please install the Pressure Reducing Valve for water supply. If the pressure higher than that, it might damage the Tank Unit.
- A Pressure Reducing Valve (field supply) with below specification is strongly advised to be installed along the line of the tube connector ⓒ of Tank Unit. Refer Typical Piping Installation section to locate both of these valves. Recommended Pressure Reducing Valve specifications:
  - Set pressure: 0.35 MPa (3.5 bars)
- Must connect a faucet to Tank Unit Tube Connector ⓓ and main water supply, in order to supply water with appropriate temperature for shower or tap usage. Failure to do so might cause scalding.
- Failure to connect the tube appropriately might causing the Tank Unit malfunction.

(C) Drain Elbow and Hose Installation

- Fix the Drain Elbow 3 and Packing 4 to the bottom of Drain Water Hole ⓗ.



- Use inner diameter 17 mm drain hose in the market, fix to Drain Elbow 3.
- This hose must to be installed in a continuously downward direction and in a frost-free environment. Improper drain piping may cause water leakage hence damage the furnitures.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Guide the drain hose to outdoor as illustrated.

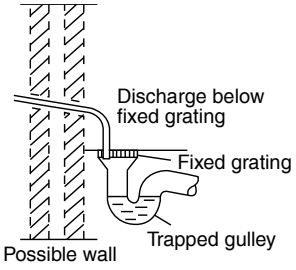


Illustration of guide drain hose to outdoor

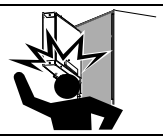
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulphuric gas etc.
- If necessary, use hose clamp to further tighten the hose at drain hose connector to prevent leakage.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.

(D) Domestic Hot Water Tank Discharge (Drain Tap) and Safety Relief Valve Pipework

- Safety Relief Valve 0.8MPa (8 bars) incorporated in Domestic Hot Water Tank.
- Drain Tap and Safety Relief Valve discharge fittings share the same drainage outlet.
- Use R½" male connector for this drainage outlet connection (Tube connector ⓖ).
- Piping must always be installed in a continuously downward direction. It must not be longer than 2m, with no more than 2 elbows, and must not allow condensation to build up or freezing to occur.
- The pipe from this drainage outlet fitting must not be shut off. The discharge must be freed.
- The end of this pipework must be in such a way so that the outlet is visible and can not cause any damage. Keep away from electrical components.
- It is recommended to fit a tundish into this ⓖ pipework. Tundish should be visible and positioned away from frost environment and electrical components.

13.2.4 Connect the Cable to the Tank Unit

⚠ WARNING
This section is for authorized and licensed electrician only. Work behind the Control Board Cover ⑤ secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

⚠ CAUTION	
Please take extra precaution when open the control board cover ⑤ and control board ⑥ for unit installation and servicing. Failure to do so may cause injury.	

13.2.4.1 Fixing of Power Supply Cable and Connecting Cable

1 Connecting cable between Tank Unit and Outdoor Unit shall be approved polychloroprene sheathed flexible cord, type designation 60245 IEC 57 or heavier cord. See below table for cable size requirement.

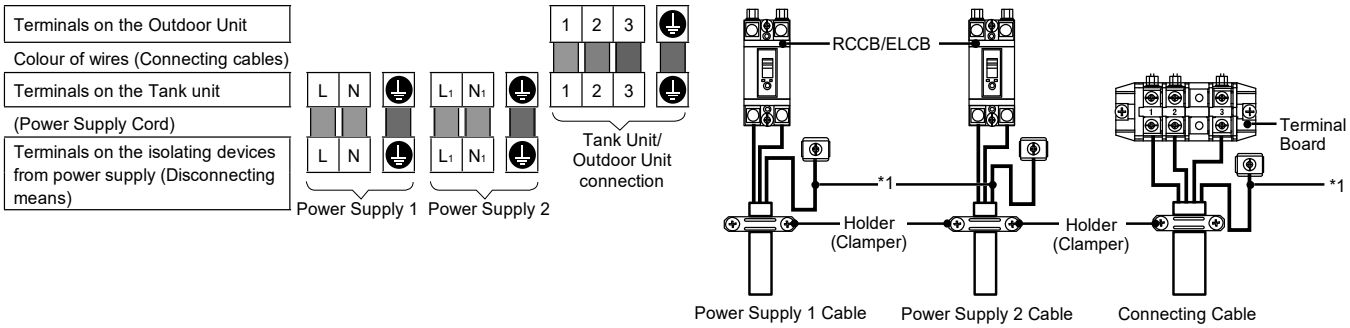
Model		Connecting Cable Size
Tank Unit	Outdoor Unit	
WH-ADC0309K3E5B	WH-UDZ03KE5*, WH-UDZ05KE5*	4 x min 1.5 mm^2
	WH-UDZ07KE5*, WH-UDZ09KE5*	4 x min 2.5 mm^2

- Ensure the colour of wires of Outdoor Unit and the terminal no. are the same to the Tank Unit respectively.
- Earth wire shall be longer than the other wires as shown in the figure for the electrical safety in case of the slipping out of the cord from the Holder (Clamper).

2 An isolating device must be connected to the power supply cable.
- Isolating device (disconnecting means) should have minimum 3.0 mm contact gap.
- Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means).
  See below table for cable size requirement.

Model		Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
Tank Unit	Outdoor Unit				
WH-ADC0309K3E5B	WH-UDZ03KE5*, WH-UDZ05KE5*	1	3 x min 1.5 mm^2	15/16A	30mA, 2P, type A
		2	3 x min 1.5 mm^2	15/16A	30mA, 2P, type AC
	WH-UDZ07KE5*, WH-UDZ09KE5*	1	3 x min 2.5 mm^2	25A	30mA, 2P, type A
		2	3 x min 1.5 mm^2	15/16A	30mA, 2P, type AC

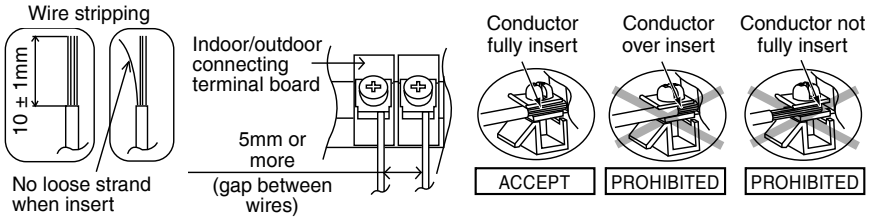
3 To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through a bushing (located at the bottom of Control Board ⑥) before terminal board. The bushing must be used and must not be removed.



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25)

*1 - Earth wire must be longer than other cables for safety reasons

13.2.4.2 Wire Stripping and Connecting Requirement



13.2.4.3 Connecting Requirement

For Tank Unit WH-ADC0309K3E5B with WH-UDZ03KE5*, WH-UDZ05KE5*, WH-UDZ07KE5*, WH-UDZ09KE5*

- The equipment's Power Supply 1 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-3 and can be connected to current supply network.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-3 and can be connected to current supply network.

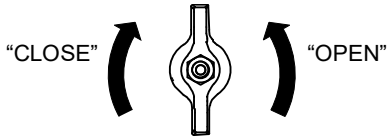
13.2.5 Charging and Discharging the Water

- Make sure all the piping installations are properly done before carry out below steps.

13.2.5.1 Charge the Water

For Domestic Hot Water Tank

1 Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

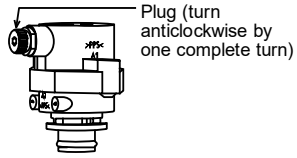


Domestic Hot Water Tank Discharge (Drain Tap) ⓖ

2 Set all Tap / Shower “OPEN”.
3 Start filling water to the Domestic Hot Water Tank via Tube Connector ⓒ. After 20~40min, water should flow out from Tap / Shower. Else, please contact your local authorized dealer.
4 Check and make sure no water leaking at the tube connecting points.
5 Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “OPEN” for 10 seconds to release air from this pipeline. Then set it “CLOSE”.
6 Turn the Safety Relief Valve ㉕ knob counterclockwise slightly and hold for 10 seconds to release air from this pipeline. Then recover the knob to original position.
7 Ensure Step 5 & 6 is carried out each time after charging water to Domestic Hot Water Tank.
8 To prevent back pressure from happening to the Safety Relief Valve ㉕, do turn the Safety Relief Valve ㉕ knob counterclockwise.

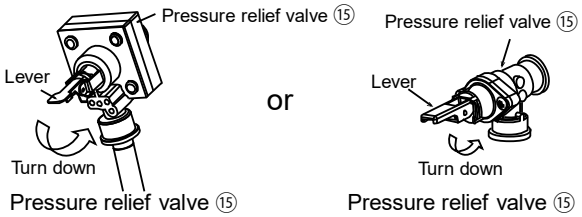
For Space Heating / Cooling

1 Turn the plug on the Air Purge Valve ⑭ outlet anticlockwise by one complete turn from fully closed position.



Air purge valve ⑭

2 Set the Pressure Relief Valve ⑮ lever “DOWN”.



Pressure relief valve ⑮ Pressure relief valve ⑮

3 Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Space Heating / Cooling circuit via Tube Connector ⓐ. Stop filling water if the free water flow through Pressure Relief Valve ⑮ discharge hose.
4 Turn ON the Tank Unit and make sure Water Pump ④ is running.
5 Check and make sure no water leaking at the tube connecting points.
6 The water may drip from this discharge hose. Therefore must guide the hose without close or block the outlet of the hose.

13.2.5.2 Discharge the Water

For Domestic Hot Water Tank

1 Turn OFF power supply.
2 Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “OPEN”.
3 Open Tap / Shower to allow air inlet.
4 Turn the Safety Relief Valve ㉕ knob counterclockwise slightly and hold it until all air is released from this pipeline. Then recover the knob to original position after ensured the pipeline is emptied.
5 After discharge, set Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

13.2.6 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings.

*13.2.6.1 Check Water Pressure *(0.1 MPa = 1 bar)*

Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure from Remote Controller). If necessary add water into Tank Unit (via Tube Connector ⓐ).

13.2.6.2 Check Pressure Relief Valve ⑮

- Check for correct operation of Pressure Relief Valve ⑮ by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keep on draining out from the Tank Unit, switch off the system, and then contact your local authorized dealer.

13.2.6.3 Expansion Vessel ⑬ Pre Pressure Checking

For Space Heating / Cooling

- Expansion Vessel ⑬ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit's piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m.

13.2.6.4 Check RCCB/ELCB

Ensure the RCCB/ELCB set to "ON" condition before check RCCB/ELCB.
Turn on the power supply to the Tank Unit.
This testing could only be done when power is supplied to the Tank Unit.

⚠ WARNING
Be careful not to touch parts other than RCCB/ELCB test button when the power is supplied to Tank Unit. Else, electrical shock may happen. Before obtaining access to terminals, all supply circuits must be disconnected.

- Push the "TEST" button on the RCCB/ELCB. The lever would turn down and indicate "0", if it functions normal.
- Contact authorized dealer if the RCCB/ELCB malfunction.
- Turn off the power supply to the Tank Unit.
- If RCCB/ELCB functions normal, set the lever to "ON" again after testing finish.

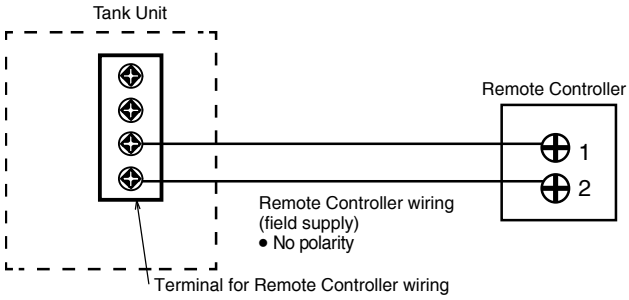
13.2.7 Installation of Remote Controller as Room Thermostat

- Remote Controller ① mounted to the Tank Unit can be moved to the room and serve as Room Thermostat.

13.2.7.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1 By the window, etc. exposed to direct sunlight or direct air.
  2 In the shadow or backside of objects deviated from the room airflow.
  3 Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4 Location near heat source.
  5 Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

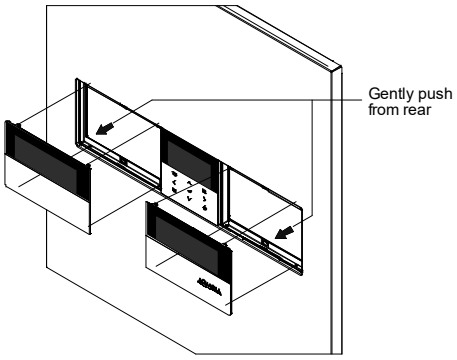
13.2.7.2 Remote Controller Wiring



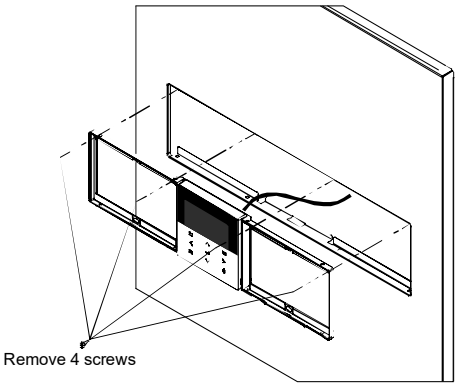
- Remote Controller cable shall be (2 x min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals of Tank Unit (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.

13.2.7.3 *Remove The Remote Controller From Tank Unit*

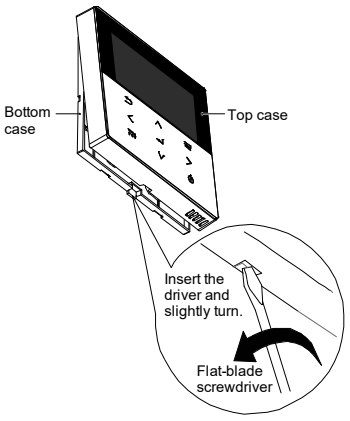
1 Remove both Left Decoration Panel ② and Right Decoration Panel ③ from Front Plate ⑱ with gently push the panels from back.



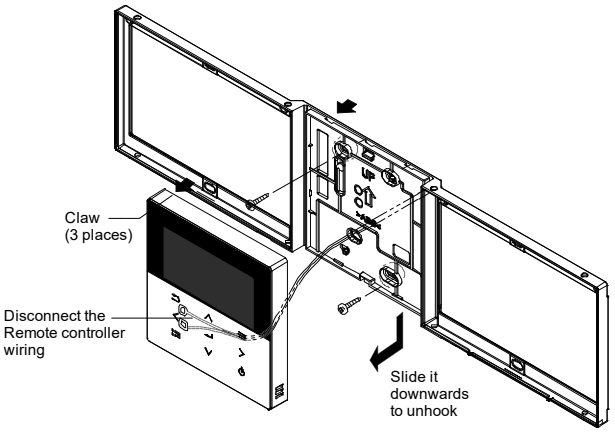
2 Remove the 4 screws and take out the holder with Remote Controller ①.



3 Remove the top case from the bottom case.



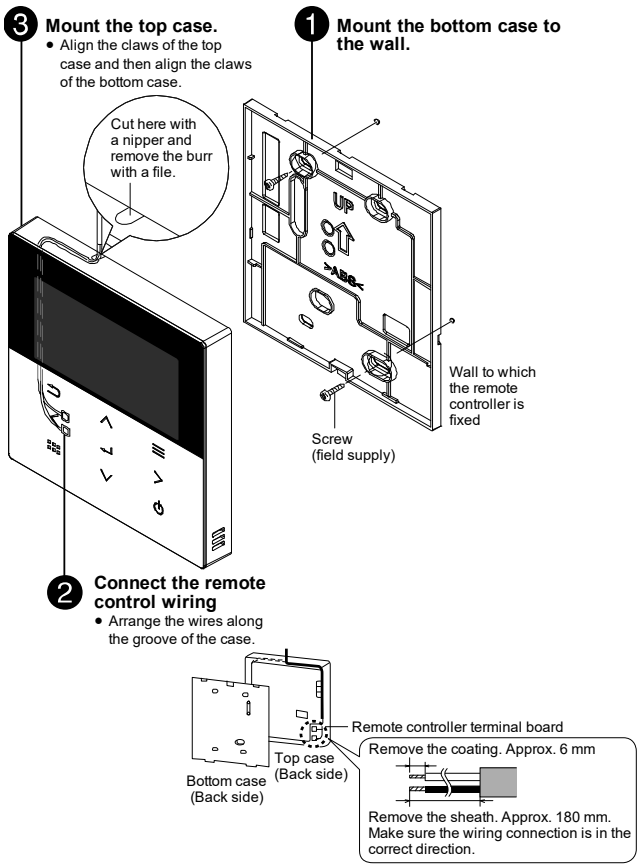
4 Remove the wiring between Remote controller ① and Tank Unit terminal.



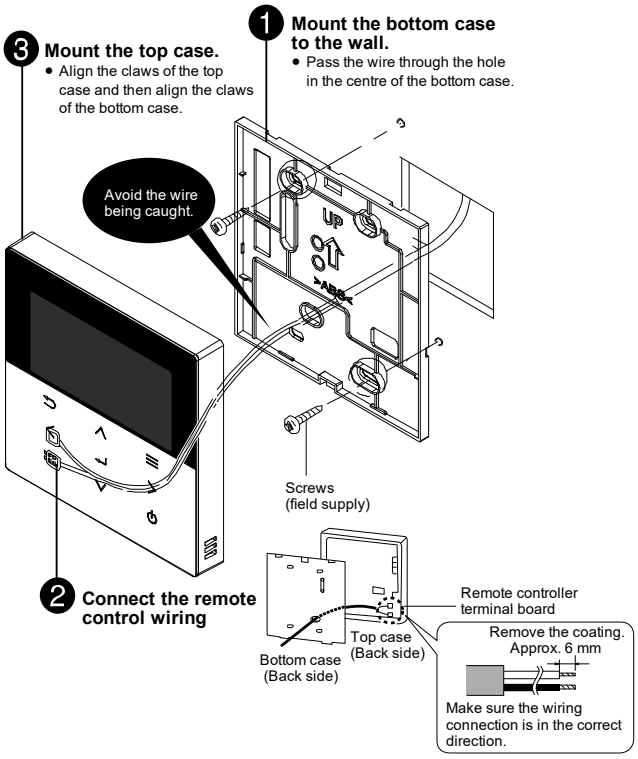
13.2.7.4 *Mounting The Remote Controller*

For exposed type

Preparation: Make 2 holes for screws using a driver.

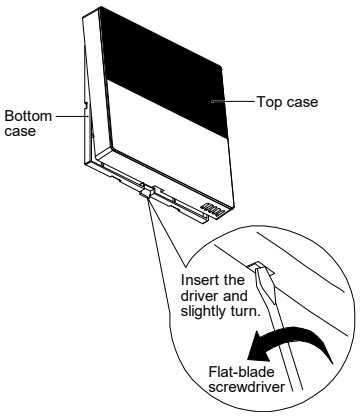


For embedded type
Preparation: Make 2 holes for screws using a driver.



13.2.7.5 Replace The Remote Controller Cover

- Replace the existing Remote Controller with Remote Controller Case [5] to close the hole left after remove the Remote Controller.
  1 Refer Section “Remove The Remote Controller From Tank Unit” for remove Remote Controller.
  2 Remove the top case from the bottom case of Remote Controller Case [5].



  3 Reverse the steps 1 to 4 of section “Remove The Remote Controller From Tank Unit” to fix Remote Controller Case [5] on Tank Unit.

13.2.8 Test Run

1 Before test run, make sure below items have been checked:-
  a) Pipework are properly done.
  b) Electric cable connecting work are properly done.
  c) Tank Unit is filled up with water and trapped air is released.
  d) Please turn on the power supply after filling the tank until full.
2 Switch ON the power supply of the Tank Unit. Set the Tank Unit RCCB/ELCB to “ON” condition. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run.
  Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

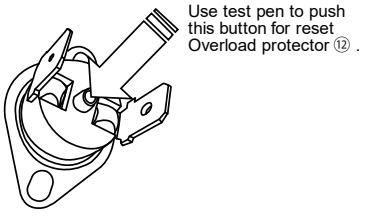
3 For normal operation, Water Pressure reading should be in between 0.05 MPa and 0.3 MPa (0.5 bar and 3 bar).
  If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4 After test run, please clean the Magnetic Water Filter Set ⑨ and Water Filter Set ㉚. Reinstall it after finish cleaning.

13.2.8.1 Check Water Flow of Water Circuit

Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]

13.2.8.2 Reset Overload Protector ⑫

Overload Protector ⑫ a serves the safety purpose to prevent the water over heating. When the Overload Protector ⑫ a trip at high water temperature, take below steps to reset it.
1 Take out the cover.
2 Use a test pen to push the centre button gently in order to reset the Overload Protector ⑫.
3 Fix the cover to the original fixing condition.

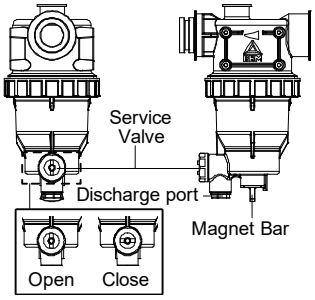


13.2.9 Maintenance

- In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

13.2.9.1 Maintenance for Magnetic Water Filter Set ⑨

1 Turn OFF power supply.
2 Place a container below Magnetic Water Filter Set ⑨.
3 Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑨.
4 By using Allen key (8mm), remove the Cap of Discharge Port.
5 By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
6 Reinstall the Cap of Discharge Port and Magnet Bar.
7 Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 13.2.5 for details.)
8 Turn ON power supply.



13.2.9.2 Maintenance for Water Filter Set ㉚

1 Turn OFF power supply.
2 Set the two valves for the Water Filter Set ㉚ to "CLOSE".
3 Drain the Space Heating / Cooling circuit water with set the Pressure Relief Valve ⑮ lever UP, so that water pressure drop below 0.5 bar.
4 Take off the clip, then gently pull out the mesh. Beware of small amount water drain out from it.
5 Clean the mesh with warm water to remove all the stain. Use soft brush if necessary.
6 Reinstall the mesh to the Water Filter Set ㉚ and set back the clip on it.
7 Set the two valves for the Water Filter Set ㉚ to "OPEN".
8 Re-charging the water to Space Heating / Cooling circuit (refer Section 13.2.5 for details.)
9 Turn ON power supply.

13.2.9.3 Maintenance for Safety Relief Valve ㉕

- It is strongly recommended to operate the valve by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

13.2.9.4 Proper Pump Down Procedure

⚠ WARNING

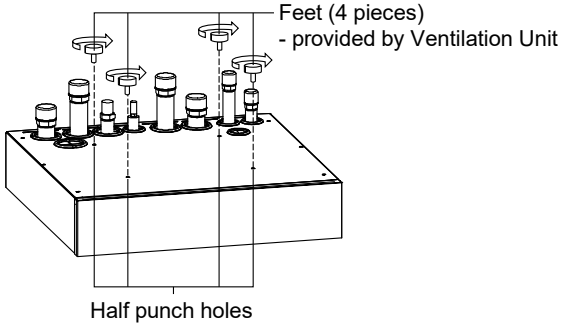
Strictly follow the steps below for proper pump down procedure. Explosion may occur if the steps are not followed as per sequence.
1. When the Tank Unit is not in operation (standby), enter the Service setup menu in the Remote Controller and select Pump down operation to turn it ON. (See APPENDIX for detail)
2. After 10~15 minutes, (after 1 or 2 minutes in case very low ambient temperatures (< 10°C)), fully close 2 way valve on Outdoor Unit.
3. After 3 minutes, fully close 3 way valve on Outdoor Unit.
4. Press the "OFF/ON" switch on the Remote Controller ① to stop pump down operation.
5. Remove the refrigerant piping.

13.2.9.5 Installation of Ventilation Unit on top of Tank Unit (Optional)

- For installation works of Ventilation Unit on top of Tank Unit, refer to the Ventilation Unit Installation Manual.

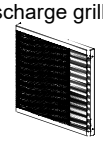
⚠ CAUTION

Before install Ventilation Unit, fix the Feet that provided by Ventilation Unit to the half punch holes on Top Panel of Tank Unit, Otherwise, heavy Ventilation Unit may fall and cause injury.



13.3 Outdoor Unit

Attached accessories

No.	Accessories part	Qty.
1	Drain elbow 	1
2	Rubber cap 	7 (For WH-UDZ03KE5*) 3 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
3	Discharge grille 	1 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
4	Metal plate 	2 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
5	Screw 	8 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)

Optional accessories

No.	Accessories part	Qty.
6	Base Pan Heater CZ-NE2P	1

- It is strongly recommended to install a Base Pan Heater (optional) if the outdoor unit is install in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.
- Applicable Piping Kit (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*) CZ-52F5,7,10BP
- Applicable Piping Kit (For WH-UDZ03KE5*) CZ-4F5,7,10BP

13.3.1 Select the Best Location

- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- For WH-UDZ03KE5*, avoid installation in areas where the ambient temperature may drop below -20°C.
- For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*, avoid installations in areas where the ambient temperature may drop below -25°C.
- Keep the spaces indicated by arrows from wall, ceiling, fence or other obstacles.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- If outdoor unit installed near sea, region with high content of sulphur or oily location (e.g. machinary oil, etc), it lifespan maybe shorten.
- If piping length is over 10 m, additional refrigerant should be added as shown in the table.

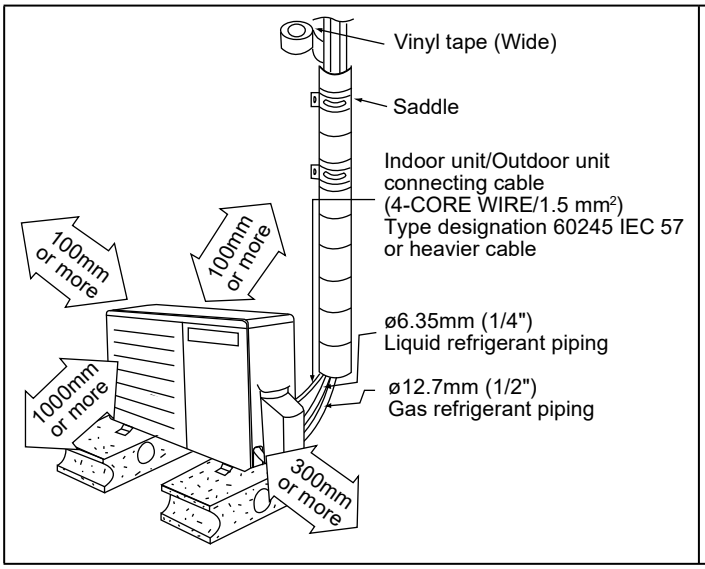
Model	Piping size		Pre-charged Refrigerant (kg)	Rated Length (m)		Max. Elevation (m)	Min. Piping Length (m)	Max. Piping Length (m)	Additional Refrigerant (g/m)
	Gas	Liquid		For Heat Pump Indoor Unit	For Hydromodule + Tank				
WH-UDZ03KE5*	ø12.7mm (1/2")	ø6.35mm (1/4")	0.90	7	7	20	3	25	20
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8")	ø6.35mm (1/4")	1.30	7	7	30	3	50	25

Example: WH-UDZ03KE5*
If piping length is 15m, the quantity of additional refrigerant should be 100g. [(15-10)m x 20 g/m = 100g]

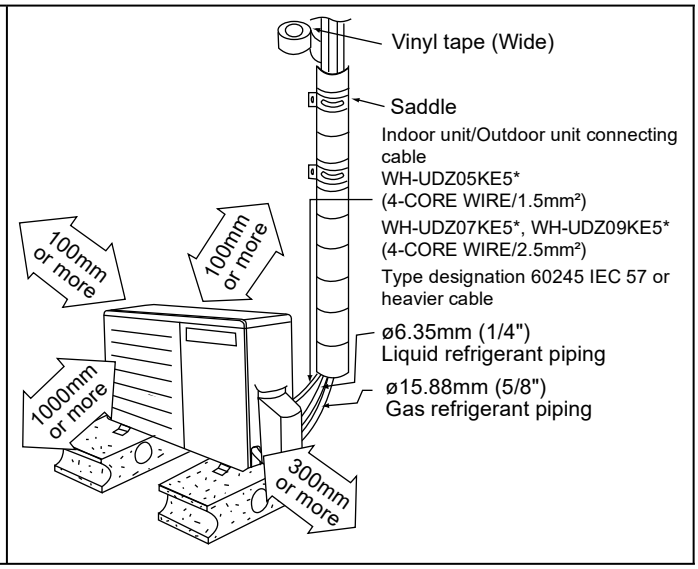
13.3.2 Install the Outdoor Unit

13.3.2.1 Installation Diagram

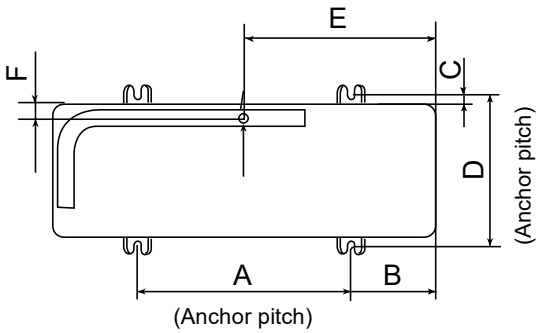
- It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.
- This illustration is for explanation purposes only.



For WH-UDZ03KE5*



For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*



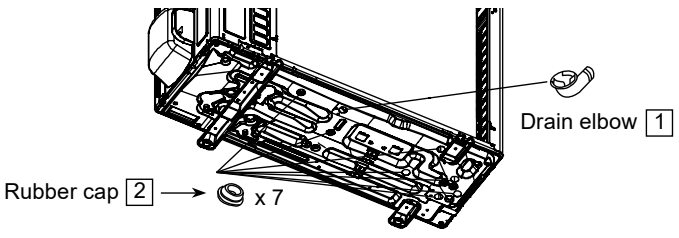
Model	A	B	C	D	E	F
WH-UDZ03KE5*	540	160	20	330	430	46
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	613	130	24	360.5	543	32

(Unit : mm)

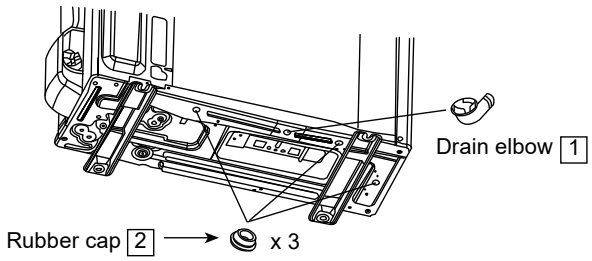
- After selecting the best location, start installation according to Installation Diagram.
  1 Fix the unit on concrete or rigid frame firmly and horizontally by bolt nut (ø10 mm).
  2 When installing at roof, please consider strong wind and earthquake. Please fasten the installation stand firmly with bolt or nails.

13.3.2.2 Disposal of Outdoor Unit Drain Water

- When a Drain elbow [1] is used, please ensure to follow below:
  - the unit should be placed on a stand which is taller than 50 mm.
  - cover the ø20mm holes with Rubber cap [2] (refer to illustration below).
  - use a tray (field supply) when necessary to dispose the outdoor unit drain water.
- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow [1] and Rubber cap [2], for the drain water freezes and the fan will not rotate.



WH-UDZ03KE5*



WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*

13.3.3 Install the Discharge Grille

- This section is for WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5* only.
- Make sure to install the outdoor unit on concrete or rigid frame by bold nut before install the discharge grille 3.
- Hook the metal plate 4 on the cabinet front plate at 2 locations and tighten with screw 5. (Figure a)
- Hook the discharge grille 3 to the cabinet front plate at 4 locations. (Figure b)
- Tighten the discharge grille to the cabinet front plate at 6 locations with screw 5. (Figure c)

⚠ CAUTION
Make sure to install the discharge grille to the outdoor unit before powering ON to protect againts a rotating fan

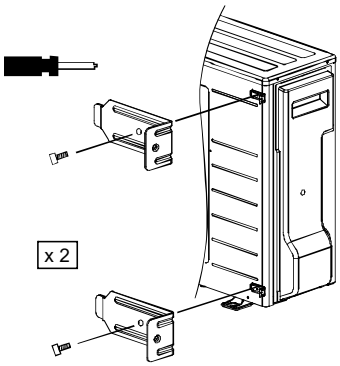


Figure a

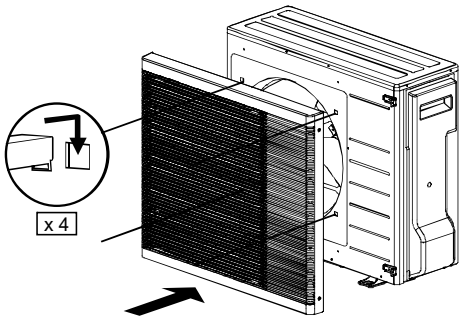


Figure b

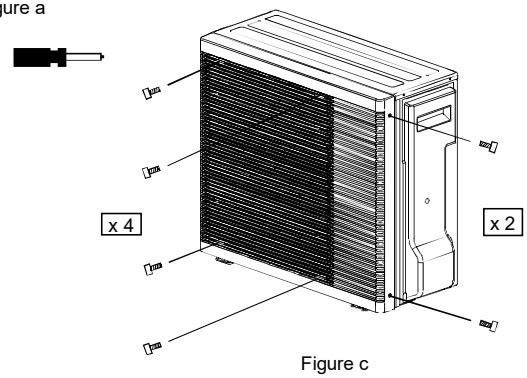
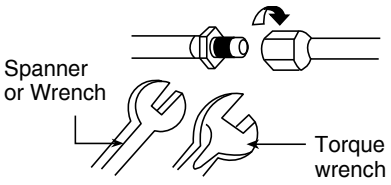


Figure c

13.3.4 Connecting the Piping

13.3.4.1 Connecting the Piping to Outdoor Unit

Decide piping length and then cut by using pipe cutter. Remove burrs from cut edge. Make flare after inserting the flare nut (locate at valve) onto the copper pipe. Align center of piping to valves and then tighten with torque wrench to the specified torque as stated in the table.

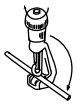


Be sure to use two spanners to tighten.
(If the nuts are overtightened, it may cause the flares to break or leak.)

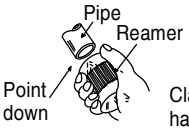
Model	Piping size (Torque)	
	Gas	Liquid
WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]

13.3.4.1.1 Cutting and Flaring the Piping

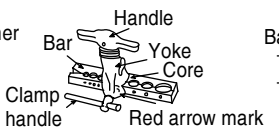
1 Please cut using pipe cutter and then remove the burrs.
2 Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3 Please make flare after inserting the flare nut onto the copper pipes.



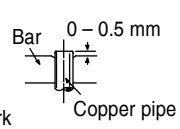
1. To cut

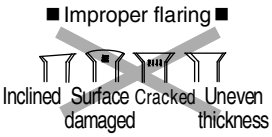


2. To remove burrs



3. To flare



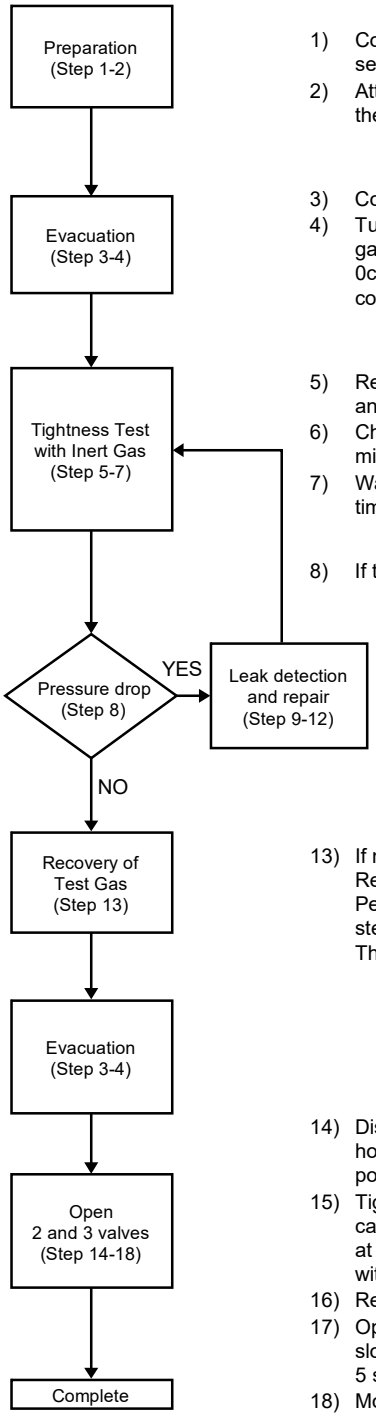


When properly flared, the internal surface of the flare will evenly shine and be of even thickness. Since the flare part comes into contact with the connections, carefully check the flare finish.

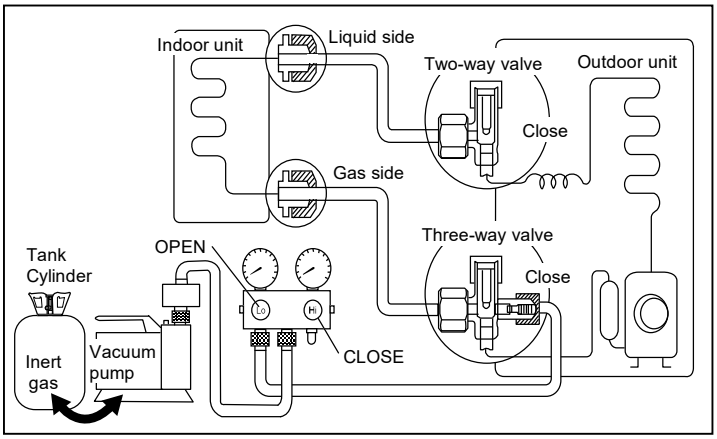
13.3.5 Air Tightness Test on the Refrigerating System

⃠	Do not purge the air with refrigerants but use a vacuum pump to vacuum the installation.
❗	There is no extra refrigerant in the outdoor unit for air purging.

- Before system is charged with refrigerant and before the refrigerating system is put into operation, below site test procedure and acceptance criteria shall be vertified by the certified technicians, and/or the installer.
- Be sure to check whole system for gas leakage.



1) Connect a charging hose with a push pin to the Low side of a charging set and the service port of the 3-way valve.
2) Attach the gauge manifold set correctly and tightly. Make sure that both valves of the manifold gauge (low pressure and high pressure) is in close position.
3) Connect the center hose of the manifold gauge to a vacuum pump.
4) Turn on the power switch of the vacuum pump, then turn open the low side manifold gauge valve and make sure that the needle in the gauge moves from 0cmHg (0 MPa) to –76 cmHg (–0.1 MPa) or vacuum until 500 microns is achieved. This process continues for approximately ten minutes. Then close the low side manifold gauge valve.
5) Remove the vacuum pump from the centre hose and connect the center hose to cylinder of any applicable inert gas as test gas.
6) Charge test gas into the system and wait until the pressure within the system to reach min. 1.04MPa (10.4barg).
7) Wait and monitor the pressure reading on the gauges. Check if there is any pressure drop. Waiting time depends on the size of the system.
8) If there is any pressure drop, perform step 9-12. If there is no pressure drop, perform step 13.
9) Use Gas Leak Detector to check for leaks. Must use the detection equipment with a sensitivity of 5 grams per year of test gas or better.
10) Move the probe along the Air-to-Water Heatpump system to check for leaks, and mark for repair.
11) Any leak detected and marked shall be repaired.
12) After repair, repeat evacuation steps 3-4 and tightness test steps 5-7. Check the pressure drop as in step 8.
13) If no leak, Recover the test gas. Perform evacuation of steps 3-4. Then proceed to step 14.



14) Disconnect the charging hose from the service port of the 3-way valve.
15) Tighten the service port caps of the 3-way valve at a torque of 18 N•m with a torque wrench.
16) Remove the valve caps of both of the 2-way valve and 3-way valve.
17) Open both of the valves, using a hexagonal wrench (4mm). It is recommended to allow refrigerant slowly flow into the refrigerant system to prevent refrigerant freezing. Slightly open 2-way valve for 5 seconds then close the valve. Repeat this action for 3 cycles then fully open the valve.
18) Mount back the valve caps onto the 2-way valve and the 3-way valve to complete this process.

Notes:

Recommended use of any of the following leak detector,

I) Universal Sniffer leak detector
II) Electronic halogen leak detector
III) Ultrasonic Leak Detector

13.3.6 Connect the Cable to the Outdoor Unit

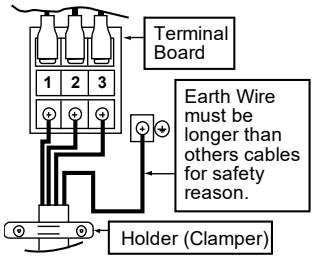
(FOR DETAIL REFER TO WIRING DIAGRAM AT UNIT)

1 Remove the control board cover from the unit by loosening the screw.

2 Connecting cable between indoor unit and outdoor unit shall be approved polychloroprene sheathed flexible cable (see below table), type designation 60245 IEC 57 or heavier cable.

Model	Flexible cable specification
WH-UDZ03KE5*, WH-UDZ05KE5*	4 x (1.5 mm^2)
WH-UDZ07KE5*, WH-UDZ09KE5*	4 x (2.5 mm^2)

Terminals on the indoor unit	1	2	3	⏚
Colour of wires				
Terminals on the outdoor unit	1	2	3	⏚



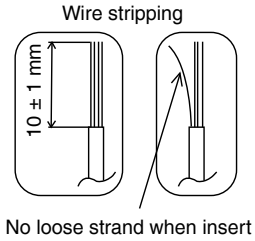
⚠ WARNING

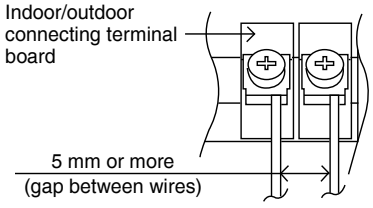
⏚ This equipment must be properly earthed.

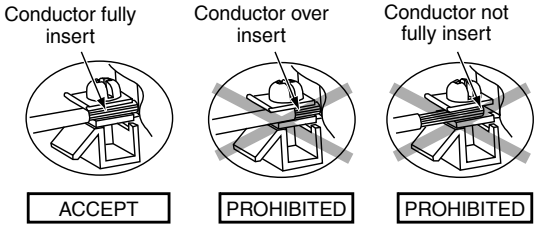
3 Secure the cable onto the control board with the holder (clamper).

4 Attach the control board cover back to the original position with screw.

13.3.6.1 Wire Stripping and Connecting Requirement







13.3.7 Pipe Insulation

1 Please carry out insulation at pipe connection portion as mentioned in Indoor/Outdoor Unit Installation Diagram. Please wrap the insulated piping end to prevent water from going inside the piping.

2 If drain hose or connecting piping is in the room (where dew may form), please increase the insulation by using POLY-E FOAM with thickness 6 mm or above.

⚠ CAUTION

If cleaning of the outdoor unit is necessary during installation or servicing, do not clean the outdoor unit with any hydrocarbon based solvent.

13.4 Appendix

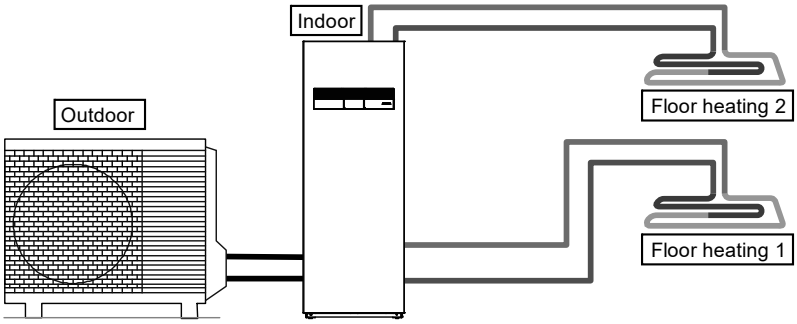
13.4.1 Variation of System

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.

13.4.1.1 Introduce Application Related to Temperature Setting

13.4.1.1.1 Temperature Setting Variation for Heating

1. Remote Controller



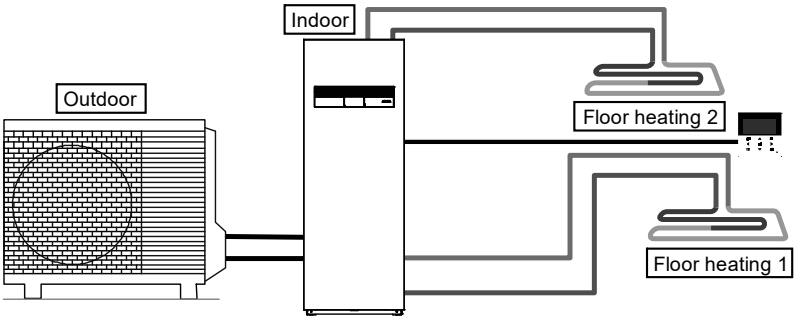
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1 Sensor:
Room
Water temperature
Zone 2 Sensor:
Room
Water temperature

Connect floor heating or radiator directly to the Tank Unit.
Remote controller is installed on Tank Unit.
This is the basic form of 2 zone system.

2. Internal Room Thermostat

Tank Unit receive Room Thermo signal (ON/OFF) from Remote controller to control HP and circulation pump.
There is a build-in thermistor in the remote controller.

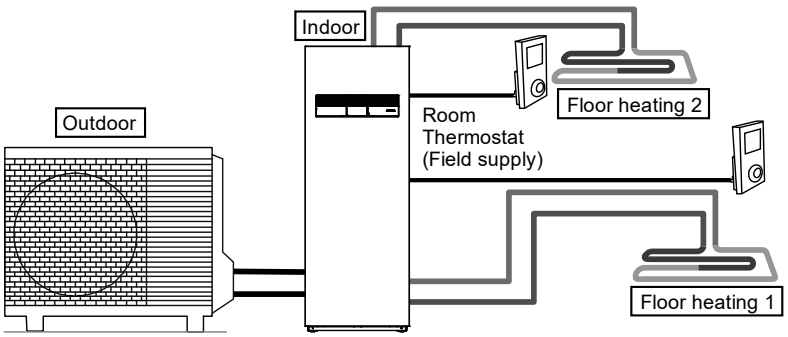


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1 Sensor:
Room thermostat
Internal
Zone 2 Sensor:
Room
Water temperature

Connect floor heating or radiator directly to the Tank Unit.
Remove remote controller from Tank Unit and only can be install it in room 1 and room 2.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat

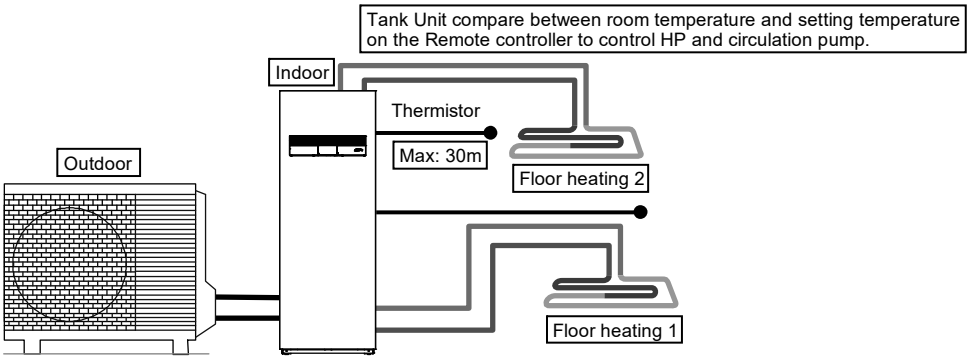


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1 Sensor:
Room thermostat
External
Zone 2 Sensor:
Room thermostat
External

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat to control room temperature.
(NOTE) For this model, external Room Thermostat of Zone 1 must be connected to main indoor PCB only.

4. Room Thermistor



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1 Sensor:
Room thermostat
Zone 2 Sensor:
Room thermostat

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor to control room temperature.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, shift compensation curve according to the thermo ON/OFF situation.

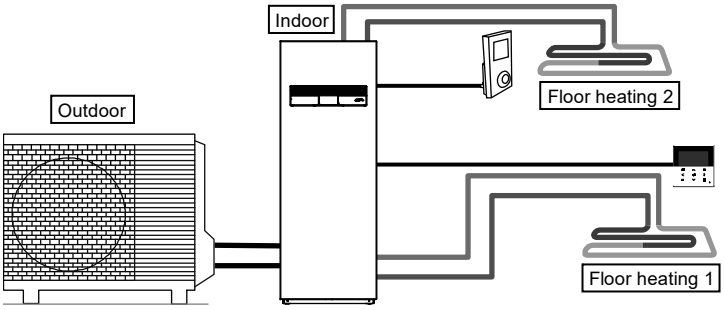
- (Example) If room temperature increasing speed is;
  very slow → shift up the compensation curve
  very fast → shift down the compensation curve

(NOTE)

- Zone 1 should always set with higher water temperature than zone 2. If not set correctly, water temperature flow into zone 1 water circuit may higher than the set temperature.
- Please adjust flow rate of zone 1 and zone 2 to be in balance by setting zone 1 only and zone 2 only water flow to equivalent water flow. If it is not adjusted correctly, it may affects the performance. (eg. If zone 2 pump flow rate is high, there is possibility that no hot water flowing to zone 1)
  a) Zone 1 only water flow can be adjusted and check through Service Setup (Pump Max Speed)
  b) Zone 2 only water flow can be adjusted by knot of pump and check through Maintenance Menu (Actuator Check) -> Activate Zone 2 pump and Switch Zone 2 Mixing Valve to “+” direction.
- For this model, external room thermistor of Zone 1 must be connected to main indoor PCB only.

13.4.1.1.2 Examples of Installations

Floor heating 1 + Floor heating 2



Setting of remote controller

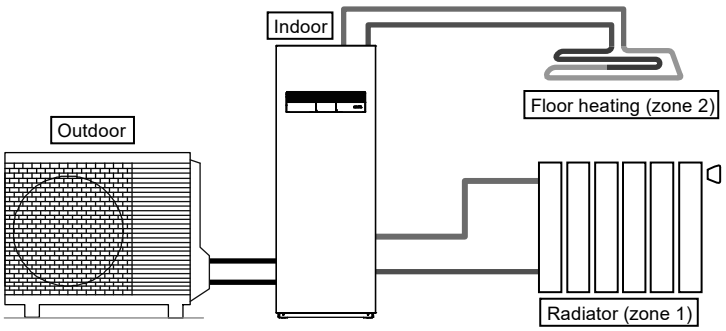
Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor - 2 Zone system
Zone 1:Sensor
Room thermostat
Internal

Zone 2:Sensor
Room
Room thermostat
(External)

Connect two floor heating circuit through indoor as shown in the figure.
Remove remote controller from Tank Unit, install it in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.

Floor heating + Radiator



Setting of remote controller

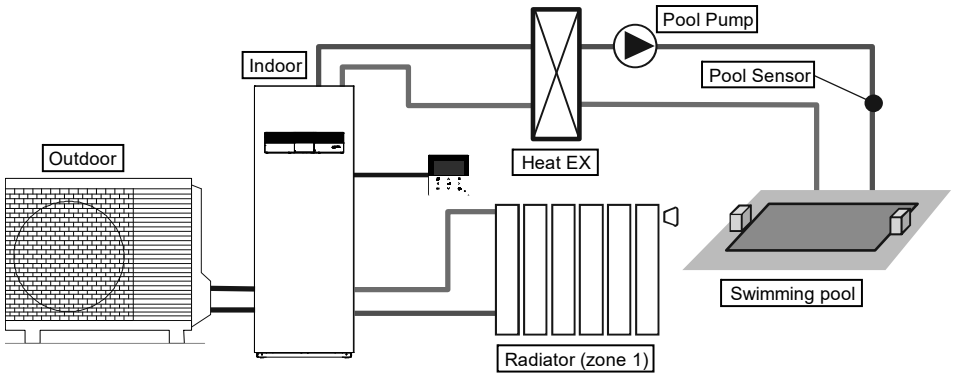
Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor- 2 Zone system
Zone 1:Sensor
Water temperature

Zone 2:Sensor
Room
Water temperature

Connect floor heating or radiator to 2 circuits through indoor.
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
(Radiator recommended to connect as zone 1 circuit for higher water circulation temperature)

Floor heating + Swimming pool



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor - 2 Zone system
Zone 1:Sensor
Room thermostat
Internal

Zone 2
Swimming pool
ΔT

Connect floor heating and swimming pool to 2 circuits through buffer Tank Unit as shown in figure.
Install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Remove remote controller from Tank Unit and install in room where floor heating is installed.
Circulation water temperature of floor heating and swimming pool can be set independently.

❇ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.

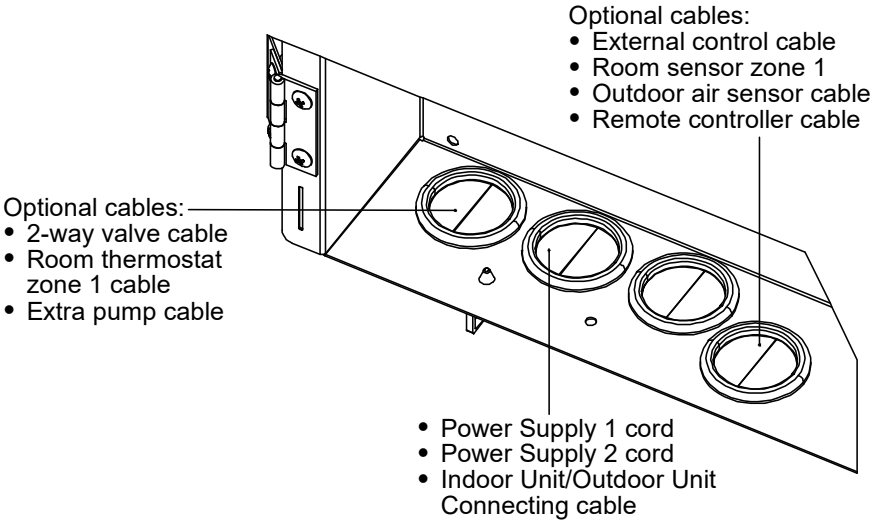
(NOTE)

- Zone 1 should always set with higher water temperature than zone 2. If not set correctly, water temperature flow into zone 1 water circuit may higher than the set temperature.
- Please adjust flow rate of zone 1 and zone 2 to be in balance. If it not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirm by “Actuator Check” from maintenance menu.

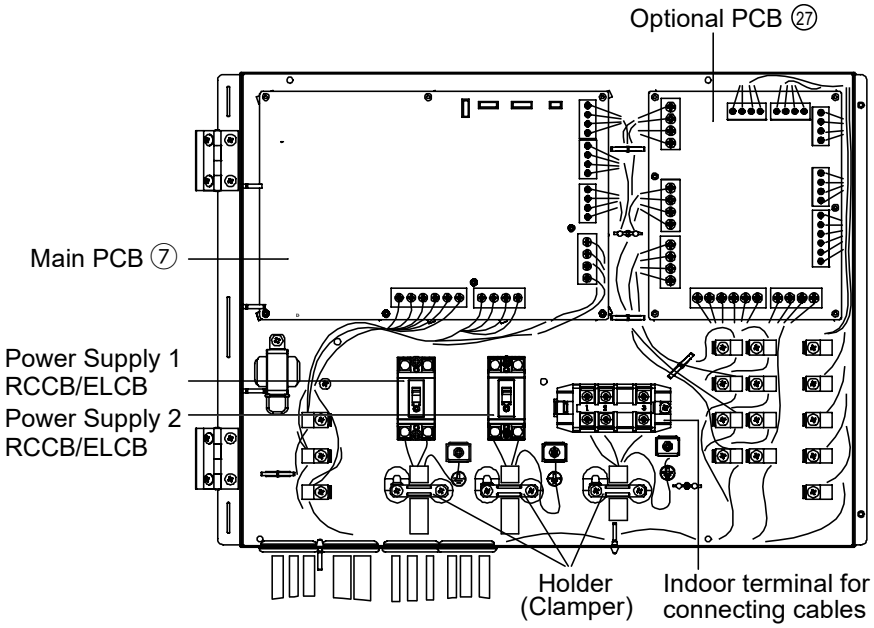
13.4.2 How to Fix Cable

13.4.2.1 Connecting with External Device (Optional)

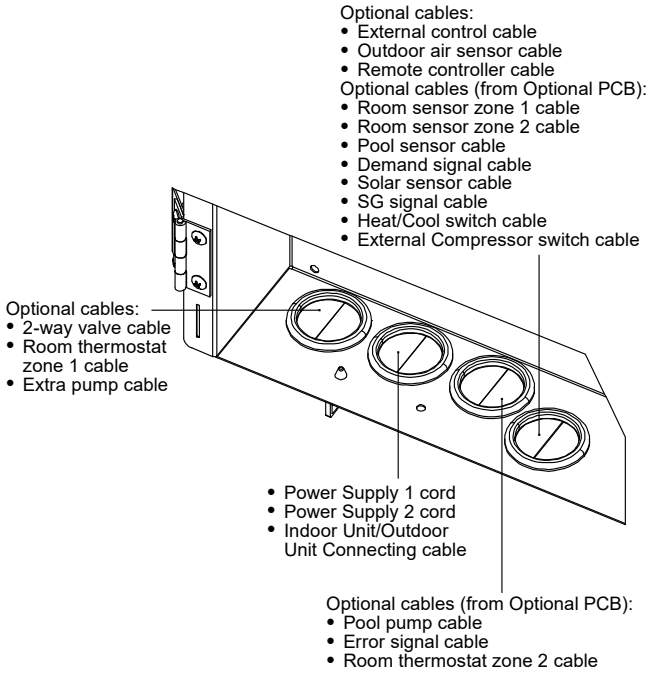
- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB ⑦
  1 Two-way valve shall be spring and electronic type, refer to “Field Supply Accessories” table for details. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
  *note: - Two-way Valve shall be CE marking compliance component.
  - Maximum load for the valve is 9.8VA.
  2 Room thermostat zone 1 cable must be (4 or 3 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
  3 Extra pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  4 External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  *note: - Switch used shall be CE compliance component.
  - Maximum operating current shall be less than $3A_{rms}$.
  5 Room sensor zone 1 and outdoor air sensor cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.



- For connection to Optional PCB ㉗
  1 Pool pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  2 Room thermostat zone 2 cable shall be (4 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
  3 Room sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  4 Pool water sensor cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  5 Demand signal cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  6 SG signal cable shall be (3 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  7 Heat/Cool switch cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  8 External compressor switch cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide the optional cables and power supply cord
(view without internal wiring)

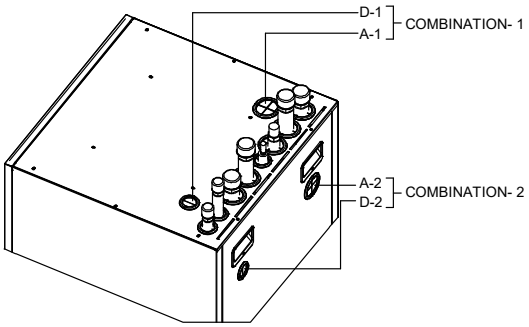


Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

13.4.2.2 Guide Optional Cables and Power Supply Cords to Bushings

⚠ CAUTION
Wire guiding shall be free from hot surfaces. Else, cable insulator damage and electrical shock may happen.
Wire ways shall be smooth and free from sharp edges. Else, cable insulator damage and electrical shock may happen.

■ Use either “COMBINATION-1” or “COMBINATION-2” for guiding Optional Cables and Power Supply Cords to Bushings.



■ A-1 and A-2 Bushings are for:

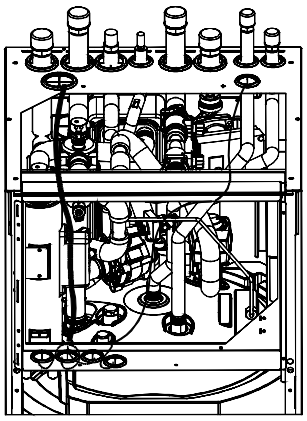
- Power Supply 1 cord
- Power Supply 2 cord
- Indoor Unit/Outdoor Unit Connecting cable
- Room thermostat zone 1 cable
- Room thermostat zone 2 cable
- 2-way valve cable
- Extra pump cable

■ D-1 and D-2 Bushings are for:

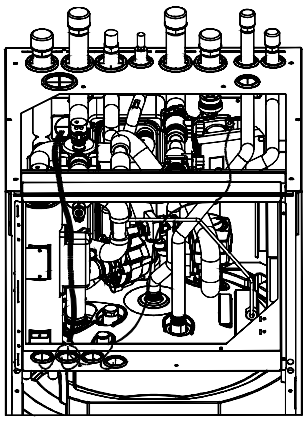
- External control cable
- Outdoor air sensor cable
- Remote controller cable
- Room sensor zone 1 cable
- Room sensor zone 2 cable
- Pool sensor cable
- Demand signal cable
- SG signal cable
- Heat/Cool switch cable
- External Compressor switch cable

■ Ensure all sensor cables are not touching with Front Panel ⑯

■ Guide the wiring inside the unit like below figure.
Once all wiring work done, tie the cable / cord with the banding strap (field supply), to prevent them touching with hot surfaces such as Heater Assembly, bare copper pipes and etc.



Wiring for “COMBINATION-1”



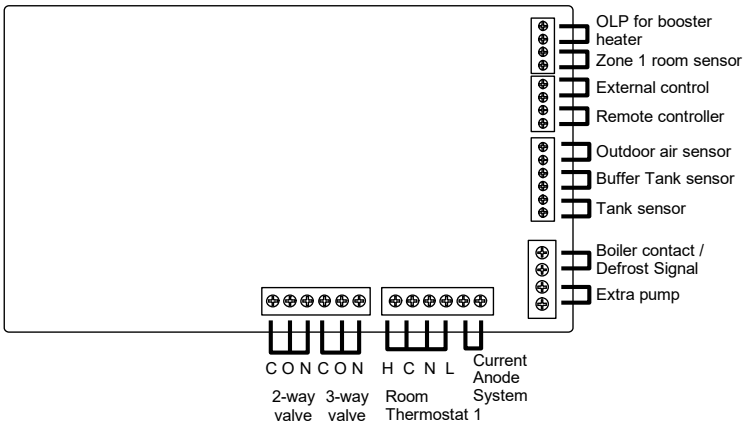
Wiring for “COMBINATION-2”

13.4.2.3 Connecting Cables Length

When connecting cables between Tank Unit and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Room thermostat	50
Extra pump	50
Pool pump	50
Pump	50
External control	50
Room sensor	30
Outdoor air sensor	30
Pool water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

13.4.2.4 Connection of the Main PCB



- Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

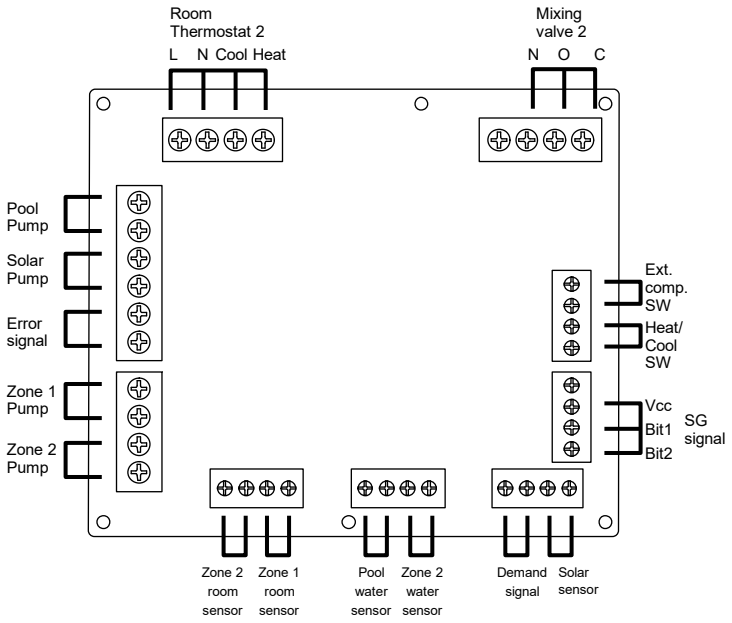
- Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)	AC230V, 12 VA
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)	AC230V, 12 VA
Extra pump	AC230V (Used when Tank Unit pump capacity is insufficient)	AC 230V, 0.6 A max

- Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT ※It does not work when using the Optional PCB
Outdoor air sensor	AW-A2W-TSOD (Total cable length shall be 30m or less)

13.4.2.5 Connection of Optional PCB ㉗



Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.OFF, Short=Comp.ON (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

- Outputs

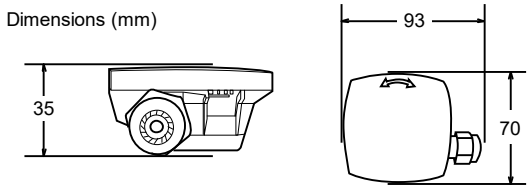
Pool pump	AC230V	AC 230V, 0.6 A max

- Thermistor inputs

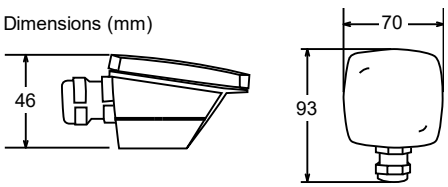
Zone room sensor	PAW-A2W-TSRT
Pool water sensor	PAW-A2W-TSHC

13.4.2.6 Recommended External Device Specification

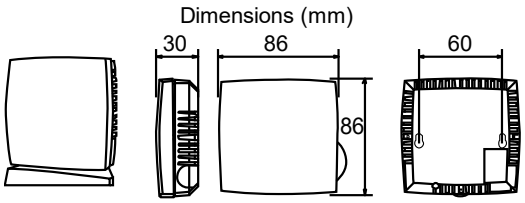
- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.
  1 Pool water sensor: PAW-A2W-TSHC
  Use to detect the water temperature of the control zone.
  Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



  2 Outdoor sensor: PAW-A2W-TSOD
  If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
  In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



  3 Room sensor: PAW-A2W- TSRT
  Install the room temperature sensor to the room which requires room temperature control.

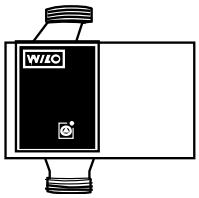


  4 Please refer to the table below for sensor characteristic of the sensors mentioned above.

Temperature (°C)	Resistance (kΩ)
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo

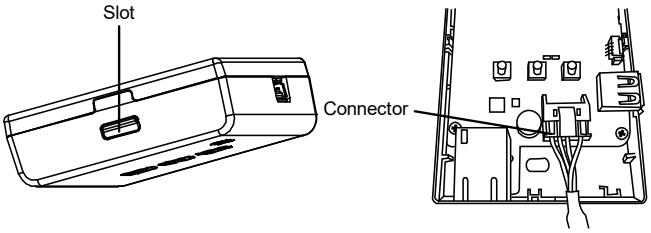


> ⚠ **WARNING**
>
> This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

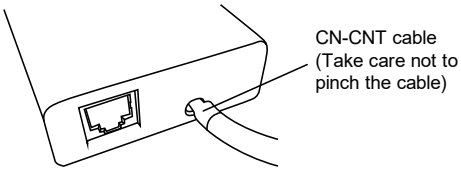
13.4.2.7 Network Adaptor 6 Installation (Optional)

1 **Remove the Control Board Cover ⑤, then connect the cable included with this adaptor to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Tank Unit so that there is no pinching.
   - If an Optional PCB has been installed in the Tank Unit, connect to the CN-CNT connector of the Optional PCB.

2 **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**

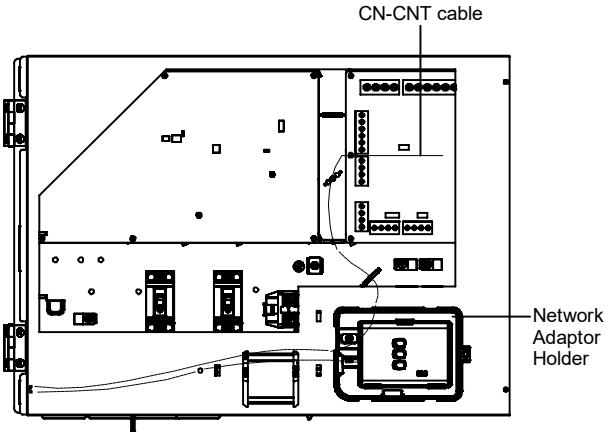


3 **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**



4 **Fix the Network Adaptor 6 to Network Adaptor Holder.**
  Guide the cable as shown in the diagram so that external forces cannot act on the connector in the adaptor.

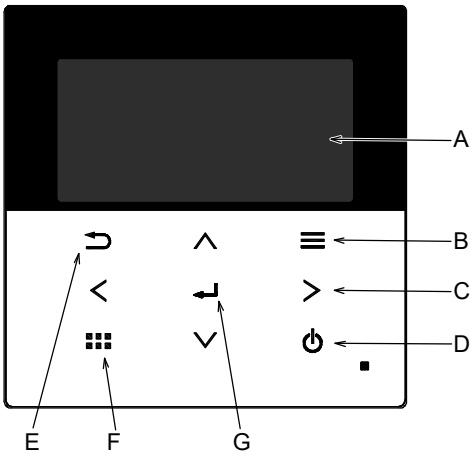
  Connection examples:



13.4.3 System Installation

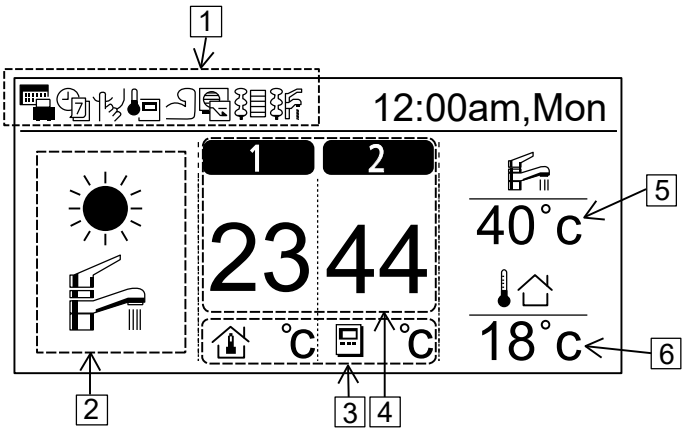
13.4.3.1 Remote Controller Outline

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.



Name	Function
A: Main screen	Display information
B: Menu	Open/Close main menu
C: Triangle (Move)	Select or change item
D: Operate	Start/Stop operation
E: Back	Back to previous item
F: Quick Menu	Open/Close Quick menu
G: OK	Confirm

LCD Display
(Actual - Dark background with white icons)



	Name	Function
1:	Function icon	Display set function/status

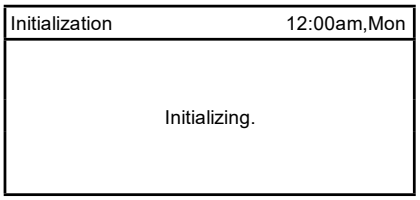
Holiday mode | Demand control
Weekly timer | Room heater
Quiet mode | Tank heater
Remote controller room thermostat
Powerful mode

2:	Mode	Display set mode/current status of mode

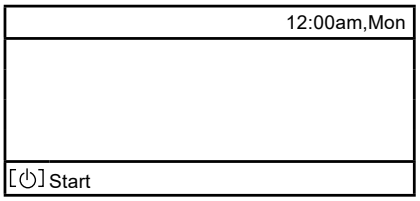
Heating | Cooling
Auto | Hot water supply | Auto heating | Auto cooling
Heat pump operating

3:	Temp setting	Set room temp / Compensation curve / Set direct water temp / Set pool temp / Room Thermostat →External
4:	Display Heat temp	Display current heating temperature (it is set temperature when enclosed by line)
5:	Display tank temp	Display current tank temperature (it is set temperature when enclosed by line)
6:	Outdoor temp	Display outdoor temp

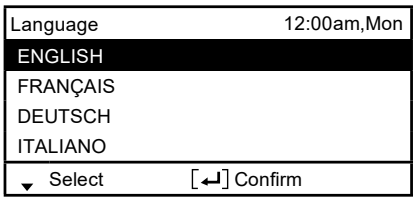
First time of power ON (Start of installation)



When power is ON, firstly initialization screen appears (10 sec)



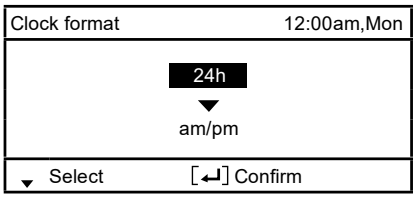
When initialization screen ends, it turns to normal screen.



When any button is pressed, language setting screen appears.
(NOTE) If initial setting is not performed, it does not go into menu.

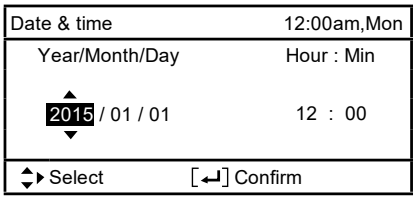
When there are two remote controllers installed from the beginning, the first remote controller to set and confirm language will be recognised as main remote controller.

Set language & confirm



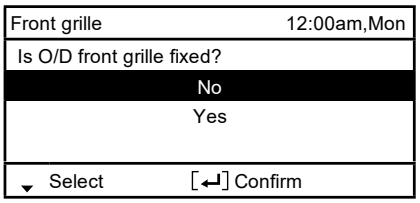
When language is set, setting screen of time display appears (24h/am/pm)

Set time display & confirm

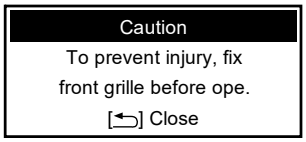


YY/MM/DD/Time setup screen appears

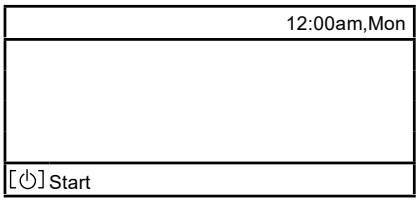
Set YY/MM/DD/Time & confirm



If set No & confirm, a caution message will be displayed to ensure outdoor front grille is installed before proceed to operate the unit.

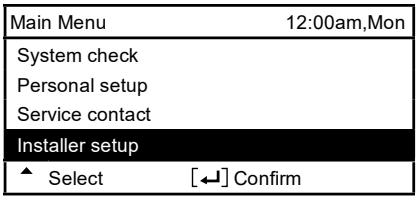


Set Yes & confirm if outdoor front grille has been installed



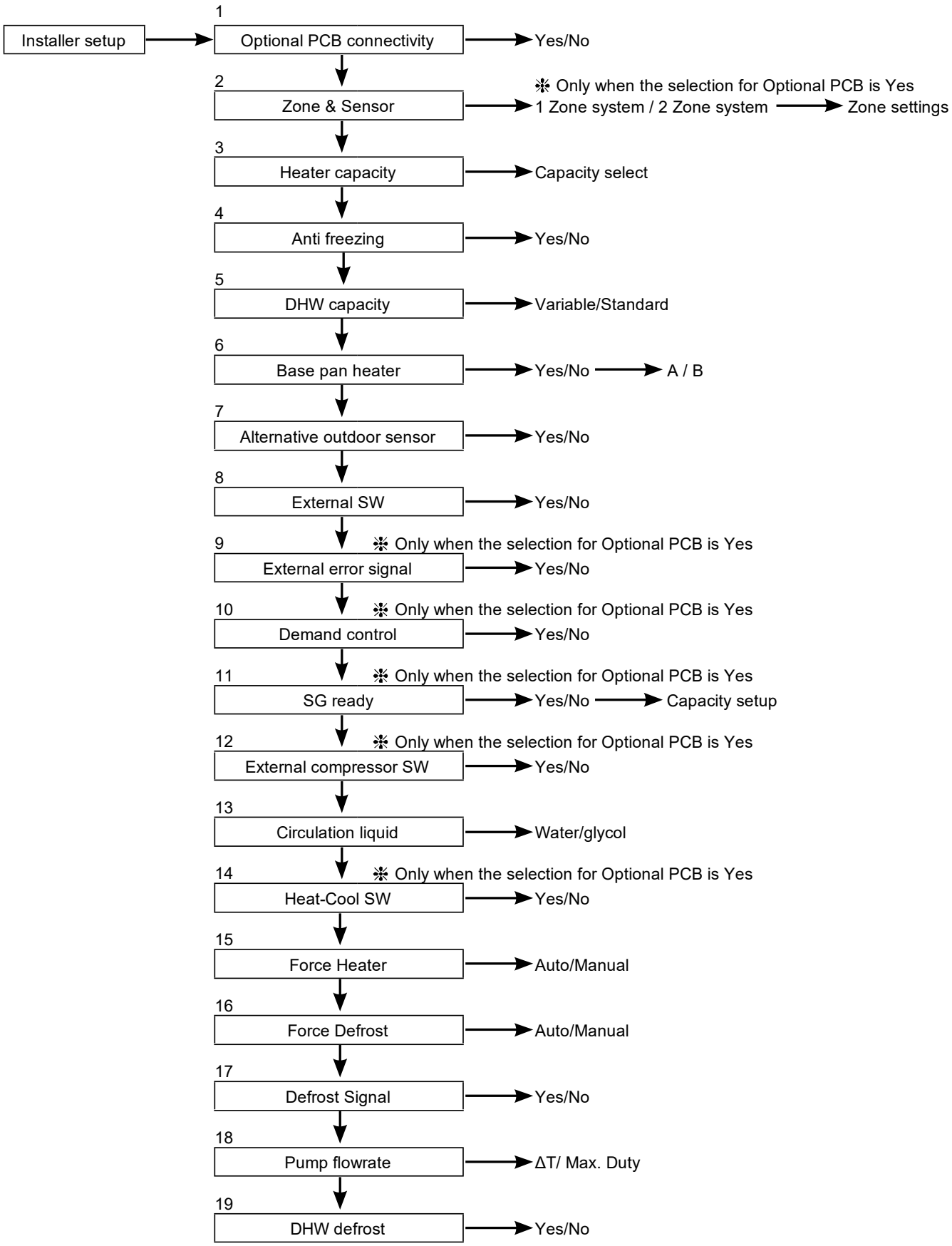
Back to initial screen

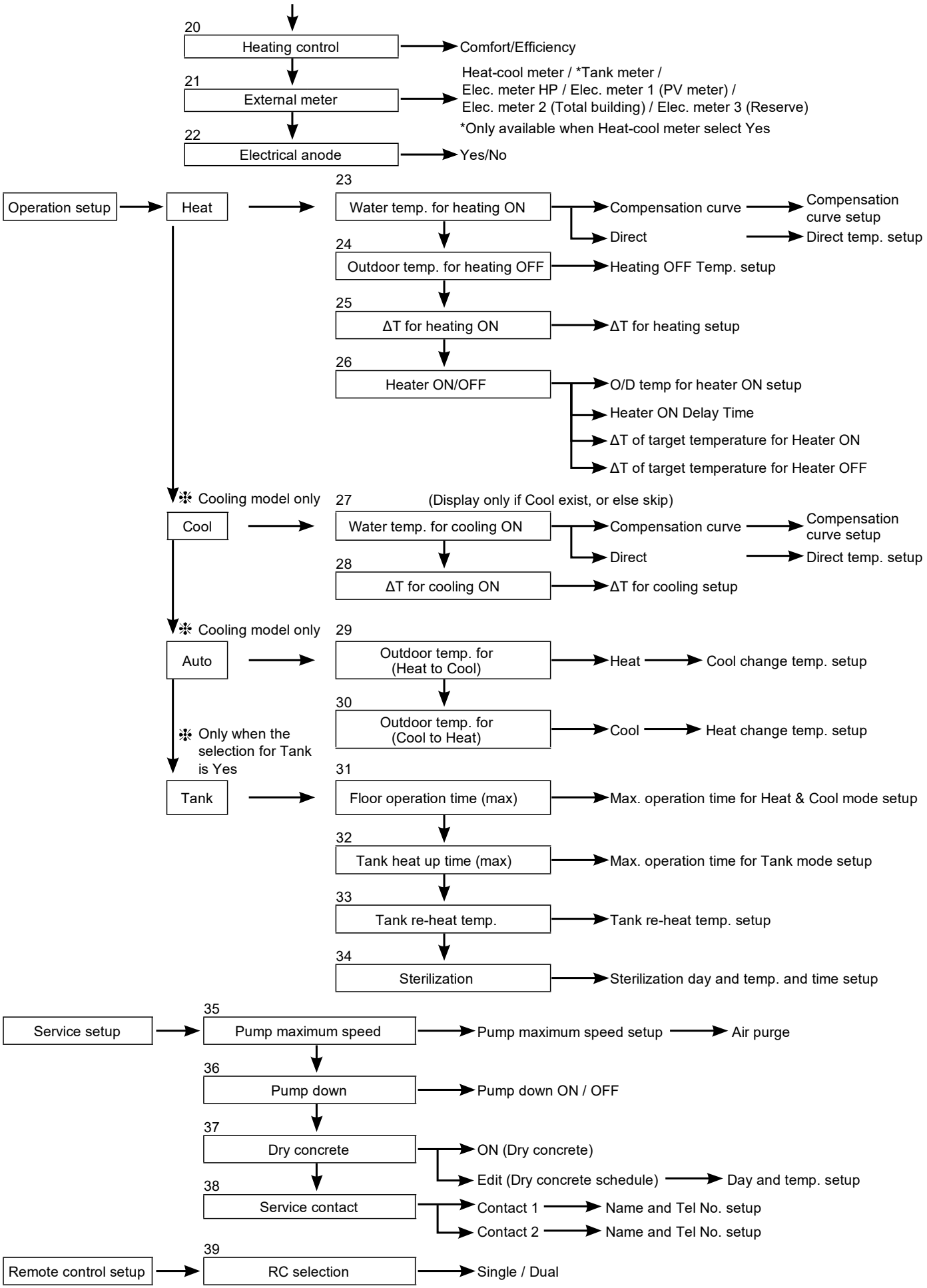
Press menu, select Installer setup



Confirm to go into Installer setup

13.4.3.2 Installer Setup



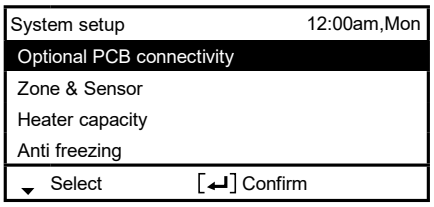

20
Heating control
Comfort/Efficiency
21
External meter
Heat-cool meter / *Tank meter /
Elec. meter HP / Elec. meter 1 (PV meter) /
Elec. meter 2 (Total building) / Elec. meter 3 (Reserve)
*Only available when Heat-cool meter select Yes
22
Electrical anode
Yes/No
Operation setup
Heat
23
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
24
Outdoor temp. for heating OFF
Heating OFF Temp. setup
25
ΔT for heating ON
ΔT for heating setup
26
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
Cooling model only
Cool
27
(Display only if Cool exist, or else skip)
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
28
ΔT for cooling ON
ΔT for cooling setup
Cooling model only
Auto
29
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
30
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
Only when the selection for Tank is Yes
Tank
31
Floor operation time (max)
Max. operation time for Heat & Cool mode setup
32
Tank heat up time (max)
Max. operation time for Tank mode setup
33
Tank re-heat temp.
Tank re-heat temp. setup
34
Sterilization
Sterilization day and temp. and time setup
Service setup
35
Pump maximum speed
Pump maximum speed setup
Air purge
36
Pump down
Pump down ON / OFF
37
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
38
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup
Remote control setup
39
RC selection
Single / Dual

13.4.3.3 System Setup

1. Optional PCB connectivity | Initial setting: Yes

If function below is necessary, please purchase and install Optional PCB.
Please select Yes after installing Optional PCB.

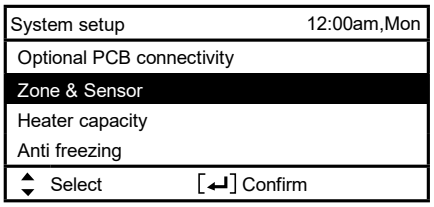
- 2-zone control
- Pool
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

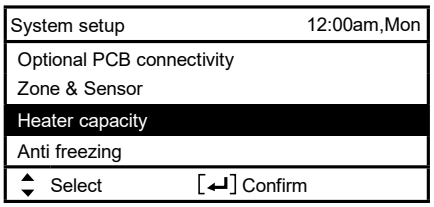
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

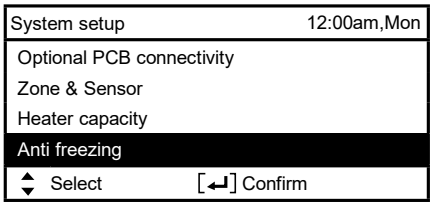
(NOTE) There are models which cannot select Heater capacity.



4. Anti freezing | Initial setting: Yes

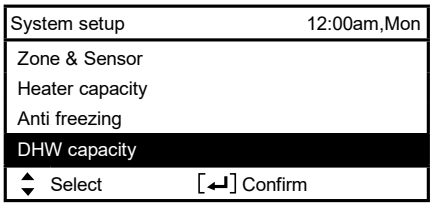
Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



5. DHW Capacity | Initial setting: Variable

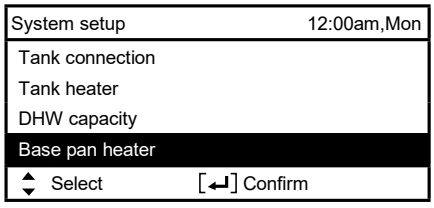
Variable DHW capacity setting normally run with efficient boiling which is energy saving heating. But while hot water usage high and tank water temperature low, variable DHW mode will run with fast heat up which heat up the tank with high heating capacity.
If standard DHW capacity setting is selected, heat pump run with heating rated capacity at tank heat up operation.



6. Base pan heater | Initial setting: No

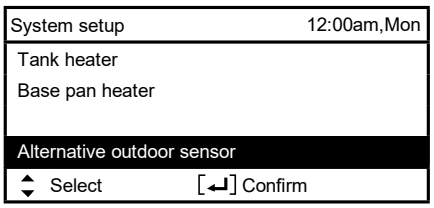
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



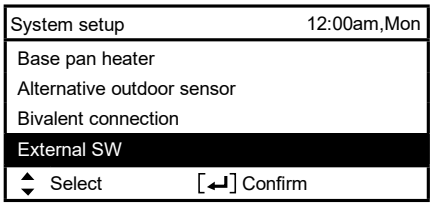
7. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



8. External SW | Initial setting: No

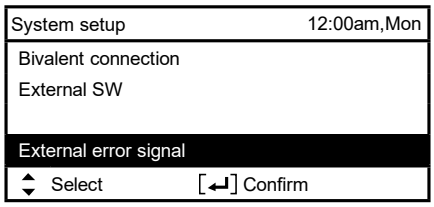
Able to turn ON/OFF the operation by external switch.



9. External error signal | Initial setting: No

Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

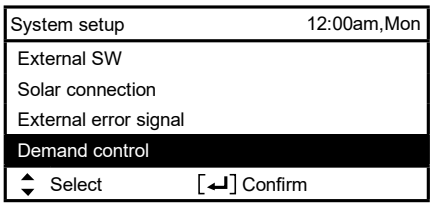
(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off "close" from the display, error signal will still remain ON.



10. Demand control | Initial setting: No

Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.



Analog input [v]	Rate [%]		
0.0	not activate		
0.1 ~ 0.6			
0.7	10	not activate	
0.8			
0.9 ~ 1.1	10		
1.2	15	10	
1.3			
1.4 ~ 1.6	15		
1.7	20	15	
1.8			
1.9 ~ 2.1	20		
2.2	25	20	
2.3			
2.4 ~ 2.6	25		
2.7	30	25	
2.8			
2.9 ~ 3.1	30		
3.2	35	30	
3.3			
3.4 ~ 3.6	35		
3.7	40	35	
3.8			

Analog input [v]	Rate [%]		
3.9 ~ 4.1	40		
4.2	45	40	
4.3			
4.4 ~ 4.6	45		
4.7	50	45	
4.8			
4.9 ~ 5.1	50		
5.2	55	50	
5.3			
5.4 ~ 5.6	55		
5.7	60	55	
5.8			
5.9 ~ 6.1	60		
6.2	65	60	
6.3			
6.4 ~ 6.6	65		
6.7	70	65	
6.8			
6.9 ~ 7.1	70		
7.2	75	70	
7.3			

Analog input [v]	Rate [%]		
7.4 ~ 7.6	75		
7.7	80	75	
7.8			
7.9 ~ 8.1	80		
8.2	85	80	
8.3			
8.4 ~ 8.6	85		
8.7	90	85	
8.8			
8.9 ~ 9.1	90		
9.2	95	90	
9.3			
9.4 ~ 9.6	95		
9.7	100	95	
9.8			
9.9 ~	100		

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

11. SG ready | Initial setting: No

Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

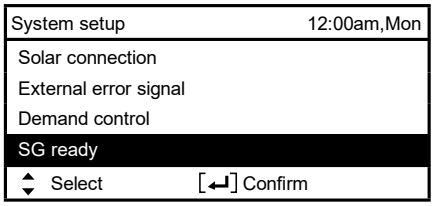
SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2

Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

} Set by SG ready setting of remote controller

(NOTE) Does not display if there is no Optional PCB.

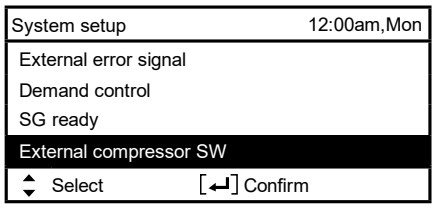


12. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

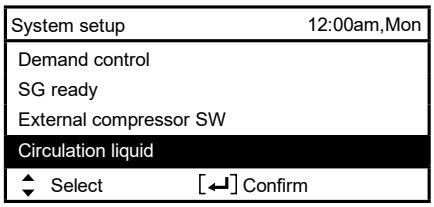


13. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

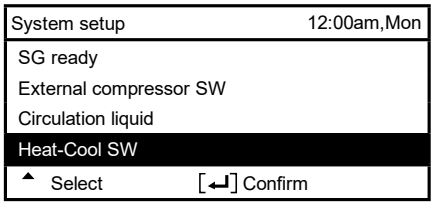


14. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

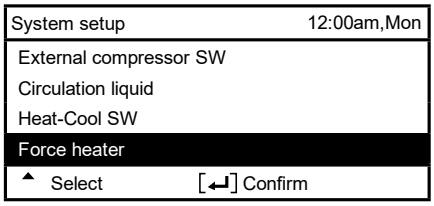


15. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

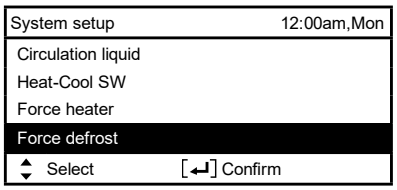
Heater source will ON during force heater mode.



16. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

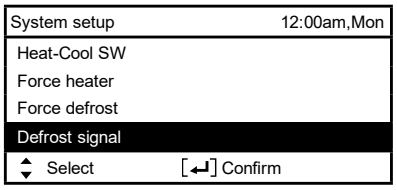
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition. (Even auto is selected, user still can turn on force defrost through quick menu)



17. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

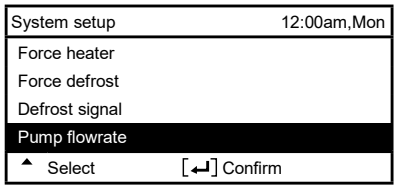
When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).



18. Pump flowrate | Initial setting: ΔT

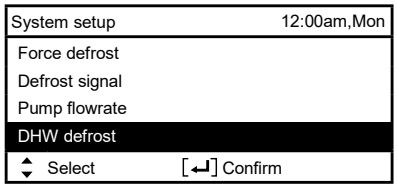
If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



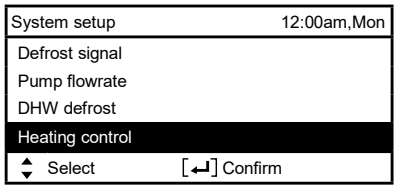
19. DHW defrost | Initial setting: Yes

When DHW defrost set to YES, hot water of domestic hot water tank will be used during defrost cycle.
When DHW defrost set to NO, hot water of floor heating circuit will be used during defrost cycle.



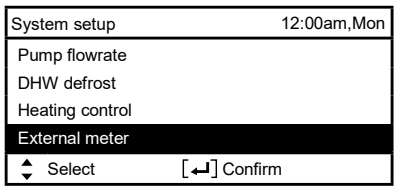
20. Heating control | Initial setting : Comfort

There are two modes to select for compressor frequency control : Comfort or Efficiency.
When set to Comfort mode, compressor will run at zone limit maximum frequency to reach set temperature faster.
When set to Efficiency mode, compressor will run at part load frequency at initial stage for energy saving.



21. External meter

Initial setting : [Heat-cool meter : No]
[Tank meter : No] *only available when Heat-cool meter select Yes
[Elec. meter HP : No]
[Elec. meter 1 (PV meter) : No]
[Elec. meter 2 (Total building) : No]
[Elec. meter 3 (Reserve) : No]



There are two systems for generation meter connection : one generation meter system (Heat-cool meter) or two generation meter system (Heat-cool meter and Tank meter)
Both systems can provide all generation data of heating, cooling and DHW directly from external meter.

If Heat-cool meter is set to Yes, it will read from external meter for heat pump's energy generation data during heating, cooling and DHW operation *1.
If Heat-cool meter is set to No, it will base on unit's calculation for heat pump's energy generation data during heating, cooling and DHW operation.
If Tank meter is set to Yes, it will read from external meter for heat pump's energy generation data during DHW operation *1.

If Elec. meter HP is set to Yes, it will read from external meter for heat pump's energy consumption data.
If Elec. meter HP is set to No, it will base on unit's calculation for heat pump's energy consumption data.

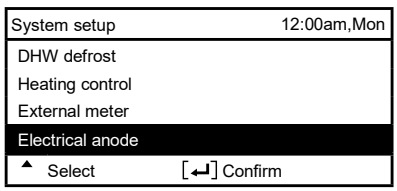
If Elec. meter 1 (PV meter) is set to Yes, it will read from external meter for energy generation data of solar system and display it on Cloud system.
If Elec. meter 2 (Building) is set to Yes, it will read from external meter for energy consumption data of the building and display it on Cloud system.
If Elec. meter 3 (Reserve) is set to Yes, it will read from external meter for energy consumption data obtained from reserved electricity meter and display it on Cloud system.

*1 Set Heat-cool meter to Yes and set Tank meter to No when 1 generation meter system is installed.
Set Heat-cool meter to Yes and set Tank meter to Yes when 2 generation meter system is installed.

Remark : Elec. meter HP refers to Electricity meter that measures Heat Pump unit's consumption.
Elec. meter 1 / 2 / 3 refers to Electricity meter no. 1 / no. 2 / no. 3

22. Electrical anode | Initial setting : No

When Electrical anode set to YES, anode will be turned on.
When Electrical anode set to NO, anode will not be turned on.



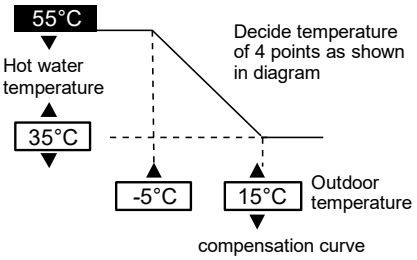
13.4.3.4 Operation Setup

Heat

23. Water temp. for heating ON | Initial setting: compensation curve

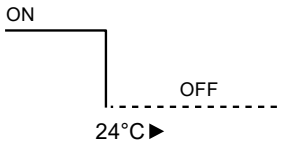
Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



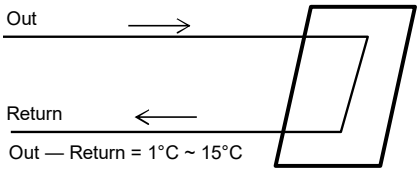
24. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C



25. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

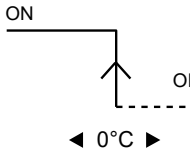


26. Heater ON/OFF

a. Outdoor temp. for heater ON | Initial setting: 0°C

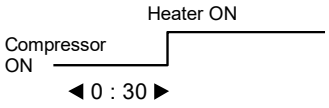
Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.



b. Heater ON delay time | Initial setting: 30 minutes

Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

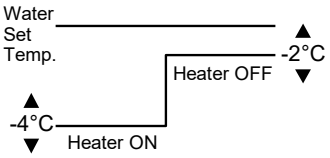


c. Heater ON: ΔT of target Temp. | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp. | Initial setting: -2°C

Set water temperature for heater to turn off at heat mode.
Setting range is -8°C ~ 0°C

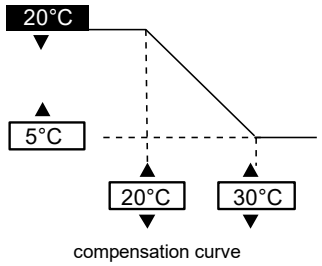


Cool

27. Water temp. for cooling ON | Initial setting: compensation curve

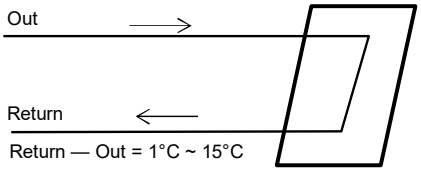
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



28. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

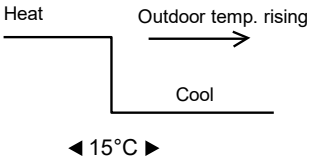


Auto

29. Outdoor temp. for (Heat to Cool) | Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 5°C ~ 25°C

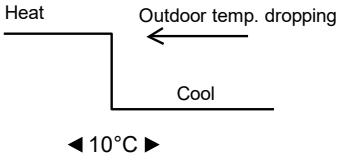
Timing of judgement is every 1 hour



30. Outdoor temp. for (Cool to Heat) | Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 25°C

Timing of judgement is every 1 hour

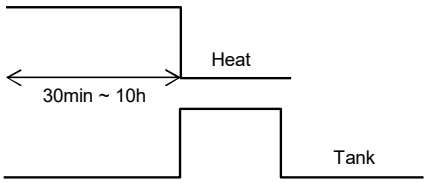


Tank

31. Floor operation time (max) | Initial setting: 8h

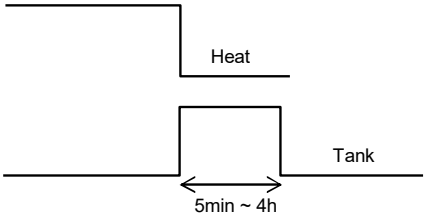
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



32. Tank heat up time (max) | Initial setting: 60min

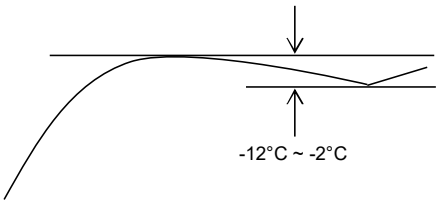
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



33. Tank re-heat temp. | Initial setting: -8°C

Set temp to perform reboil of tank water.
(When boiled by heat pump only, (51°C – Tank re-heat temp) shall become max temp.)

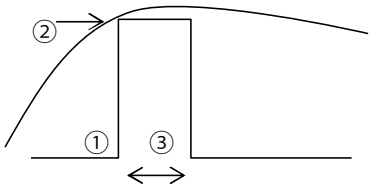
Setting range is -12°C ~ -2°C



34. Sterilization | Initial setting: 65°C 10min

Set timer to perform sterilization.
① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~75°C ❊ If use back-up heater, it is 65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.

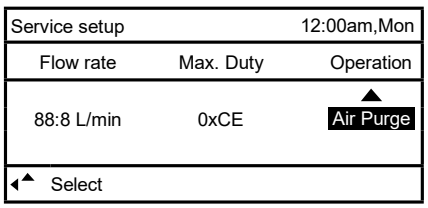


13.4.3.5 Service Setup

35. Pump maximum speed | Initial setting: Depend on model

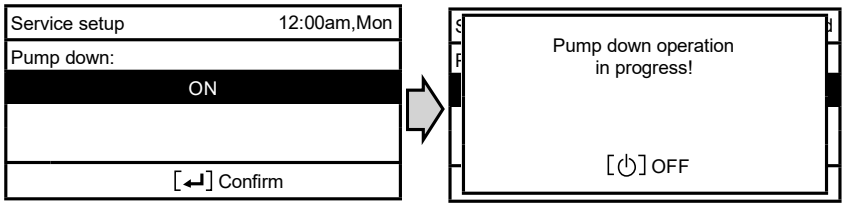
Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.



36. Pump down

Operate pump down operation

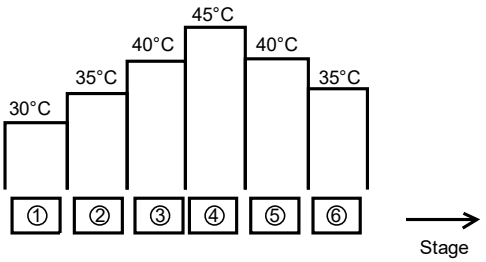


37. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

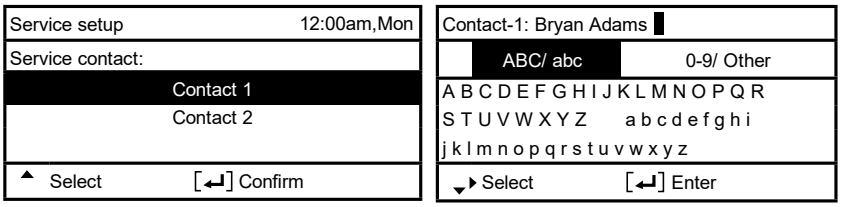
When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



38. Service contact

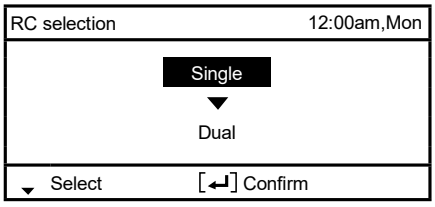
Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)



13.4.3.6 Remote Control Setup

39. RC selection | Initial setting : Single

Set to "Single" when only one remote controller is installed.
Set to "Dual" when two remote controllers are installed.



13.5 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

13.5.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

13.5.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it. In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer. When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

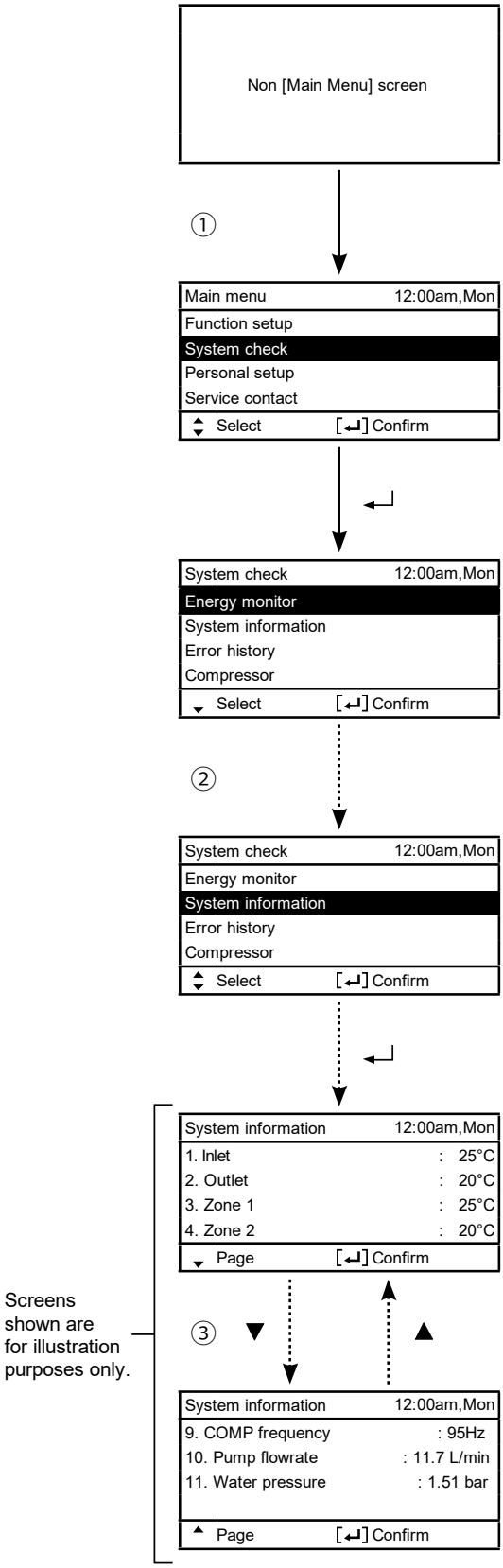
③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

13.5.3 Check Water Pressure from Remote Controller

1. Press ☰ SW and scroll to "System check".
2. Press ↵ and scroll to "System information".
3. Press ↵ and search for "Water pressure".



14. Installation Instruction

(For WH-ADC0309K3E5UK only)

14.1 Indoor Floor Area Requirement

- **If the total refrigerant charge in the system is <1.84 kg**, no additional minimum floor area is required.
- **If the total refrigerant charge in the system is ≥1.84 kg**, additional minimum floor area requirements is complied as described below:

Symbol	Description	Unit
m_c	Total refrigerant charge in system	kg
m_{max}	Maximum refrigerant charge allowed	kg
m_{excess}	m_c - m_{max}	kg
H	Installation height	m
VA_{min}	Minimum ventilation opening area	cm^2

Total refrigerant charge in system, m_c (kg)
= Pre-charged refrigerant amount in unit (kg)
+ Additional refrigerant amount after installation (kg)

A) Determine **Maximum refrigerant charge allowed, m_{max}**

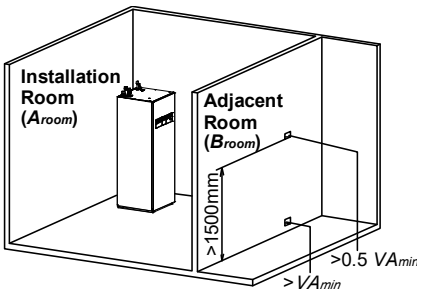
1. Calculate Installation Room Area, A_{room}.
2. Based on Table I, select m_{max} which corresponds to the calculated A_{room} value.
3. If $m_{max} \geq m_c$, the unit can be installed in the installation room with the specified installation height (***H*=600mm**) in Table I and without additional room area or any additional ventilation.
4. Else, proceed to B) and C).

B) Determine **Total Floor Area of A_{room} and B_{room} compliance to $A_{min\ total}$**

1. Calculate the B_{room} area adjacent to the A_{room}.
2. Determine the $A_{min\ total}$ based on the Total Refrigerant Charge, m_c from Table II.
3. The total floor area of both A_{room} and B_{room} must exceed $A_{min\ total}$.

C) Determine **Minimum Venting Opening Area**, **VA_{min}** for natural ventilation

1. From Table III, calculate m_{excess}.
2. Then determine VA_{min} corresponding to the calculated m_{excess} for natural ventilation between A_{room} and B_{room}.
3. The unit can be installed at specific room only when the following conditions are fulfilled:
   - Two permanent openings (cannot be closed), one at bottom, another at top, for ventilation purposes are made between A_{room} and B_{room}.
   - **Bottom opening:** - Must comply to the minimum area requirement of VA_{min}.
     - Opening must be located ≤300mm from the floor.
     - At least 50% of required opening area must be ≤200mm from the floor.
     - The bottom of the opening shall not be higher than the point of release when the unit is installed and must be situated ≤100mm above the floor.
     - Must be as close as possible to the floor and lower than H.
   - **Top opening:** - The total size of the Top opening must be more than 50% of VA_{min}.
     - Opening must be located ≥1500mm above the floor.



   - The height of the openings must more than 20mm.
   - A direct ventilation opening to outside is **NOT** encouraged for ventilation opening (the user can block the opening when it is cold).
   - The value of H is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.

Table I - Maximum refrigerant charge allowed in a room

A_{room} (m²)	Maximum refrigerant charge in a room (m_{max}) (kg)
	***H*=0.6m**
1	0.138
2	0.276
3	0.414
4	0.553
5	0.691
6	0.829
7	0.907
8	0.970
9	1.028
10	1.084
11	1.137
12	1.187
13	1.236
14	1.283
15	1.328
16	1.371
17	1.413
18	1.454
19	1.494
20	1.533
21	1.571
22	1.608
23	1.644
24	1.679
25	1.714
26	1.748
27	1.781
28	1.814
29	1.846
30	1.877
31	1.909
32	1.939
33	1.969
34	1.999
35	2.028
36	2.057
37	2.085
38	2.113
39	2.141
40	2.168
41	2.195
42	2.221
43	2.248
44	2.274
45	2.299

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate A_{room} values, the value that corresponds to the lower A_{room} value from the table is considered.
  Example:
  For A_{room} = 10.5 m², the value that corresponds to "A_{room} = 10 m²" is considered.

Table II - Minimum floor area

m_c (kg)	Minimum floor area ($A_{min\ total}$ (m²))
	***H*=0.6m**
1.84	28.81
1.86	29.44
1.88	30.08
1.90	30.72
1.92	31.37
1.94	32.03
1.96	32.70
1.98	33.37
2.00	34.04
2.02	34.73
2.04	35.42
2.06	36.12
2.08	36.82
2.10	37.53
2.12	38.25
2.14	38.98
2.16	39.71
2.18	40.45
2.20	41.19
2.22	41.94
2.24	42.70
2.26	43.47
2.28	44.24
2.30	45.02

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate m_c values, the value that corresponds to the higher m_c value from the table is considered.
  Example:
  If m_c = 1.85 kg, the value that corresponds to "m_c = 1.86 kg" is considered.
- Systems with total refrigerant charge lower than 1.84 kg are not subjected to any room area requirements.
- Charges above 2.30 kg are not allowed in the unit.

Table III - Minimum venting opening area for natural ventilation

m_c (kg)	m_{max} (kg)	m_{excess} (kg) = m_c - m_{max}	Minimum venting opening area (VA_{min}) (cm²)
			***H*=0.6m**
2.3	0.1	2.20	890
2.3	0.3	2.00	809
2.3	0.5	1.80	728
2.3	0.7	1.60	647
2.3	0.9	1.40	583
2.3	1.1	1.20	552
2.3	1.3	1.00	500
2.3	1.5	0.80	430
2.3	1.7	0.60	343
2.3	1.9	0.40	242
2.3	2.1	0.20	127
2.3	2.3	0.00	0

- For *H* values lower than 0.6m, the value of *H* is considered as 0.6m to comply to IEC 60335-2-40:2018 Clause GG2.
- For intermediate m_{excess} values, the value that corresponds to the higher m_{excess} value from the table is considered.
  Example:
  m_{excess} = 1.45 kg, the value that corresponds to "m_{excess} = 1.6 kg" is considered.

Attached accessories

No.	Accessory part	Qty.	No.	Accessory part	Qty.
1	Adjustable Feet 	4	3	Drain Elbow 	1
2	Reducing Adapter 	1	4	Packing 	1

Optional Accessories

No.	Accessories part	Qty.
5	Remote Controller Case	1
6	Network Adaptor (CZ-TAW1B) and Extension Cable (CZ-TAW1-CBL)	1
7	Optional PCB (CZ-NS5P)	1

Field Supply Accessories (Optional)

No.	Part		Model	Specifications	Maker
i	2-way valve kit *Cooling model	Electromotoric Actuator	SFA21/18	AC230V, 12 VA	Siemens
		2-port Valve	VXI46/25		Siemens
ii	Room thermostat	Wired	PAW-A2W-RTWIRED	AC230V	-
		Wireless	PAW-A2W-RTWIRELESS		
iii	Mixing valve	-	167032	AC230V, 6VA	Caleffi
iv	Pump	-	Yonos 25/6	AC 230V, 0.6 A max	Wilo
v	Buffer tank sensor	-	PAW-A2W-TSBU	-	-
vi	Outdoor sensor	-	PAW-A2W-TSOD	-	-
vii	Zone water sensor	-	PAW-A2W-TSHC	-	-
viii	Zone room sensor	-	PAW-A2W-TSRT	-	-
ix	Solar sensor	-	PAW-A2W-TSSO	-	-

- It is recommended to purchase the field supply accessories listed in above table.

14.2 Indoor Unit

14.2.1 Select the Best Location

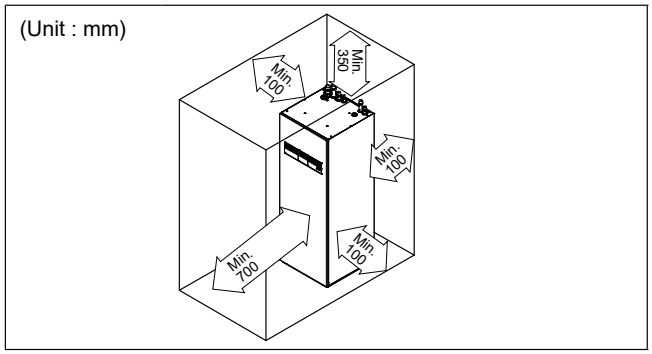
Before choosing the installation site, obtain user approval.

- Install the Tank Unit in indoors with frost free weather proof location only.
- Must install on a flat horizontal and solid hard surface.
- There should not be any heat source or steam near the Tank Unit.
- A place where air circulation in the room is good.
- A place where drainage can be easily done (e.g. Utility room).
- A place where Tank Unit's operation noise will not cause discomfort to the user.
- A place where Tank Unit is far from door way.
- A place where accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated below from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- Secure the Tank Unit to prevent it being knocked over accidentally or during earthquakes.

Please avoid installations which expose the Tank Unit to any of the following conditions:

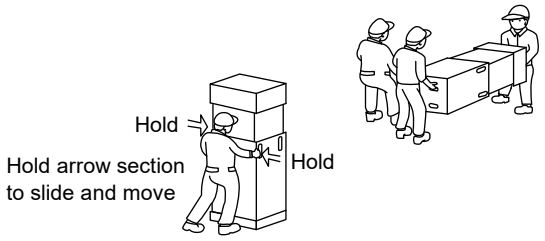
- Extraordinary environment conditions; installation in frost or exposure to unfavorable weather conditions.
- Voltage input exceeding the specified voltage.

14.2.1.1 Required Space for Installation



14.2.1.2 Transport and Handling

- Be careful during transportating the unit so that it is not damaged by impact.
- Only remove the packaging material once it has reached it is desired installation location.
- It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.
- The Tank Unit can be transported either in vertical or horizontal.
  - If it transported in horizontal, make sure Front of packaging material (printed with "FRONT") must facing upwards.
  - If it transported in vertical, use the hand holes on sides, slide and move to the desired location.
- Fix the Adjustable Feet [1], if the Tank unit installed on a uneven surface.

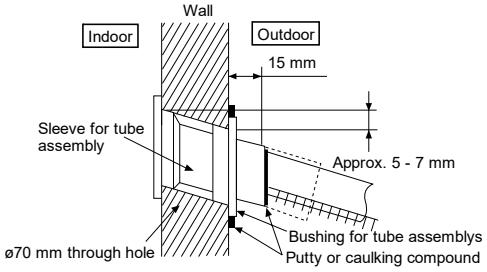


14.2.2 To Drill a Hole in the Wall and Install a Sleeve of Piping

1 Make a Ø70 mm through hole.
2 Insert the piping sleeve to the hole.
3 Fix the bushing to the sleeve.
4 Cut the sleeve until it extrudes about 15 mm from the wall.

⚠ CAUTION
When the wall is hollow, please be sure to use the sleeve for tube assembly to prevent dangers caused by mice biting the connection cable.

5 Finish by sealing the sleeve with putty or caulking compound at the final stage.

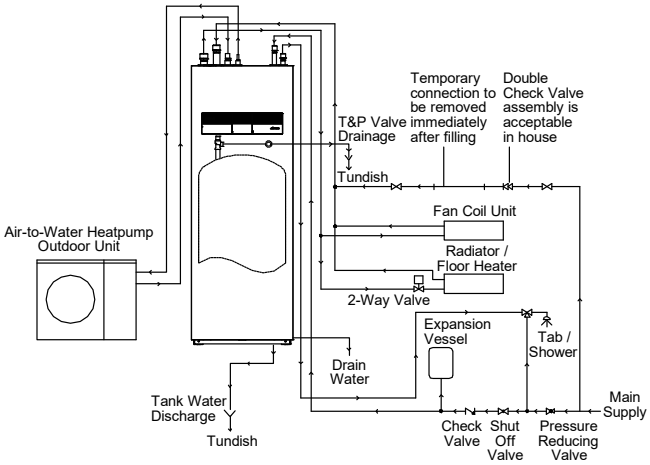


14.2.3 Piping Installation

14.2.3.1 Water Quality Requirement

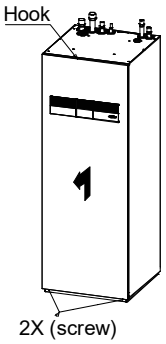
Must use water that complies with European water quality standard 98/83 EC. The lifespan of the Tank Unit will be shorter if groundwater (include spring water and well water) is used.
The Tank Unit shall not be used with the tap water containing contaminants such as salt, acid, and other impurities which may corrode the tank and its component.

14.2.3.2 Typical Piping Installation



14.2.3.3 Access to Internal Components

⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

Open and Close Front Plate ⑱

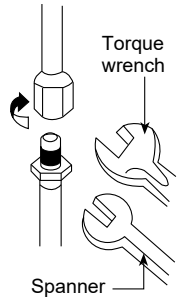
1 Remove the 2 mounting screws of Front Plate ⑱.
2 Slide it upwards to unhook the Front Plate ⑱ hook.
3 Reverse above steps 1~2 for close it.

14.2.3.4 Refrigerant Piping Installation

This Tank Unit is designed for combination with Panasonic Air-to-Water Heat Pump Outdoor Unit. If Outdoor Unit from other manufacturer are being used in combination with Panasonic Tank Unit, optimum operation and reliability of the system is not guaranteed. Thus warranty cannot be given in such case.

1 Connect Tank Unit to Air-to-Water Heatpump Outdoor Unit with correct piping size. Use Reducing Adapter [2] for Outdoor Unit WH-UDZ03KE5* Refrigerant Gas ⓔ piping connection.

Model		Piping size (Torque)		Use Reducing Adapter [2]
Tank Unit	Outdoor Unit	Gas	Liquid	
WH-ADC0309K3E5UK	WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]	Yes
	WH-UDZ05KE5*, WH-UDZ07KE5*, WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]	No



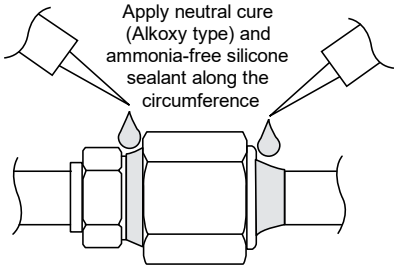
⚠ CAUTION
Do not overtighten, overtightening may cause gas leakage.
Do not pull and push refrigerant piping excessively, deformed pipe may cause refrigerant leak.

2 Please make flare after inserting flare nut (located at joint portion of tube assembly) onto the copper pipe. (In case of using long piping)
3 Do not use pipe wrench to open refrigerant piping. Flare nut may be broken and cause leakage. Use proper spanner or ring wrench.
4 Connect the piping:
   - Align the centre of piping and sufficiently tighten the flare nut with fingers.
   - Further tighten the flare nut with torque wrench in specified torque as stated in the table.

<u>Additional Precautions For R32 Models when connecting by flaring at indoor side</u>

❗ Ensure to do the re-flaring of pipes before connecting to units to avoid leaking.
❗ Connections made between components of refrigerant system shall be accessible for ease of maintenance.

Seal sufficiently the flare nut (both gas and liquid sides) with neutral cure (Alkoxy type) & ammonia-free silicone sealant and insulation material to avoid the gas leak caused by freezing.



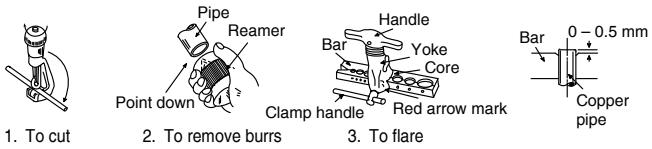
Neutral cure (Alkoxy type) & ammonia-free silicone sealant is only to be applied after pressure testing and cleaning up by following instructions of sealant, only to the outside of the connection. The aim is to prevent moisture from entering the connection joint and possible occurrence of freezing. Curing sealant will take some time. Make sure sealant will not peel off when wrapping the insulation.

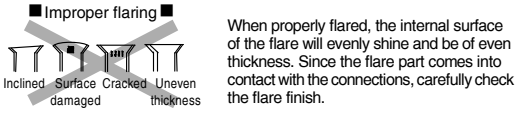
14.2.3.4.1 Checking for Gas Leakage

- Check for leakage of gas after air purging.
- See the in the installation manual for the outdoor.

14.2.3.5 Cutting and Flaring the Piping

1 Please cut using pipe cutter and then remove the burrs.
2 Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3 Please make flare after inserting the flare nut onto the copper pipes.

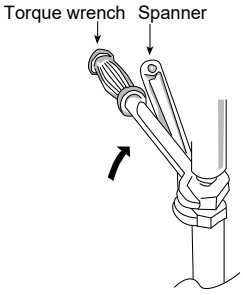




14.2.3.6 Water Piping Installation

- Please engage a licensed water circuit installer to install this water circuit.
- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not use worn out tube or detachable hose-set.
- Do not apply excessive force to pipes that may damage the pipes.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Further tighten the nuts with torque wrench in specified torque as stated in the table.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall.
- If non-brass metallic piping is used for installation, make sure to insulate the pipes to prevent galvanic corrosion.
- Do not connect galvanised pipes, this will cause galvanic corrosion.
- Use correct nut for all Tank Unit tube connections and clean all tubes with tap water before installation. See Tube Position Diagram for detail.

Tube Connector	Nut Size	Torque
ⓐ & ⓑ	RP 1¼"	117.6 N•m
ⓒ & ⓓ	RP ¾"	58.8 N•m



⚠ CAUTION

Do not overtighten, overtightening may cause gas leakage.

- Make sure to insulate the water circuit pipes to prevent reduction of heating capacity.
- After installation, check the water leakage condition in connection area during test run.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Protection From Frost:
  If the Tank Unit is being exposed to frost while power supply failure or pump operating failure, drain the system. When water is idle inside the system, freezing up is very likely to happen which could damage the system. Make sure the power supply is turned off before draining. Heater Assembly ⑩ may be damaged under dry heating.
- Corrosion Resistance:
  Duplex stainless steel is naturally corrosion resistant to mains water supply. No specific maintenance is required to maintain this resistance. However, please note that Tank Unit is not guaranteed for use with a private water supply.
- It is recommended to use a tray (field supply) to collect water from the Tank Unit if water leakage occur.

Recommended piping installation sequence:
(a) → (c) → (e) → (f) → (b) → (d)

(A) Space Heating/Cooling Pipework

- Connect Tank Unit Tube Connector ⓐ to outlet connector of Zone 1 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓑ to inlet connector of Zone 1 Panel/Floor heater.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Refer below table for the rated flow rate of each particular Outdoor Unit.

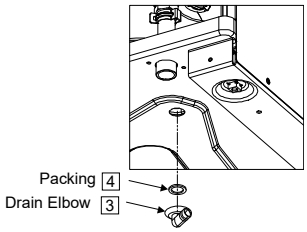
Model		Rated Flow Rate (l/min)	
Tank Unit	Outdoor Unit	Cool	Heat
WH-ADC0309K3E5UK	WH-UDZ03KE5*	9.2	9.2
	WH-UDZ05KE5*	14.3	14.3
	WH-UDZ07KE5*	19.2	20.1
	WH-UDZ09KE5*	23.5	25.8

(B) Domestic Hot Water Tank Pipework

- It's strongly recommended to install an expansion vessel (field supply) in the Domestic Hot Water Tank circuit. Refer Typical Piping Installation section to locate the expansion vessel.
  - Expansion Vessel type and specifications:
    - Size : Not more than ¾"
    - Pre-charge pressure : 0.35MPa (3.5 bars)
- If secondary return circuits are used then an additional expansion vessel may be required.
- In high water pressure or water supply is above 500kPa, please install the Pressure Reducing Valve for water supply. If the pressure higher than that, it might damage the Tank Unit.
- A Pressure Reducing Valve (field supply) and Pressure Relief Valve (field supply) with below specification must be installed along the line of the tube connector ⓒ of Tank Unit. Refer Typical Piping Installation section to locate both of these valves.
  - Pressure Reducing Valve type and specifications:
    - Size : Not more than ¾"
    - Set Pressure : 0.35MPa (3.5 bars)
  - Pressure Relief Valve type and specifications:
    - Size : Not more than ¾"
    - Set Pressure : 0.8MPa (8.0 bars)
- The pressure after pressure reducing valve is less than 0.35MPa (3.5 bars).
- Must connect a faucet to Tank Unit Tube Connector ⓓ and urban water supply, in order to supply water with appropriate temperature for shower or tap usage. Failure to do so might cause scalding.
- Failure to connect the tube appropriately might causing the Tank Unit malfunction.

(C) Drain Elbow and Hose Installation

- Fix the Drain Elbow 3 and Packing 4 to the bottom of Drain Water Hole ⓗ.



- Use inner diameter 17 mm drain hose in the market, fix to Drain Elbow 3.
- This hose must to be installed in a continuously downward direction and in a frost-free environment. Improper drain piping may cause water leakage hence damage the furnitures.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Guide the drain hose to outdoor as illustrated.

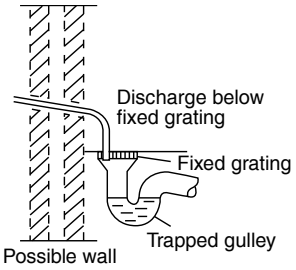


Illustration of guide drain hose to outdoor

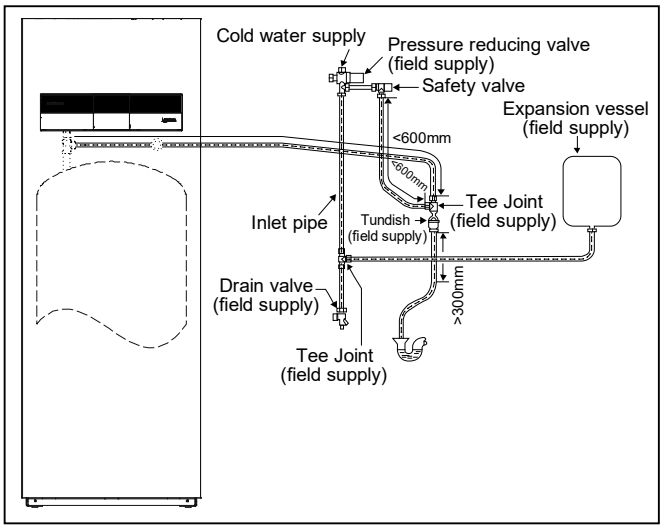
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulphuric gas etc.
- If necessary, use hose clamp to further tighten the hose at drain hose connector to prevent leakage.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.

(D) Domestic Hot Water Tank Discharge (Drain Tap)

- Use R½" male connector for this drainage outlet connection (Tube connector ⓖ).
- Piping must always be installed in a continuously downward direction. It must not be longer than 2m, with no more than 2 elbows, and must not allow condensation to build up or freezing to occur.
- The pipe from this drainage outlet fitting must not be shut off. The discharge must be freed.
- The end of this pipework must be in such a way so that the outlet is visible and can not cause any damage. Keep away from electrical components.
- It is recommended to fit a tundish into this ⓖ pipework. Tundish should be visible and positioned away from frost environment and electrical components.

(E) Temperature and Pressure Relief Valve

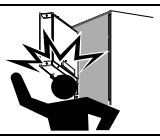
The Temperature and Pressure Relief Valve need appropriate discharge pipework. In accordance with Building Regulations a tundish must be fitted into the pipework within 600mm of the safety device. Due to the distance between the two safety devices it may be necessary to fit each safety device with its own tundish before run the pipework together to a safety discharge. The Rear Plate (22) has a Bushing so that the connection can be made to the factory fitted Temperature and Pressure Relief Valve. Then connect the Temperature and Pressure Relief Valve to the discharge pipework (Ø15mm). The following instructions are a requirement of UK Building Regulations and must be adhere to. For the other countries please refer to local legislation. If there is any doubt the insulation procedure, always contact local building office.



1. Connect the tundish and route the discharge pipe as shown above. Tundish should be visible and positioned away from frost environment and electrical components.
2. The tundish should be fitted vertically and as close to the safety device as possible and within 600mm of the device.
3. The tundish should be visible to users and positioned away from electrical devices.
4. The discharge pipe from the tundish should be terminate in a safe place where there is no risk to person nearby to the discharge, be of metal construction and:
   A) Be at least one pipe size larger than the nominal outlet size of the safety device unless its total equivalent hydraulic resistance exceeds that of a straight pipe 9m long. Bends must be taken into account in calculating the flow resistance.
   B) Have a vertical section of pipe at least 300mm long, below the tundish before any elbows or bends in the pipework.
   C) Be installed with a continuous fall.
   D) Have discharges visible at both the tundish and final point of discharge.

14.2.4 Connect the Cable to the Tank Unit

⚠ WARNING
This section is for authorized and licensed electrician only. Work behind the Control Board Cover ⑤ secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

⚠ CAUTION	
Please take extra precaution when open the control board cover ⑤ and control board ⑥ for unit installation and servicing. Failure to do so may cause injury.	

14.2.4.1 Fixing of Power Supply Cable and Connecting Cable

1 Connecting cable between Tank Unit and Outdoor Unit shall be approved polychloroprene sheathed flexible cord, type designation 60245 IEC 57 or heavier cord. See below table for cable size requirement.

Model		Connecting Cable Size
Tank Unit	Outdoor Unit	
WH-ADC0309K3E5UK	WH-UDZ03KE5*, WH-UDZ05KE5*	4 x min 1.5 mm^2
	WH-UDZ07KE5*, WH-UDZ09KE5*	4 x min 2.5 mm^2

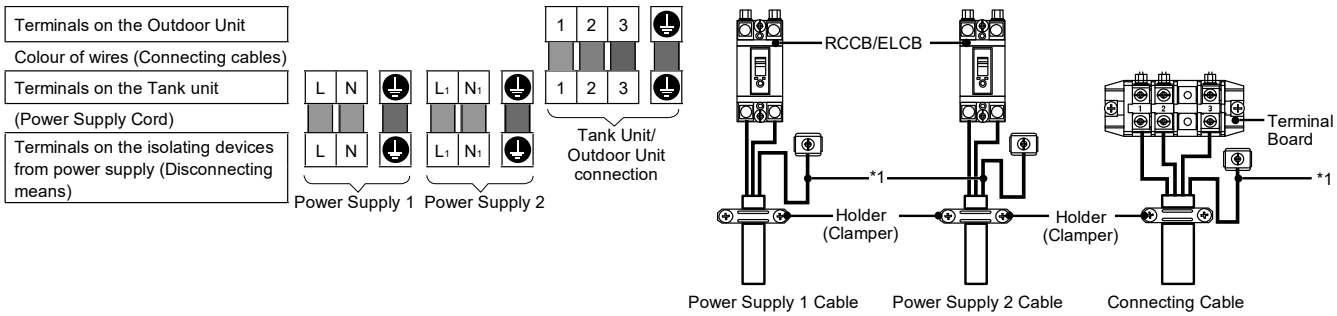
- Ensure the colour of wires of Outdoor Unit and the terminal no. are the same to the Tank Unit respectively.

- Earth wire shall be longer than the other wires as shown in the figure for the electrical safety in case of the slipping out of the cord from the Holder (Clamper).

2 An isolating device must be connected to the power supply cable.
- Isolating device (disconnecting means) should have minimum 3.0 mm contact gap.
- Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means).
  See below table for cable size requirement.

Model		Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
Tank Unit	Outdoor Unit				
WH-ADC0309K3E5UK	WH-UDZ03KE5*, WH-UDZ05KE5*	1	3 x min 1.5 mm^2	15/16A	30mA, 2P, type A
		2	3 x min 1.5 mm^2	15/16A	30mA, 2P, type AC
	WH-UDZ07KE5*, WH-UDZ09KE5*	1	3 x min 2.5 mm^2	25A	30mA, 2P, type A
		2	3 x min 1.5 mm^2	15/16A	30mA, 2P, type AC

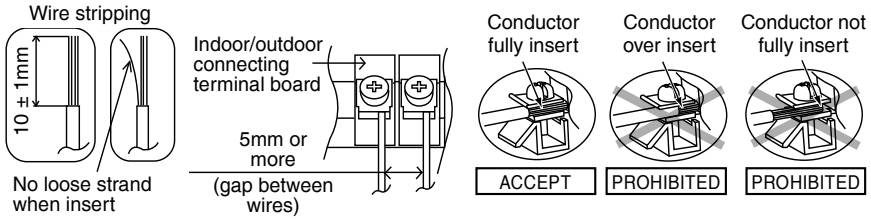
3 To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through a bushing (located at the bottom of Control Board ⑥) before terminal board. The bushing must be used and must not be removed.



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25)

*1 - Earth wire must be longer than other cables for safety reasons

14.2.4.2 Wire Stripping and Connecting Requirement



14.2.4.3 Connecting Requirement

For Tank Unit WH-ADC0309K3E5UK with WH-UDZ03KE5*, WH-UDZ05KE5*, WH-UDZ07KE5*, WH-UDZ09KE5*
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-3 and can be connected to current supply network.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-3 and can be connected to current supply network.

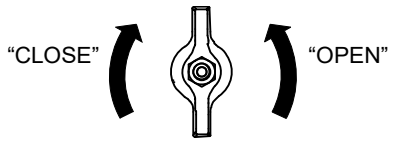
14.2.5 Charging and Discharging the Water

- Make sure all the piping installations are properly done before carry out below steps.

14.2.5.1 Charge the Water

For Domestic Hot Water Tank

1 Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

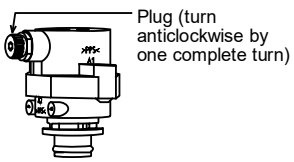


Domestic Hot Water Tank Discharge (Drain Tap) ⓖ

2 Set all Tap / Shower “OPEN”.
3 Start filling water to the Domestic Hot Water Tank via Tube Connector ⓒ. After 20~40min, water should flow out from Tap / Shower. Else, please contact your local authorized dealer.
4 Check and make sure no water leaking at the tube connecting points.

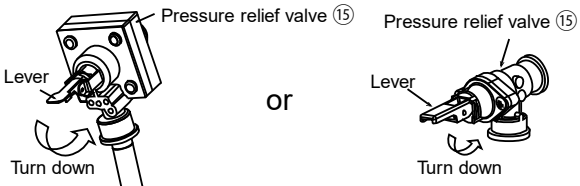
For Space Heating / Cooling

1 Turn the plug on the Air Purge Valve ⑭ outlet anticlockwise by one complete turn from fully closed position.



Air purge valve ⑭

2 Set the Pressure Relief Valve ⑮ lever “DOWN”.



Pressure relief valve ⑮ Pressure relief valve ⑮

3 Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Space Heating / Cooling circuit via Tube Connector ⓐ. Stop filling water if the free water flow through Pressure Relief Valve ⑮ discharge hose.
4 Turn ON the Tank Unit and make sure Water Pump ④ is running.
5 Check and make sure no water leaking at the tube connecting points.
6 The water may drip from this discharge hose. Therefore must guide the hose without close or block the outlet of the hose.

14.2.5.2 Discharge the Water

For Domestic Hot Water Tank

1 Turn OFF power supply.
2 Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “OPEN”.
3 Open Tap / Shower to allow air inlet.
4 After discharge, set Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

14.2.6 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings.

*14.2.6.1 Check Water Pressure *(0.1 MPa = 1 bar)*

Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure from Remote Controller). If necessary add water into Tank Unit (via Tube Connector ⓐ).

14.2.6.2 Check Pressure Relief Valve ⑮

- Check for correct operation of Pressure Relief Valve ⑮ by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keep on draining out from the Tank Unit, switch off the system, and then contact your local authorized dealer.

14.2.6.3 Expansion Vessel ⑬ Pre Pressure Checking

For Space Heating / Cooling

- Expansion Vessel ⑬ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit’s piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m.

14.2.6.4 Check RCCB/ELCB

Ensure the RCCB/ELCB set to “ON” condition before check RCCB/ELCB.
Turn on the power supply to the Tank Unit.
This testing could only be done when power is supplied to the Tank Unit.

> **⚠ WARNING**
>
> Be careful not to touch parts other than RCCB/ELCB test button when the power is supplied to Tank Unit. Else, electrical shock may happen. Before obtaining access to terminals, all supply circuits must be disconnected.

- Push the “TEST” button on the RCCB/ELCB. The lever would turn down and indicate “0”, if it functions normal.
- Contact authorized dealer if the RCCB/ELCB malfunction.
- Turn off the power supply to the Tank Unit.
- If RCCB/ELCB functions normal, set the lever to “ON” again after testing finish.

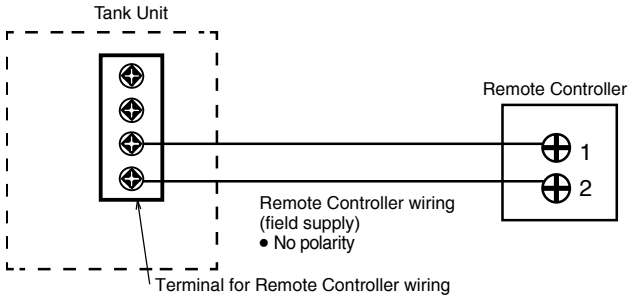
14.2.7 Installation of Remote Controller as Room Thermostat

- Remote Controller ① mounted to the Tank Unit can be moved to the room and serve as Room Thermostat.

14.2.7.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1 By the window, etc. exposed to direct sunlight or direct air.
  2 In the shadow or backside of objects deviated from the room airflow.
  3 Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4 Location near heat source.
  5 Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

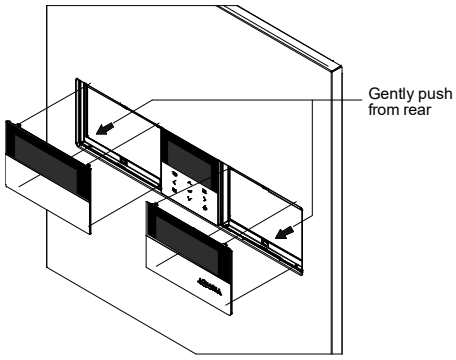
14.2.7.2 Remote Controller Wiring



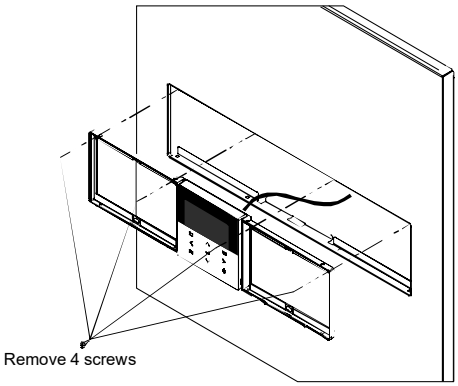
- Remote Controller cable shall be (2 x min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals of Tank Unit (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.

14.2.7.3 Remove The Remote Controller From Tank Unit

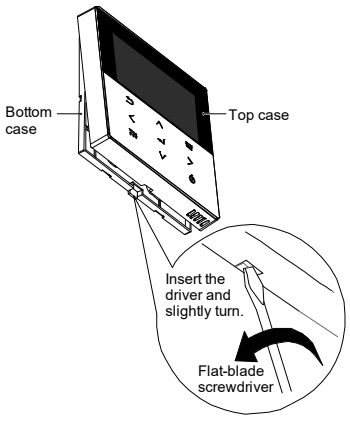
1 Remove both Left Decoration Panel ② and Right Decoration Panel ③ from Front Plate ⑱ with gently push the panels from back.



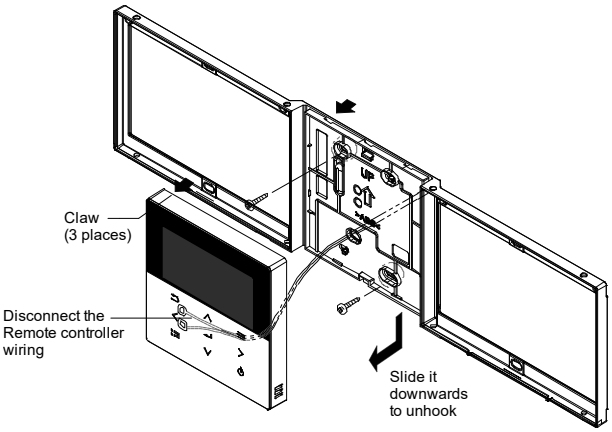
2 Remove the 4 screws and take out the holder with Remote Controller ①.



3 Remove the top case from the bottom case.



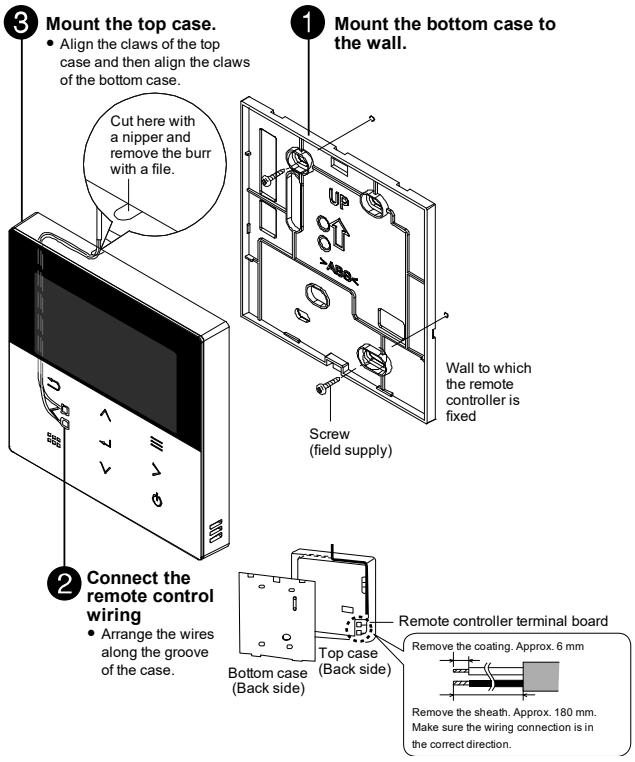
4 Remove the wiring between Remote controller ① and Tank Unit terminal.



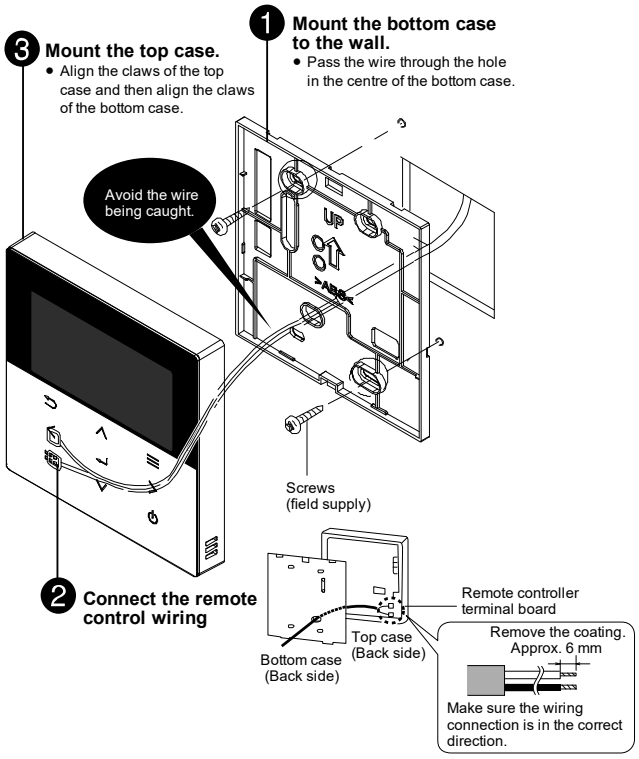
14.2.7.4 Mounting The Remote Controller

For exposed type

Preparation: Make 2 holes for screws using a driver.

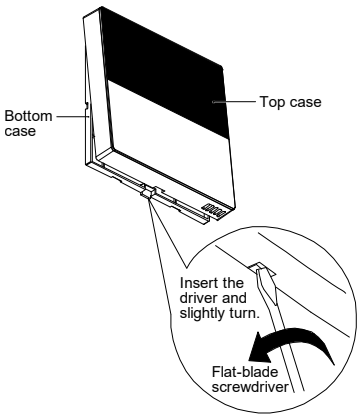


For embedded type
Preparation: Make 2 holes for screws using a driver.



14.2.7.5 Replace The Remote Controller Cover

- Replace the existing Remote Controller with Remote Controller Case [5] to close the hole left after remove the Remote Controller.
  1 Refer Section “Remove The Remote Controller From Tank Unit” for remove Remote Controller.
  2 Remove the top case from the bottom case of Remote Controller Case [5].



  3 Reverse the steps 1 to 4 of section “Remove The Remote Controller From Tank Unit” to fix Remote Controller Case [5] on Tank Unit.

14.2.8 Test Run

1 Before test run, make sure below items have been checked:-
  a) Pipework are properly done.
  b) Electric cable connecting work are properly done.
  c) Tank Unit is filled up with water and trapped air is released.
  d) Please turn on the power supply after filling the tank until full.
2 Switch ON the power supply of the Tank Unit. Set the Tank Unit RCCB/ELCB to “ON” condition. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run.
  Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

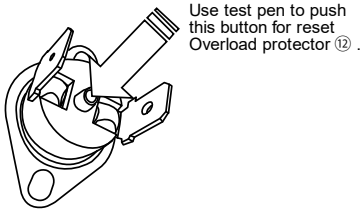
3 For normal operation, Water Pressure reading should be in between 0.05 MPa and 0.3 MPa (0.5 bar and 3 bar). If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4 After test run, please clean the Magnetic Water Filter Set ⑨ and Water Filter Set ㉚. Reinstall it after finish cleaning.

14.2.8.1 Check Water Flow of Water Circuit

Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]

14.2.8.2 Reset Overload Protector ⑫

Overload Protector ⑫ a serves the safety purpose to prevent the water over heating. When the Overload Protector ⑫ a trip at high water temperature, take below steps to reset it.
1 Take out the cover.
2 Use a test pen to push the centre button gently in order to reset the Overload Protector ⑫.
3 Fix the cover to the original fixing condition.

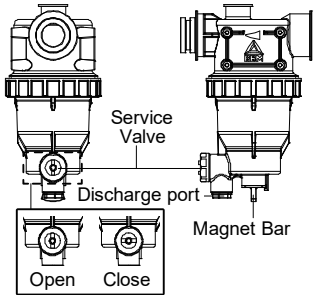


14.2.9 Maintenance

- In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

14.2.9.1 Maintenance for Magnetic Water Filter Set ⑨

1 Turn OFF power supply.
2 Place a container below Magnetic Water Filter Set ⑨.
3 Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑨.
4 By using Allen key (8mm), remove the Cap of Discharge Port.
5 By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
6 Reinstall the Cap of Discharge Port and Magnet Bar.
7 Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 14.2.5 for details.)
8 Turn ON power supply.



14.2.9.2 Maintenance for Temperature and Pressure Relief Valve ㉖

- Manually operate the Temperature and Pressure Relief Valve ㉖ by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

14.2.9.3 Proper Pump Down Procedure

⚠ WARNING

Strictly follow the steps below for proper pump down procedure. Explosion may occur if the steps are not followed as per sequence.

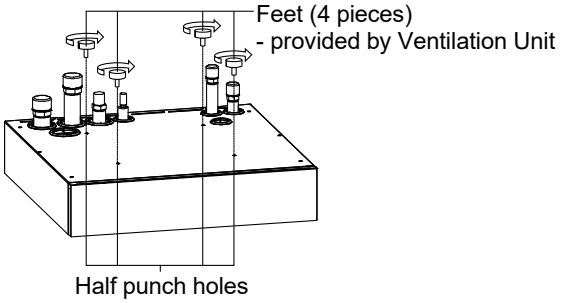
1. When the Tank Unit is not in operation (standby), enter the Service setup menu in the Remote Controller and select Pump down operation to turn it ON. (See APPENDIX for detail)
2. After 10~15 minutes, (after 1 or 2 minutes in case very low ambient temperatures (< 10°C)), fully close 2 way valve on Outdoor Unit.
3. After 3 minutes, fully close 3 way valve on Outdoor Unit.
4. Press the “OFF/ON” switch on the Remote Controller ① to stop pump down operation.
5. Remove the refrigerant piping.

14.2.9.4 Installation of Ventilation Unit on top of Tank Unit (Optional)

- For installation works of Ventilation Unit on top of Tank Unit, refer to the Ventilation Unit Installation Manual.

⚠ CAUTION

Before install Ventilation Unit, fix the Feet that provided by Ventilation Unit to the half punch holes on Top Panel of Tank Unit, Otherwise, heavy Ventilation Unit may fall and cause injury.



14.2.10 Technical Data

• Model	WH-ADC0309K3E5UK
• Tank Capacity – Nominal – Actual	 200 L 185 L
• Rate Heat Exchanger Volume	8.175 L
• Weight – Empty – Full	 98 kg 283 kg
• Maximum operating Pressure – Primary – Secondary	 3.0 bar 3.5 bar
• Maximum Operating Temperature	65°C
• Operating Pressure – Tank Unit – Expansion Relief Valve	 3.5 bar 8.0 bar
• Maximum Working / Design Pressure – Space Heating / Cooling – Tank Circuit	 3.0 bar 10.0 bar
• Standing Heat Loss	1.200kWh/24h
• Reheat Time	87m 54s
• Temperature and Pressure Relief Valve – Size – Pressure Relief – Temperature Relief	 ½" x 15mm 10 bar 90°C - 95°C
• Primary Heater Pressure Drop	0.2 bar
• Primary Flow Rate (Nominal)	9.2 - 25.8 L/min
• Primary Heating Power Input / Flow Rate	32.2 kW/ 15 L/min
• Hot Water Capacity as per EN 12897:2016	177 L

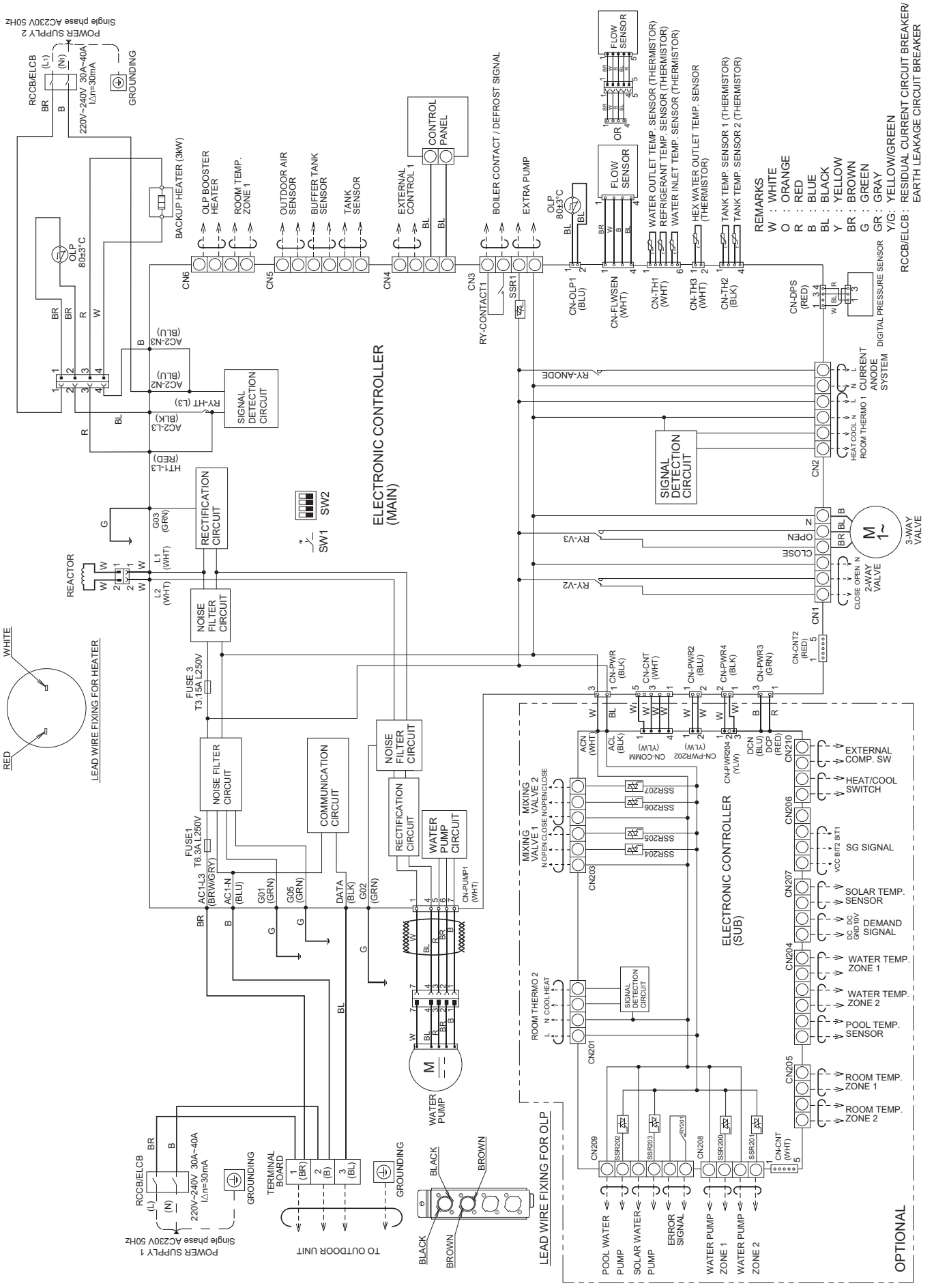
This Tank Unit comply with EN 12897:2016.

14.2.11 PAW-G3KIT-1 - Panasonic G3 22MM Unvented Kit

Part Code	Description	WRAS Approval Number
533117H CST	Inlet Control Set – 3.5 Bar PRV and 6 Bar SRV	1907367
AI-002402	½" Drain Cock	1811800
TI-10396151504	15mm x ½" x 15mm Tee	1804353
TI-1130031	22mm X ¾" MI DZR Elbow	1804353
PA-2822362151001	¾"F x ¾" M 1000mm Hose	2201803
DE 18 (7303030)	18 Litre Potable Vessel with Integral Bracket	2209753
2209753	15 x 22mm Straight Tundish	1901349

14.2.12 Control of Water Temperature : PCB, Limit Thermostat

Wiring Diagram



14.3 Outdoor Unit

Attached accessories

No.	Accessories part	Qty.
1	Drain elbow 	1
2	Rubber cap 	7 (For WH-UDZ03KE5*) 3 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
3	Discharge grille 	1 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
4	Metal plate 	2 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)
5	Screw 	8 (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*)

Optional accessories

No.	Accessories part	Qty.
6	Base Pan Heater CZ-NE2P	1

- It is strongly recommended to install a Base Pan Heater (optional) if the outdoor unit is install in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.
- Applicable Piping Kit (For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*) CZ-52F5,7,10BP
- Applicable Piping Kit (For WH-UDZ03KE5*) CZ-4F5,7,10BP

14.3.1 Select the Best Location

- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- For WH-UDZ03KE5*, avoid installation in areas where the ambient temperature may drop below -20°C.
- For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*, avoid installations in areas where the ambient temperature may drop below -25°C.
- Keep the spaces indicated by arrows from wall, ceiling, fence or other obstacles.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- If outdoor unit installed near sea, region with high content of sulphur or oily location (e.g. machinary oil, etc), it lifespan maybe shorten.
- If piping length is over 10 m, additional refrigerant should be added as shown in the table.

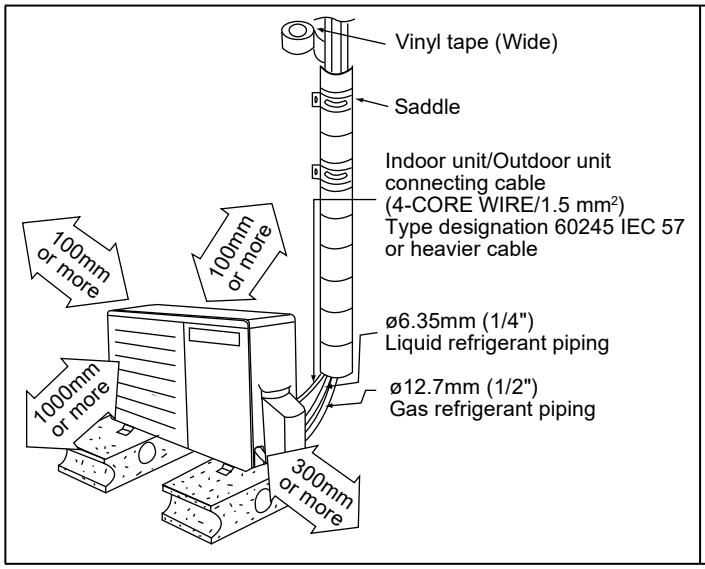
Model	Piping size		Pre-charged Refrigerant (kg)	Rated Length (m)		Max. Elevation (m)	Min. Piping Length (m)	Max. Piping Length (m)	Additional Refrigerant (g/m)
	Gas	Liquid		For Heat Pump Indoor Unit	For Hydromodule + Tank				
WH-UDZ03KE5*	ø12.7mm (1/2")	ø6.35mm (1/4")	0.90	7	7	20	3	25	20
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8")	ø6.35mm (1/4")	1.30	7	7	30	3	50	25

Example: WH-UDZ03KE5*
If piping length is 15m, the quantity of additional refrigerant should be 100g. [(15-10)m x 20 g/m = 100g]

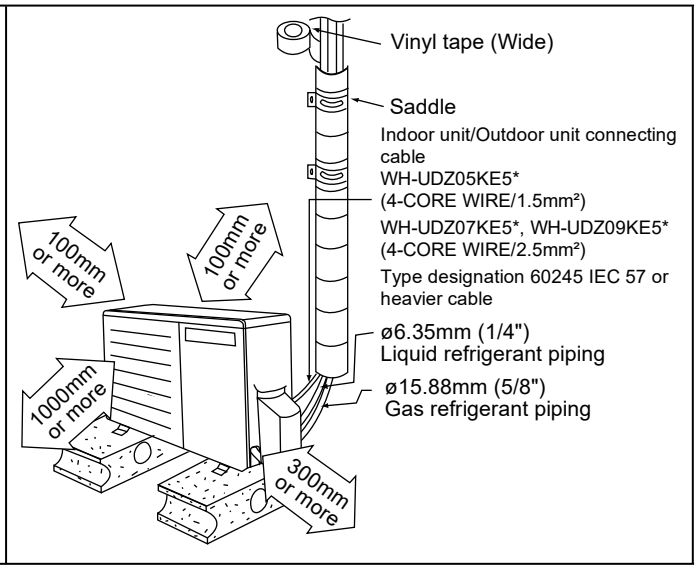
14.3.2 Install the Outdoor Unit

14.3.2.1 Installation Diagram

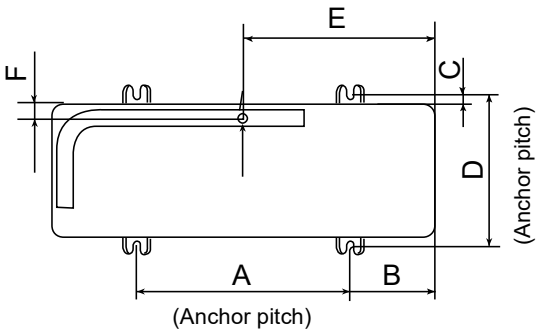
- It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.
- This illustration is for explanation purposes only.



For WH-UDZ03KE5*



For WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*



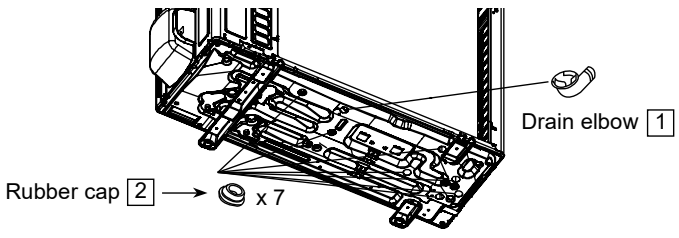
Model	A	B	C	D	E	F
WH-UDZ03KE5*	540	160	20	330	430	46
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	613	130	24	360.5	543	32

(Unit : mm)

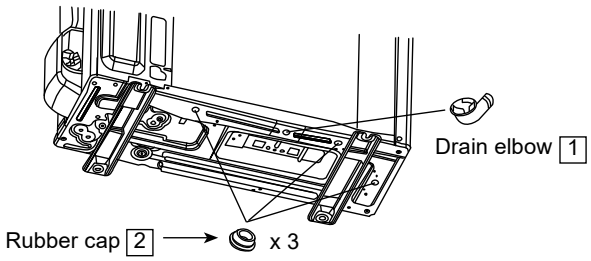
- After selecting the best location, start installation according to Installation Diagram.
  1 Fix the unit on concrete or rigid frame firmly and horizontally by bolt nut (ø10 mm).
  2 When installing at roof, please consider strong wind and earthquake. Please fasten the installation stand firmly with bolt or nails.

14.3.2.2 Disposal of Outdoor Unit Drain Water

- When a Drain elbow [1] is used, please ensure to follow below:
  - the unit should be placed on a stand which is taller than 50 mm.
  - cover the ø20mm holes with Rubber cap [2] (refer to illustration below).
  - use a tray (field supply) when necessary to dispose the outdoor unit drain water.
- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow [1] and Rubber cap [2], for the drain water freezes and the fan will not rotate.



WH-UDZ03KE5*



WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*

14.3.3 Install the Discharge Grille

- This section is for WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5* only.
- Make sure to install the outdoor unit on concrete or rigid frame by bold nut before install the discharge grille 3.
- Hook the metal plate 4 on the cabinet front plate at 2 locations and tighten with screw 5. (Figure a)
- Hook the discharge grille 3 to the cabinet front plate at 4 locations. (Figure b)
- Tighten the discharge grille to the cabinet front plate at 6 locations with screw 5. (Figure c)

⚠ CAUTION
Make sure to install the discharge grille to the outdoor unit before powering ON to protect againts a rotating fan

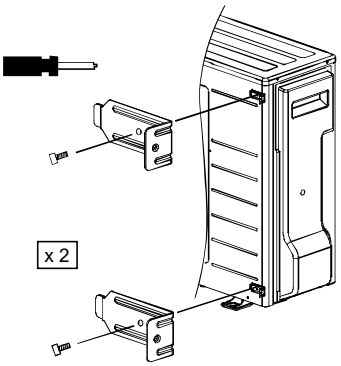


Figure a

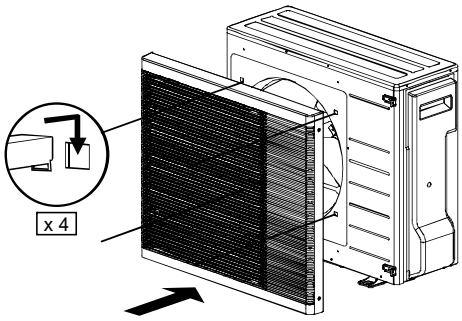


Figure b

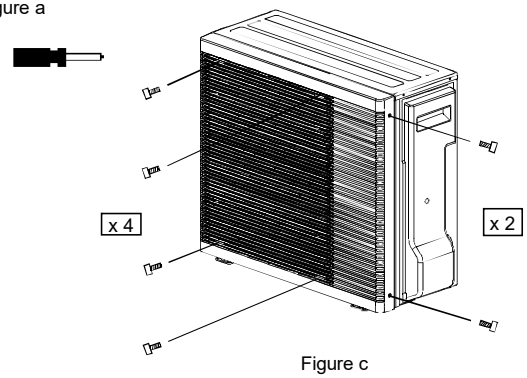
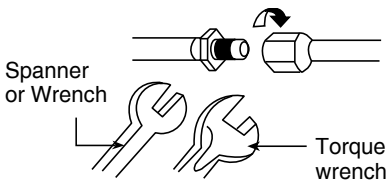


Figure c

14.3.4 Connecting the Piping

14.3.4.1 Connecting the Piping to Outdoor Unit

Decide piping length and then cut by using pipe cutter. Remove burrs from cut edge. Make flare after inserting the flare nut (locate at valve) onto the copper pipe. Align center of piping to valves and then tighten with torque wrench to the specified torque as stated in the table.

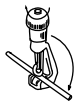


Be sure to use two spanners to tighten.
(If the nuts are overtightened, it may cause the flares to break or leak.)

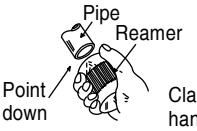
Model	Piping size (Torque)	
	Gas	Liquid
WH-UDZ03KE5*	ø12.7mm (1/2") [55 N•m]	ø6.35mm (1/4") [18 N•m]
WH-UDZ05KE5*, WH-UDZ07KE5* and WH-UDZ09KE5*	ø15.88mm (5/8") [65 N•m]	ø6.35mm (1/4") [18 N•m]

14.3.4.1.1 Cutting and Flaring the Piping

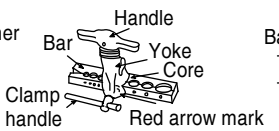
1 Please cut using pipe cutter and then remove the burrs.
2 Remove the burrs by using reamer. If burrs is not removed, gas leakage may be caused. Turn the piping end down to avoid the metal powder entering the pipe.
3 Please make flare after inserting the flare nut onto the copper pipes.



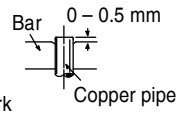
1. To cut

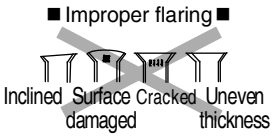


2. To remove burrs



3. To flare



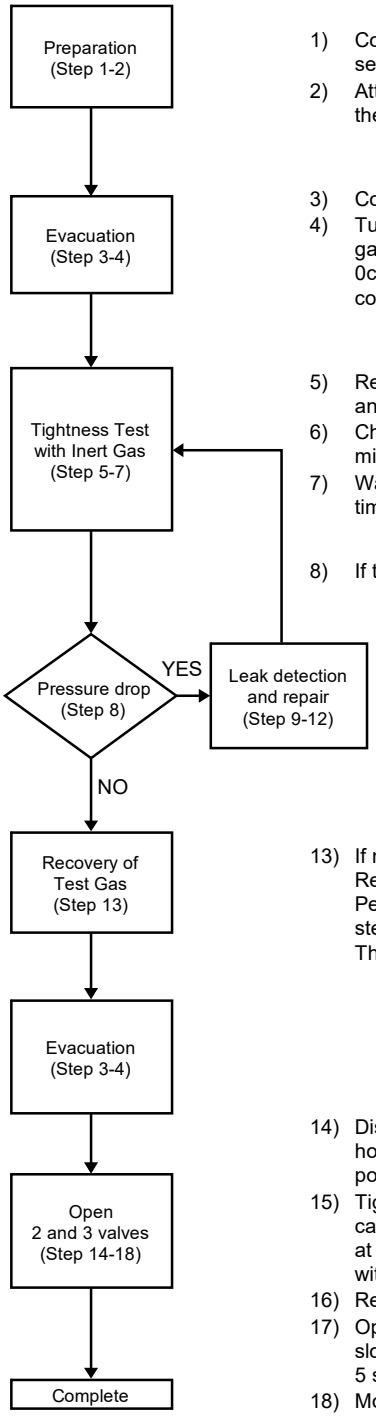


When properly flared, the internal surface of the flare will evenly shine and be of even thickness. Since the flare part comes into contact with the connections, carefully check the flare finish.

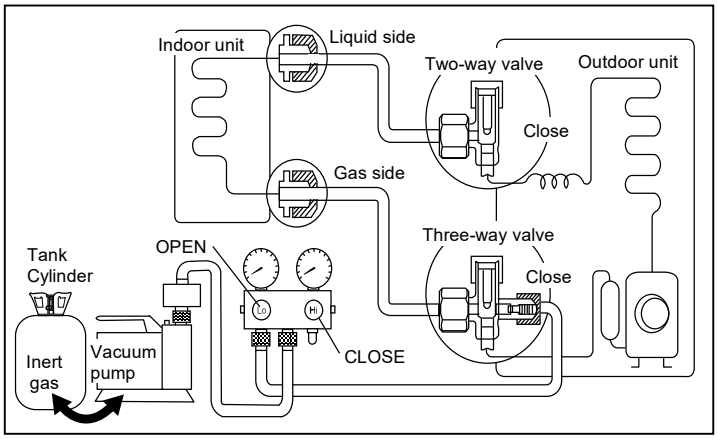
14.3.5 Air Tightness Test on the Refrigerating System

⊘	Do not purge the air with refrigerants but use a vacuum pump to vacuum the installation.
❗	There is no extra refrigerant in the outdoor unit for air purging.

- Before system is charged with refrigerant and before the refrigerating system is put into operation, below site test procedure and acceptance criteria shall be vertified by the certified technicians, and/or the installer.
- Be sure to check whole system for gas leakage.



1) Connect a charging hose with a push pin to the Low side of a charging set and the service port of the 3-way valve.
2) Attach the gauge manifold set correctly and tightly. Make sure that both valves of the manifold gauge (low pressure and high pressure) is in close position.
3) Connect the center hose of the manifold gauge to a vacuum pump.
4) Turn on the power switch of the vacuum pump, then turn open the low side manifold gauge valve and make sure that the needle in the gauge moves from 0cmHg (0 MPa) to –76 cmHg (–0.1 MPa) or vacuum until 500 microns is achieved. This process continues for approximately ten minutes. Then close the low side manifold gauge valve.
5) Remove the vacuum pump from the centre hose and connect the center hose to cylinder of any applicable inert gas as test gas.
6) Charge test gas into the system and wait until the pressure within the system to reach min. 1.04MPa (10.4barg).
7) Wait and monitor the pressure reading on the gauges. Check if there is any pressure drop. Waiting time depends on the size of the system.
8) If there is any pressure drop, perform step 9-12. If there is no pressure drop, perform step 13.
9) Use Gas Leak Detector to check for leaks. Must use the detection equipment with a sensitivity of 5 grams per year of test gas or better.
10) Move the probe along the Air-to-Water Heatpump system to check for leaks, and mark for repair.
11) Any leak detected and marked shall be repaired.
12) After repair, repeat evacuation steps 3-4 and tightness test steps 5-7. Check the pressure drop as in step 8.
13) If no leak,
Recover the test gas.
Perform evacuation of steps 3-4.
Then proceed to step 14.



14) Disconnect the charging hose from the service port of the 3-way valve.
15) Tighten the service port caps of the 3-way valve at a torque of 18 N•m with a torque wrench.
16) Remove the valve caps of both of the 2-way valve and 3-way valve.
17) Open both of the valves, using a hexagonal wrench (4mm). It is recommended to allow refrigerant slowly flow into the refrigerant system to prevent refrigerant freezing. Slightly open 2-way valve for 5 seconds then close the valve. Repeat this action for 3 cycles then fully open the valve.
18) Mount back the valve caps onto the 2-way valve and the 3-way valve to complete this process.

Notes:

Recommended use of any of the following leak detector,
I) Universal Sniffer leak detector
II) Electronic halogen leak detector
III) Ultrasonic Leak Detector

14.3.6 Connect the Cable to the Outdoor Unit

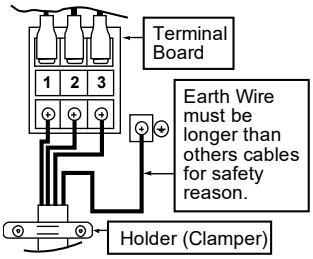
(FOR DETAIL REFER TO WIRING DIAGRAM AT UNIT)

1 Remove the control board cover from the unit by loosening the screw.

2 Connecting cable between indoor unit and outdoor unit shall be approved polychloroprene sheathed flexible cable (see below table), type designation 60245 IEC 57 or heavier cable.

Model	Flexible cable specification
WH-UDZ03KE5*, WH-UDZ05KE5*	4 x (1.5 mm^2)
WH-UDZ07KE5*, WH-UDZ09KE5*	4 x (2.5 mm^2)

Terminals on the indoor unit	1	2	3	⏚
Colour of wires				
Terminals on the outdoor unit	1	2	3	⏚

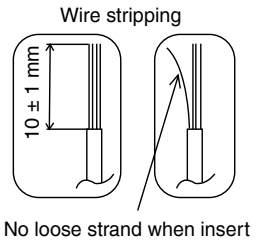


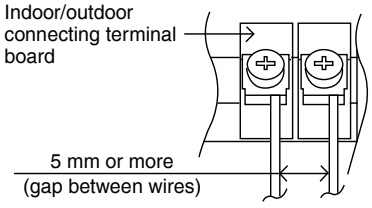
⚠ WARNING
⏚ This equipment must be properly earthed.

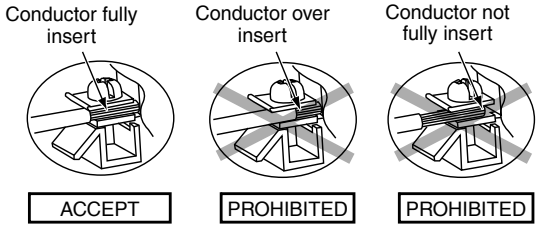
3 Secure the cable onto the control board with the holder (clamper).

4 Attach the control board cover back to the original position with screw.

14.3.6.1 Wire Stripping and Connecting Requirement







14.3.7 Pipe Insulation

1 Please carry out insulation at pipe connection portion as mentioned in Indoor/Outdoor Unit Installation Diagram. Please wrap the insulated piping end to prevent water from going inside the piping.

2 If drain hose or connecting piping is in the room (where dew may form), please increase the insulation by using POLY-E FOAM with thickness 6 mm or above.

⚠ CAUTION
If cleaning of the outdoor unit is necessary during installation or servicing, do not clean the outdoor unit with any hydrocarbon based solvent.

14.4 Appendix

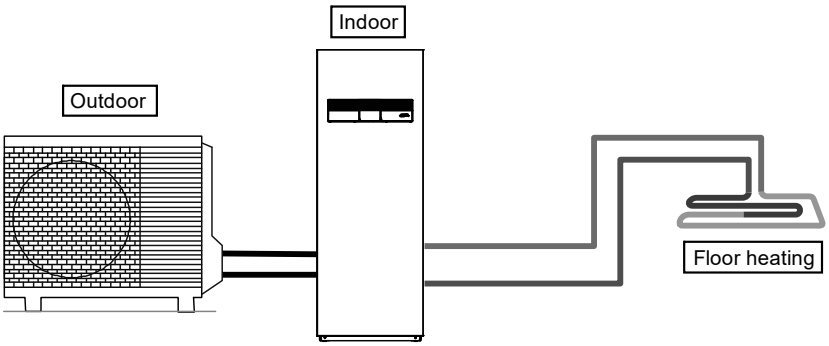
14.4.1 Variation of System

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.
(NOTE) : For this model, both external room thermistor of Zone 1 and external room thermostat of Zone 1 must always be connected to main indoor PCB only regardless of Optional PCB (CZ-NS5P) connection.

14.4.1.1 Introduce Application Related to Temperature Setting

14.4.1.1.1 Temperature Setting Variation for Heating

1. Remote Controller



Setting of remote controller

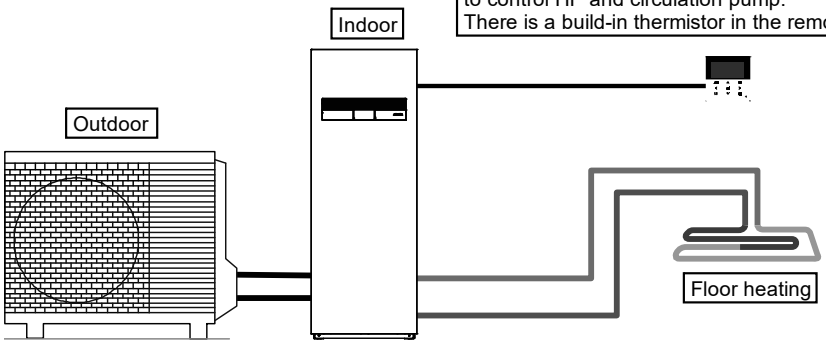
Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Water temperature

Connect floor heating or radiator directly to the Tank Unit.
Remote controller is installed on Tank Unit.
This is the basic form of the most simple system.

2. Room Thermostat

Tank Unit receive Room Thermo signal (ON/OFF) from Remote controller to control HP and circulation pump.
There is a build-in thermistor in the remote controller.



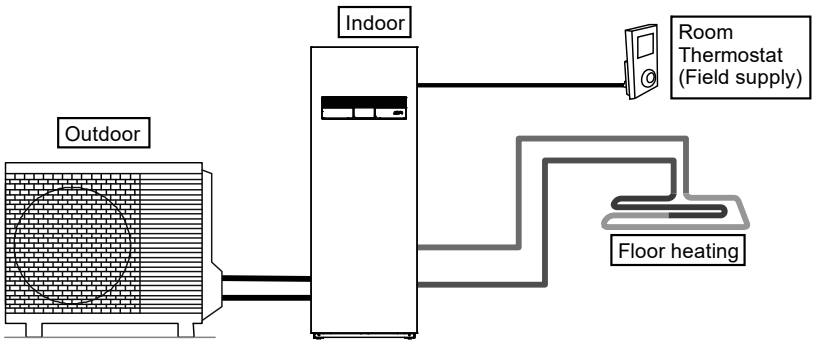
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
Internal

Connect floor heating or radiator directly to the Tank Unit.
Remove remote controller from Tank Unit and install it in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat

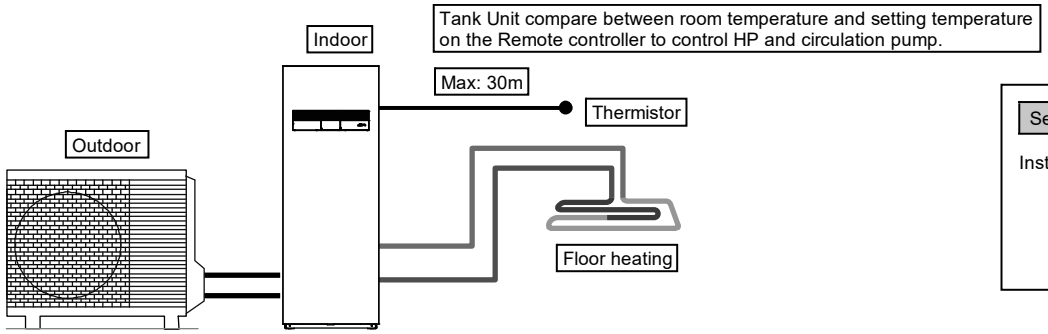


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No
Zone & Sensor:
Room thermostat
(External)

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No
Zone & Sensor:
Room thermistor

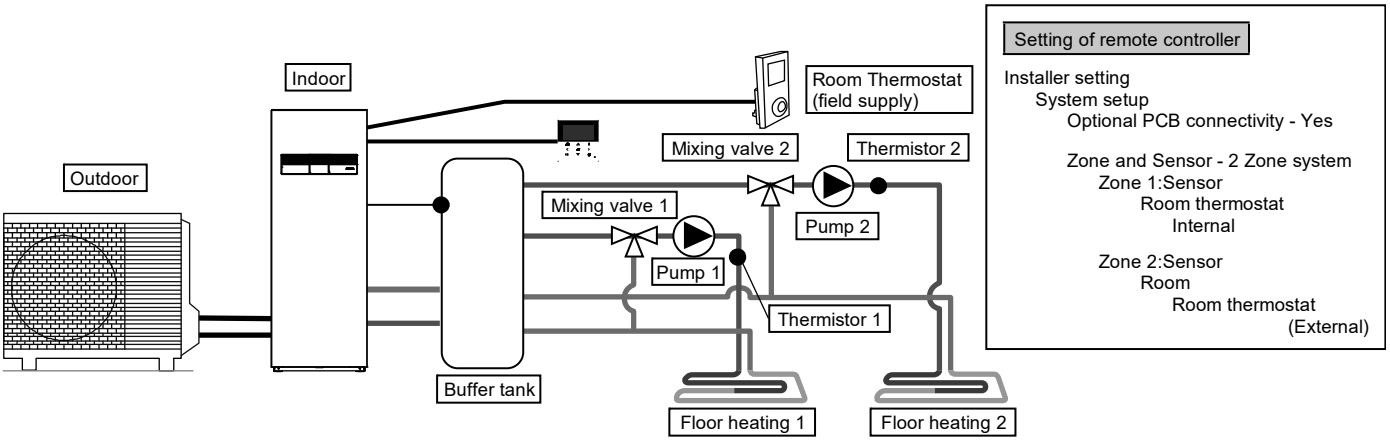
Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, compensation curve is shifted according to the thermo ON/OFF situation.

- (Example) If room temperature increasing speed is;
  very slow → shift up the compensation curve
  very fast → shift down the compensation curve

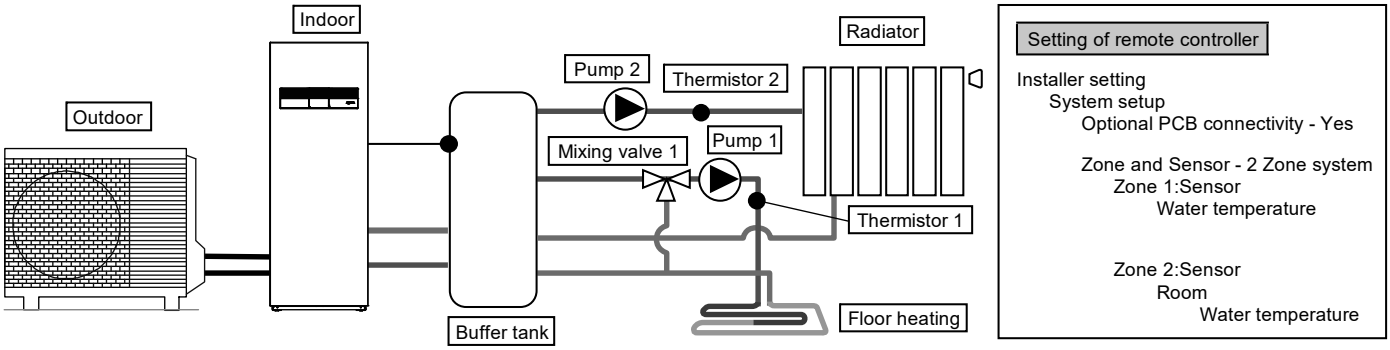
14.4.1.1.2 Examples of Installations

Floor heating 1 + Floor heating 2



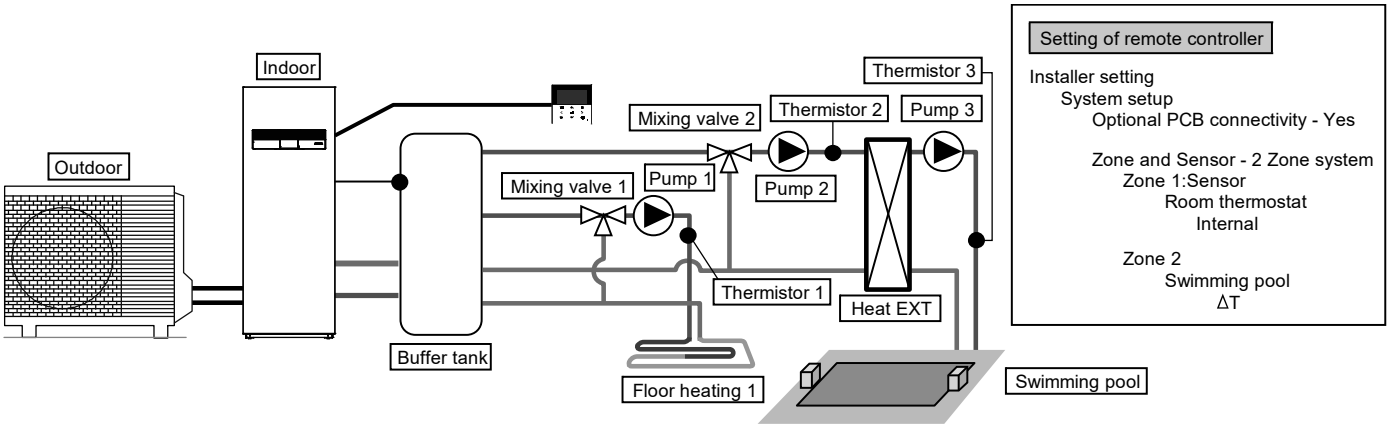
Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Remove remote controller from Tank Unit, install it in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Floor heating + Radiator



Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
Remote controller is installed on Tank Unit.
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the Optional PCB (CZ-NS5P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

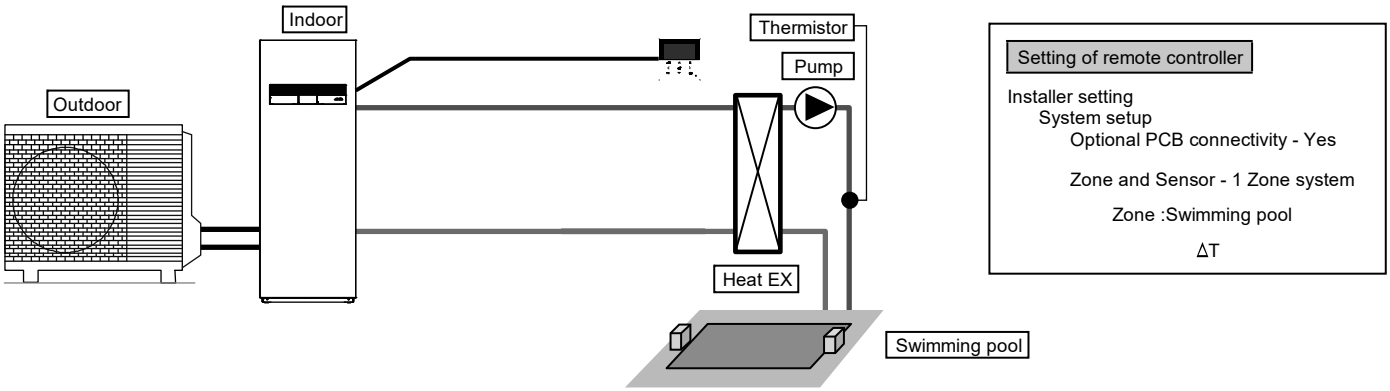
Floor heating + Swimming pool



Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Remove remote controller from Tank Unit and install in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the Optional PCB (CZ-NS5P).

❇ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

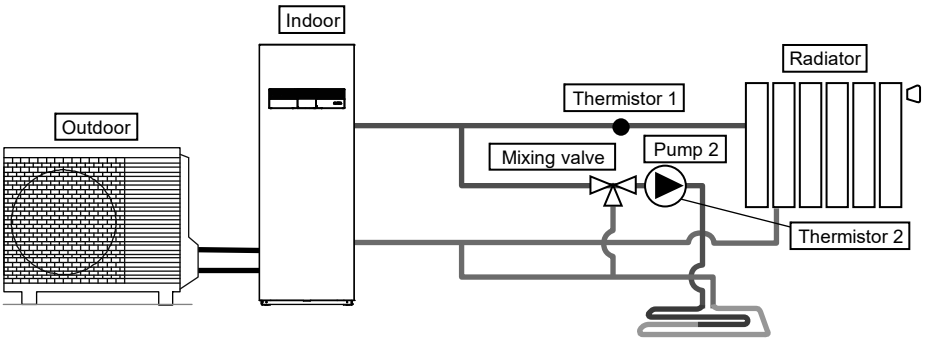
Swimming pool only



This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Tank Unit without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Remove remote controller from Tank Unit and install in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the Optional PCB (CZ-NS5P).

In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor - 2 Zone system
Zone 1:Sensor
Water temperature

Zone 2:Sensor
Room
Water temperature

Operation setup
Heat
ΔT for heating ON – 1°C

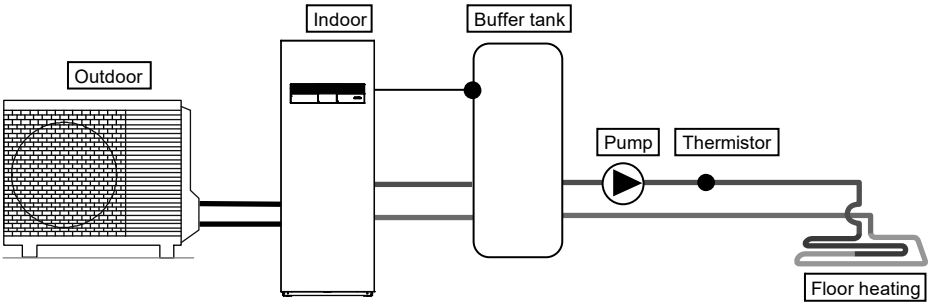
Cool
ΔT for cooling ON – 1°C

This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Tank Unit served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the Optional PCB (CZ-NS5P).

(NOTE)

- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
- Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

Buffer tank connection



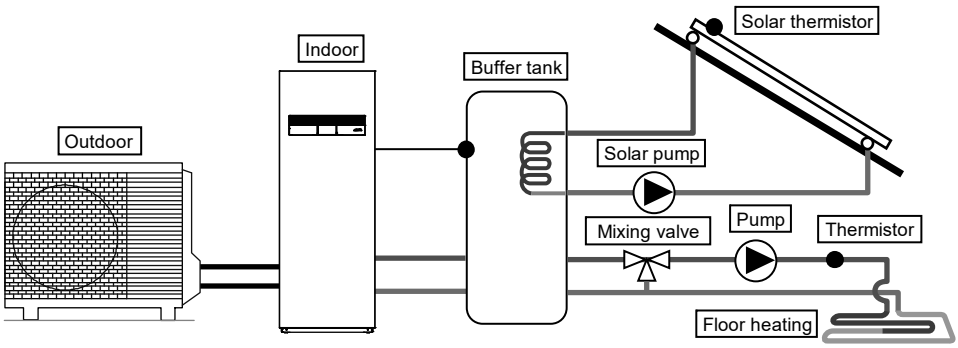
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Buffer Tank connection - Yes
ΔT for buffer tank

This is an application that connects the buffer tank to the Tank Unit.
Buffer tank’s temperature is detected by buffer tank thermistor (specified by Panasonic).
Without connection of Optional PCB, external pump can be used for circulation in the floor heating circuit.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Buffer tank + Solar

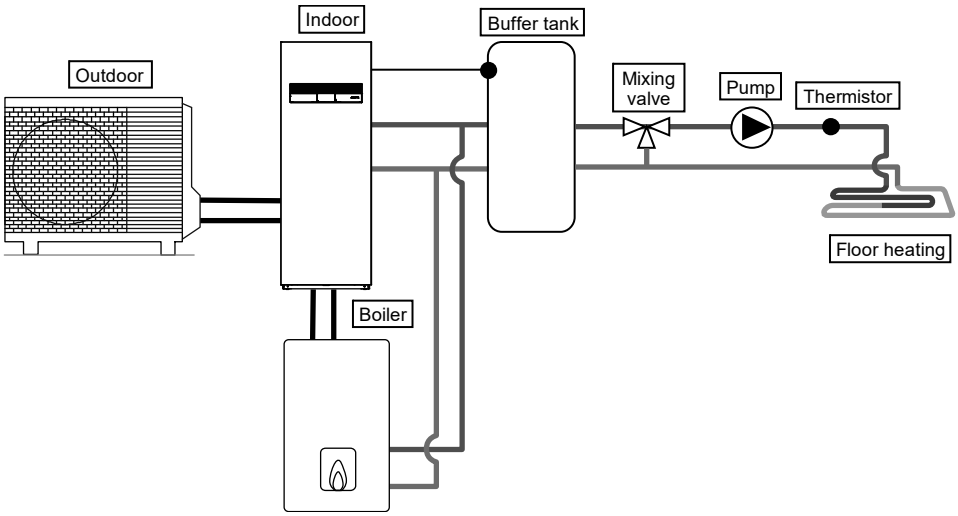


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank
Solar connection - Yes
Buffer tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the buffer tank to the Tank Unit before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Boiler connection



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Bivalent - Yes
Turn ON: outdoor temp
Control pattern

This is an application that connects the boiler to the Tank Unit, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
Besides that, an application that connects to the DHW tank's circuit to heat up tank 's hot water is also possible.
Boiler output can be control by either SG ready input from optional PCB or Auto control by 3 modes selection pattern. (Operation setting of boiler shall be responsible by installer.)
This system requires Optional PCB (CZ-NS5P) for SG ready input control.
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when select Advanced Parallel setting.)
Note : Buffer tank thermistor must be connected to main indoor PCB only.

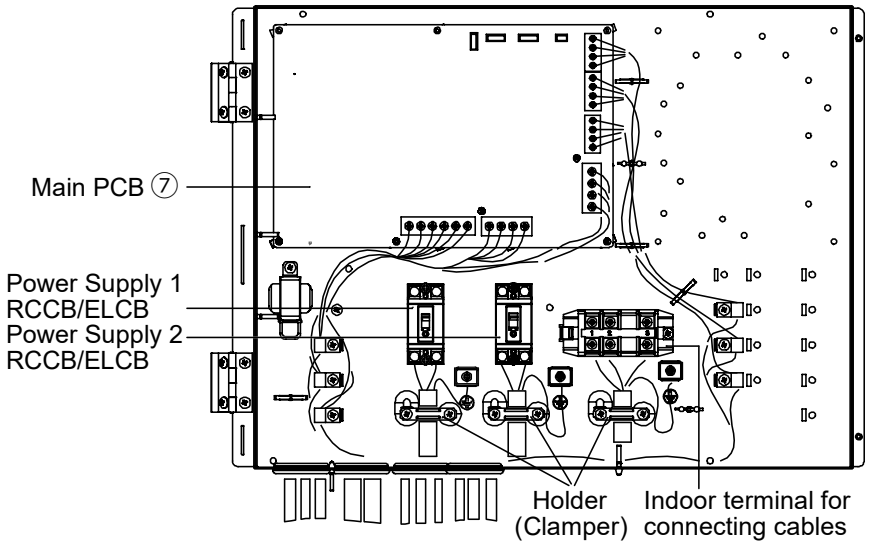
⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the Tank Unit does NOT exceed 55°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

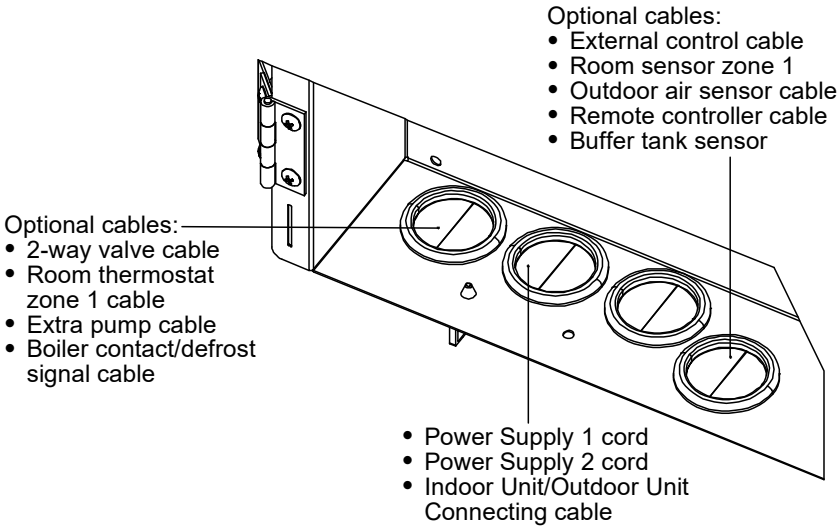
14.4.2 How to Fix Cable

14.4.2.1 Connecting with External Device (Optional)

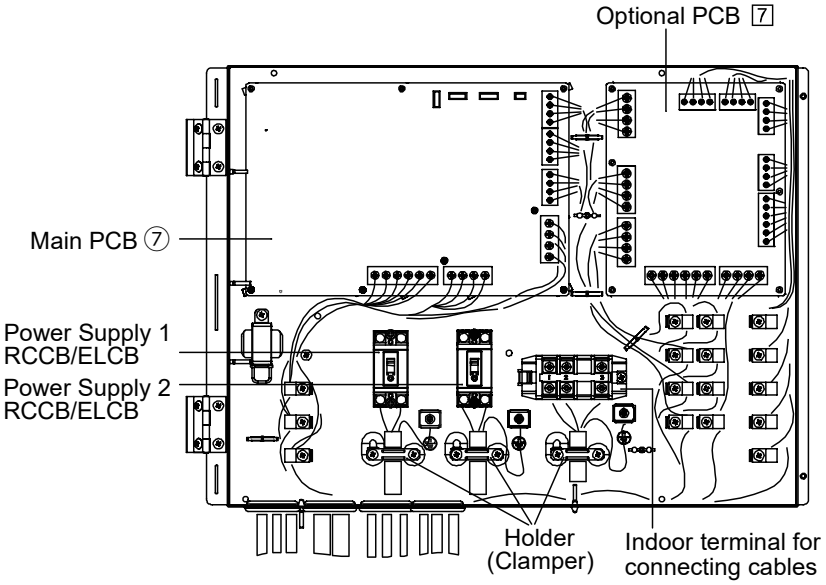
- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB ⑤
  1 Two-way valve shall be spring and electronic type, refer to “Field Supply Accessories” table for details. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
  *note: - Two-way Valve shall be CE marking compliance component.
  - Maximum load for the valve is 9.8VA.
  2 Room thermostat zone 1 cable must be (4 or 3 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
  3 Extra pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  4 Boiler contact/defrost signal cable shall be (2 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
  5 External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  *note: - Switch used shall be CE compliance component.
  - Maximum operating current shall be less than $3A_{rms}$.
  6 Room sensor zone 1, outdoor air sensor and buffer tank sensor cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.



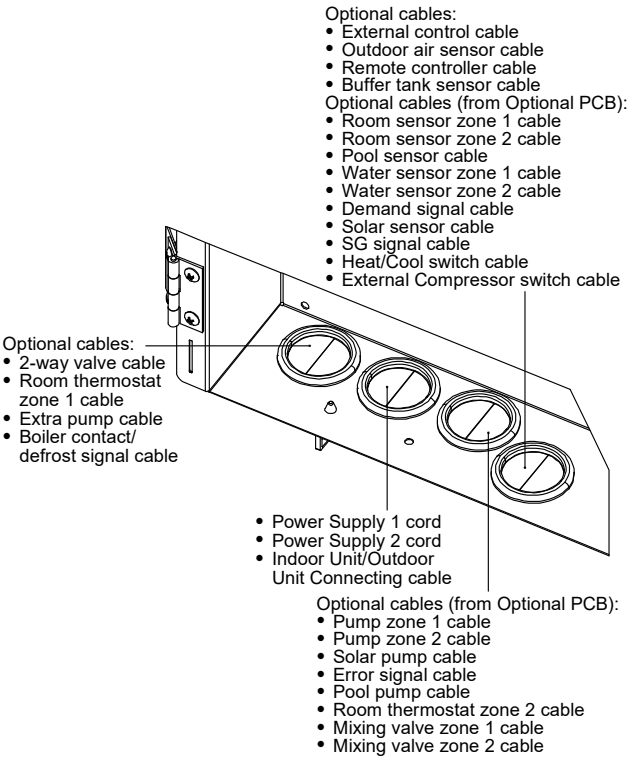
How to guide the optional cables and power supply cord
(view without internal wiring)



- For connection to Optional PCB [7]
  1 By connecting Optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in Optional PCB. Temperature of each zone can be controlled independently by remote controller.
  2 Pump zone 1 and zone 2 cable shall be (2 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  3 Solar pump cable shall be (2 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  4 Pool pump cable shall be (2 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  5 Room thermostat zone 2 cable shall be (4 x min 0.5 mm²), of type designation 60245 IEC 57 or heavier.
  6 Mixing valve zone 1 and zone 2 cable shall be (3 x min 1.5 mm²), of type designation 60245 IEC 57 or heavier.
  7 Room sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm²), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  8 Pool water sensor and solar sensor cable shall be (2 x min 0.3 mm²), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  9 Water sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  10 Demand signal cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  11 SG signal cable shall be (3 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  12 Heat/Cool switch cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  13 External compressor switch cable shall be (2 x min 0.3 mm²), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide the optional cables and power supply cord
(view without internal wiring)

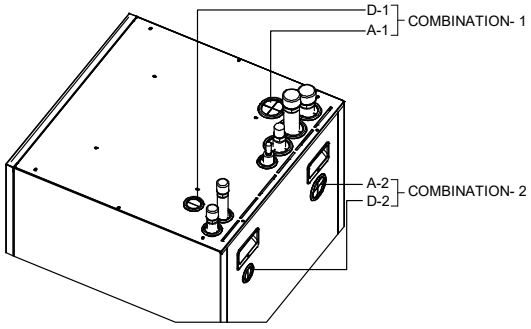


Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

14.4.2.2 Guide Optional Cables and Power Supply Cords to Bushings

⚠ CAUTION
Wire guiding shall be free from hot surfaces. Else, cable insulator damage and electrical shock may happen.
Wire ways shall be smooth and free from sharp edges. Else, cable insulator damage and electrical shock may happen.

■ Use either “COMBINATION-1” or “COMBINATION-2” for guiding Optional Cables and Power Supply Cords to Bushings.



■ A-1 and A-2 Bushings are for:

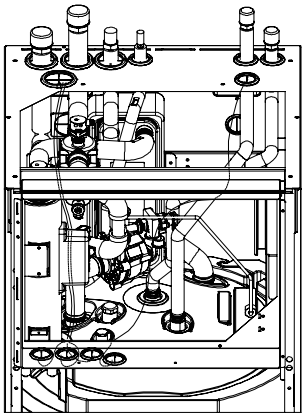
- Power Supply 1 cord
- Power Supply 2 cord
- Indoor Unit/Outdoor Unit Connecting cable
- Pump zone 1 cable
- Pump zone 2 cable
- Solar pump cable
- Room thermostat zone 1 cable
- Room thermostat zone 2 cable
- Mixing valve zone 1 cable
- Mixing valve zone 2 cable
- 2-way valve cable
- Extra pump cable
- Boiler contact cable

■ D-1 and D-2 Bushings are for:

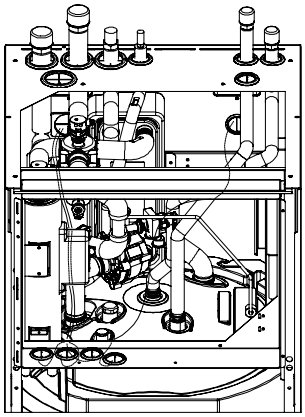
- External control cable
- Outdoor air sensor cable
- Remote controller cable
- Room sensor zone 1 cable
- Room sensor zone 2 cable
- Buffer tank sensor cable
- Pool sensor cable
- Water sensor zone 1 cable
- Water sensor zone 2 cable
- Demand signal cable
- Solar sensor cable
- SG signal cable
- Heat/Cool switch cable
- External Compressor switch cable

■ Ensure all sensor cables are not touching with Front Panel ⑯

■ Guide the wiring inside the unit like below figure.
Once all wiring work done, tie the cable / cord with the banding strap (field supply), to prevent them touching with hot surfaces such as Heater Assembly, bare copper pipes and etc.



Wiring for “COMBINATION-1”



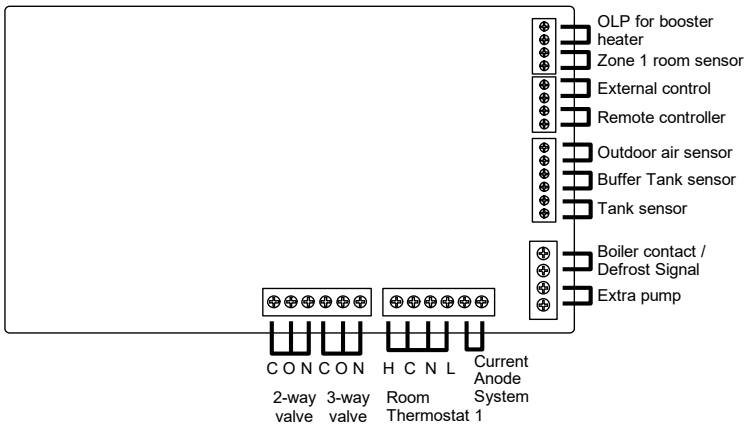
Wiring for “COMBINATION-2”

14.4.2.3 Connecting Cables Length

When connecting cables between Tank Unit and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Mixing valve	50
Room thermostat	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Room sensor	30
Outdoor air sensor	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

14.4.2.4 Connection of the Main PCB



- Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

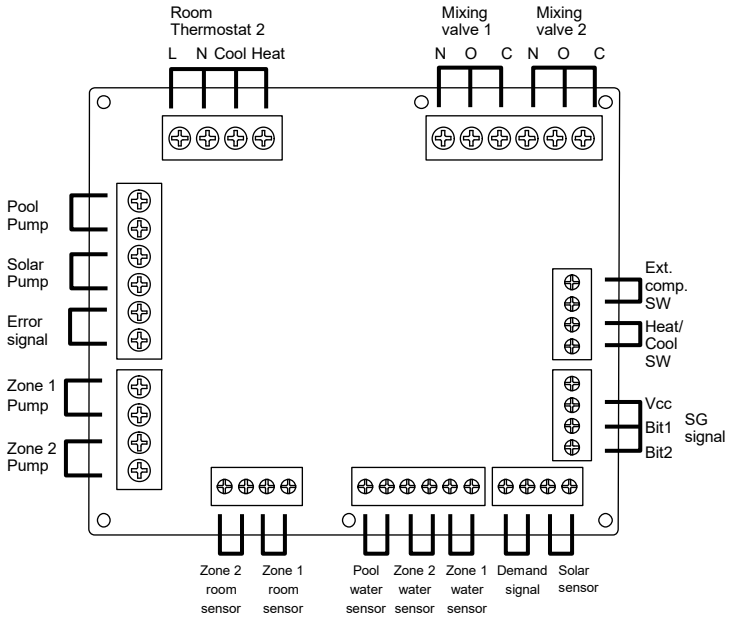
- Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)	AC230V, 12 VA
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)	AC230V, 12 VA
Extra pump	AC230V (Used when Tank Unit pump capacity is insufficient)	AC 230V, 0.6 A max
Boiler contact / Defrost signal	Dry contact (System setup necessary)	

- Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT ❇It does not work when using the Optional PCB
Outdoor air sensor	AW-A2W-TSOD (Total cable length shall be 30m or less)
Buffer tank sensor	PAW-A2W-TSBU

14.4.2.5 Connection of Optional PCB (CZ-NS5P)



Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.OFF, Short=Comp.ON (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

- Outputs

Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s	AC230V, 6VA
Pool pump	AC230V	AC 230V, 0.6 A max
Solar pump	AC230V	AC 230V, 0.6 A max
Zone pump	AC230V	AC 230V, 0.6 A max

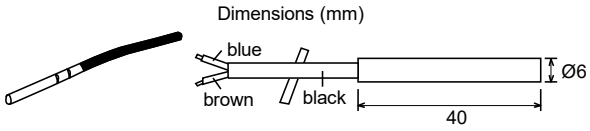
- Thermistor inputs

Zone room sensor	PAW-A2W-TSRT
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

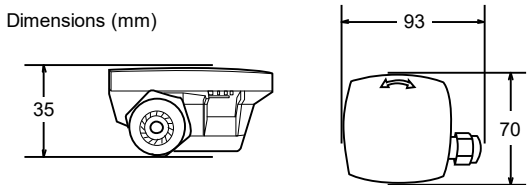
14.4.2.6 Recommended External Device Specification

- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.

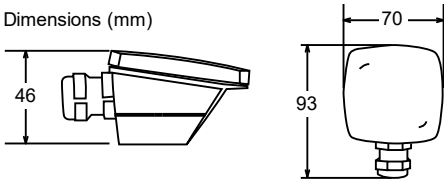
1 Buffer tank sensor: PAW-A2W-TSBU
Use for measurement of the buffer tank temperature.
Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



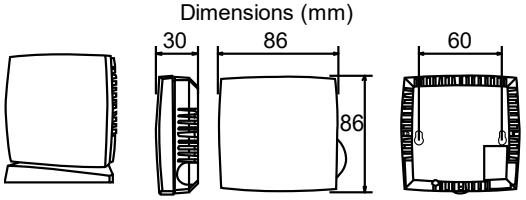
2 Zone water sensor: PAW-A2W-TSHC
Use to detect the water temperature of the control zone.
Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



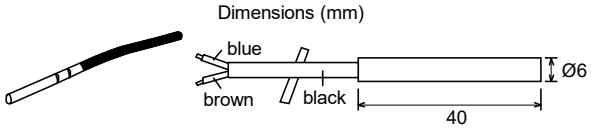
3 Outdoor sensor: PAW-A2W-TSOD
If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



4 Room sensor: PAW-A2W-TSRT
Install the room temperature sensor to the room which requires room temperature control.



5 Solar sensor: PAW-A2W-TSSO
Use for measurement of the solar panel temperature.
Insert the sensor into the sensor pocket and paste it on the solar panel surface.

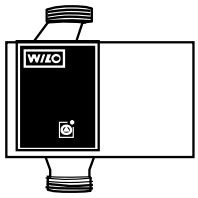


6 Please refer to the table below for sensor characteristic of the sensors mentioned above.

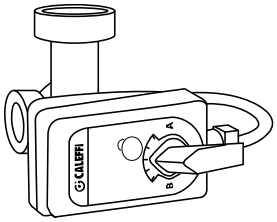
Temperature (°C)	Resistance (kΩ)
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo



- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

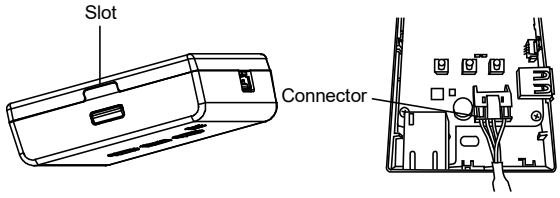


⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

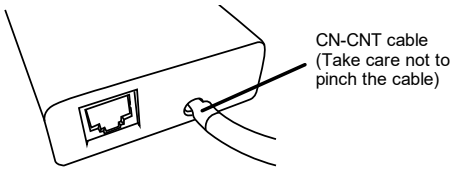
14.4.2.7 Network Adaptor 6 Installation (Optional)

1 **Remove the Control Board Cover ⑤, then connect the cable included with this adaptor to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Tank Unit so that there is no pinching.
   - If an Optional PCB has been installed in the Tank Unit, connect to the CN-CNT connector of the Optional PCB.

2 **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**

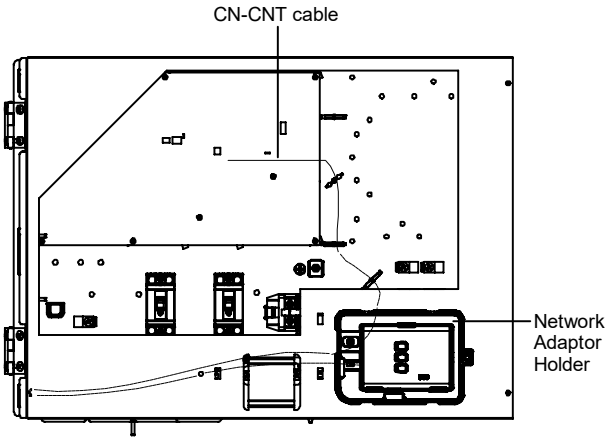


3 **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**

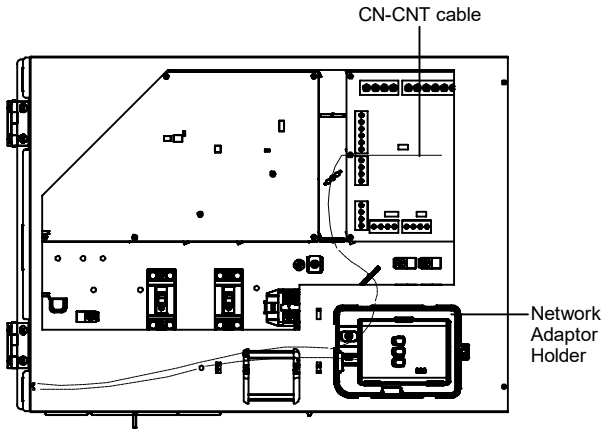


4 **Fix the Network Adaptor 6 to Network Adaptor Holder.**
Guide the cable as shown in the diagram so that external forces cannot act on the connector in the adaptor.

Connection examples:



Without Optional PCB

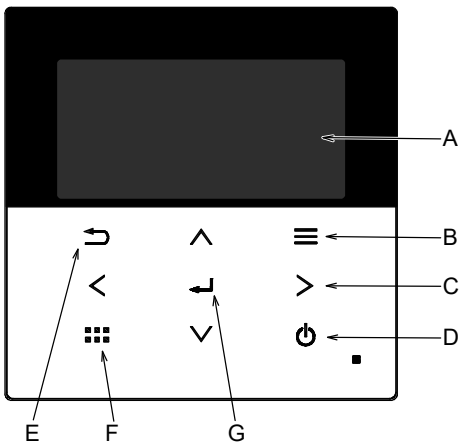


With Optional PCB

14.4.3 System Installation

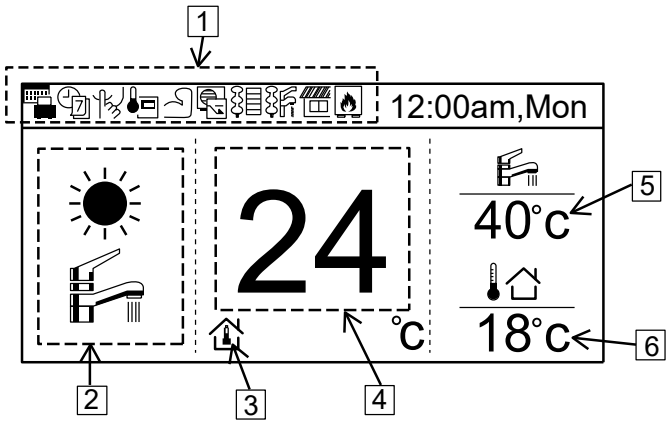
14.4.3.1 Remote Controller Outline

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.



LCD Display
(Actual - Dark background with white icons)

Name	Function
A: Main screen	Display information
B: Menu	Open/Close main menu
C: Triangle (Move)	Select or change item
D: Operate	Start/Stop operation
E: Back	Back to previous item
F: Quick Menu	Open/Close Quick menu
G: OK	Confirm



	Name	Function
1:	Function icon	Display set function/status

Holiday mode | Demand control
Weekly timer | Room heater
Quiet mode | Tank heater
Remote controller room thermostat | Solar
Powerful mode | Boiler

2: Mode — Display set mode/current status of mode

Heating | Cooling
Auto | Hot water supply | Auto heating | Auto cooling
Heat pump operating

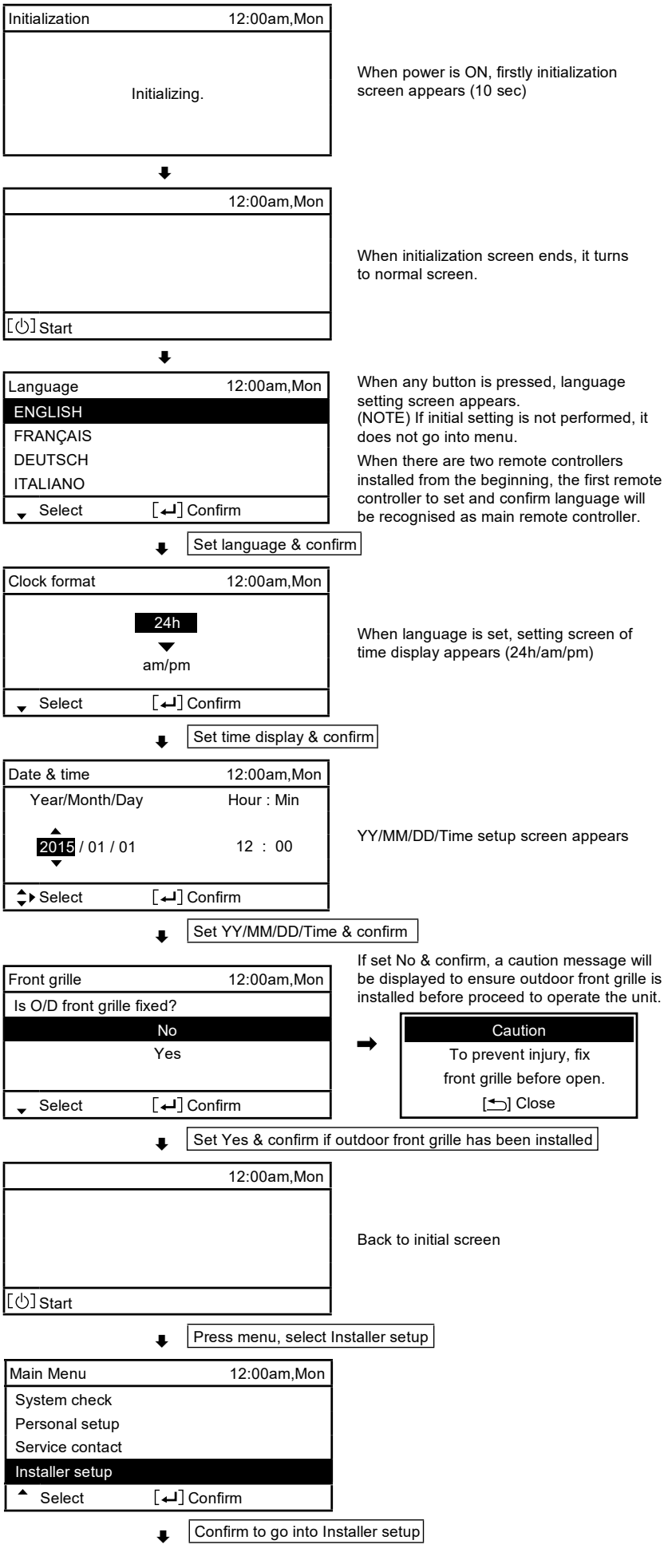
3: Temp setting — Set room temp | Compensation curve | Set direct water temp | Set pool temp

4: Display Heat temp — Display current heating temperature (it is set temperature when enclosed by line)

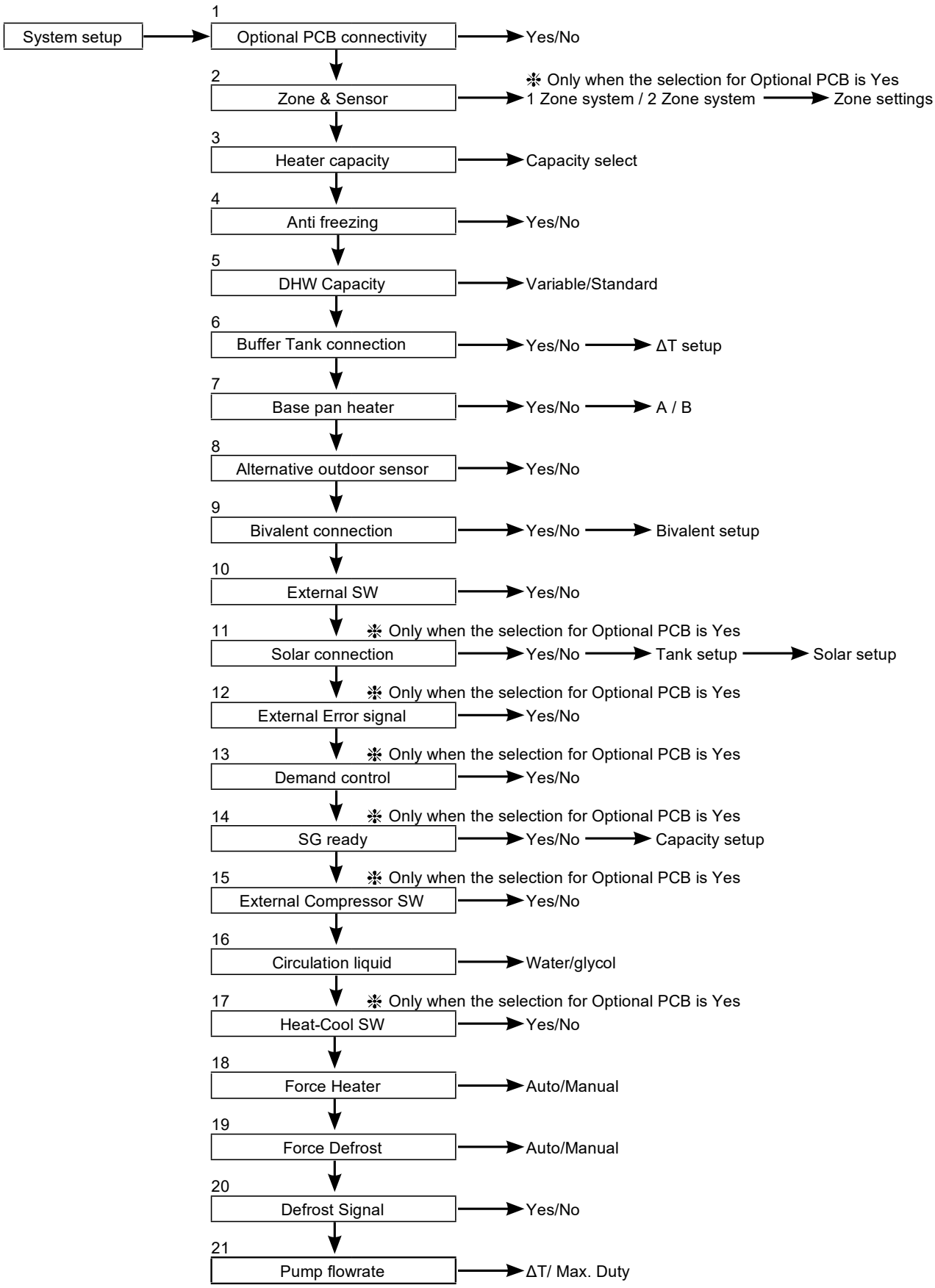
5: Display tank temp — Display current tank temperature (it is set temperature when enclosed by line)

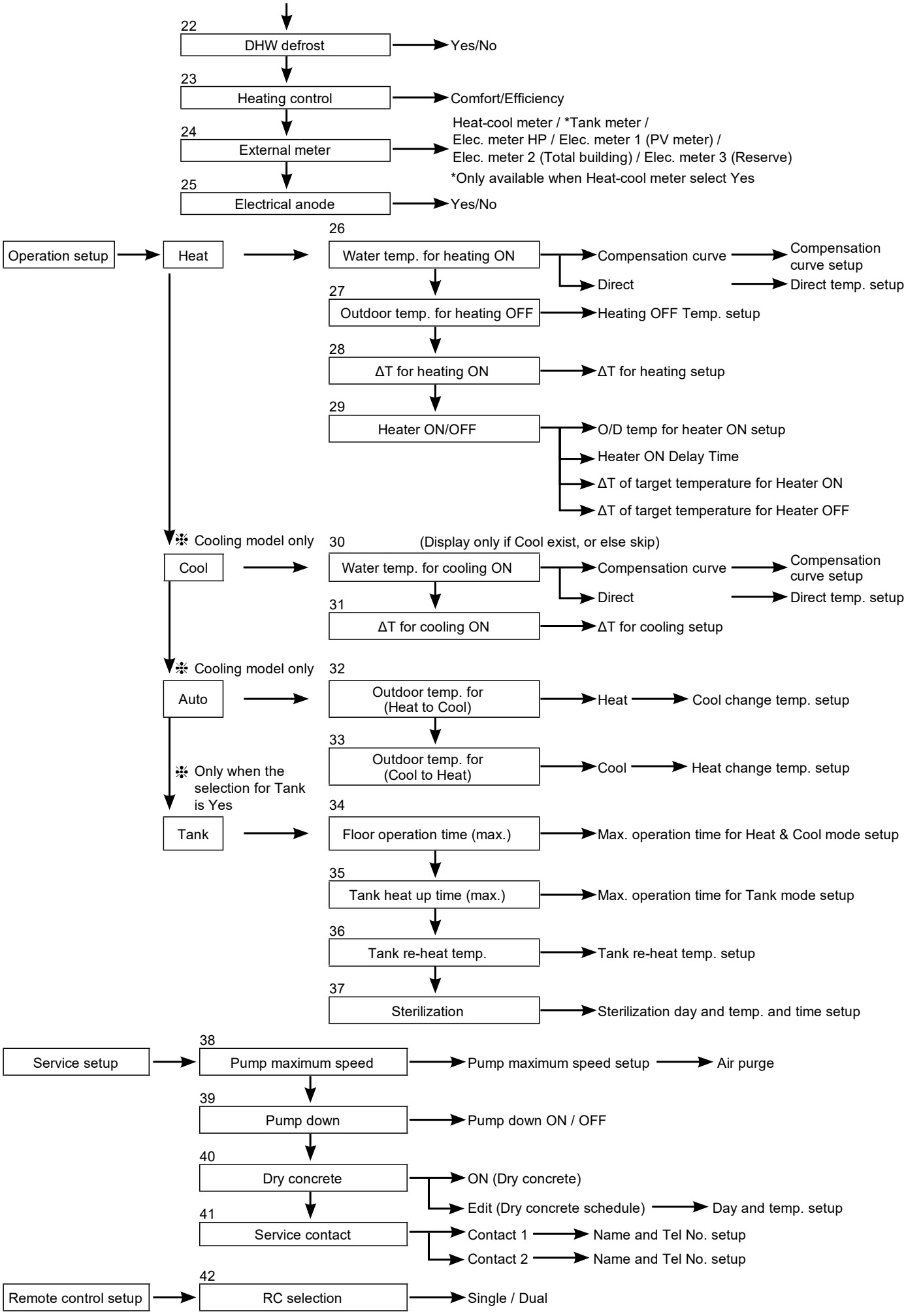
6: Outdoor temp — Display outdoor temp

First time of power ON (Start of installation)



14.4.3.2 Installer Setup



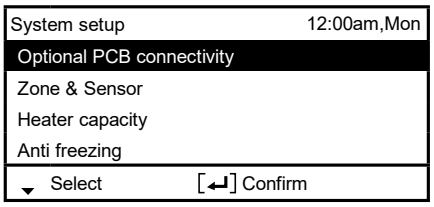

22
DHW defrost
Yes/No
23
Heating control
Comfort/Efficiency
24
External meter
Heat-cool meter / *Tank meter /
Elec. meter HP / Elec. meter 1 (PV meter) /
Elec. meter 2 (Total building) / Elec. meter 3 (Reserve)
*Only available when Heat-cool meter select Yes
25
Electrical anode
Yes/No
Operation setup
Heat
26
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
27
Outdoor temp. for heating OFF
Heating OFF Temp. setup
28
ΔT for heating ON
ΔT for heating setup
29
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
❄ Cooling model only
Cool
30
(Display only if Cool exist, or else skip)
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
31
ΔT for cooling ON
ΔT for cooling setup
❄ Cooling model only
Auto
32
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
33
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
❄ Only when the selection for Tank is Yes
Tank
34
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
35
Tank heat up time (max.)
Max. operation time for Tank mode setup
36
Tank re-heat temp.
Tank re-heat temp. setup
37
Sterilization
Sterilization day and temp. and time setup
Service setup
38
Pump maximum speed
Pump maximum speed setup
Air purge
39
Pump down
Pump down ON / OFF
40
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
41
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup
Remote control setup
42
RC selection
Single / Dual

14.4.3.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install Optional PCB.
Please select Yes after installing Optional PCB.

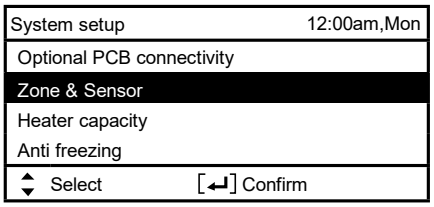
- 2-zone control
- Pool
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

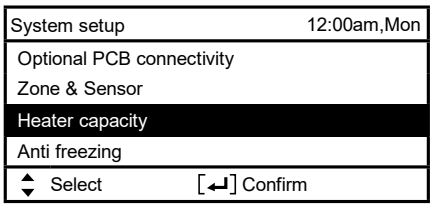
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

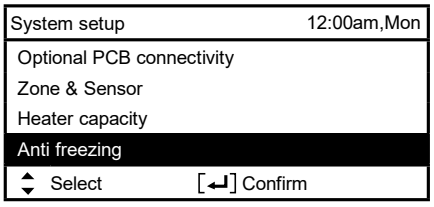
(NOTE) There are models which cannot select Heater capacity.



4. Anti freezing | Initial setting: Yes

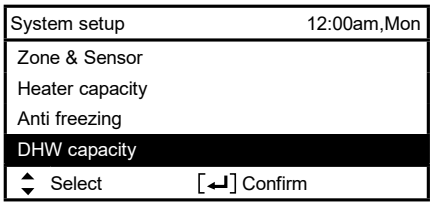
Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



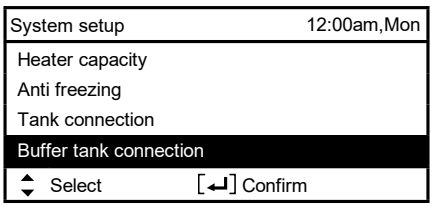
5. DHW Capacity | Initial setting: Variable

Variable DHW capacity setting normally run with efficient boiling which is energy saving heating. But while hot water usage high and tank water temperature low, variable DHW mode will run with fast heat up which heat up the tank with high heating capacity.
If standard DHW capacity setting is selected, heat pump run with heating rated capacity at tank heat up operation.



6. Buffer Tank connection | Initial setting: No

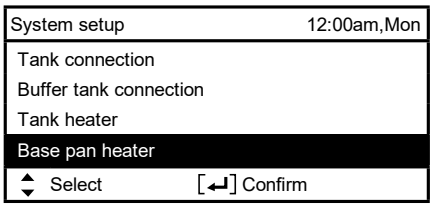
Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
If the buffer tank capacity is not so large, please set larger value for ΔT.



7. Base pan heater | Initial setting: No

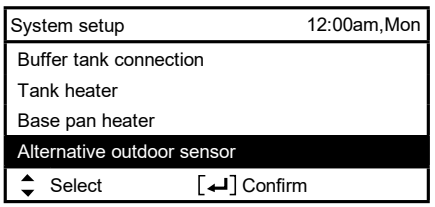
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



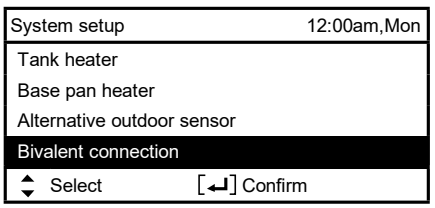
8. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



9. Bivalent connection | Initial setting: No

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.



After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)
1) SG ready (Only available to set when optional PCB set to YES)
- SG Ready input from optional PCB terminal control ON/OFF of boiler and heat pump as below condition

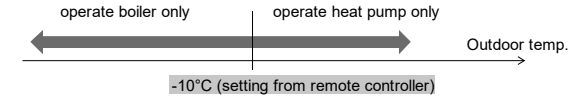
SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

* This bivalent SG ready input is sharing same terminal as [14. SG ready] connection. Only one of these two setting can be set at the same time.
When one is set, another setting will reset to not set.

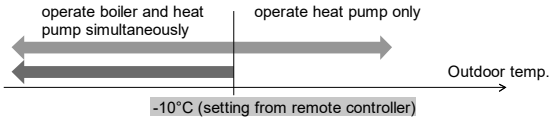
2) Auto
There are 3 different modes in the boiler auto pattern operation. Movement of each modes are shown below.
② Alternative (switch to boiler operation when drops below setting temperature)
③ Parallel (allow boiler operation when drops below setting temperature)
④ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is “ON”, “boiler contact” is “ON”, “_”(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

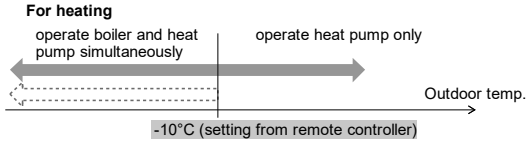


Parallel mode

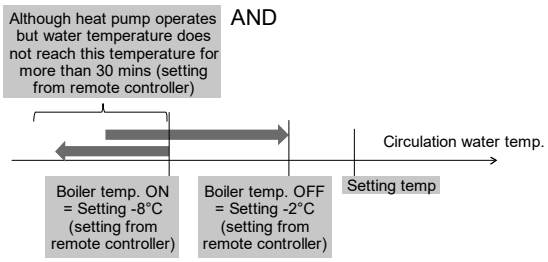


Advanced Parallel mode

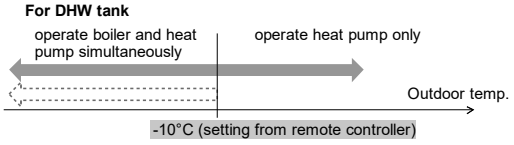
For heating



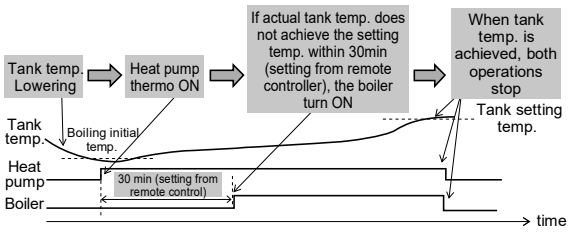
AND



For DHW tank



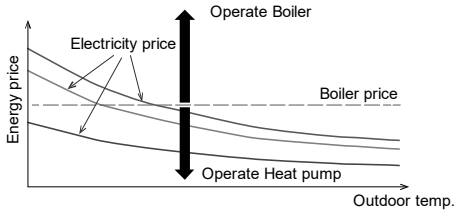
AND



In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of “Heating/Tank” mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

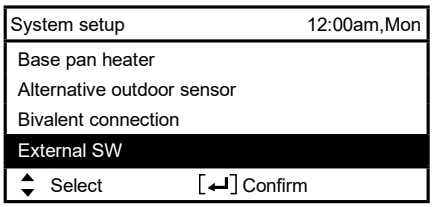
3) Smart

There are Energy price (both electricity and boiler) and Schedule to be set on remote controller.
Operation setting of Energy price and Schedule shall be responsible by installer.
Based on these settings, system will calculate the final price for both electricity and boiler.
When final price of Electricity is lower than Boiler's, heat pump will operate.
When final price of Electricity is higher than Boiler's, boiler will operate.



10. External SW | Initial setting: No

Able to turn ON/OFF the operation by external switch.

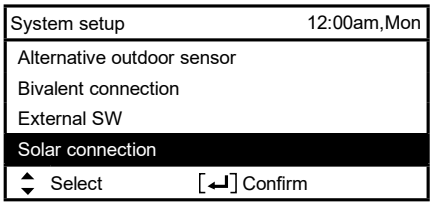


11. Solar connection | Initial setting: No

Set when solar water heater is installed.

Setting include items below.

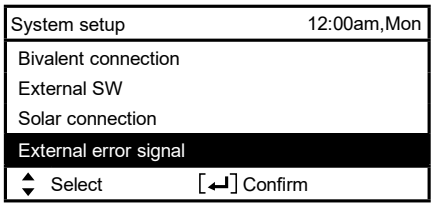
① Set either buffer tank or DHW tank for connection with solar water heater.
② Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to operate the solar pump.
③ Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to stop the solar pump.
④ Anti-freezing operation start temperature (please change setting based on usage of glycol.)
⑤ Solar pump stop operation when it exceeds high limit temperature (when tank temperature exceed designated temperature (70~90°C))



12. External Error Signal | Initial setting: No

Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

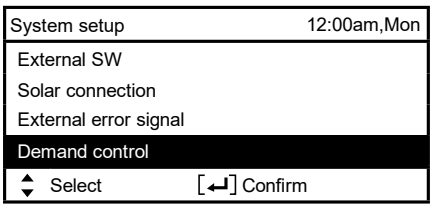
(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off “close” from the display, error signal will still remain ON.



13. Demand control | Initial setting: No

Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.



Analog input [v]	Rate [%]
0.0	not activate
0.1 ~ 0.6	
0.7	10
0.8	
0.9 ~ 1.1	10
1.2	15
1.3	
1.4 ~ 1.6	15
1.7	20
1.8	
1.9 ~ 2.1	20
2.2	25
2.3	
2.4 ~ 2.6	25
2.7	30
2.8	
2.9 ~ 3.1	30
3.2	35
3.3	
3.4 ~ 3.6	35
3.7	40
3.8	

Analog input [v]	Rate [%]
3.9 ~ 4.1	40
4.2	45
4.3	
4.4 ~ 4.6	45
4.7	50
4.8	
4.9 ~ 5.1	50
5.2	55
5.3	
5.4 ~ 5.6	55
5.7	60
5.8	
5.9 ~ 6.1	60
6.2	65
6.3	
6.4 ~ 6.6	65
6.7	70
6.8	
6.9 ~ 7.1	70
7.2	75
7.3	

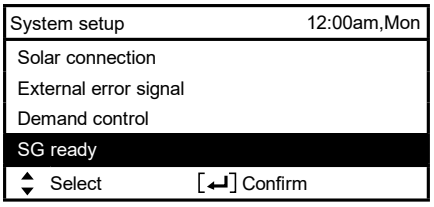
Analog input [v]	Rate [%]
7.4 ~ 7.6	75
7.7	80
7.8	
7.9 ~ 8.1	80
8.2	85
8.3	
8.4 ~ 8.6	85
8.7	90
8.8	
8.9 ~ 9.1	90
9.2	95
9.3	
9.4 ~ 9.6	95
9.7	100
9.8	
9.9 ~	100

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

14. SG ready | Initial setting: No

Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2



Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

} Set by SG ready setting of remote controller

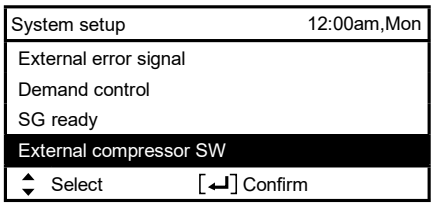
(When SG ready set to YES, Bivalent control pattern will set to Auto.)
(NOTE) Does not display if there is no Optional PCB.

15. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

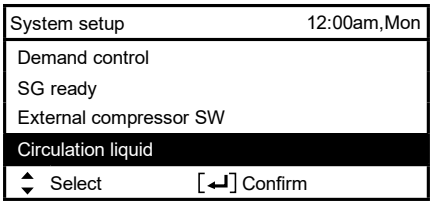


16. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

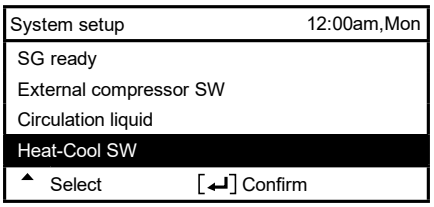


17. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

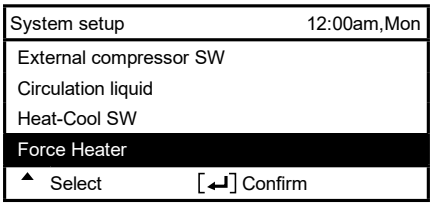


18. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

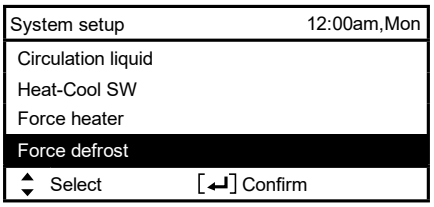
Heater source will ON during force heater mode.



19. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

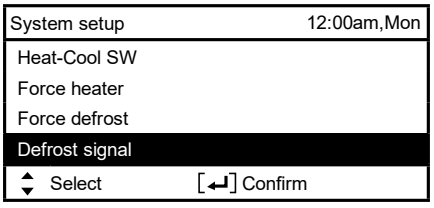
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition. (Even auto is selected, user still can turn on force defrost through quick menu)



20. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

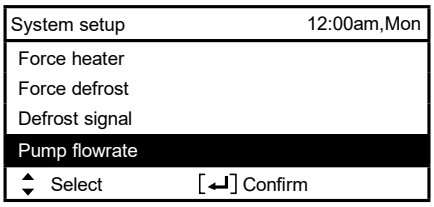
When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).



21. Pump flowrate | Initial setting: ΔT

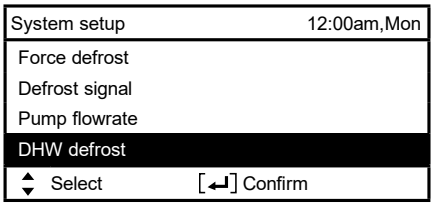
If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



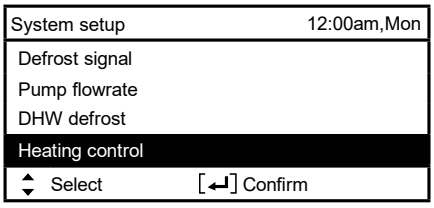
22. DHW defrost | Initial setting: Yes

When DHW defrost set to YES, hot water of domestic hot water tank will be used during defrost cycle.
When DHW defrost set to NO, hot water of floor heating circuit will be used during defrost cycle.



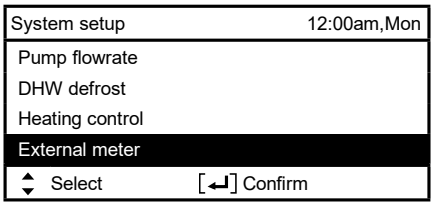
23. Heating control | Initial setting : Comfort

There are two modes to select for compressor frequency control : Comfort or Efficiency.
When set to Comfort mode, compressor will run at zone limit maximum frequency to reach set temperature faster.
When set to Efficiency mode, compressor will run at part load frequency at initial stage for energy saving.



24. External meter

Initial setting : [Heat-cool meter : No]
[Tank meter : No] *only available when Heat-cool meter select Yes
[Elec. meter HP : No]
[Elec. meter 1 (PV meter) : No]
[Elec. meter 2 (Total building) : No]
[Elec. meter 3 (Reserve) : No]



There are two systems for generation meter connection : one generation meter system (Heat-cool meter) or two generation meter system (Heat-cool meter and Tank meter)
Both systems can provide all generation data of heating, cooling and DHW directly from external meter.

If Heat-cool meter is set to Yes, it will read from external meter for heat pump's energy generation data during heating, cooling and DHW operation [*1].
If Heat-cool meter is set to No, it will base on unit's calculation for heat pump's energy generation data during heating, cooling and DHW operation.
If Tank meter is set to Yes, it will read from external meter for heat pump's energy generation data during DHW operation [*1].

If Elec. meter HP is set to Yes, it will read from external meter for heat pump's energy consumption data.
If Elec. meter HP is set to No, it will base on unit's calculation for heat pump's energy consumption data.

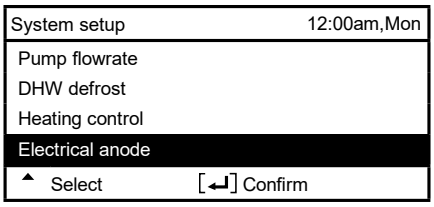
If Elec. meter 1 (PV meter) is set to Yes, it will read from external meter for energy generation data of solar system and display it on Cloud system.
If Elec. meter 2 (Building) is set to Yes, it will read from external meter for energy consumption data of the building and display it on Cloud system.
If Elec. meter 3 (Reserve) is set to Yes, it will read from external meter for energy consumption data obtained from reserved electricity meter and display it on Cloud system.

[*1] Set Heat-cool meter to Yes and set Tank meter to No when 1 generation meter system is installed.
Set Heat-cool meter to Yes and set Tank meter to Yes when 2 generation meter system is installed.

Remark : Elec. meter HP refers to Electricity meter that measures Heat Pump unit's consumption.
Elec. meter 1 / 2 / 3 refers to Electricity meter no. 1 / no. 2 / no. 3

25. Electrical anode | Initial setting : No

When Electrical anode set to YES, anode will be turned on.
When Electrical anode set to NO, anode will not be turned on.



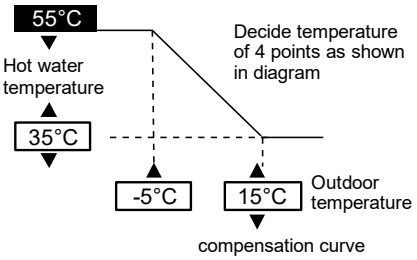
14.4.3.4 Operation Setup

Heat

26. Water temp. for heating ON | Initial setting: compensation curve

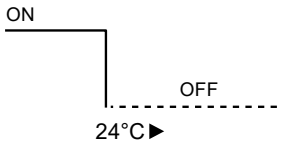
Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



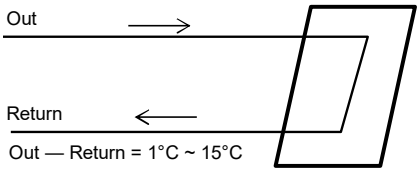
27. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C



28. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

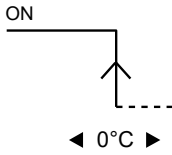


29. Heater ON/OFF

a. Outdoor temp. for heater ON | Initial setting: 0°C

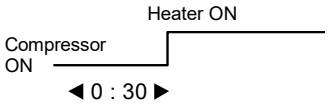
Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.



b. Heater ON delay time | Initial setting: 30 minutes

Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

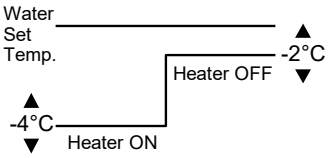


c. Heater ON: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp | Initial setting: -2°C

Set water temperature for heater to turn off at heat mode.
Setting range is -8°C ~ 0°C

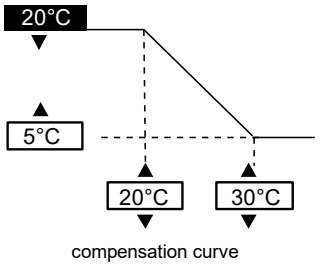


Cool

30. Water temp. for cooling ON | Initial setting: compensation curve

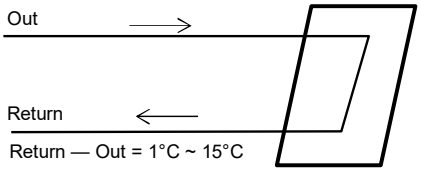
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



31. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

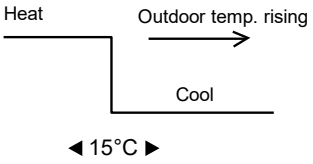


Auto

32. Outdoor temp. for (Heat to Cool) | Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 5°C ~ 25°C

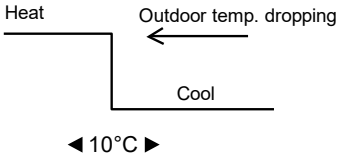
Timing of judgement is every 1 hour



33. Outdoor temp. for (Cool to Heat) | Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 25°C

Timing of judgement is every 1 hour

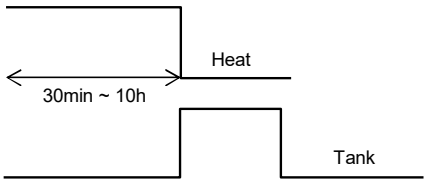


Tank

34. Floor operation time (max) Initial setting: 8h

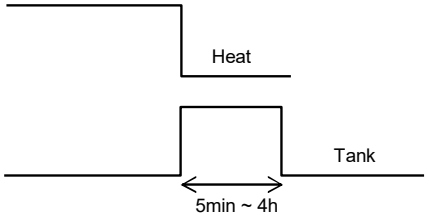
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



35. Tank heat up time (max) Initial setting: 60min

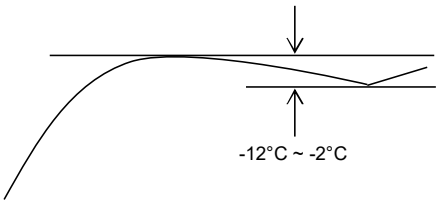
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



36. Tank re-heat temp. Initial setting: -8°C

Set temp to perform reboil of tank water.
(When boiled by heat pump only, (51°C – Tank re-heat temp) shall become max temp.)

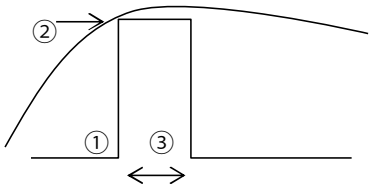
Setting range is -12°C ~ -2°C



37. Sterilization Initial setting: 65°C 10min

Set timer to perform sterilization.
① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~75°C ❋ If use back-up heater, it is 65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.

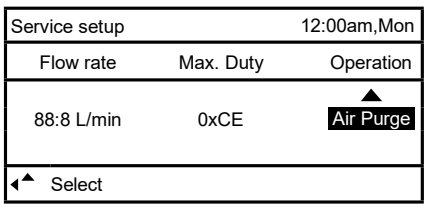


14.4.3.5 Service Setup

38. Pump maximum speed | Initial setting: Depend on model

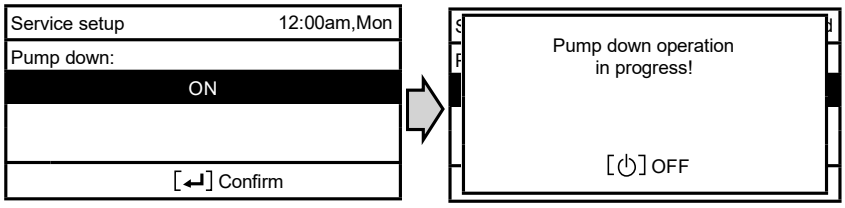
Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.



39. Pump down

Operate pump down operation

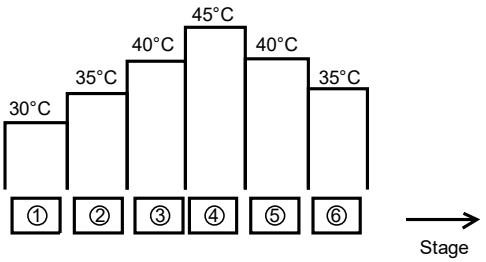


40. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

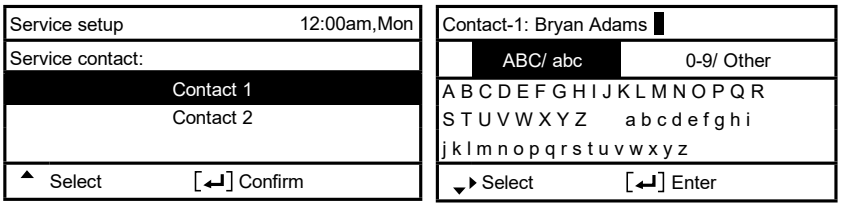
When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



41. Service contact

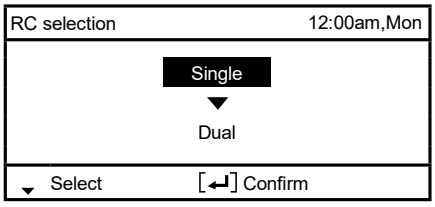
Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)



14.4.3.6 Remote Control Setup

42. RC selection | Initial setting : Single

Set to "Single" when only one remote controller is installed.
Set to "Dual" when two remote controllers are installed.



14.5 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

14.5.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

14.5.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it. In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer. When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

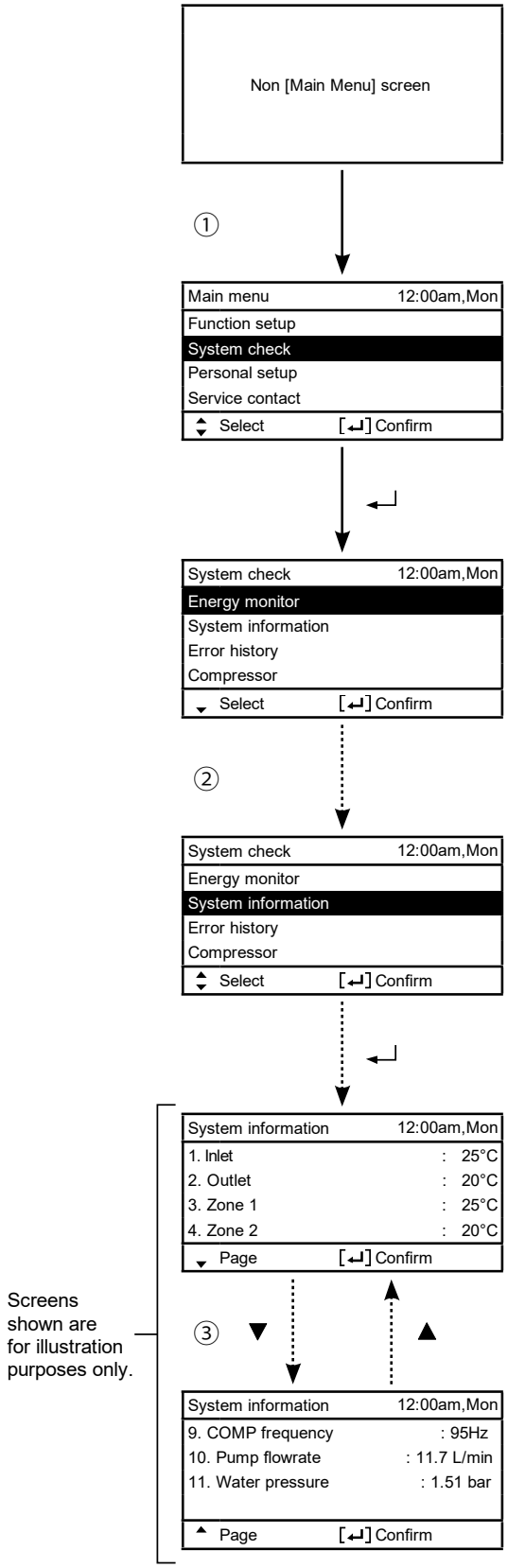
③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

14.5.3 Check Water Pressure from Remote Controller

1. Press ☰ SW and scroll to "System check".
2. Press ↵ and scroll to "System information".
3. Press ↵ and search for "Water pressure".



15. Installation and Servicing Heat Pump using R32

15.1 About R32 Refrigerant

For heat pump refrigerants such as R410A, the refrigerants were collected back in order to prevent their air dissipation, to curbe the global warming impact, in case they were released into the atmosphere. In the “4th Environmental Basic Plan”, 80% reduction of greenhouse gas emissions by 2050 is required, and due to this requirement, further reduction in the emission of high greenhouse effect gas, such as CFCs, is required. Therefore, the conversion of heat pump refrigerant into the ones who has smaller greenhouse effect, even if it is dissipated into the atmosphere, became our responsibility.

Nevertheless, in case of heat pump refrigerant, it would be the best if there is a refrigerant which has smaller impact on global warming, but ensures good energy efficiency and performance, and is safe; however, there is no such refrigerant which satisfies all these conditions. As a result, we have been considering the practical usage, within the safety frame-work, of R32 refrigerant which has short lifetime in the atmosphere, and has smaller effect of global warming, but is slightly flammable.

In 2004, due to the revision of heat pump safety standards by the International Electro-safety Commission (IEC), the safety standards of heat pump using slightly flammable refrigerant was issued. In 2010, the regulations of American Society of Heating, Refrigerating and Air-Conditioning Engineers in the United States (ANSI/ASHRAE34) was issued adopting the grades for refrigerants which are difficult to inflame due to their slow burning rates, and as a result have smaller damages in cases of fire. The burning rate of R32 is lower by 10cm / per second, and safety standardization for various usage is now being processed.

15.2 Characteristics of R32 Refrigerant

1. Chemical Characteristics

R32 is one of the refrigerants used in R410A, has almost no toxicity, and chemically stable compound formed by hydrogen, carbon and fluorine.

R32 has short lifetime of 4 to 9 years in case of being released into the atmosphere; therefore, it has smaller greenhouse gas effect but has slight inflammability because of the large proportion of hydrogen.

Chemical Characteristic Table of R32, R410A and R22.

	R32	R410A	R22
Chemical Formula	CH2F2	CH2F2 / CHF2CF3	CHCLF2
Composition (mixture ratio wt.%)	Single Composition	R32 / R125A (50 / 50 wt.%)	Single Composition
Boiling Point (°C)	-51.7	-51.5	-40.8
Pressure (physical) *1	3.14	3.07	1.94
Capacity (physical) *2	160	141	100
COP (physical) *3	95	91	100
Ozone Depletion Potential (ODP)	0	0	0.055
Global Warming Potential (GWP) *4	675	2090	1810
Inflammability *5	Slightly Inflammable (A2L)	Non-inflammable (A1)	Non-inflammable (A1)
Toxicity	None	None	None

*1：Physical property of temperature condition 50°C
*2：Relative value of temperature condition 0/50°C, providing R22=100
*3：Te/Tc/SC/SH=5/50/3/0°C
*4：GWP=Global Warming Potential, each figure is based on “4th IPCC4 Report”
*5：Based on ANSI / ASHRAE std. 34-2010

2. Characteristic of Pressure

As shown in Table 2, R32 does not have much difference in vapor pressure at the same refrigerant temperature comparing to R410A, but comparing to R22, it is higher at 1.6 times more. Thus, the same as in case of R410A, it is necessary to do installation and service using high-pressure tools and components.

Table 2. Saturated vapor pressure comparison table

(Unit: MPa)

Temperature	Refrigerant		
	R32	R410A	R22
-20	0.30	0.30	0.14
0	0.71	0.70	0.40
20	1.37	1.35	0.81
40	2.38	2.32	1.43
60	3.84	3.73	2.33
65	4.29	4.17	2.60

Reference： Thermal properties table of Japan Society of Refrigerating and Air Conditioning Engineers (60, 65°C)
NIST REFPROP V8.0 (-20 ~ 40°C)

15.3 Refrigerant piping installation • Tools used in services

15.3.1 Required Tools

R32 refrigerant heat pump use the common parts as R410A heat pump for two-way valves and three-way valves (diameters of service ports); thus, they maintain commonality in the maintenance of the compressive strength, the size of pipe flaring, and the size of flare nuts as R410A. Therefore, for refrigerant pipe installation and services, you can use tools for R410A.

However, mixing of refrigerants is not allowed, so that you have to separate the cylinders for the recovery of refrigerants.

Tools used for installation • relocation • replacement of heat pump units

Works	R32	R410A	R22
Flaring	Flare tools for R410A (clutch type)		Flare tools for R22 (clutch type)
Connection of pipes	Torque wrench (diameter 1/4 3/8)		
	Torque wrench (diameter 1/2 5/8) *1		Toque wrench (diameter 1/2 5/8)
Manifold gauge charging hose	R32 & R410A Common (As at November 2013)		R22 Only
Air purging	Vacuum pump + Reducer / expander		Vacuum pump
Gas leakage test	Detection liquid or soup water, HFC detector		

*1. Nut diameters of 1/2 5/8, the size of torque wrench common with R410A
For other installation, you can use general tools such as screw drivers (+, -), metal saws, electric drills, long-nose pliers, hole core drills (ø70 or ø65), linen tape, levels, temperature gauges, clamp meters, electric knives, nippers, pipe cutters, reamers or scrapers, spring benders, (diameters 1/4 3/8 1/2 5/8), monkey wrenches, fixing wrenches (17 or 12 mm), feeler gauges, hexagon wrenches (4 mm), testers, megohm testers, etc.

Tools used for services

Works	R32	R410A	R22
Insertion of refrigerant	Digital scale for refrigerant charging, refrigerant cylinders, cylinder adopters and packing *a		
Recovery of refrigerant	Refrigerant recovery devices, refrigerant cylinders, manifold gauges, charging hoses *b		

*a. Use cylinder for each refrigerant, cylinder adopter and packing.
*b. Use refrigerant recovery cylinder separately for each refrigerant **(no mixture of refrigerant allowed)**. Please be aware that there are some refrigerant collection devices which do not have self-certification.

15.3.2 Tools for R32 (common with R410A)

1. Flare gauges
Use flare gauges when you perform flaring with flare tools (crutch type). Flare gauges are used to set the pipe ends at 0.5 ~ 1.5 mm from clump bars of flare tools.

Flare gauges



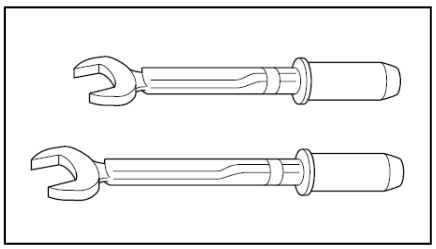
2. Flare tools (clutch type)
Flare tools have larger holes of clump bars in order to set the pipe end at 0 ~ 0.5 mm, and have stronger springs inside to ensure solid flaring torques. These flare tools can be used commonly for R22.

Flare tools (clutch type)



3. Torque wrenches (diameters 1/2, 5/8)
In order to strengthen the compressive strength, the diameters of wrenches change depending on the flare nut sizes.

Torque wrenches



Differences in torque wrenches

	R32 (common R410A)	R22
1/2 (diameter × torque)	26 mm × 55 N•m (550 kgf•cm)	24 mm × 55 N•m (550 kgf•cm)
5/8 (diameter × torque)	29 mm × 65 N•m (650 kgf•cm)	27 mm × 65 N•m (650 kgf•cm)

4. Manifold gauges
R22 gauges cannot be used because of the high pressures.

Each port of manifold has different shapes in order to prevent inserting wrong refrigerant.
*However, the port shape for R410A and R32 is the same; therefore, attention need to be paid not to insert wrong refrigerant.

Differences in high/low pressure gauges

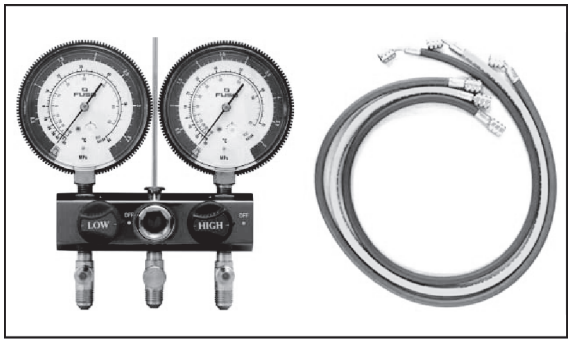
	R32 (common R410A)	R22
High pressure gauges (red)	-0.1 ~ 5.3 MPa -76 cmHg ~ 53 kgf / cm^2	-76 cmHg ~ 35 kgf / cm^2
Low pressure gauges (blue)	-0.1 ~ 3.8 MPa -76 cmHg ~ 38 kgf / cm^2	-76 cmHg ~ 17 kgf / cm^2

Difference in manifold port sizes

	R32 (common R410A)	R22
Port sizes	1/2 UNF20	7/16 UNF20

5. Charging hoses
The pressure resistance of charge hoses is increased. At the same time, the material is changed to HFC resistant, and the size of each manifold adopter is changed, as the port size of manifold gauge itself. Further, some hoses are with anti-gas pressure backflow valves placed near the adopters. (hoses with the valves recommended)

Manifold gauges / Charging hoses

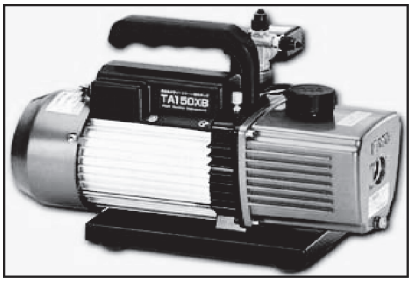


Differences in charging hoses

		R32 (common R410A)	R22
Pressure Resistance	Normal operation pressure	5.1 MPa (52 kgf / cm^2)	3.4 MPa (35 kgf / cm^2)
	Burst pressure	27.4 MPa (280 kgf / cm^2)	17.2 MPa (175 kgf / cm^2)
Material		HNBR rubber Internal nylon coating	NBR rubber

6. Vacuum pump and Vacuum pump adopter
When using a vacuum pump, it is necessary to set a solenoid valve in order to prevent backflow of vacuum pump oil into the charge hoses, and use a vacuum pump with oil backflow prevention function, or use the vacuum pump with vacuum pump adopter.
If vacuum pump oil（mineral oil-based）mixes with R410A (R32), it may cause damage to the machine.

Vacuum pump



Vacuum pump adopter



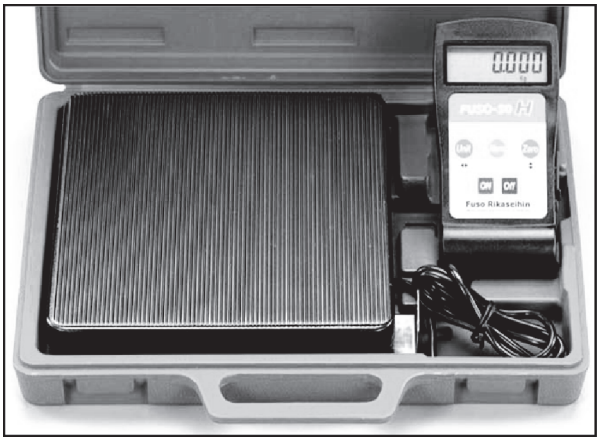
7. HFC refrigerant_Electric gas leakage tester
R32 refrigerant is often used for other mixed refrigerant (R410A, R404A, R407C etc.). Therefore, the usage of existing HFC detectors is possible, but in order to detect more accurately, we recommend to use detectors specially set and adjusted for R32 detection.

HFC refrigerant_Electric gas leakage tester



8. Digital scale for refrigerant charging
R32 and R410A have high pressure level and their evaporation speed is high.
Thus, if you recover the refrigerant by cylinder charging method, the refrigerant evaporates within the weighing scale glass, which makes reading the scale difficult, rather than liquidating the refrigerant into the cylinder. (Charging cylinders for R22 have different pressure resistance, scale, connection port size; therefore, they are not usable) At the same time, the digital scale for refrigerant charging is strengthened by receiving the weight of the refrigerant cylinders with four pillars at the corners. The connection ports of charging hoses have two separate ports for R22 (7/16 UNF20) and R32/R410A (1/2 UNF20) therefore, they can be used for the insertion of the existing refrigerants.

Digital scale for refrigerant charging



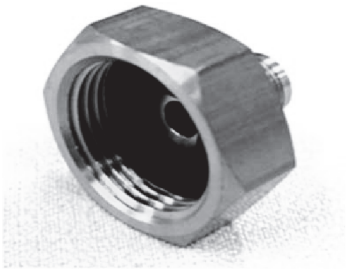
9. Refrigerant cylinders
Refrigerant cylinders for R410A are painted in pink, and the ones for R32 are painted in other colors that might subject to change according to the international standards. R32 is a single refrigerant, so that both liquid and gas insertion are possible. Additional charging is also possible.
(R410A is a mixed refrigerant, so only liquid insertion is possible)

Refrigerant cylinders



10. Connection ports of refrigerant cylinders and packing
Charging ports which fit to the charging hose connection port size (1/2 UNF20) is needed. At the same time, the packing has to be of HFC resistant materials.

Connection ports and packing



11. Tools used for refrigerant piping installations and services

	Tools for R410A	Common with R32	Possibility of usage for R22
1.	Pipe cutters, reamers or scrapers	○	○
2.	Flare tools (clutch type)	○	○
3.	Torque wrench (1/4, 3/8)	○	○
4.	Torque wrench (1/2, 5/8)	○	×
5.	Manifold gauges・charging hoses	○	×
6.	Vacuum pumps, vacuum pump adopters	○ Connection 5/16	○ Connection 1/4
7.	Electric gas leakage testers for HFC *1	○	△
8.	Digital scale for refrigerant charging	○	○
9.	HCF recovery devices (connection port 5/16) *2	○ Connection 5/16	○ Connection 1/4
10.	Refrigerant cylinders (pressure resistant: FC3)	Same specs ×	×
11.	Refrigerant cylinders (pink)	Other (colors that might subject to change according to the international standards).	×
12.	Refrigerant cylinder connection ports and packing	○	×
13.	Allen wrench (4 mm) Electric knives	○	○
*1	Those testers only for HCFC22 (R22), but not for HCF32 (R32) and HCF410A (R410A) cannot be for common use.		
*2	Recovery devices which are self-certified for each HCF type can be used.		

[Knowledge for the common usage of tools for R410A & R32]

- R410A and R32 machines use different compressor oils.
- If unregulated compressor oil gets mixed into, it may cause damage to the machine function.
- Careful pump down will ensure the recovery of compressor oil, and it will minimize the remaining amount of the oil in the manifold gauge and charging hose.
- If you only perform the recovery of refrigerant and not be able to perform pump down, you have to dispose the compressor oil in the charging hose.

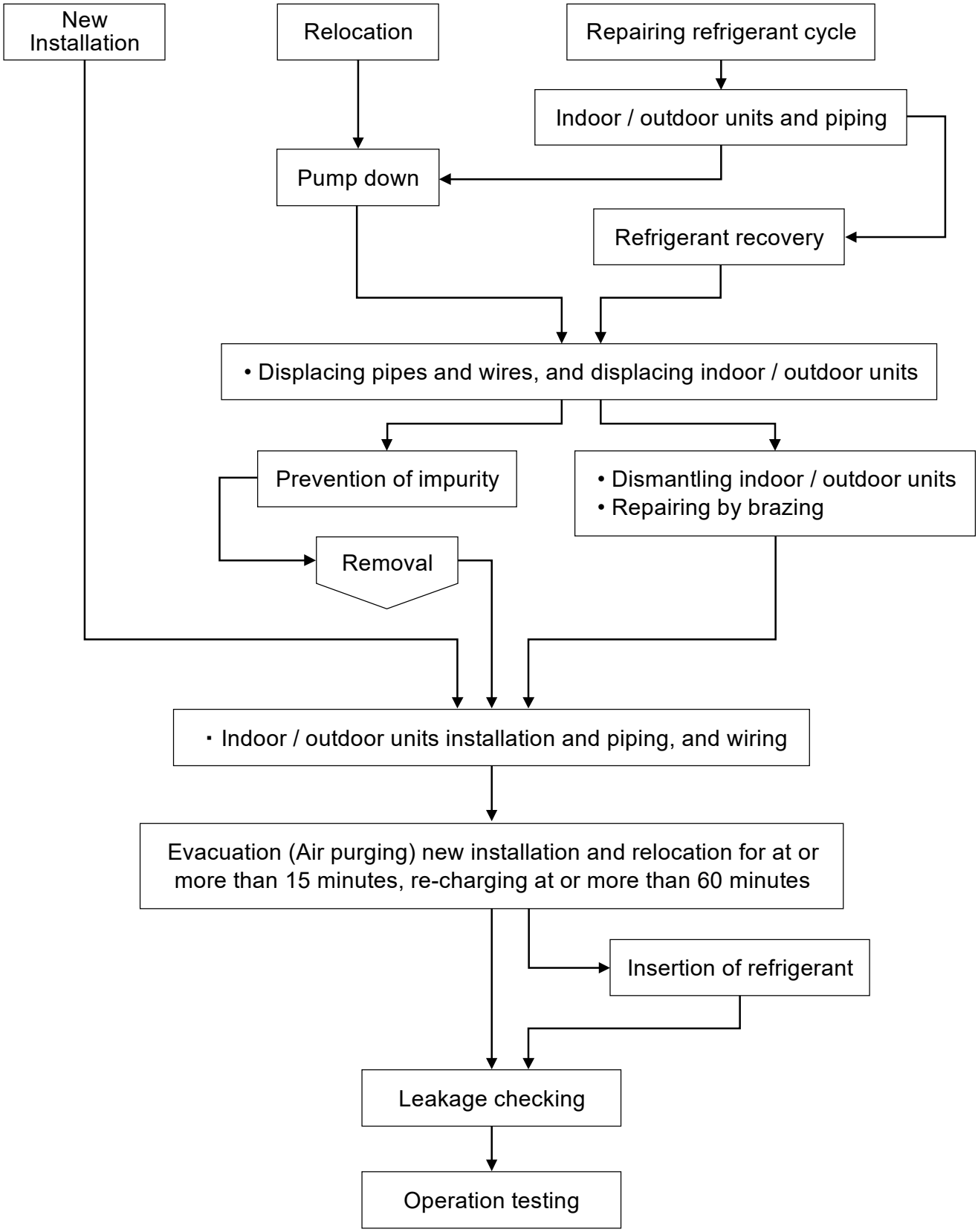
[Precaution of repairing refrigerant cycle]

- In the brazing, open 2-way and 3-way valves, and make sure the refrigerant is completely recovered back and not remaining the system.
- When repairing outside, make sure no refrigerant is in the air, ensure good air flow, and perform the brazing.

[Inserting wrong refrigerant]

- It may cause “not cooling” and “not heating” customer claims because each component (expansion valve, compressor, PCB) of the refrigeration cycle is specially adjusted for R32.
- At the same time, it is not subject to product warranty, if wrong refrigerant was inserted into the system.

15.4 New installation, Relocation, Repairing of Refrigerant Cycle System The Procedures



15.5 Piping installation of R32

15.5.1 Pipe materials used and flaring

Copper pipes are used for refrigerant piping. Pipes which comply with JIS Regulations need to be used. Room heat pump which use R410A and R32 have higher pressure; thus, using pipes which comply with the Regulations is important.
The pipe thickness is regulated by revised JIS B 8607 “Flaring and brazing fittings for refrigerant” and the pipe thickness for R410A, R32 is shown in the table.

Pipe thickness

O and OL materials		Thickness (mm)		
Diameter	Diameter (mm)	R410A	R32	R22
1/4	6.35	0.80		
3/8	9.52	0.80		
1/2	12.70	0.80		
5/8	15.88	1.00		

Caution

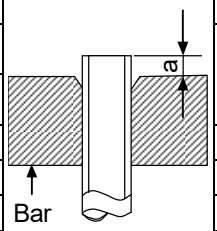
- For connection piping, use copper phosphate seamless pipes (1220T) as regulated in “JIS H 3300” and the pipe thickness is 0.8 mm.
- In the market, there are some pipes of 0.7 mm thickness, but do not use these pipes (0.8 mm thickness has to be strictly followed).
- It is recommended to use pipes whose adhesion amount of oil is at or less than 40 mg / 10 m.
  At the same time, do not use pipes with dent, de-shape, and color change (especially inside).

15.5.2 Processing and connection of pipes

For refrigerant pipe installation, be aware of moisture and dirt do not get into the pipes, and make sure of no refrigerant leakage.
◎ The procedure of flaring and precautions

a) Cutting of pipes：use pipe cutter and cut the pipe slowly not to de-shape the pipe.
b) Removal of burrs on the edge of pipe (reamer or scraper)
   If the condition of pipe edge after the deburring is no good or if burrs attaches on the flaring, it may cause refrigerant leakage. Turn the pipe end down and perform deburring carefully.
c) Insert the flare nut (use the nut which is a part of the CZ parts)
d) Flaring
   Ensure the cleanliness of clump bar and pipe, and perform flaring carefully.
   Use the existing flare tools or flare tools for R410A. Be aware that the sizes and dimensions of flaring is different in each flaring tool. If you use the existing flaring tools, use flaring gauge to measure the length of the flaring part.

Piping size (mm)	Dimensions “a” (mm) R22			Dimensions “a” (mm) R410A/R32			Nut outer diameter (mm)	
	Flare tools		Flare tools for R410A	Flare tools		Flare tools for R410A	R22	R410A / R32
	Clutch type	Wing nut type	Clutch type	Clutch type	Wing nut type	Clutch type		
6.35 (1/4")	0.5 ~ 1.0	1.0 ~ 1.5	0 ~ 0.5	1.0 ~ 1.5	1.5 ~ 2.0	0 ~ 0.5	17	17
9.52 (3/8")	0.5 ~ 1.0	1.0 ~ 1.5	0 ~ 0.5	1.0 ~ 1.5	1.5 ~ 2.0	0 ~ 0.5	22	22
12.70 (1/2")	0.5 ~ 1.0	1.5 ~ 2.0	0 ~ 0.5	1.0 ~ 1.5	2.0 ~ 2.5	0 ~ 0.5	24	26
15.88 (5/8")	0.5 ~ 1.0	1.5 ~ 2.0	0 ~ 0.5	1.0 ~ 1.5	2.0 ~ 2.5	0 ~ 0.5	27	29

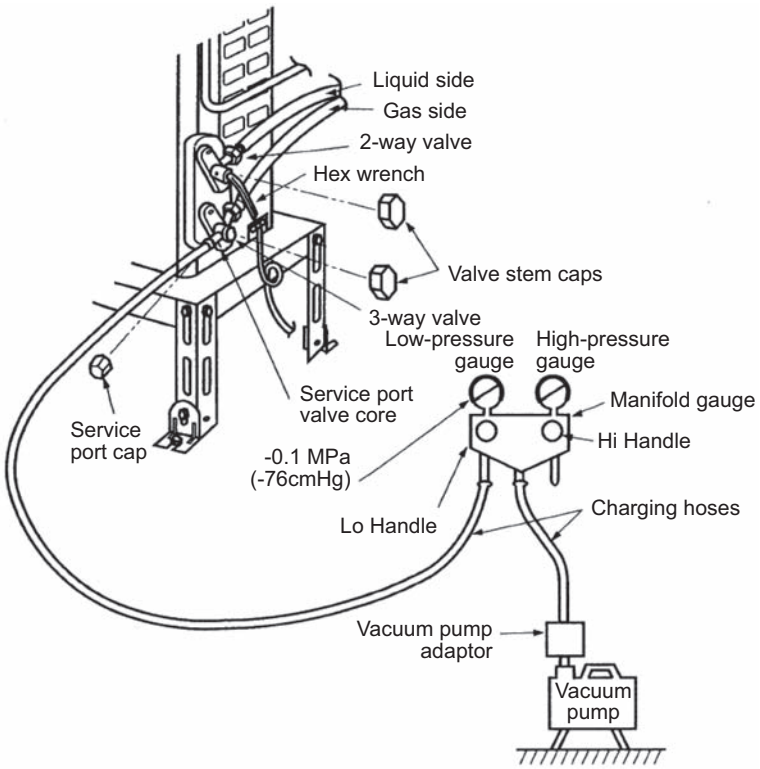


15.6 Installation and Service

15.6.1 Air purge and gas leak test for new installation (using new refrigerant pipes) using vacuum pump

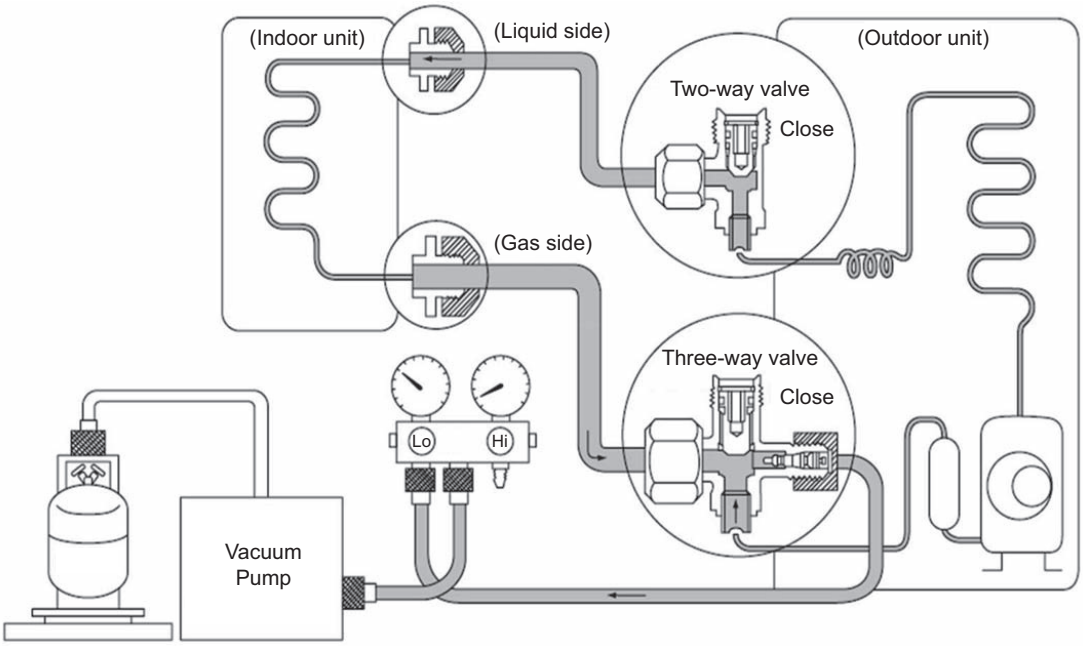
(From the point of view of global environment protection, do not release CFCs into the atmosphere during installation work)

1. Connect the charging hose of manifold gauge to the service port of 3-way valve (pushing insect pin).
2. Fully open the handle Lo of manifold gauge and operate vacuum pump.
   (If the needle of the low-pressure gauge reaches the vacuum immediately, check 1 procedure again)
3. Perform vacuuming 15 minutes or more, and make sure low pressure gauge reaches to −0.1 MPa (-76cmHg).
   When the vacuuming completes, fully open the handle Lo of manifold gauge and stop the operation of vacuum pump, and leave it for 1 ~ 2 minutes. Then, remove the connection side of the charging hose of vacuum pump adopter after checking the needle of manifold gauge does not turn back.
4. Open the stem of 2-way valve to 90° in anti-clock wise, and close the 2-way valve after 10 seconds, and perform gas leakage test.
5. Remove the charge hose from the service port of 3-way valve, and open the stems of 2-way and 3-way valves (open the valves to anti-clock wise carefully, do not use full strength to open)
6. Tighten the service port cap with torque wrench 18 N•m (1.8 kgf•m)
   Tighten the caps of 2-way and 3-way valves with torque wrench 18 N•m (1.8kg f•m)
7. After the tightening of each cap, check gas leakage around the cap.



15.6.2 Process of refrigerant recovery

1. Connect the center charging hose of manifold gauge to the in-let side of recovery device.
2. Connect the valves of the discharge side of recovery device and liquid side of refrigerant cylinder with red hose (charging hose).
3. Connect the yellow float switch cable of the recovery device to the refrigerant cylinder.
4. Open the low pressure side valve of manifold gauge.
5. Slightly loosen the charging hose of in-let connecting side of recovery device and perform air purge.
6. Open the liquid valve of refrigerant cylinder and slightly loosen the charging hose in discharging side of recovery device, and perform air purge (the recovery cylinder needs slight inside pressure).
7. Insert electric plug of recovery device into electrical outlet (the fan operation starts).
8. Turn the valve 1 and 2 of recovery device to pressure equalization point.
9. After a few seconds, turn back the valve 1 and 2 to the original position.
10. Turn the switch of the recovery device to “ON”. (the compressor operation starts)
11. When the low pressure of manifold gauge is close to “0”, close the low pressure side valve, turn “OFF” the recovery device switch.
12. Remove the center charging hose of manifold gauge from the recovery device.



15.6.3 Replacement of heat pump units and evacuation (when re-using the existing pipes)

When replacing the heat pump units, you might use the existing pipes, but it is recommended to perform flaring again. In case of unit replacement, even if the unit is new refrigerant heat pump, if the refrigerant oil is different, it may cause problem. Further, when re-using the existing refrigerant pipes, it is recommended to evacuate the pipes as much as possible, due to the reason that much refrigerant oil may be attached on the surface of the pipes. If the pipes are used without evacuation, the remaining refrigerant oil may cause under-performance and abnormal refrigerant cycle caused by non-compatibility of those oils.

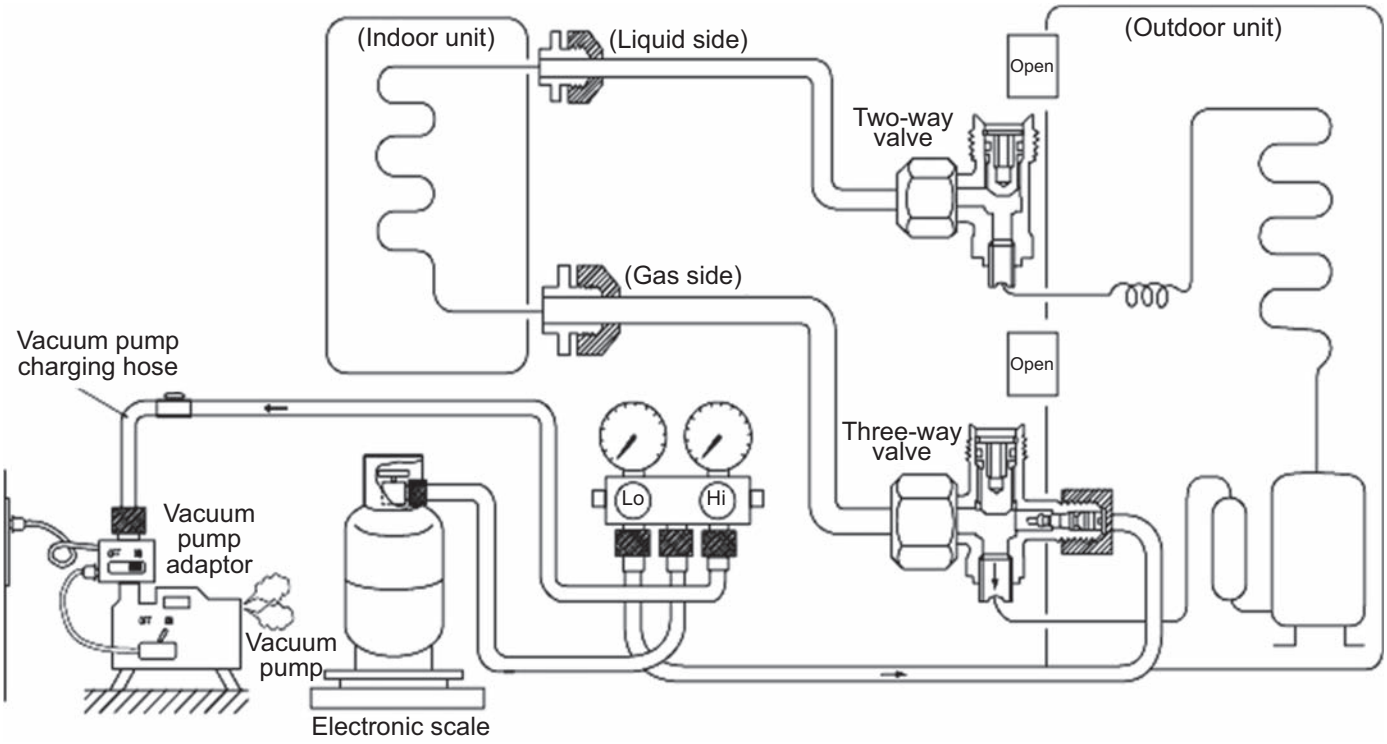
15.6.4 Inter-changeability of refrigerant

Do not operate heat pump units inserting wrong (or mixed) refrigerant (R22, R410A, R32). It may cause mal-function of the units, and at the same time, may cause serious incident such as rupture of the refrigerant cycle.

15.6.5 Re-insertion of refrigerant in service

When re-insertion is needed, follow the procedures to ensure the insertion of new refrigerant at correct amount.

1. Attach charging hose (blue) to the service port of the outdoor unit.
2. Attach charging hose (red) to the vacuum pump. Fully open the 2-way and 3-way valves or both 3 way valves.
3. Place the refrigerant cylinder on the digital scale for refrigerant charging and connect the charge hose (yellow) to the connection port of the vacuum pump and the digital scale. Leave the cylinder valve fully open.
4. Fully open the handles Lo and Hi of the manifold gauge, and switch on the vacuum pump, and then perform evacuation for at or more than one hour.
5. Confirm the compound gauge of -0.1 MPa (-76cmHg) and fully open the handles of Lo and Hi, and switch off the vacuum pump. Leave it for about 1 ~ 2 minutes and confirm the needle of the compound gauge does not turn back.
   Refer to the picture below to follow the procedures below.
6. Remove the charging hose (red) of the manifold gauge from the vacuum pump adopter.
7. After adjusting the digital scale to zero, open the cylinder valve and the valve Lo of the manifold gauge, and insert the refrigerant.
8. If it is not possible to insert the refrigerant at regulated amount at once, operate the cooling mode and gradually insert the refrigerant (recommended amount approx. 150 g / 1 time)
   *Do not insert much refrigerant at once.
9. Close the open/close valve and insert the refrigerant in the charging hose to the outdoor unit.
   *Perform this procedure during operating cooling operation. Close the stem of the two-way valve, and when the pressure of the manifold gauge becomes zero (0), quickly remove the charging hose (blue). Immediately open the 2-way valve, and stop the cooling operation.
10. Final checking • • • Confirm the 2-way and 3-way valves are fully open.
    Attach the caps of the service port and control valve, and then check the gas leakage around the caps.



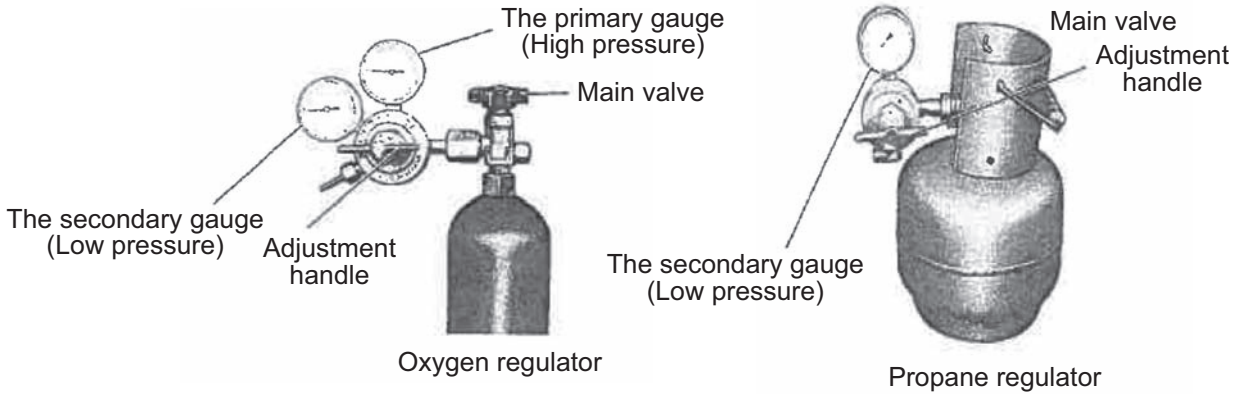
15.7 Repairing of refrigerant cycle / Brazing point

15.7.1 Preparation for repairing of refrigerant cycle / brazing

Brazing which is a technique needed for repairing refrigerant cycle requires advanced technique and experience, and this brazing procedure can only be performed by the workers who completed “Gas Welding Skill Training” regulated by the Occupational Safety and Health Act, and went through the training programs of refrigerant operations. Dismantling and re-connecting (assembling) refrigerant system requires working space, and the space has to ensure good air flow and fire prevention (water bucket and fire extinguisher). Moreover, the worker has to ensure the wearing of goggles, grabs, safety shoes, and long sleeve shirts, and be aware of work safety and attempt to prevent secondary defect (quality assurance of products). For brazing the indoor / outdoor unit structural components (heat exchangers, compressors, expansion valves, four-way valve blocks), after the recovery of all refrigerant, confirm that no refrigerant remains in the system, and fully open the 2-way and 3-way valves. When the brazing is conducted outside, check and make sure no refrigerant is contained in the air (be careful with vaporized refrigerant). Furthermore, protect the compressor terminal with metal plates, and heat but use wet clothes to cool down (releasing the heat) the expansion valves, and four way valves (prevent destruction of parts). In brazing, it is important to pour the brazing material without melting the base metal based on capillary action principle. In case of holes and oxidizing caused by overheating, do not perform re-brazing or alteration but replace the parts.

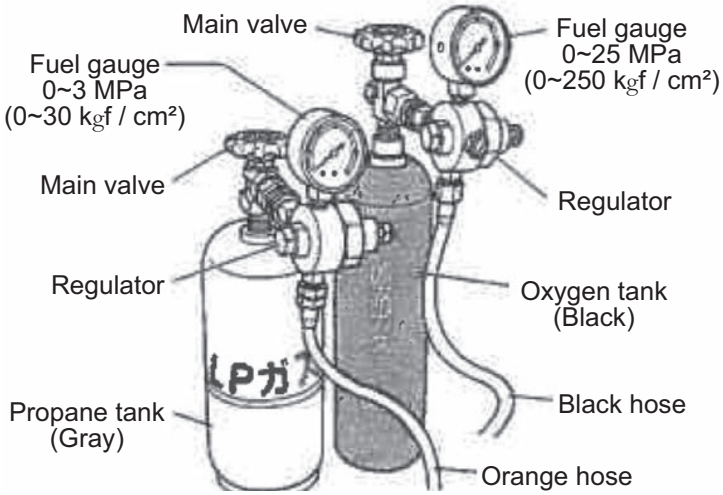
15.7.2 Adjustment of vacuum pump pressure

1. Cylinder with adjustment handle
    1. Check and confirm the adjustment handle of the 1st pressure adjuster is loosen (anticlockwise). If cylinder valve is opened when the 1st gauge pressure adjust handle is closed, the 2nd gauge might get broken.



    2. Open the cylinder valve, and check the remaining amount with the first t side pressure gauge.
    3. Check the pressure of 2nd gauge and turn the adjustment handle to clock-wise direction to adjust the pressure.
        ◎ Oxygen 2nd side gauge pressure ・・・・・・・・・・・0.5 MPa (5.0 kgf / cm^2)
        ◎ Propane 2nd side gauge pressure ・・・・・・・・・0.05 MPa (0.5 kgf / cm^2)

2. Cylinder without adjustment valve
   2nd side gauge pressure is adjusted by the adjuster.
   Check the both side valves of the torch and open the cylinder valve to check the remaining refrigerant in the cylinder.
   Caution: Do not attach oil component on the connection port of the adjuster.
   Especially, use an oxygen cylinder adjuster which is no oil substance type. Do not dismantle or repair the adjuster and pressure gauge.



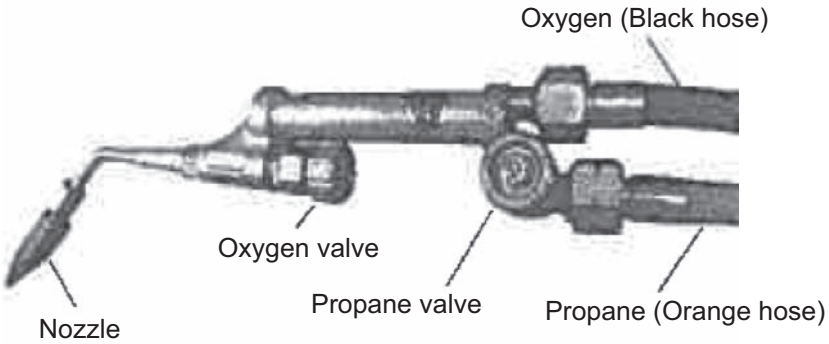
15.7.3 Checking of gas provision

Checking there is no fire around the torch, and then confirm the provision of gas.

1. Slightly open the “propane valve” of the torch, and make sure the gas comes out from the torch crater and then close the “propane valve”.
2. Slightly open the “oxygen valve” of the torch and make sure the gas comes out from the torch crater and then close the “oxygen valve”.
   Check there is no gas leakage around the hose connection.

15.7.4 Adjustment of flame

1. Slightly open the “propane valve” of the torch and lit with spark lighter.
   This moment, the flame is only by propane and the color is red.
2. Gradually open the “oxygen valve” of the torch to mix oxygen, and adjust the amount of propane and oxygen with the valve to make the flame suitable for brazing work.
   If the white core flame splits into two, the torch crater might be clogged. In this case, remove the crater from the torch and check.



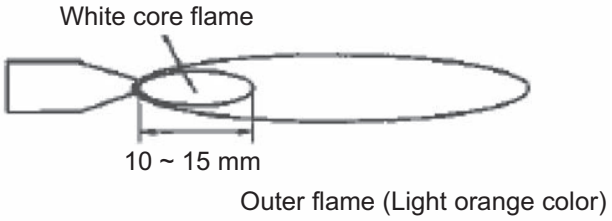
15.7.5 Types of flame

Types of flame change based on the proportion of propane and oxygen.

[Neutral Flame]

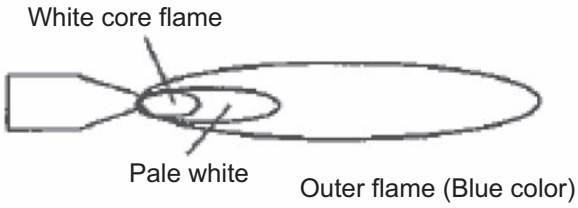
Perform brazing with this flame

(This is a flame when oxygen and propane are mixed at proper proportion, and has lesser effect on the brazed metals)



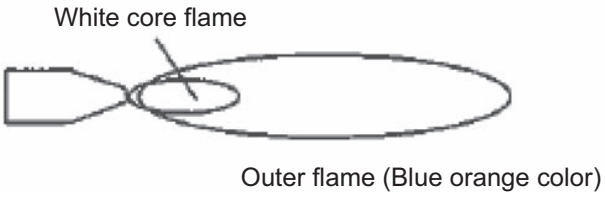
[Carbide Flame]

When propane is excessive, the flame has white color flame in between the white core flame and outer flame.
(This is due to the lack of oxygen and the proportion of unburned propane is excessive.
The black carbon created during the brazing work may contaminate the surface of the brazed metal).



[Oxidizing Flame]

Oxygen is more compared to the neutral flame. Although the flame size is small, this has the highest flame heat. However, due to the excessive oxygen contained in the flame, the brazing point gets oxidized. (This flame may cause holes, due to the high heat. The pipe may get melt）



15.7.6 Closing the flame

[In case of short break]

1. Close the “propane valve” of the torch.
2. Close the “oxygen valve” of the torch.

[In case of finishing work]

1. As above, close the flame following the procedure of “In case of short break”.
2. Completely close the valves of oxygen and propane cylinders.
3. Release the remaining gas inside the hose by opening the “oxygen valve” and “propane valve” of the torch.

Confirm the 1st and 2nd side gauge pressures of “oxygen” and “propane” cylinder pressure adopter are “zero”.

15.7.7 Selection of brazing material

Use BAg brazing material (silver solder) to increase the welding performance.

Category	JIS Standard Number	Composition of ingredients (%)						Temperature (°C)			Tensile strength (Reference)		Characteristics and applications
		Ag	Cu	Zu	Cd	Ni	P	Solidus	Liquidus	Brazing temp	Kgf•cm^2	Base material	
BAg	BAg • 1A	49.0 ~ 51.0	14.5 ~ 16.5	14.5 ~ 18.5	17.0 ~ 19.0	—	—	approx. 625	approx. 635	635 ~ 760	45.5	S20C	Liquidity is good at low temperature, it is preferable to a small junction of the gap in the universal form.
	BAg • 1	44.0 ~ 56.0	14.0 ~ 16.0	14.0 ~ 18.0	23.0 ~ 25.0	—	—	approx. 605	approx. 620	620 ~ 760	45.5	S20C	It has similar performance to the BAg • 1A, and suitable for every base material except the light weight metal.
	BAg • 2	34.0 ~ 36.0	25.0 ~ 27.0	19.0 ~ 23.0	17.0 ~ 19.0	—	—	approx. 605	approx. 700	700 ~ 845	45.5	S20C	It is a brazing filler metal in universal form, suitable for a slightly larger gap junction.
	BAg • 3	48.0 ~ 51.0	14.5 ~ 16.5	13.5 ~ 17.5	15.0 ~ 17.0	2.5 ~ 3.5	—	approx. 630	approx. 690	690 ~ 815	35 ~ 70	SS ~ SUS	It has good corrosion resistance in stainless steel-based brazing, suitable for brazing tungsten carbide, aluminum bronze and copper.
BCuP	BCuP-2	—	remain	—	—	—	6.8 ~ 7.5	approx. 710	approx. 785	690 ~ 815	21 ~ 24.5	Cu	Good liquidity, suitable for brazing copper tube.
	BCuP-3	4.8 ~ 5.2	remain	—	—	—	5.8 ~ 6.7	approx. 645	approx. 815	720 ~ 815	21 ~ 24.5	Cu	Suitable for brazing when the joint spacing is not constant
	BCuP-5	14.5 ~ 15.5	remain	—	—	—	4.8 ~ 5.3	approx. 645	approx. 800	705 ~ 815	21 ~ 24.5	Cu	When brazing of copper and copper, it is used without a flux, but not possible for brazing basic materials

Caution
BCuP (phosphorus copper wax) is easy to react with sulfur, and makes a brittle compound water soluble, and causes gas leakage. In hot spring areas, use other brazing materials or paint the surface for protection.

15.7.8 Need of flux

Use flux to protect the base materials.

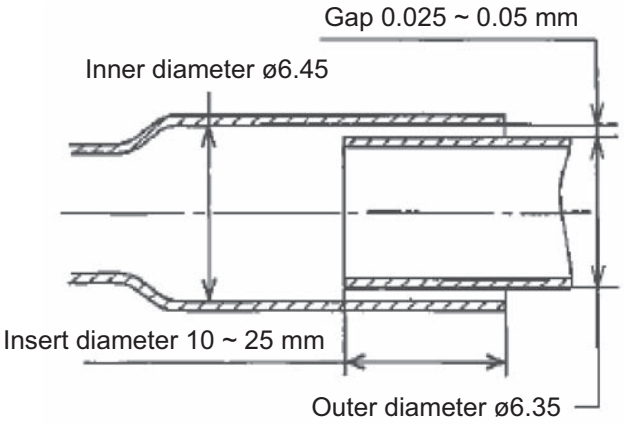
1. Remove impurity and oxide film on the metal base, and improve the flow of the brazing material.
2. Prevent oxidation of the metal surface in brazing.
3. Reduce the surface tension of the brazing material.

15.7.9 Need of nitrogen gas

In order to prevent oxidation in the pipe, perform the brazing operation in nitrogen gas flow.
Flow rate 0.05 m^3 / h, or pressure reducing valve at 0.02 MPa (0.2kgf / cm^2) below.

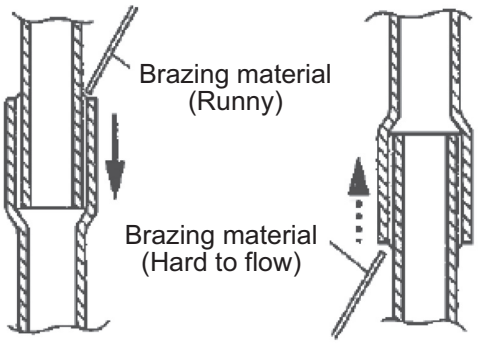
15.7.10 Checking of brazing (insert) points

1. No impurity on the brazing point
   If dirt or oil is attached on the brazing point, the brazing filler metal does not reach to junction, and it may cause poor welding.

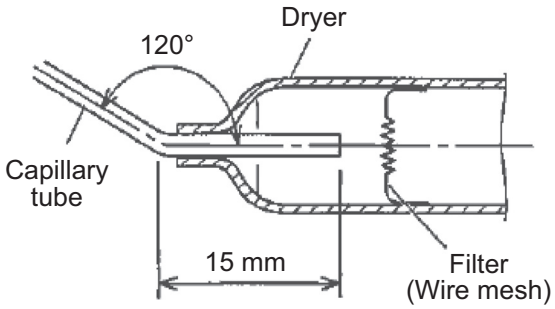


2. Adequate gap space in the brazing point
   The advantage of capillary current situation is used in brazing. If the gap space is too large, this phenomenon may not occur and it may cause poor welding because brazing filler metal does not flow to join the front part.

3. Appropriate size for insertion
   The guideline for pipe insertion dimensions is to three times the diameter of the base material, but you need to decide the insertion size in consideration of the clogging of the brazing material. Generally, for thin pipes, you need to increase the insert size, and for thick pipe vice versa.

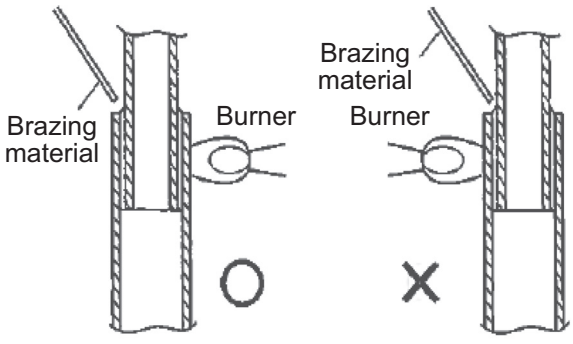


4. Brazing material to flow from top to bottom
   Brazing filler metal will easily flow to the connecting portion by capillary action.
   Further, by bending the brazing portion of [dryer side] of the capillary tube at 15 mm from the tube top to the angle of about 120°, you can prevent the damage of dryer inside and the clogging of brazing material caused by the excessive insertion of capillary tube.



15.7.11 Brazing and heating

1. Place the flame to a pipe which has more heat capacity in order to let the brazing material melt by the pipe heat. Heat the pipe up to the melting temperature of the brazing material, but when it is overheating, assess the temperature by pipe color in order not to melt the pipe.



The pre-heating is to heat the base material until the melting temperature, and requires certain training to distinguish the color of the heated base material in order not to melt the material.
The color and temperature of copper tube

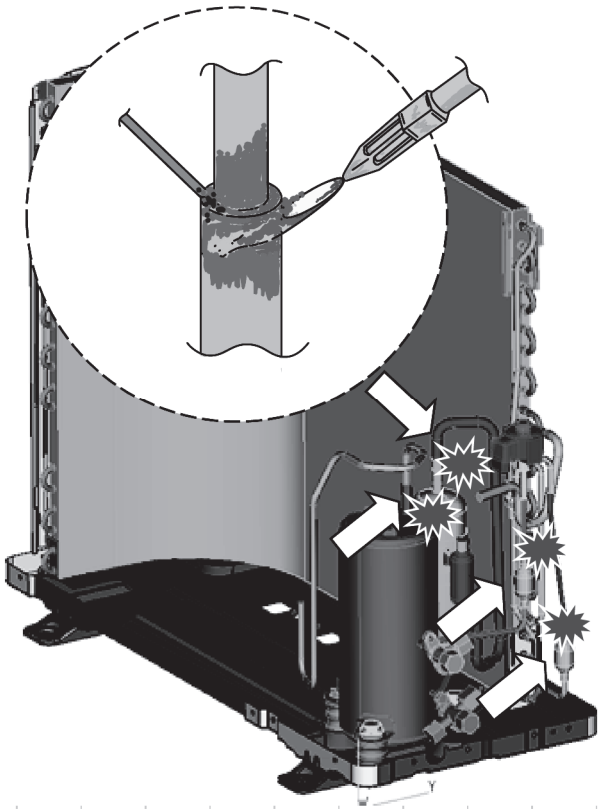
- Becoming red color • • • • • • • 480°C
- Dull red • • • • • • • 650°C
- Cherish red • • • • • • • 760°C
- Brightening cherish red • • • • • • • 870°C

(Reference)
Melting temperature of copper • • • • • • • Approx. 1083°C
Maximum temperature obtained in propane and oxygen • • • • • • • Approx. 1083°C
The important point is to heat the bonding part uniformly within a short period of time until reaching to the brazing temperature in the following manner.

2. Apply the flame on to the side with better heat transmission. If the pipe thickness is consistent, by heating like 30% iron and 70% copper, the copper pipe inside reaches to brazing temperature.
   Iron pipes have low heat transmission and only the part the flame is applied get high temperature, and this causes oxidization of the pipe. The flow of the brazing filler is affected negatively.

3. Apply the flame on to the side of larger heat capacity.
   When brazing a thin tubes such as capillary tube and dryer, etc., caution has to be taken to apply the flame to the dryer side (thick pipe side), in order to prevent burn out by the heat.

4. When brazing the compressor connection pipes (suction and discharge), remove the sound insulation plate and the fan, and place the compressor stand vertically (to prevent the leakage of compressor refrigerating machine oil), and apply the flame from the compressor body side.



15.7.12 Terminologies of brazing

Pin holes → Small holes are generated on the surface of the brazing metal.
Wet temperature → Liquidus temperature at which the brazing material starts flowing out by heating, generally it is the liquidus-line temperature.
Blow holes → Hollows made by gas in the brazing material of brazing portion (gas reservoirs).
Pits → As a result of blow holes, small dents generated on the outside surface of welding.
Voids → The blazing material does not reach completely to the brazing part. It cannot be identified from outside.

16. Operation and Control

16.1 Basic Function

Inverter control, which equipped with a microcomputer in determining the most suitable operating mode as time passes, automatically adjusts output power for maximum comfort always. In order to achieve the suitable operating mode, the microcomputer maintains the set temperature by measuring the temperature of the environment and performing temperature shifting. The compressor at outdoor unit is operating following the frequency instructed by the microcomputer at indoor unit that judging the condition according to internal water setting temperature and water outlet temperature.

16.1.1 Internal Water Setting Temperature

Once the operation starts, control panel setting temperature will be taken as base value for temperature shifting processes. These shifting processes are depending on the Air-to-Water Heat pump settings and the operation environment. The final shifted value will be used as internal water setting temperature and it is updated continuously whenever the electrical power is supplied to the unit.

16.1.2 Heating Operation

16.1.2.1 Thermostat Control

- Compressor is OFF when Water Outlet Temperature – Internal Water Setting Temperature > 2°C for continuously 3 minutes.
- Compressor is ON after waiting for 3 minutes, if the Water Outlet Temperature – Water Inlet Temperature (temperature at thermostat OFF is triggered) <-3°C.

16.1.2.2 Thermostat Control (Outdoor Ambient Temperature)

Stops provide heating to room side during high outdoor ambient condition.
Control content:

- Heating operation and water pump will turn OFF when outdoor ambient temperature > outdoor thermo off temperature + 3°C.
  (Outdoor thermo off set temperature is set by control panel. Thermo off set temperature is between 5°C ~ 35°C)
- Heating operation will resume when Outdoor ambient temperature < Outdoor thermo OFF set temperature + 1°C.

16.1.2.3 Heat Mode Operation

Operation of heat pump provide heating capacity to room side by hot water through heating panel, floor heating or fan coil unit.

1. 3 ways valve control:
   - 3 ways valve switch and fix to heating side.
2. Heat pump operates follow normal heating operation.
3. Back up heater operate follow normal operation.
4. 2 ways valve control:
   - 2 ways valve opens.

16.1.2.4 Compressor Maximum Frequency Control During Heating Operation

Purpose:- To ensure that the compressor does not directly operate at the maximum frequency from beginning. Installer can activate this control by setting on remote control : “Efficiency” under “Heating control” in the Installer setup menu.

Start condition
When all of the below conditions achieved

- Remote control setting = Efficiency
- It is not test mode
- Compressor run time < 1 hour
- Present water outlet temperature - Water outlet set temperature ≥ -10°C
- Outdoor ambient ≥ -10°C

Under this control, heating operation frequency will be based on heating coefficient calculation.

Compressor maximum Hz = Heat mode zone limit max Hz X Coefficient

Time from start (minutes)	Below 20 minutes	Below 40 minutes	Below 60 minutes	Above 60 minutes
Coefficient	0.5	0.5 + (0.5/3)	0.5 + 2*(0.5/3)	1.0

The operating frequency will gradually increase when the capacity is deemed insufficient by referring to dRange value:-

- dRange = Current water outlet temperature - Water outlet set temperature
  dRange value is checked every 10 minutes.
- If current dRange < previous dRange, the new coefficient value will increase by 0.1.
  Maximum frequency after change is limited by Zone limit maximum Hz.

Stop condition
When any one of below conditions achieved

- Remote control setting = Comfort
- It is test mode
- Compressor run time ≥ 1 hour
- Present water outlet temperature - Water outlet set temperature < -10°C
- Outdoor ambient < -10°C

16.1.3 Cooling Operation

16.1.3.1 Thermostat Control

- Compressor is OFF when Water Outlet Temperature – Internal Water Setting Temperature > -1.5°C for continuously 3 minutes.
- Compressor is ON after waiting for 3 minutes, if the Water Outlet Temperature – Water Inlet Temperature (temperature at thermostat OFF is triggered) >3°C.

16.1.3.2 Cool Mode Operation

1. 3 ways valve control:
   - 3 ways valve switch and fix to cooling side.
2. Heat pump operates follow normal cooling operation.
3. Room heater DOES NOT operate during cool mode.
4. 2 ways valve control:
   - 2 ways valve is closed.

16.1.4 Target Water Temperature Setting

16.1.4.1 Target Water Temperature Control of Standard System (Optional PCB not connected)

There are 2 types of temperature control selection which are Compensation and Direct.

- Temperature control type selection by installer:
  1 Compensation : Wlo, WHi, ODLo, ODHi can be set at installer menu.
  2 Direct : Direct Water Temperature Set
- Remote control setting by user:
  1 Compensation : Shift value ±5°C from the compensation curve
  2 Direct : Direct water temperature set change

*This setting only able to set when room sensor select as Water Temperature.
*Instead of water temperature, user will set target room temperature when room sensor select as Room Thermistor OR Internal Room Thermostat.

- Target water temperature is calculated as below condition.
  - Target water temperature = A (Base temperature) + B (shift temperature)

A (Base Temperature)	Compensation	Direct
	Value from the curve + User shift value set	Direct value from user setting

- B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

B (Shift Temp.)	B shift value depend on the room sensor selection at remocon as table below
Sensor selection	
Water temperature	B = 0
External Room thermostat	B = 0
Internal Room thermostat & Room Thermistor	**Cool Mode:** B = 0 ; when Zone OFF or Zone Room Thermo OFF B = B = 1* (room set temp (R/C) – actual room temp) Max/Min Regulation of B: (Max = 5 ; Min = -5)
	Heat Mode: B = 0 ; when Zone OFF or Zone Room Thermo OFF B = follow Heating Room Temperature PI control logic

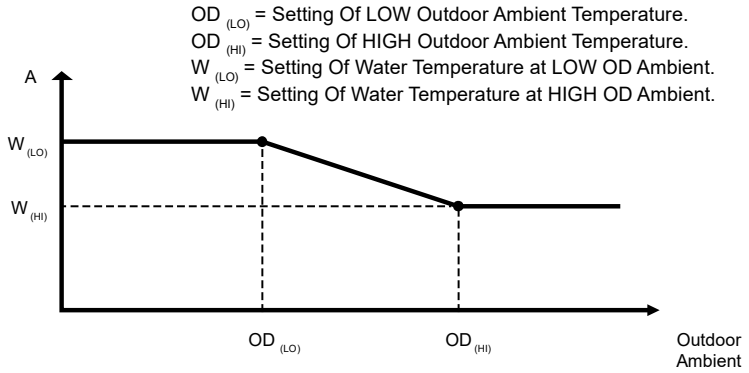
- Maximum/minimum regulation of Target Water Temperature

	Heating	**Cooling**
Max	55°C / 60°C*	20°C
Min	20°C	5°C

* Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C

Compensation Type: (Operation under Heat Mode and Cool Mode)

- The set temperature defines the parameters for the ambient (Outdoor temperature) dependent operation of the unit. The water temperature is determined automatically depending on the outdoor temperature. Default setting is the colder outdoor temperature will result in warmer water and vice versa. The user has the possibility to shift up and shift down the target water by remote control setting.

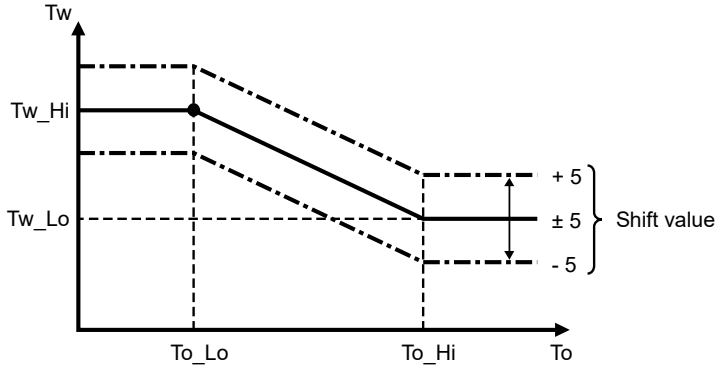


- Outdoor ambient is updated every 5 minutes when operation ON.
- Setting water outlet temperature always follow $W_{(LO)}$ or $W_{(HI)}$ whenever is higher if outdoor ambient sensor or indoor communication error happen.

However, when powerful mode is requested by remote control during heating mode, the higher value of HLo or Whi will be used for A calculation.

* There are 2 compensation curves (for heating and cooling). During heating mode, the heating curve is used and during cooling mode, the cooling curve is use.

- Compensation curve set shift value:



16.1.5 Target Water Temperature at Extension System (Optional PCB is connected)

Target water temperature is calculated as below.

- Heat Mode:
  - Target water temperature setting:
  - Max= 55°C / 60°C* Min= 20°C

* Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C

- When buffer tank selection is “YES:”
  Target water temperature = Target buffer tank temperature + [2°C]

- When buffer tank selection is “NO”
  - If both zone 1 and zone 2 is active
    Target Water Temperature = Higher zone target water temperature of Zone 1 and Zone 2.
  - If only one zone is active
    Target Water Temperature = Zone target water temperature of active zone.

- Cool mode:
  - Target water temperature setting:
    Max = 20°C Min= 5°C

- When buffer tank selection is “YES”
  - If both zone 1 and zone 2 active
    Target Water Temperature = Lower Zone Target Water Temperature of Zone 1 and Zone 2
  - If only one zone is active
    Target Water Temperature = Zone Target Water Temperature of active zone

- When buffer tank selection is “NO”
  - If both zone 1 and zone 2 active
    Target Water Temperature = Lower Zone Target Water Temperature of Zone 1 and Zone 2
  - If only one zone is active
    Target Water Temperature = Zone Target Water Temperature of active zone

16.1.6 Target Zone Water Temperature Control

Purpose:- To control zone mixing and zone pump according to the zone sensor temperature

16.1.6.1 Target Zone 1 water temperature setting control

- Start condition
  - Heating zone 1 is ON by remote control or Timer or Auto Mode **OR**
  - Cooling zone 1 is ON by remote control or Timer or Auto Mode.

- Cancel condition
  - Heating zone 1 is OFF by remote control or Timer or Auto mode **AND**
  - Cooling zone 1 is OFF by remote control or Timer or Auto mode.

- Target Zone 1 water temperature is calculated as below condition.
  - Target Zone 1 water temperature = A (Base temperature) + B (shift temperature)

A (Base Temperature)	Compensation	Direct
	Value from the curve + User shift value set	Direct value from user setting

* During heat mode and compensation select, if powerful mode is activated, higher value of WLo or WHi will be use as curve value.

- B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

B (Shift Temp.)	B shift value depend on the room sensor selection at remocon as table below
Sensor selection	
Water temperature	B = 0
External Room thermostat	B = 0
Internal Room thermostat & Room Thermistor	**Cool Mode:** B = 0 ; when Zone OFF or Zone Room Thermo OFF B = B = 1* (room set temp (R/C) – actual room temp) Max/Min Regulation of B: (Max = 5 ; Min = -5)
	Heat Mode: B = 0 ; when Zone OFF or Zone Room Thermo OFF B = follow Heating Room Temperature PI control logic
Pool Function Selected	B = Delta value setting from remocon

* B = 0 regardless of which sensor selection, if SHP control bit is enable except Pool function select (maintain Pool "B" value)

** Pool function also can be select at Zone 1 when optional PCB is connected and Zone 1 system is select.

- Maximum/minimum regulation of Target Water Temperature.

	Heating	**Cooling**
Max	55°C / 60°C*	20°C
Min	20°C	5°C

* Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C

- Target Zone 1 Water Temperature (Heat mode) during SG ready control
  - If buffer tank selection is "NO" then following shift is carried out.
    - While digital input is "10" or "11" then,
      Final Target Zone 1 water temperature
      = Target Zone 1 water temperature* (SG ready % setting (remote control menu))%
  - If buffer tank selection is "YES"
    - No shift of Target Zone 1 Water Temperature. Target Buffer Tank Temperature will change accordingly.
      * Refer to "Buffer tank temperature control"

- Target Zone 1 Water Temperature (Cool mode) during SG ready control
  - Final Target Zone 1 water temperature = Target Zone 1 water temperature + SG Cool Setting

16.1.6.2 Target Zone 2 water temperature setting control

- Start condition
  - Heating zone 2 is ON by remote control or Timer or Auto Mode **OR**
  - Cooling zone 2 is ON by remote control or Timer or Auto Mode.

- Cancel condition
  - Heating zone 2 is OFF by remote control or Timer or Auto mode **AND**
  - Cooling zone 2 is OFF by remote control or Timer or Auto mode.

- Target Zone 2 water temperature is calculated as below condition.
  - Target Zone 2 water temperature = A (Base temperature) + B (shift temperature)

A (Base Temperature)	**Compensation**	**Direct**
	Value from the curve + User shift value set	Direct value from user setting

* During heat mode and compensation select, if powerful mode is activated, higher value of WLo or WHi will be use as curve value.